DEPARTMENT OF DEFENSE
Small Business Innovation Research (SBIR) Program

SBIR 25.2 Annual Program Broad Agency Announcement (BAA)

IMPORTANT DATES

April 2, 2025: Topics Pre-release
April 23, 2025: Topics Open; DoD begins accepting proposals in DSIP
May 7, 2025: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
May 21, 2025: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

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1.0 PROGRAM DESCRIPTION

1.1 Objectives and Context

The Defense Small Business Innovation Research (SBIR) Program's objectives include stimulating technological innovation, strengthening the role of small business to meet DoD research and development (R&D) needs, fostering and encouraging minority and disadvantaged persons' participation in technological innovation, and increasing the commercial application of DoD-supported research or R&D results. DoD invites proposing SBCs with the capability to conduct R&D and commercialize the results in any of the defense-related topic areas described in this SBIR Program BAA to submit proposals.

The Small Business Administration (SBA), through its SBIR/STTR Policy Directive, purposely departs from normal government solicitation formats and requirements, which simplifies the SBIR/STTR award process and minimizes the regulatory burden on small business. Consistent with the SBA SBIR/STTR Policy Directive, DoD is soliciting proposals as a broad agency announcement (BAA). The guidelines in this BAA incorporate and make use of the SBA SBIR/STTR Policy Directive's flexibility to encourage scientific and technical approaches proposals most likely to yield significant results for DoD and the private sector.

This BAA is for research topics accepting Phase I or Direct to Phase II proposals. A separate BAA will not be issued requesting Phase II proposals, and unsolicited proposals will not be accepted. All proposing SBCs that receive a Phase I award from this BAA will be eligible to participate in Phase II competitions and potential Phase III awards. DoD Services/Components will notify Phase I awardees of the Phase II proposal submission requirements.

DoD is not obligated to make any awards under Phase I, Phase II, or Phase III, and all awards are subject to both a risk-based due diligence security review and funds availability. DoD is not responsible for any monies the proposing small business concern (SBC) spends before any award issuance. Proposals must conform to this announcement's terms. DoD is under no obligation to fund any proposal or any specific number of proposals in each topic. It also may elect to fund several or none of the proposed approaches to the same topic.

1.2 A Three Phased Program

The SBIR Program has three phases, Phases I, II, and III. Phase I determines, to the extent possible, an idea's scientific, technical, and commercial merit and feasibility within the SBIR program. Phase I and II awards are made adhering to current SBA Policy Directive guidelines.

The Phase I period of performance is generally between six to twelve months. Proposals should focus on Research or Research & Development (R/R&D)to prove the proposed effort's scientific and technical feasibility, and commercialization potential, the successful completion of which is a prerequisite for further DoD support in Phase II. Proposing SBCs are encouraged to consider whether the research or R&D being proposed to DoD Services/Components also has private sector potential, either for the proposed application or as a base for other applications.

Phase II awards will be made to proposing SBCs based on results of Phase I awards and the Phase II proposal's scientific merit, technical merit, and commercialization potential. The period of performance is generally 24 months. The objective of Phase II is to continue and further develop the R/R&D effort from the completed Phase I award.

Phase III refers to work that derives from, extends, or completes an effort made under prior SBIR funding agreements, but is funded by sources other than the SBIR Program. Under Phase III, the SBC should

focus on commercializing previously SBIR-funded technology and is required to obtain funding from either the private sector, a non-SBIR federal source, or both, to develop the prototype into a viable product or non-R&D service for sale in military or private sector markets.

1.3 OUSD(R&E) Critical Technology Areas

Although each DoD Service/Component develops SBIR and STTR topics tailored to their mission needs, topics generally align with the Office of the Under Secretary of Defense, Research & Engineering (OUSD(R&E)) critical technology areas. While many technologies may cross between these categories, these areas represent the broad and different approaches required to advance technologies crucial to the Department, with a focus on accelerating key capabilities' transitions to the Military Services and Combatant Commands.

OUSD(R&E) critical technology areas include:

- FutureG
- Trusted AI and Autonomy
- Biotechnology
- Advanced Computing and Software
- Integrated Sensing and Cyber
- Directed Energy (DE)
- Hypersonics
- Microelectronics
- Integrated Network Systems-of-Systems
- Quantum Science
- Space Technology
- Renewable Energy Generation and Storage
- Advanced Materials
- Human-Machine Interfaces

Below are additional technology areas supporting DoD Services/Component-specific mission-critical areas:

- Advanced Infrastructure & Advanced Manufacturing
- Combat Casualty Care
- Emerging Threat Reduction
- Military Infectious Diseases
- Military Operational Medicine
- Mission Readiness & Disaster Preparedness
- Nuclear
- Sustainment & Logistics

Full descriptions of the above technology areas can be reviewed here.

1.4 Eligibility and Performance Requirements

Each proposing SBC must qualify as an SBC as defined in 13 C.F.R §§ 701-705 at time of award and certify to this in the proposal's cover sheet. The eligibility requirements for the SBIR/STTR programs are unique and do not correspond to those of other small business programs.

a. Proposing SBC must meet eligibility requirements for Small Business Ownership and Control (see 13 CFR § 121.702).
b. The proposing SBC must conduct a minimum of two-thirds of the Phase I research and/or analytical work. For Phase II, the proposing SBC must perform no less than 50 percent of the research and/or analytical work. The work percentage is measured via direct and indirect costs. Occasionally, deviations from these SBIR requirements may occur with the Funding Agreement officer's written approval after consultation with the agency SBIR/STTR program manager/coordinator. For more information on the percentage of work calculation during proposal submission, refer to section 3.7.
c. For both Phase I and II, the principal investigator's primary employment must be with the proposing SBC at the time of the award and during the conduct of the proposed effort. Primary employment means that more than one-half of the principal investigator's time is spent with the

small business (based on a 40-hour work week). Primary employment with an SBC precludes full-time employment at another organization. Deviations from this requirement or changes to the principal investigator are subject to the Funding Agreement officer approval.

d. For both Phase I and Phase II, the SBC and its subcontractors must perform all research or R&D work in the United States.
e. Joint ventures and limited partnerships are permitted, provided that the entity qualifies as small business in accordance with the ownership requirements in 13 CFR 121.702(a)(1)(iii) and the size requirements in 13 CFR 121.702(c)(6). Proposing SBC must disclose joint ventures with existing (or planned) relationships/partnerships with any foreign entity or any foreign government-controlled companies. See sections 2.6 and 3.7 for more detail.

1.5 Majority Ownership in Part by Multiple Venture Capital, Hedge Fund, and Private Equity Firms

Unless otherwise noted in the participating Service/Component instructions, proposing SBCs that are multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds majority owned are **ineligible** to submit applications to or receive awards under this BAA. If a Service/Component authorizes such participation, any proposing VCOC, hedge fund, and/or private equity fund-owned SBC, whether in part or in whole, must identify each foreign national, foreign entity, or foreign government holding or controlling greater than a 5 percent, either directly or indirectly held, equity stake in the proposing SBC. The proposing SBC must also identify any ultimate parent owner(s) and other entities and/or individuals owning more than a 5 percent equity stake in its ownership chain.

In accordance with the requirements of 13 CFR 121.702(a)(2), no single venture capital operating company, hedge fund, or private equity firm may own more than 50 percent of the concern unless that single venture capital operating company, hedge fund, or private equity firm qualifies as a small business concern that is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States.

1.6 Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms

Proposing SBCs with multiple prior SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The Phase I to Phase II Transition Rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The Commercialization Benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' application for more experienced firms. Find detailed information on benchmark calculations, increased performance standards for more experienced firms and consequence of failure to meet benchmarks here. SBA will notify companies failing either benchmark and the relevant officials at the participating agencies.

The SBIR/STTR Policy Directive defines the Departments of the Army, Navy, and Air Force each as its own federal agency, and the remaining DoD Components as an executive agency of the Department of Defense. Therefore, companies that fail to meet either of the benchmarks under the Increased Performance Standards for More Experienced Firms may not receive more than an overall total of 80 awards from DoD, as detailed in the breakdown below:

Army – 20 total Phase I and Direct to Phase II awards
Navy – 20 total Phase I and Direct to Phase II awards

Air Force – 20 total Phase I and Direct to Phase II awards
All other DoD Components combined – 20 total Phase I and Direct to Phase II awards

1.7 Direct to Phase II Program

15 U.S.C. §638 (cc), as amended by NDAA FY2012, Sec. 5106, and further amended by NDAA FY2019,Sec. 854, PILOT TO ALLOW PHASE FLEXIBILITY, allows DoD to make a SBIR Phase II award to an SBC with respect to a project, without regard to whether the SBC was provided a SBIR program Phase I award with respect to such project. DoD does not guarantee Direct to Phase II opportunities will be offered in future BAAs.

Each eligible topic requires proposing SBCs provide documentation to demonstrate feasibility described in the Phase I section of the topic has been met. **Feasibility documentation cannot be based upon or logically extend from any prior or ongoing federally funded SBIR or STTR work.** The proposing SBC and/or the principal investigator must have substantially performed the work submitted in the feasibility documentation. If technology in the feasibility documentation is subject to intellectual property (IP), the proposing SBC must demonstrate ownership or licensure of the IP associated with such technology prior to proposal submission to enable it and its subcontractors to legally carry out the proposed work.

If the proposing SBC fails to demonstrate technical merit and feasibility equivalent to the Phase I level as described in the associated topic, the related Phase II proposal will not be accepted or evaluated, in accordance with the Service/Component-specific Direct to Phase II instructions.

Please refer to the Service/Component-specific Direct to Phase II instructions for full details regarding Service/Component Direct to Phase II processes and proposal preparation requirements.

1.8 Program on Innovation Open Topics

15 U.S.C. §638 (ww) requires DoD establish innovation open topic activities to:

a. increase the transition of commercial technology to the DoD;
b. expand the small business nontraditional industrial base;
c. increase commercialization derived from DoD investments; and
d. expand the ability for qualifying SBCs to propose technology solutions to meet DoD needs.

Unlike conventional topics, which specify the desired technical objective and output, open topics use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

Open topics released under this BAA will be clearly identified as such in the title and topic objective. Proposal preparation instructions for open topics may vary significantly across DoD Services/Components. Proposing SBCs must carefully read and follow all instructions from the DoD Service/Component for the open topic of interest. Unless specifically noted in the Service/Component instructions, all requirements outlined in this BAA remain in effect for open topics.

An SBC may only submit one proposal to each open topic. If an SBC submits more than one proposal for a single open topic, only the most recent certified proposal submitted prior to the submission deadline will receive an evaluation. All previously submitted proposals for the same open topic will be marked nonresponsive and will not receive an evaluation.

1.9 Discretionary Technical and Business Assistance (TABA)

DoD has not mandated the use of discretionary technical and business assistance (TABA). The proposing SBCs should review individual Service/Component-specific instructions to determine if TABA is offered by the Service/Component and follow instructions for requesting TABA funding.

1.10 Phase II Enhancement Policy

To further encourage the transition of SBIR research into both DoD acquisition programs and the private sector, certain DoD Services/Components developed their own Phase II Enhancement policies. Under this policy, the Service/Component will provide a Phase II awardee with additional Phase II SBIR funding if the proposing SBC can match the additional SBIR funds with non-SBIR funds from DoD acquisition programs or the private sector.

See Service/Component instructions for more details on Phase II Enhancement opportunities.

1.11 Commercialization Readiness Program (CRP)

The SBIR/STTR Reauthorization Act of 2011 established the Commercialization Pilot Program (CPP) as a long-term program called the Commercialization Readiness Program (CRP). Each Military Department (Army, Navy, and Air Force) has a CRP; please check the Service/Component instructions for further information.

The Defense SBIR/STTR Program also established the OSD Transitions SBIR Technology (OTST) Pilot Program as an interim technology maturity phase (Phase II) inserted into the SBIR development. For more information contact osd.ncr.ousd-r-e.mbx.sbir-sttr-tech-transition@mail.mil.

1.12 State and Other Available Assistance

Many states have established programs to provide services to those proposing SBCs and individuals wishing to participate in the Federal SBIR Program. These services vary from state to state, but may include:

a. Information and technical assistance;
b. Matching funds to SBIR recipients;
c. Assistance in obtaining Phase III funding.

Contact your State SBIR/STTR Support office for further information. SBCs may seek general administrative guidance from small and disadvantaged business utilization specialists located in various defense contract management activities throughout the continental United States.

1.13 Fraud and Fraud Reporting

Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine of up to $10,000, up to five years in prison, or both.

The DoD Office of Inspector General Hotline ("Defense Hotline") is an important avenue for reporting fraud, waste, abuse, and mismanagement within the DoD. The Office of Inspector General operates this hotline to receive and investigate complaints or information from contractor employees, DoD civilians, Service members, and the public. Individuals who wish to report fraud, waste or abuse may contact the Defense Hotline at (800) 424-9098 between 8:00 a.m. and 5:00 p.m. Eastern Time or visit their website to submit a complaint. Mailed correspondence should be addressed to the Defense Hotline, The Pentagon, Washington, DC 20301-1900, or email addressed to hotline@dodig.mil.

2.0 CERTIFICATIONS AND REGISTRATIONS

2.1 System for Award Management (SAM) Registration

The System for Award Management (SAM) allows proposing SBCs to provide basic information on business structure, capabilities, and financial and payment information with the Federal Government. Proposing SBCs must register in SAM here. Registration in SAM will generate the Unique Entity ID (UEI) number and the Commercial and Government Entry (CAGE) code. The UEI is required for registration in the U.S. Small Business Administration's (SBA) Company Registry. A proposing SBC that is already registered in SAM should verify the registration is active, and its representations and certifications are current to avoid award delay.

2.2 SBA Company Registry

Proposing SBCs must be registered in the SBA Company Registry. SBCs will be required to verify registration by providing the SBC Control ID and Proof of Registration/Certification during proposal submission.

2.3 Defense SBIR/STTR Innovation Portal (DSIP) Registration

Individuals from proposing SBCs must be registered in the DSIP to prepare and submit proposals. Proposing SBCs submitting through this site for the first time will be asked to register. All users are required to have an individual user account to access DSIP. It is recommended proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process.

DSIP user accounts are authenticated by Login.gov. Users who do not already have a Login.gov account will be required to create one. Users who already have a Login.gov account can link their existing Login.gov account with their DSIP account. Job Aids and Help Videos to walk you through the process are in the Learning & Support section of DSIP.

Be advised the sharing of accounts and passwords is a violation of the Terms of Use for Login.gov and DoD policy.

Registered SBCs will have a designated DSIP Firm Admin responsible for creating the Firm PIN, controlling access for other users in the SBC and completing and maintaining the Firm-level forms, which must be completed before any proposals can be submitted.

Users should complete their account registrations as soon as possible to avoid any delays in proposal submissions.

NOTE: The DSIP application is only accessible from within the United States, which is defined as the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

2.3.1 DSIP Assistance and Support

For assistance with the DSIP application, please visit the Learning & Support section of DSIP. Email DSIP Support at DoDSBIRSupport@reisystems.com only for further assistance with issues pertaining directly to the DSIP application. Questions submitted to DSIP Support will be addressed in the order received during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET). Please include information on your SBC, a proposal number (if applicable), and screenshots of any pertinent errors or issues encountered.

DSIP Support cannot provide updates to proposal status after submission, such as proposal selection/non-selection status or contract award status. Contact the DoD Service/Component that originated the topic following the Service/Component-specific instructions given at the beginning of that Service/Component-specific instructions.

2.4 Required Certifications

At the time of proposal submission, each SBC must certify via the Cover Sheet of the proposal that it meets the size, ownership, and other requirements of the SBIR Program. In addition, the Policy Directive includes certification requirements set forth in the SBIR and STTR Extension Act of 2022 (Public Law 117-183). SBCs are required to certify that they are meeting the Program's requirements during the life cycle of the funding agreement: at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the total Phase II award amount, and prior to final payment on the Phase II award.

2.5 Due Diligence Program to Assess Security Risks

15 U.S.C. §638 (vv) requires the DoD, in coordination with the SBA, to establish and implement a due diligence program to assess and, when possible, mitigate national security risks from SBCs seeking a federally funded award. The Department will use the proposal and information in response to the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (proposal submission Volume 7) to conduct a risk-based due diligence review of the following areas: cybersecurity practices; patent analysis; employee analysis and foreign ownership, including the SBC's financial ties and obligations (which shall include surety, equity, and debt obligations); and SBC employees' ties to a foreign country, foreign person, or foreign entity. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the purpose of confirming the accuracy of the information provided as well as determining if the proposing SBC failed to disclose the information set forth in 15 U.S.C. 638(g)(13).

After reviewing the proposing SBC's responses to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries, if DoD determines it appropriate the Department may ask the SBC to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint-venture like arrangement with an enterprise owned by a foreign state or any foreign entity in effect during the five-year period preceding the SBC's proposal submission.

The DoD may not make awards that pose an unacceptable risk to national security. If the risk-based due diligence review concludes that an SBC should not be eligible for the specific SBIR or STTR award due to a national security risk that cannot be adequately mitigated, the proposal will not receive consideration for possible award regardless of the results of the technical review of the proposal. Attachment 2: Defense SBIR and STTR Due Diligence Program Common Risk Matrix of the memo titled "Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program", dated 13 May 2024, provides details on the factors for assessing SBC risk during the due diligence review.

2.5.1 Training for Understanding FOCI

DoD has partnered with Project Spectrum to provide an online course on Understanding Foreign Ownership, Control, or Influence (FOCI). This course defines FOCI, explains what it means to be under FOCI, and details FOCI's effect on a company seeking initial or continued eligibility for access to a federally funded award. Small business concerns can register and access this course by following the instructions below:

1. Go to projectspectrum.io

2. Click “Profile/Dashboard” in the top right and then click “Sign Up” from the dropdown menu.
3. Follow the instructions to sign up for an account. Descriptions of the account types are provided below each option.
4. Verify your email by entering the code sent to the email address you provided when signing up.
5. Log in to Project Spectrum by clicking “Profile/Dashboard > Login” in the top right.
6. Hover over “Courses” in the Navigation Bar, and then select “FOCI” from the dropdown listing.
7. Copy the provided password.
8. Click on the “Understanding Foreign Ownership, Control, or Influence (FOCI)” course, which will open a new browser tab.
9. From the new tab, log in to Encite.io using your email address and the copied password.
10. Enroll in the course and click “Enter” to begin.

For Project Spectrum registration or access assistance, please email support@projectspectrum.io.

2.6 Joint Ventures

A small business joint venture entity must submit, with its proposal, the representation required in paragraph (c) of FAR solicitation provision 52.212-3, Offeror Representations and Certifications-Commercial Products and Commercial Services, and paragraph (c) of FAR provision 52.219-1, Small Business Program Representations, in accordance with 52.204-8(d) and 52.212-3(b) for the following categories:

a. Small business;
b. Service-disabled veteran-owned small business;
c. Women-owned small business (WOSB) under the WOSB Program;
d. Economically disadvantaged women-owned small business under the WOSB Program; or
e. Historically underutilized business zone small business.

These representations can be found here and must be uploaded to Volume 5, Supporting Documents of the proposal submission in DSIP, if applicable.

2.7 Conflicts of Interest

Contract awards to an SBC owned by or employing current, or previous, Federal Government employees could create conflicts of interest for those employees, which may be a violation of federal law.

2.8 Organizational Conflicts of Interest (OCI)

FAR 9.5 Requirements

In accordance with FAR 9.5, proposing SBCs are required to identify and disclose all facts relevant to potential organizational conflicts of interest (OCIs) involving the proposing SBC’s organization and any proposed team member (sub-awardee, consultant). The proposing SBC is responsible for providing this disclosure with each submitted proposal. The disclosure must include the proposing SBC’s, and as applicable, proposed team member’s OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposing SBC has taken, or intends to take, to prevent the existence of conflicting roles that might bias the proposing SBC’s judgment, and to prevent the proposing SBC from having unfair competitive advantage. The OCI mitigation plan will specifically discuss the disclosed OCI in the context of each of the OCI limitations outlined in FAR 9.505-1 through FAR 9.505-4.

Agency Supplemental OCI Policy
DoD Services/Components also may have a supplemental OCI policy prohibiting contractors/performers from concurrently providing scientific engineering technical assistance (SETA), advisory and assistance services (A&AS), or similar support services, and being a technical performer. As part of the FAR 9.5 disclosure requirement, a proposing SBC must affirm whether the proposing SBC or any proposed team member (sub-awardee, consultant) is providing SETA, A&AS, or similar support to any DoD Service/Component office(s) under: (a) a current award or sub-award; or (b) a past award or sub-award that ended within one calendar year prior to the proposal's submission date. If SETA, A&AS, or similar support is or was provided to any DoD Service/Component office(s), the proposal must include:

a. The name of the DoD Service/Component office receiving the support;
b. The prime contract number;
c. Identification of proposed team member (sub-awardee, consultant) providing the support; and
d. An OCI mitigation plan in accordance with FAR 9.5.

Government Procedures
In accordance with FAR 9.503, 9.504 and 9.506, the Government will evaluate OCI mitigation plans to avoid, neutralize, or mitigate potential OCI issues before award and determine whether it is in the government's interest to grant a waiver. The U.S. Government will only evaluate OCI mitigation plans for proposals determined selectable under the BAA evaluation criteria and funding availability.

The government may require proposing SBCs provide additional information to support evaluation of the proposing SBC's OCI mitigation plan.

If the government determines a proposer failed to fully disclose an OCI; or failed to provide a government waiver as described above; or failed to reasonably provide additional information the government requested when evaluating the proposer's OCI mitigation plan, the government may reject the proposal and withdraw it from consideration for award.

2.9 Research Involving Human Subjects/Human Subject Research (RIHS/HSR)

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws, and agency policy/guidelines for human subject protection (see Section 5.2 and Appendix B).

Institutions receiving funding for research involving human subjects must provide documentation of a current federal assurance of compliance with federal regulations for human subject protection; for example a Department of Health and Human Services, Office for Human Research Protections federal-wide assurance (http://www.hhs.gov/ohrp). The awarding DoD Service/Component may also request additional federal assurance documentation. All institutions engaged in human subject research, to include subcontractors, must also have a valid assurance. In addition, personnel involved in human subjects research must provide documentation of completed appropriate training for the protection of human subjects. Institutions proposing to conduct human subject research that meets one of the exemption criteria in 32 CFR 219.101 are not required to have a federal assurance of compliance. Proposing SBCs should clearly segregate research activities involving human subjects from other R&D activities in their proposal.

If selected, institutions must also provide documentation of institutional review board (IRB) approval, or a determination from an appropriate official in the institution, that the work meets one of the exemption criteria with 32 CFR 219. As part of the IRB review process, evidence that all investigators are appropriately trained should accompany the protocol. The protocol, separate from the proposal, must

include a detailed description of the research plan, study population, risks and benefits of study participation, recruitment and consent process, and data collection and analysis.

The amount of time required for the IRB to review and approve the protocol will vary based on the IRB's procedures, the complexity of the research, the level of risk to study participants and the responsiveness of the investigator. The average IRB approval process can last between one and three months. Once the IRB has approved the research, the awarding DoD Service/Component will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with both DoD and Service/Component policies. The DoD review process can last between three to six months. Ample time should be allotted to complete both the IRB and DoD approval processes prior to recruiting subjects. **No funding can be used towards human subject research until ALL approvals are granted. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.10 Research Involving Animal Subjects

All research, development, testing, experimentation, education, or training involving the use of animals shall comply with the applicable federal and agency rules on animal acquisition, transport, care, handling, and use (see Section 5.2 and Appendix B).

For submissions containing animal use, proposals should briefly describe plans for their institutional animal care and use committee (IACUC) review and approval.

All recipients must receive their IACUC's approval, as well as secondary or headquarters-level approval from a DoD veterinarian trained or experienced in laboratory animal medicine and science. **No animal research may be conducted using DoD funding until all appropriate DoD office(s) grant approval. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.11 Research Involving Recombinant DNA Molecules

All research involving recombinant DNA molecules shall comply with the applicable federal and state law, regulation, and additional agency guidance. An institutional biosafety committee must approve the research.

In addition to the standard federal and DoD procurement certifications, the SBA SBIR Policy Directive requires the proposing business concerns provide certain information at time of award and during the award life cycle. Each proposing SBC must provide this additional information at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the Phase II total award amount, and prior to final payment on the Phase II award.

2.12 Federal Acquisition Supply Chain Security Act Orders

FAR 52.204-29 Federal Acquisition Supply Chain Security Act (FASCA) Orders—Representation and Disclosures and FAR 52.204-30 FASCA Orders—Prohibition are included in this solicitation. In accordance with FAR 52.204-29 and FAR 52.204-30, proposing SBCs must review FASCSA orders here for covered articles, or any products or services produced or provided by a source, that an applicable FASCSA order prohibits.

During contract performance, the contractor shall review SAM.gov at least once every three months, or as the contracting officer advises, to check for covered articles, or products or services produced subject as part of any new FASCSA order(s) that could impact their supply chain, and report to the contracting officer any covered article, or product or service produced or provided by a source provided to the

government or used during the contract performance.

The proposing SBC represents that, via proposal submission under this BAA, it conducted a reasonable inquiry, and it does not propose to provide or use any covered article, or any products or services produced or provided by a source, if an applicable FASCSA prohibited the covered article or the source effective this BAA's issue date.

3.0 PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

3.1 Introduction

The proposal must provide sufficient information to demonstrate to the evaluator(s) that the proposed work represents an innovative approach to an important scientific or engineering problem and is worthy of support under the stated criteria. The proposed research or R&D must be responsive to the chosen topic, although it does not need to use the exact approach specified in the topic. SBCs should consider the following:

a. Does the technical approach have a reasonable chance of meeting the topic objective?
b. Is this approach innovative, not routine, with potential for commercialization?
c. Does the proposing SBC have the capability to implement the technical approach or can it obtain the appropriate people and equipment for the task?

DSIP provides a structure for providing the following proposal volumes:

a. Volume 1: Proposal Cover Sheet
b. Volume 2: Technical Volume
c. Volume 3: Cost Volume
d. Volume 4: Company Commercialization Report
e. Volume 5: Supporting Documents
f. Volume 6: Fraud, Waste and Abuse Training
g. Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Each Service/Component guidance on allowable proposal content may vary. A completed proposal submission in DSIP does NOT indicate that each proposal volume has been completed in accordance with the Service/Component-specific instructions. Accordingly, it is the proposing SBC's responsibility to consult the Service/Component-specific instructions for detailed guidance, including required proposal documentation and structure, cost and duration limitations, budget structure, TABA allowance and proposal page limits.

3.2 Export-Controlled Topic Requirements

For proposals submitted under export-controlled topics, either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions and FAQs may be found at the United States/Canada Joint Certification Program website, https://www.dla.mil/Logistics-Operations/Services/JCP/DD23%2045Instructions/.

DD Form 2345 approval will be required if a proposal submitted to an ITAR/EAR-marked topic receives an SBIR award.

3.3 Classified Proposals

Classified proposals will not be accepted under the DoD SBIR Program. If topics require classified work during Phase II, the proposing SBC must have a facility clearance to perform the work. For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency (DCSA) website at: https://www.dcsa.mil/mc/ctp/fc/.

3.4 Promotional Materials

Promotional and non-project related discussion is discouraged, and additional information provided via website links or on computer disks, CDs, DVDs, video tapes or any other medium will not be accepted or considered in the proposal evaluation.

3.5 Prior, Current, or Pending Support of Similar Proposals or Awards

While it is permissible, with proposal notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program BAAs or solicitations, it is unlawful to enter negotiation for contracts requiring essentially equivalent effort. If there is any question concerning prior, current, or pending support of similar proposals or awards, it must be disclosed to the soliciting agency or agencies as early as possible and declared on the proposal cover sheet.

3.6 Marking Proprietary Proposal Information

Proposing SBCs that include data in their proposals they do not want disclosed to the public for any purpose, or only used for government evaluation purposes, shall:

a. Mark the first page of each volume of the proposal submission with the following legend:

   "This proposal includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this proposing SBC as a result of-or in connection with-the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]"; and

b. Mark each sheet of data it wishes to restrict with the following legend:

   "Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

The DoD assumes no liability for disclosure or use of unmarked data and may use or disclose such data for any purpose.

Restrictive notices notwithstanding, support contractors may handle proposals and final reports submitted through DSIP for administrative purposes only; they are required to adhere to appropriate non-disclosure agreements.

3.7 Phase I Proposal Instructions

a. **Proposal Cover Sheet (Volume 1)**

   The proposal cover sheet is prepared on DSIP. The cover sheet must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters.

Do not include proprietary or classified information in the proposal cover sheet. If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the cover sheet is saved. You may modify the cover sheet as needed until the BAA closes.

NOTE: the amounts listed in the percentage of work (POW) certification question on the proposal cover sheet are derived from SBC-entered information in the cost volume (Volume 3). Details on the calculation can be viewed in DSIP during proposal submission.

If the POW calculations fall below eligibility requirements, the funding agreement officer must upload either an explanatory letter or approval to the certification question to complete the submission. Some DoD Services/Components will not accept any deviations from the POW minimum requirements. Please refer to the Service/Component instructions regarding deviations acceptance to the POW requirements.

b. **Technical Volume Format (Volume 2)**

1. **File Type.** The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the technical volume file. If a virus is detected, the proposal may be rejected. **Do not lock, password protect or encrypt the uploaded file. Do not include or embed active graphics, such as videos, moving pictures, or other similar media, in the document.**

2. **Length.** It is the proposing SBC's responsibility to verify that the technical volume does not exceed the page limit after upload to DSIP. Please refer to Service/Component-specific instructions for how a technical volume is handled if the stated page count is exceeded. Some Services/Components will reject the entire technical proposal if the proposal exceeds the stated page count.

3. **Layout.** Number all proposal pages consecutively. Submit a direct, concise, and informative research or R&D proposal (no type smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins, including the header). Each header on each page in the technical volume should contain the proposing SBC's name, topic number, and the DSIP-assigned proposal number from the cover sheet.

c. **Technical Volume Content (Volume 2)**

The Technical Volume should cover the following items in the order given below:

1. Identification and Significance of the Problem or Opportunity
2. Phase I Technical Objectives
3. Phase I Statement of Work
4. Related Work
5. Relationship with Future Research or Research and Development
6. Commercialization Strategy
7. Key Personnel
8. Foreign Citizens
9. Facilities/Equipment
10. Subcontractors/Consultants
11. Prior, Current, or Pending Support of Similar Proposals or Awards
12. Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software

A Phase I technical volume template is available in Appendix A to provide details and helpful guidelines for completing each section of your Phase I technical proposal.

Refer to the Service/Component-specific Direct to Phase II instructions for details on proposal preparation and technical volume content requirements.

d. **Cost Volume Content (Volume 3)**

Complete the cost volume using the DSIP cost volume form. Some items in the cost breakdown may not apply to the proposed project. There is no need to provide information on each individual item; make sure to provide enough information for evaluators to understand the requested funds' planned use if a contract is awarded.

1. List all key personnels' names and include their individual hours dedicated to the project as direct labor.
2. While special tooling and test equipment and material cost may be included under Phase I, equipment and material inclusion will be carefully reviewed relative to need and appropriateness for the work proposed. Special tooling and test equipment purchases must, in the Service/Component contracting officer's opinion, be advantageous to the U.S. Government and should relate directly to the specific topic. These may include such items as innovative instrumentation or automatic test equipment. Title to property the U.S. Government furnished or acquired with government funds will be vested with the DoD Service/Component, unless it is determined that title transfer to the contractor would be more cost effective than the DoD Service/Component equipment recovery.
3. Cost for travel funds must be justified and related to the project needs.
4. Cost sharing is permitted for proposals under this BAA; cost sharing is not required, nor will it be an evaluation factor in the Phase I proposal consideration.
5. A Phase I option (if applicable) should be fully costed separately from the Phase I (base) approach.
6. All subcontractor costs and consultant costs, such as labor, travel, equipment, materials, must be detailed at the same level as prime contractor costs. Provide detailed subcontractor costs substantiation in your cost proposal. Supporting Documents (Volume 5) may be used if additional space is needed.

If a proposal is selected for negotiation and possible award, you must be prepared to submit further documentation to the Service/Component contracting officer to substantiate costs (e.g., a cost estimates explanation for equipment, materials, and consultants or subcontractors). For more information about cost proposals and accounting standards, see visit DCAA's website.

e. **Company Commercialization Report (Volume 4)**

The company commercialization report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. SBA requires SBIR and STTR awardees to update and maintain their organization's CCR on SBIR.gov. Commercialization information is required upon the last deliverable's completion under the funding agreement. Thereafter, SBIR and STTR awardees are requested to voluntarily update the information in the database annually for a minimum 5-year period.

If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, regardless of whether the project has any commercialization to date, the firm admin must download the CCR's PDF copy from SBIR.gov and upload it to DSIP's "Firm Forms" section. The DSIP firm admin completes the firm forms are completed and are applies them to all proposals the proposing SBC submits. To fulfill the DSIP CCR requirement complete the following:

1. Log into the firm account at https://www.sbir.gov/.
2. Navigate to My Dashboard > My Documents to view or print the information currently contained in the "Company Registry Commercialization Report."
3. Create or update the commercialization record, from the company dashboard, scroll to the "My Commercialization" section and click the "Create/Update Commercialization" tab under "Current Report Version." Please refer to the "Instructions" and "Guide" documents contained on dashboard for more detail on completing and updating the CCR. **Ensure the report is certified and submitted.**
4. Click the "Company Commercialization Report" PDF under the dashboard's "My Documents" section to download the CCR PDF.
5. Upload the CCR PDF (downloaded from SBIR.gov in previous step) to the "Company Commercialization Report" in DSIP's "Firm Forms" section. The firm admin must complete this upload action.

In Volume 4 of the DSIP proposal submission, the proposing SBC will be prompted to answer: "Do you have a new or revised Company Commercialization Report to upload?" There are three possible courses of action:

a. If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards and **has a new or revised CCR from SBIR.gov to upload to DSIP**, select YES.
   1. If the user is the firm admin, they can upload the CCR PDF from SBIR.gov directly on this page. It will also be updated in the "Firm Forms" and be associated with all new or in-progress proposals the proposing SBC submitted. If the user is not the firm admin, they will receive a message that they do not have access and must contact the firm admin to complete this action.
   2. **WARNING:** Uploading a new CCR under the DSIP "Firm Forms" section or clicking "Save" or "Submit" in one proposal's Volume 4 is considered a change for ALL proposals under any open BAAs or CSOs. If a proposing SBC has previously certified and submitted any Phase I or Direct to Phase II proposals under ***any*** BAA or CSO ***still open***, those proposals will be automatically reopened. Proposing SBCs will have to recertify and resubmit affected proposals. If a proposing SBC does not recertify or resubmit affected proposals, they will not be considered fully submitted and will not be evaluated.

b. If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, and **no new or revised CCR from SBIR.gov to upload to DSIP**, select NO.
   1. If a prior CCR was uploaded to the "Firm Forms", the proposing SBC will see a file dialog box at the bottom of the page and can view the previously uploaded CCR. This read-only access allows the proposing SBC to confirm the firm admin uploaded the CCR.
   2. If no file dialog box appears at the bottom of the page **there is no previously uploaded CCR in the DSIP "Firm Forms**." To fulfill the DSIP CCR requirement the firm admin must follow steps 1-5 listed above to download a CCR PDF from

SBIR.gov and upload it to the DSIP "Firm Forms" to be included with all proposal submissions.

c. If the proposing SBC has **NO** prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, the CCR upload from SBIR.gov is not required and SBC will select NO. The proposal's CCR section will be marked complete.

Please refer to the Service/Component-specific instructions for details on how the CCR information will be considered during proposal evaluations.

f. **Supporting Documents (Volume 5)**

Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3).

The following documents may be included in Volume 5, if applicable to the proposal. Refer to Service/Component-specific instructions for additional Volume 5 requirements. **Reminder: A completed proposal submission in DSIP does NOT indicate the mandatory supporting documents have been uploaded in accordance with the Service/Component-specific instructions.**

1. Letters of support
2. Additional cost information
3. Funding agreement certification
4. Technical data rights (assertions)
5. Lifecycle certification
6. Allocation of rights
7. Verification of Eligibility of Small Business Joint Ventures, if applicable
8. DD Form 2345, Militarily Critical Technical Data Agreement, if applicable (see section 3.2)

g. **Fraud, Waste, and Abuse Training (Volume 6)**

The fraud, waste, and abuse (FWA) training is **required** for DoD SBIR/STTR proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA. The training currently consists of a 3-page PDF, consistent with the tutorial provided by the SBA. This training material must be thoroughly reviewed once per year and can be found here and in the DSIP proposal submission module for Volume 6. Plan time to review the tutorial during completion of Volume 6, prior to the proposal submission deadline. The individual designated as the Proposal Owner must complete this training on behalf of the proposing small business.

h. **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**

In accordance with 15 U.S.C. §638 (vv) and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this BAA to assess security risks of SBCs seeking a federally funded award. SBCs must complete the DSIP Volume 7 webform "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (NOTE: PDF uploads are no longer accepted). The corporate official cannot certify and submit the full proposal until the Volume 7 webform is fully completed and submitted.

Please be aware that the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" form WILL NOT be accepted as a supporting document in DSIP's Volume 5 proposal submission. Do not upload any previous versions of this form to Volume 5.

For additional details, please refer to Section 2.5. The disclosure questions are below:

1. Is any owner or covered individual of the applicant or awardee party to any malign foreign talent recruitment program? If yes, disclose the first and last name of each owner or covered individual, identify their role (i.e., owner or covered individual), and the malign foreign talent recruitment program.
2. Is there a parent company, joint venture, or subsidiary, of the applicant or awardee that is based in or receives funding from, any foreign country of concern? If yes, disclose the name, full address, applicant or awardee relationships (i.e., parent company, joint venture, or subsidiary) of each entity based in, or funded by, any foreign country of concern.
3. Does the applicant or awardee have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an enterprise owned by a foreign state or any foreign entity? If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (*i.e.,* contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).
4. Is the applicant or awardee wholly owned in a foreign country? If yes, disclose the foreign country.
5. Does the applicant or awardee have any venture capital or institutional investment? If yes, proceed to question 5a. If no, proceed to question 6.

   5a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with any foreign country of concern? If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).
6. During the previous 5-year period, did the applicant or awardee have any technology licensing or intellectual property sales or transfers, to a foreign country of concern? If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.
7. Is there any foreign business entity, offshore entity, or entity outside the United States related to the applicant or awardee? If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.
8. Does the applicant or awardee have an owner, officer, or covered individual that has a foreign affiliation with a research institution located in a foreign country of concern? If yes, disclose the first and last name of each owner, officer, or covered individual that has a foreign affiliation with a foreign country of concern, identify their role (i.e., owner, officer, or covered individual), and the name of the foreign research institution and the foreign country of concern where it is located.

3.8 Phase II Proposal Information

Only Phase I awardees may submit Phase II proposals. Phase II proposals submission must follow individual Service/Component instructions. Awarding DoD Services/Component, either in the Phase I

award or via subsequent notification, will provide details on Phase II proposal due date, content, and submission requirements. If a proposing SBC submits their Phase II proposal prior to the individual Service/Component's dates, it may be rejected without evaluation.

Due to specific limitations on the amount of funding and number of awards awarded to a particular proposing SBC per topic using SBIR/STTR program funds, head of agency determinations are now required before a different agency may make an award using another agency's topic. This limitation does not apply to Phase III funding. Please contact your original sponsoring agency before submitting a Phase II proposal to an agency other than the one who sponsored the original topic.

SBIR/STTR Policy Directive Section 4(b)(1)(i) allows that, at the agency's discretion, projects awarded a Phase I under a solicitation for SBIR may transition in Phase II to STTR and vice versa. A proposing SBC wishing to transfer from one program to another must contact their designated technical monitor to discuss the reasons for the request and the agency's ability to it. The transition may be proposed prior to award or during the Phase II effort performance. Agency disapproval of a request to change programs shall not be grounds for granting relief from any contractual performance requirement. All approved transitions between programs must be noted in the contracting officer-signed Phase II award or award modification that indicates the removal or addition of the research institution and the revised percentage of work requirements.

3.8.1 Phase II Commercialization Strategy

At a minimum, the commercialization strategy must address the following five questions:

a. What will be the first product to use this technology?
b. Who will be the customers, and what is the estimated market size?
c. How much money is needed to bring the technology to market, and how will that money be raised?
d. Does the proposing SBC contain marketing expertise and, if not, how will the SBC acquire that expertise?
e. Who are the proposing SBC's competitors, and what is the price and/or quality advantage over those competitors?

The commercialization strategy must also include a schedule showing the anticipated quantitative commercialization results at 1) one year after the Phase II project starts, 2) at Phase II completion, and 3) after Phase II completion (i.e., additional investment amount, sales revenue, etc.). After Phase II award, the proposing SBC is required to report actual sales and investment data in its SBA company commercialization report via SBIR.gov's "My Dashboard" on a minimum annual basis. Please refer to the Service/Component-specific instructions for guidance on formatting, page count and other details.

3.8.2 Phase II Adequate Accounting System

To reduce the small business's risk and avoid potential contracting delays, companies interested in pursuing Phase II SBIR contracts and other contracts of similar size with the DoD, must have an adequate accounting system in place per General Accepted Accounting Principles, Generally Accepted Government Auditing Standards, Federal Acquisition Regulation (FAR) and Cost Accounting Standards. The Defense Contract Audit Agency (DCAA) will audit the accounting system. See DCAA's website for requirements and standards, the audit process overview, and a pre-award system adequacy checklist.

4.0 METHOD OF SELECTION AND EVALUATION CRITERIA

4.1 Evaluation Process

All proposals will be evaluated and judged on a competitive basis in terms of technical capability and technical value. Proposals will be initially screened to determine responsiveness to the topic objective. Proposals passing this initial screening will be technically evaluated by engineers, scientists, or subject matter experts to determine the most promising technical and scientific approaches. As a common statement of work does not exist, each proposal will be assessed on the merit of the approach in achieving the technical objectives established in the topic.

4.2 Evaluation Criteria

Proposals will be evaluated based on the criteria outlined below, unless otherwise specified in the Service/Component-specific instructions. Selections will be based on a determination of the overall technical value of each proposal and an evaluation of the cost volume, with the appropriate method of analysis given the contract type to be awarded, for selection of the proposal(s) most advantageous to the Government, considering the following factors which are listed in descending order of importance:

a. The soundness, technical merit, and innovation of the proposed approach and its incremental progress toward topic or subtopic solution.
b. The qualifications of the proposed principal/key investigators, supporting staff, and consultants. Qualifications include not only the ability to perform the R&D but also the ability to commercialize the results.
c. The potential for commercial (government or private sector) application and the benefits expected to accrue from this commercialization.

Cost or budget data submitted with the proposals will be considered during evaluation.

Technical reviewers will base their conclusions only on information contained in the proposal. Do not assume reviewers are acquainted with the proposing SBC, key individuals, or any referenced experiments. Relevant supporting data such as journal articles, literature, including Government publications, etc., should be included based on requirements listed in Service/Component-specific instructions.

4.3 Proposal Status & Feedback

Proposing SBCs will be notified of selection or non-selection status for an award by the DoD Service/Component that originated the topic <u>no later than 90 days</u> of the closing date for this BAA. Please refer to the Service/Component-specific instructions for details.

After final selection decisions have been announced, the SBC may be provided proposal feedback in the form of a written debrief. This debriefing process varies across the DoD Services/Components. Please refer to the Service/Component-specific instructions for details on the debriefing processes.

4.4 Award Denials

The DoD will not make an award under the SBIR program if it determines:

a. The SBC submitting the proposal
    i. has an owner or covered individual that is party to a malign foreign talent recruitment program;
    ii. has a business entity, parent company, or subsidiary located in the People's Republic of China or another foreign country of concern; or
    iii. has an owner or covered individual that has a foreign affiliation with a foreign entity located in the People's Republic of China or another foreign country of concern; and

b. The relationships and commitments described in clauses (i) through (iii) of subparagraph (A)
   i. interfere with the capacity for DoD-supported activities to be carried out;
   ii. create duplication with DoD-supported activities;
   iii. present concerns about conflicts of interest;
   iv. were not appropriately disclosed to the DoD;
   v. violate Federal law or terms and conditions of DoD-awarded contracts or other agreements; or
   vi. pose a risk to national security.

4.5 Pre-Award and Post Award BAA Protests

Interested parties have the right to protest via procedures in FAR Subpart 33.1; protests exclusively related to this BAA's terms must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil

For protests filed with the Government Accountability Office (GAO), a copy of the protest shall be submitted to the email address listed above (pre-award ONLY) or DoD Service/Component POC (post-selection/award decision ONLY) within one day of filing with the GAO. Protests of small business status of a selected proposing SBC may also be made to the SBA via the procedures in FAR § 19.302.

For the purposes of a protest related to a particular topic selection, non-selection or award decision, protests should be served to the point-of-contact (POC) listed in the instructions of the DoD Component that authored the topic.

5.0 ADDITIONAL CONSIDERATIONS

5.1 Award Information

The number of awards will be consistent with the Component's RDT&E budget. No contracts will be awarded until evaluation of all qualified proposals for a specific topic is completed.

Each proposal selected for negotiation and possible award will be funded under negotiated contracts, purchase orders, or Other Transactions and will include a reasonable fee or profit consistent with normal profit margins provided to profit-making proposing SBCs for R/R&D work. Firm-Fixed-Price, Firm-Fixed-Price Level of Effort, Labor Hour, Time & Material, or Cost-Plus-Fixed-Fee type contracts can be negotiated and are at the discretion of the Component Contracting Officer.

Contract value varies among the DoD Services/Components; it is important for proposing SBCs to review Service/Component-specific instructions regarding award size.

The SBA SBIR/STTR Policy Directive, Section 7(c)(1)(ii), states agencies should issue the award no more than 180 days after the closing date of the BAA.

5.2 Contract Requirements

Upon contract award, the contractor will be required to make certain legal commitments through acceptance of U.S. Government contract clauses in the Phase I contract. The examples below are illustrative of the types of provisions the Federal Acquisition Regulation requires in the Phase I contract. This is not an exhaustive provisions list that could be included in Phase I contracts, nor does it contain specific clause wording. Appendix C of this BAA contains additional potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Copies of complete general provisions will be made available prior to award.

Examples of general provisions:

a. **Standards of Work**. Work performed under the Funding Agreement must conform to high professional standards.
b. **Inspection**. Work performed under the Funding Agreement is subject to Government inspection and evaluation at all times.
c. **Examination of Records**. The Comptroller General (or a duly authorized representative) must have the right to examine any pertinent records of the Awardee involving transactions related to this Funding Agreement.
d. **Default**. The Federal Government may terminate the Funding Agreement if the contractor fails to perform the work contracted.
e. **Termination for Convenience**. The Funding Agreement may be terminated at any time by the Federal Government if it deems termination to be in its best interest, in which case the Awardee will be compensated for work performed and for reasonable termination costs.
f. **Disputes.** Any dispute concerning the Funding Agreement that cannot be resolved by agreement must be decided by the contracting officer with right of appeal.
g. **Contract Work Hours**. The Awardee may not require an employee to work more than 8 hours a day or 40 hours a week unless the employee is compensated accordingly (for example, overtime pay).
h. **Equal Opportunity**. The Awardee will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin.
i. **Equal Opportunity for Veterans**. The Awardee will not discriminate against any employee or application for employment because he or she is a disabled veteran or veteran of the Vietnam era.
j. **Equal Opportunity for People with Disabilities**. The Awardee will not discriminate against any employee or applicant for employment because he or she is physically or intellectually disabled.
k. **Officials Not to Benefit**. No Federal Government official may benefit personally from the SBIR/STTR Funding Agreement.
l. **Covenant Against Contingent Fees**. No person or agency has been employed to solicit or secure the Funding Agreement upon an understanding for compensation except bona fide employees or commercial agencies maintained by the Awardee for the purpose of securing business.
m. **Gratuities**. The Funding Agreement may be terminated by the Federal Government if any gratuities have been offered to any representative of the Government to secure the award.
n. **Patent Infringement**. The Awardee must report each notice or claim of patent infringement based on the performance of the Funding Agreement.
o. **American Made Equipment and Products**. When purchasing equipment or a product under the SBIR/STTR Funding Agreement, purchase only American-made items whenever possible.

5.3 Agency Recovery Authority and Ongoing Reporting

In accordance with Section 5 of the SBIR and STTR Extension Act of 2022, the DoD will:

a. require an SBC receiving an award under its SBIR program to repay all amounts received from the federal agency under the award if,
    1. the SBC makes a material misstatement that the federal agency determines poses a risk to national security; or
    2. there is a change in the SBC's ownership, entity structure, or other substantial change in circumstances that the federal agency determines poses a risk to national security; and

b. require an SBC receiving an award under its SBIR program to regularly report to the federal agency and the administration throughout the duration of the award on

1. any change to a disclosure required under the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form;
2. any material misstatement made under paragraph (A) above; and
3. any change described in paragraph (B) above.

5.4 Copyrights

With prior written permission of the contracting officer, the awardee may copyright (consistent with any appropriate national security considerations) material developed with DoD support. DoD receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.5 Patents

SBCs normally may retain the principal worldwide patent rights to any invention developed with U.S. Government support. The government receives a royalty-free license for its use, reserves the right to require the patent holder to license others in certain limited circumstances, and requires that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. 35 U.S.C. § 205 authorizes that the government will not make public any information disclosing a government-supported invention for a period of five years to allow the awardee to pursue a patent. See also Section 6.8, Invention Reporting.

5.6 Invention Reporting

SBIR awardees must report inventions to the Service/Component within two months of the inventor's report to the awardee, via either paper documentation submission, including fax, or through the Edison Invention Reporting System at www.iedison.gov for participating agencies.

5.7 Technical Data Rights

Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this BAA generally remain with the contractor, except that the U.S. Government obtains a royalty-free license to use such technical data only for government purposes during the period commencing with contract award and ending not less than twenty years after that date. This data should be marked with the restrictive legend specified in DFARS 252.227-7018 Class Deviation 2020-O0007. Upon expiration of the twenty-year restrictive license, the government has government purpose rights in the SBIR data. During the license period, the U.S. Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) for evaluation purposes; (2) as expressly permitted by the contractor; or (3) a use, release, or disclosure that is necessary for emergency repair or overhaul government-operated items. See DFARS clause 252.227-7018 Class Deviation 2020-O0007 "Rights in Noncommercial Technical Data and Computer Software – Small Business Innovation Research (SBIR) Program."

If a proposing SBC plans to submit assertions in accordance with DFARS 252.227-7017 Class Deviation 2020-O0007, those assertions must be identified and assertion of use, release, or disclosure restriction must be included with your proposal submission, at the end of the technical volume. The contract cannot be awarded until assertions are approved.

5.8 Final Technical Reports - Phase I through Phase III

a. **Content**: A final report is required for each project phase. The reports must contain in detail the project objectives, work performed, results obtained, and estimates of technical feasibility. A completed SF 298, "Report Documentation Page," will be used as the first page of the report. The DoD Service/Component also may require monthly status and progress reports.

b. **SF 298 Form "Report Documentation Page" Preparation**:

a. If desirable, the proposing SBC may also use language from its Phase II proposal to cover Phase I progress in the final report.

b. For each unclassified report, the proposing SBC submitting the report should fill in Block 12 (Distribution/Availability Statement) of the SF 298, "Report Documentation Page," with the following statement: "Distribution authorized to U.S. Government only; Proprietary Information, (Date of Determination). Other requests for this document shall be referred to the Service/Component SBIR Program Office."

   Note: Data developed under a SBIR contract is subject to SBIR Data Rights, under which DFARS 252.227-7018 Class Deviation 2020-O0007 (see Section 5.7, Technical Data Rights) provides protection. The sponsoring DoD activity, after reviewing the proposing SBC's entry in Block 12, has final responsibility for assigning a distribution statement.

   For additional information on distribution statements see the following Defense Technical Information Center (DTIC) website.

c. Block 14 (Abstract) of the SF 298, "Report Documentation Page" must include as the first sentence, "Report developed under SBIR contract for topic [insert BAA topic number. [Follow with the topic title, if possible.]" The abstract must identify the work's purpose and briefly describe the work conducted, the findings or results, and the effort's potential applications. **Since DoD will publish the abstract, it must not contain any proprietary or classified data, and type "UU" in Block 17.**

d. Block 15 (Subject Terms) of the SF 298 must include the term "SBIR Report".

c. **Submission**: In accordance with DFARS 252.235-7011, submit an electronic copy of the approved final scientific or technical report, not a summary, delivered under the contract to the Defense Technical Information Center (DTIC) through the web-based input system at https://discover.dtic.mil/submit-documents/ as required by DoD Instruction 3200.12, DoD Scientific and Technical Information Program (STIP). Include a completed Standard Form (SF) 298, Report Documentation Page, in the document, or complete the web-based SF 298. Additional submission resources are available here.

   Delivery will normally be within 30 days after completion of the Phase I technical effort.

   Other requirements regarding reports and/or other deliverables submission will be defined in each contract's contract data requirements list (CDRL). Special instructions for submitting CLASSIFIED reports will be defined in the contract's delivery schedule.

   DO NOT email classified or controlled unclassified reports, or reports containing SBIR Data Rights protected under DFARS 252.227-7018 Class Deviation 2020-O0007.

6.0 PROPOSAL SUBMISSION

6.1 Submission Details

DSIP is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed guidance on DSIP proposal submission is found here.

Deadline for Receipt: Complete proposals must be certified and submitted in DSIP no later than the close date listed on the cover page of this BAA. Proposals cannot be submitted in DSIP after the deadline is reached and will not be accepted or evaluated.

The final proposal submission includes successful completion of all firm level forms, all required proposal volumes, and electronic corporate official certification. Although signatures are not required on the electronic forms at the time of submission, the proposal must be certified electronically by the corporate official for it to be considered submitted. If the proposal is selected for negotiation and possible award, the DoD Component program will contact the proposing SBC for signatures prior to award. **Please plan to submit proposals as early as possible to allow time for troubleshooting any possible issues before the BAA close. DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. DoD is not responsible for missed proposal submission due to system latency.**

If the proposal status is "In Progress" or "Ready to Certify" it will NOT be considered submitted, even if all volumes are added prior to the BAA close date. The proposing SBC may modify all proposal volumes prior to the BAA close date.

6.2 Technical Questions

a. **Direct Contact with Topic Authors**. During the pre-release period, the names of the topic authors, their phone numbers and/or email addresses are published with the topic on the DSIP Topics and Topic Q&A page. During this time, proposing SBCs can contact topic authors via telephone or email to ask technical questions about specific BAA topics. Questions must be limited to specific information related to understanding a particular topic's requirements. Proposing SBCs may not ask for advice or guidance on solution approach and may not submit additional material to the topic author.

   If information provided during an exchange with the topic author is deemed necessary for proposal preparation, that information will be made available to all parties through DSIP Topic Q&A.

   After the pre-release period, questions must be asked through DSIP Topic Q&A as described below. No further direct contact is allowed between proposing SBCs and topic authors, unless the topic author is responding to a question submitted during the pre-release period.

b. **DSIP Topic Q&A.** Proposing SBCs may submit written questions through DSIP Topic Q&A here, where all questions and answers are posted on a non-attribution basis for public viewing. DSIP Topic Q&A opens on the pre-release date and closes two weeks prior to the topic close date.

   Proposing SBCs may use the topic search feature on DSIP to locate a topic of interest. Use the form at the bottom of the topic description, enter and submit the question. Answers are generally posted within seven business days of question submission and also e-mailed directly to the inquirer.

   Questions submitted through the DSIP Topic Q&A are limited to technical information focused on understanding a topic's requirements. Any other questions, such as asking for advice or guidance on solution approach, or administrative questions, such as SBIR or STTR program eligibility, technical proposal/cost proposal structure and page count, budget and

duration limitations, or proposal due date are not appropriate and will not receive a response; for administrative questions, refer to a topic's Service/Component-specific instructions.

Once the BAA proposal submission closes, no communication of any kind is allowed either with the topic author or through topic Q&A regarding submitted proposals.

Throughout the BAA period, proposing SBCs should frequently monitor DSIP for updates and amendments to the topics and DSIP Topic Q&A for questions and answers.

7.0 Participating Component Instructions & Research Topics

The following section contains all Component-specific proposal preparation instructions and research topics this BAA.

DoD SBIR 25.2 BAA

April 2, 2025: Topics Pre-release
April 23, 2025: Topics Open; DoD begins accepting proposals in DSIP
May 7, 2025: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
May 21, 2025: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

Participating Services/Components:

- Department of the Navy (Navy)
- ~~Department of the Air Force (Air Force)~~ [Air Force topics have been removed]
- Defense Health Agency (DHA)
- Defense Threat Reduction Agency (DTRA)
- Office of the Secretary of Defense (OSD) – National Geospatial-Intelligence Agency (NGA)

Version 4

DEPARTMENT OF THE NAVY (DON)
25.2 Small Business Innovation Research (SBIR)
Phase I Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to topics:**
  - **N252-074 through N252-121**

- Submitting small business concerns are encouraged to thoroughly review the DoD SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoD Program BAA is located at: https://www.dodsbirsttr.mil/submissions/login. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

- The information provided in the DON Proposal Submission Instructions takes precedence over the DoD Instructions posted for this BAA.

- **DON Phase I Technical Volume (Volume 2) page limit is not to exceed 10 pages.**

- Proposing small business concerns that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF) or any combination of these are eligible to submit proposals in response to DON topics advertised in this BAA. Information on Majority Ownership in Part and certification requirements at time of submission for these proposing small business concerns are detailed in the section titled ADDITIONAL SUBMISSION CONSIDERATIONS.

- Phase I Technical Volume (Volume 2) and Supporting Documents (Volume 5) templates, specific to DON topics, are available at https://www.navysbir.com/links_forms.htm.

- The DON provides notice that Basic Ordering Agreements (BOAs) may be used for Phase I awards, and BOAs or Other Transaction Agreements (OTAs) may be used for Phase II awards.

- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing small business concern.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DON. More information on the programs can be found on the DON SBIR/STTR website at www.navysbir.com. Additional information on DON's mission can be found on the DON website at www.navy.mil.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	When	Contact Information
Program and administrative	Always	Navy SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoD SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
	BAA Open	DoD SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
Electronic submission to the DoD SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMANDS (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
N252-074 to N252-078	Mr. Jeffrey Kent	Marine Corps Systems Command (MCSC)	smb_mcsc_sbir_admins@usmc.mil
N252-079 to N252-094	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil
N252-095 to N252-098	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil
N252-099 to N252-113	Ms. Lore-Anne Ponirakis	Office of Naval Research (ONR)	usn.pentagon.cnr-arlington-va.mbx.onr-sbir-sttr@us.navy.mil
N252-114 to N252-121	Dr. Scott Steward	Strategic Systems Programs (SSP)	ssp.sbir@ssp.navy.mil

PHASE I SUBMISSION INSTRUCTIONS

The following section details requirements for submitting a compliant Phase I proposal to the DoD SBIR/STTR Programs.

(NOTE: Proposing small business concerns are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoD SBIR/STTR Innovation Portal (DSIP). Proposing small business concerns are required to submit proposals via the DoD SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoD SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing small business concerns submitting through DSIP for the first time will be asked to register. It is recommended that small business concerns register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DON. Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoD SBIR/STTR Program BAA for further information.

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoD SBIR/STTR Program BAA.

- **Technical Proposal (Volume 2)**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - Not to exceed ten (10) pages, regardless of page content
    - Single column format, single-spaced typed lines
    - Standard 8 ½" x 11" paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point
    - Include, within the ten-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified. Phase I Options are exercised upon selection for Phase II.
    - Work proposed for the Phase I Base must be exactly six (6) months.
    - Work proposed for the Phase I Option must be exactly six (6) months.

  - Additional information:
    - A Phase I proposal template specific to DON to meet Phase I requirements is available at https://navysbir.com/links_forms.htmhttps://navysbir.com/links_forms.htm
    - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing small business concerns are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).**
  - Cost Volume (Volume 3) must meet the following requirements or the proposal will be REJECTED:
    - The Phase I Base amount must not exceed $140,000.
    - Phase I Option amount must not exceed $100,000.
    - Costs for the Base and Option must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.
    - For Phase I, a minimum of two-thirds of the work is performed by the proposing small business concern. The two-thirds percentage of work requirement must be met in the Base costs as well as in the Option costs. DON will not accept deviations from the minimum percentage of work requirements for Phase I. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of work for the proposing small business concern the sum of all direct and indirect costs attributable to the proposing small business concern represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The subcontractor percentage is calculated by taking the sum of all costs attributable to the subcontractor (Total Subcontractor Costs (TSC)) as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator.
      - Proposing Small Business Concern Costs (included in numerator for calculation of the small business concern):
        - Total Direct Labor (TDL)
        - Total Direct Material Costs (TDM)
        - Total Direct Supplies Costs (TDS)
        - Total Direct Equipment Costs (TDE)
        - Total Direct Travel Costs (TDT)
        - Total Other Direct Costs (TODC)
        - General & Administrative Cost (G&A)

        NOTE: G&A, if proposed, will only be attributed to the proposing small business concern.
      - Subcontractor Costs (numerator for subcontractor calculation):
        - Total Subcontractor Costs (TSC)
      - Total Cost (i.e., Total Cost before Profit Rate is applied, denominator for either calculation)
    - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. If a value above or below $0.00 is entered in the Cost Sharing field the proposal will be deemed non-compliant and will be REJECTED by DON.**

  - Additional information:
    - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
    - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
    - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3. When a proposal is selected for award, be prepared to submit further documentation to the SYSCOM Contracting Officer to substantiate costs (e.g., an explanation of cost estimates for equipment, materials, and consultants or subcontractors).

- **Company Commercialization Report (Volume 4)**. DoD collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may or will require to process a proposal, if selected, for contract award.

  - Proposing small business concerns must review and submit the following items, as applicable:
    - **Majority Ownership in Part.** Proposing small business concerns that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DON topics advertised within this BAA. Complete the certification as detailed under ADDITIONAL SUBMISSION CONSIDERATIONS.

  - Additional information:
    - Proposing small business concerns may include the following administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing small business concern may want to include in Volume 5:
      - Additional Cost Information to support the Cost Volume (Volume 3)
      - SBIR/STTR Funding Agreement Certification
      - Data Rights Assertion
      - Disclosure of Information (DFARS 252.204-7000)
      - Prior, Current, or Pending Support of Similar Proposals or Awards
      - Foreign Citizens
    - Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
    - Do not include documents or information to substantiate the Technical Volume (Volume 2) in Volume 5 (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
    - A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing small business concerns are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoD requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

PHASE I EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoD and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing small business concern has met eligibility requirements and followed the instructions for the Proposal Cover Sheet as specified in the DoD SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I proposals using the evaluation criteria specified in the Method of Selection and Evaluation Criteria section of the DoD SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the evaluation process; the DON will only do a compliance review of Volume 3. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

  The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:
  - Not to exceed ten (10) pages, regardless of page content
  - Single column format, single-spaced typed lines
  - Standard 8 ½" x 11" paper
  - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
  - No font size smaller than 10-point, except as permitted in the instructions above.
  - Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified.
  - Work proposed for the Phase I Base must be exactly six (6) months.
  - Work proposed for the Phase I Option must be exactly six (6) months.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will only undergo a compliance review to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base ($140,000) and Option ($100,000).
  - Must meet minimum percentage of work; a minimum of two-thirds of the work is performed by the proposing small business concern. The two-thirds percentage of work requirement must be met in the Base costs as well as in the Option costs. DON will not accept deviations from the minimum percentage of work requirements for Phase I.
  - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. If a value above or below $0.00 is entered in the Cost Sharing field the proposal will be deemed non-compliant and will be REJECTED by DON.**

- **Company Commercialization Report (CCR) (Volume 4).** The CCR (Volume 4) will not be evaluated by the DON nor will it be considered in the award decision. However, all proposing small

business concerns must refer to the DoD SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing small business concern has included items in accordance with the PHASE I SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoD SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing small business concerns to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of Defense, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by small business concerns seeking a Federally-funded award. Please review the Certifications and Registrations section of the DoD SBIR/STTR Program BAA for details on how DoD will assess security risks presented by small business concerns. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DON to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing small business concerns may request, in their Phase I Cost Volume (Volume 3) and Phase II Cost Volume, to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase I TABA amount is up to $6,500 and is in addition to the award amount. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount, and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). As with Phase I, the amount proposed for TABA cannot include any profit/fee by the proposing small business concern and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the SBIR proposing small business concern. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. A small business concern receiving TABA will be required to submit a report detailing the results and benefits of the service received. This TABA report will be due at the time of submission of the final report.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.

- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service
- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must NOT:

- Be subject to any indirect costs, profit, or fee by the SBIR proposing small business concern
- Propose a TABA provider that is the SBIR proposing small business concern
- Propose a TABA provider that is an affiliate of the SBIR proposing small business concern
- Propose a TABA provider that is an investor of the SBIR proposing small business concern
- Propose a TABA provider that is a subcontractor or consultant of the requesting small business concern otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:

- Phase I:
  - Online DoD Cost Volume (Volume 3) – the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.
- Phase II:
  - DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:

- Phase I: A total of $6,500
- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

If a proposing small business concern requests and is awarded TABA in a Phase II contract, the proposing small business concern will be eliminated from participating in the Navy SBIR Transition Program (STP), the DON Forum for SBIR/STTR Transition (FST), and any other Phase II assistance the DON provides directly to awardees.

All Phase II awardees not receiving funds for TABA in their awards must participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing small business concern shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design,

production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). A small business concern whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Majority Ownership in Part. Proposing small business concerns that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, **are eligible** to submit proposals in response to DON topics advertised within this BAA.

For proposing small business concerns that are a member of this ownership class the following must be satisfied for proposals to be accepted and evaluated:

a. Prior to submitting a proposal, small business concerns must register with the SBA Company Registry Database.
b. The proposing small business concern within its submission must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://navysbir.com/links_forms.htm. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
c. Should a proposing small business concern become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposing small business concern must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification, which can be found on https://navysbir.com/links_forms.htm.

System for Award Management (SAM). It is strongly encouraged that proposing small business concerns register in SAM, https://sam.gov, by the Close date of this BAA, or verify their registrations are still active and will not expire within 60 days of BAA Close. Additionally, proposing small business concerns should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. A small business concern selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.204-7019, in order to be considered for award, a small business concern is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items, please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. Due to the short timeframe associated with Phase I of the SBIR/STTR process, the DON does **not** recommend the submission of Phase I proposals that require the use of Human Subjects, Animal Testing, or Recombinant DNA. For example,

the ability to obtain Institutional Review Board (IRB) approval for proposals that involve human subjects can take 6-12 months, and that lengthy process can be at odds with the Phase I goal for time-to-award. Before the DON makes any award that involves an IRB or similar approval requirement, the proposing small business concern must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human, animal, or recombinant DNA protocols. It will not impact the DON's evaluation, but requiring IRB approval may delay the start time of the Phase I award and if approvals are not obtained within two months of notification of selection, the decision to award may be terminated. If the use of human, animal, and recombinant DNA is included under a Phase I or Phase II proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

Government Furnished Equipment (GFE). Due to the typical lengthy time for approval to obtain GFE, it is recommended that GFE is not proposed as part of the Phase I proposal. If GFE is proposed, and it is determined during the proposal evaluation process to be unavailable, proposed GFE may be considered a weakness in the technical merit of the proposal.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing small business concern within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For

protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct compliance review of Volume 3 to confirm eligibility of the proposing small business concern, and to take other relevant steps necessary prior to making an award.

Contract Types. The DON typically awards a Firm Fixed Price (FFP) contract or a small purchase agreement for Phase I. In addition to the negotiated contract award types listed in the section of the DoD SBIR/STTR Program BAA titled Additional Considerations, for Phase II awards the DON may (under appropriate circumstances) propose the use of an Other Transaction Agreement (OTA) as specified in 10 U.S.C. 4021/10 U.S.C. 4022 and related implementing policies and regulations. The DON may choose to use a Basic Ordering Agreement (BOA) for Phase I and Phase II awards.

Funding Limitations. In accordance with the SBIR and STTR Policy Directive section 4(b)(5), there is a limit of one sequential Phase II award per small business concern per topic. The maximum Phase I proposal/award amount including all options is $240,000. The Phase I Base amount must not exceed $140,000 and the Phase I Option amount must not exceed $100,000. The maximum Phase II proposal/award amount including all options (including TABA) is $2,000,000 (unless non-SBIR/STTR funding is being added). Individual SYSCOMs may award amounts, including Base and all Options, of less than $2,000,000 based on available funding. The structure of the Phase II proposal/award, including maximum amounts as well as breakdown between Base and Option amounts will be provided to all Phase I awardees either in their Phase I award or a minimum of 30 days prior to the due date for submission of their Initial Phase II proposal.

Contract Deliverables. Contract deliverables for Phase I are typically a kick-off brief, progress reports, and a final report. Required contract deliverables (as stated in the contract) must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Payments. The DON makes three payments from the start of the Phase I Base period, and from the start of the Phase I Option period, if exercised. Payment amounts represent a set percentage of the Base or Option value as follows:

Days from Start of Base Award or Option	Payment Amount
15 Days	50% of Total Base or Option
90 Days	35% of Total Base or Option
180 Days	15% of Total Base or Option

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

PHASE II GUIDELINES

Evaluation and Selection. All Phase I awardees may submit an **Initial** Phase II proposal for evaluation and selection. The evaluation criteria for Phase II is the same as Phase I (as stated in this BAA). The Phase I Final Report and Initial Phase II Proposal will be used to evaluate the small business concern's potential to progress to a workable prototype in Phase II and transition the technology to Phase III. Details

on the due date, content, and submission requirements of the Initial Phase II Proposal will be provided by the awarding SYSCOM either in the Phase I contract or by subsequent notification.

Awards. The DON typically awards a Cost Plus Fixed Fee contract for Phase II; but, may consider other types of agreement vehicles. Phase II awards can be structured in a way that allows for increased funding levels based on the project's transition potential. To accelerate the transition of SBIR/STTR-funded technologies to Phase III, especially those that lead to Programs of Record and fielded systems, the Commercialization Readiness Program was authorized and created as part of section 5122 of the National Defense Authorization Act of Fiscal Year 2012. The statute set-aside is 1% of the available SBIR/STTR funding to be used for administrative support to accelerate transition of SBIR/STTR-developed technologies and provide non-financial resources for the small business concerns (e.g., the Navy STP).

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

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Version 4

N252-074 TITLE: Materials Optimization for 81mm M299 Ignition Cartridge System

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a suitable material that meets the M299 Ignition Cartridge (IC) Tube Assembly Technical Data Package (TDP) requirements and an engineering solution to optimize the current design while aligning with historical pressure characteristics and current range precision requirements.

DESCRIPTION: The current M299 IC system consists of the following critical components:

a. Flash Tube Assembly
b. Tube Assembly
c. Firing Plug
d. Head
e. Black Powder Pellet
f. Primer 150D
g. Body (drawing 9293428)
h. Cap (drawing 9294756)

The Tube Assembly contains the Tube as the base structure and the Heat Shrinkable Tubing that is applied to the exterior. The current Tube material is a sulfate paper produced using the Kraft wood pulping process and coated with paraffin wax. The current configuration of the Tube is spiral wound such that each layer of the Cartridge Paper is secured with adhesive. The Tube Assembly is positioned over the Flash Tube Assembly, secured to the IC body with adhesive, loaded with M9 flake propellant, and sealed with the Cap and applied adhesive.

The current Tube dimensions are 4.770 +/- 0.015 inches in length, an inner diameter of 0.562 +0.005/-0.000 inches, and a thickness of 0.041 +0.000/-0.010 inches. These dimensions are required to ensure there is proper spacing within the ignition cartridge to provide a cavity for the propellant.

The M299 IC specification (MIL-DTL-32247) allows for a wide range in Tube bursting strength. MIL-DTL-32247 paragraph 3.11 defines the bursting strength requirement as the following:

a. "The tube shall not burst at a compression load of 57 pounds, minimum, but shall burst at a compression load of 192 pounds, maximum.

However, the Army is currently analyzing historical test data to optimize pressure characteristics and tighten the bursting strength requirement. Variation in the Tube wall thickness directly affects bursting strength and therefore the pressure characteristics during the IC deflagration event. IC paper tube materials, design, and manufacturing control processes are critical to maintaining quality and reliability." Currently, the Tube Assembly dimensional requirements allow for variation in the quantity of Cartridge Paper layers and subsequently affects the bursting strength, causing variability in pressure output. The thickness of the Sulfate Paper is a critical dimension as functional data appears to show a thicker paper and fewer layers causes increased variability in the bursting strength. Adhering to the historical TDP

dimensional requirements is considered a significant factor in aligning with historical pressure characteristics and maintaining range precision requirements.

The purpose of this SBIR topic is to prototype an engineering change to the M299 IC system to optimize the form, fit, and function of the Tube Assembly, minimize variability in the bursting strength, and align with historical range precision performance.

The solution does not need to be made from paper. Alternative materials may be considered.
Current Paper Requirements:

1. Paper Properties:
    a. Standard Commercial Grade Paraffined Cartridge Paper
    b. Color: Yellow No. 33538 of FED-STD-595
    c. Basis Weight (gsm): 104 – 117 (Grams per Square Meter)
    d. Grade: P03-05
    e. Thickness (inch): .0047 to .0053; thickness of one spiral wrap of material
    f. Tensile Strength: 75 lbs minimum along the grain; 20 lbs minimum across grain
    g. Stretchability: 2.0% minimum along grain; 9.0% minimum across grain
    h. Weight: 70 lbs per 500 24" x 36" sheets
    i. Bursting Strength (psi): 70+

2. Bursting Strength:
MIL-DTL-32247 paragraph 4.58 defines the Bursting Strength Verification as the following:
    a. "The tube shall be cut in half lengthwise and one of the halves placed in the fixture. The stop shall be completely removed or backed off so that it does not touch the punch. Place fixture in universal tester set for a travel speed of 0.125 inches per minute. Proceed with Test Number 1 and 2."
    i. TEST NUMBER 1: NO BURST
        1. Apply 57-pound compression load, remove tube. Item is rejected if evidence of bursting is observed. Tearing or puncturing of material shall be considered evidence of bursting.
    ii. TEST NUMBER 2: BURSTING
        1. If item is not rejected, replace it in fixture, using opposite end of tube and apply 192 pounds compression load. If item shows no evidence of bursting (punch travels through material), it shall be rejected.

3. Costs
The Flash Tube Assembly, of which the tube is a component, is estimated to cost $18. The redesigned tube should not increase the cost by more than 10%.

PHASE I: Develop concepts for the paper tube technology that meet the requirements described above. Demonstrate the feasibility of the concepts in meeting Marine Corps requirements by material testing and analytical modeling, as appropriate. Establish that the concepts can be developed into a useful product for the Marine Corps. Provide a Phase II development plan with performance goals and key technical milestones, and that will address technical risk reduction.

PHASE II: Develop a prototype Tube to be assembled to the Tube Assembly for evaluation to determine its capability in meeting the performance goals defined in the Description. Demonstrate technology performance through prototype evaluation and modeling over the required range of parameters. Use evaluation results to refine the prototype into an initial design that will meet Marine Corps requirements; and for evaluation to determine its effectiveness in an operationally relevant environment approved by the Government. Provide thirty (30) complete Tube assemblies that will be used in a live fire mortar testing

event. Prepare a TDP for submittal to the Joint Safety Review Board. Provide test support during the live fire testing. Prepare a Phase III development plan to transition the technology into Marine Corps use.

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the technology for Marine Corps use. Support the Marine Corps for test and validation to certify and qualify the system for Marine Corps use.

REFERENCES:

1. Department of Defense. "MIL-DTL-32247, CARTRIDGE, IGNITION, M299 FOR M821A1 HE CARTRIDGE FOR 81 MM MORTAR." 18 December 2023. https://www.thinkpdfs.com/standards/MIL-MIL-DTL-32247/
2. Sun, V. et al. "120mm Mortar Ignition Cartridge Design Improvement and Black Powder Replacement [Presentation]." National Defense Industrial Association Conference, 41st Annual Armament Systems: Gun and Missile Systems Conference & Exhibition 2006, March 30. https://ndia.dtic.mil/wp-content/uploads/2006/garm/thursday/sun.pdf
3. Department of Defense. "MIL-A-2550B, GENERAL SPECIFICATION FOR AMMUNITION." 8 February 1973. http://everyspec.com/MIL-SPECS/MIL-SPECS-MIL-A/MIL-A-2550B_8534/
4. Department of Defense. "MIL-STD-810H CHANGE 1, ENVIRONMENTAL ENGINEERING CONSIDERATIONS AND LABORATORY TESTS." 18 May 2022. https://cvgstrategy.com/wp-content/uploads/2023/04/MIL-STD-810H-change-1.pdf

KEYWORDS: Ignition; Cartridge; Paper; Tube; Burst; Mortar

Version 4

N252-075 TITLE: Outboard Engine Replacement

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment;Sustainment

OBJECTIVE: Modify a commercial four-stroke outboard engine capable of powering the Marine Corps Enhanced - Combat Rubber Raiding Craft (E-CRRC) and re-starting after immersion.

DESCRIPTION: This SBIR effort is to replace the Marine Corps Non-gasoline Burning Outboard Engine (NBOE) with a modified four-stroke commercial off the shelf (COTS) engine to power the E-CRRC. The modified COTS outboard must meet performance requirements for speed and payload when integrated with the E-CRRC, be capable of starting after immersion, integrate carrying handle(s), be capable of horizontal transport and storage, and meet noise and weight limitations. The immersion requirement is the most difficult and unique requirement to meet and is currently met using two-stroke outboard engines. Two-stroke outboard engines are no longer produced in the United States, presenting challenges with importing and authorization for use by the Environmental Protection Agency (EPA), as well as parts availability. Modification of a commercial four-stroke outboard will significantly increase reliability, fuel efficiency, and parts availability while still meeting the unique requirements of the Marine Corps.

The outboard engine must meet the following performance and integration requirements for use with the E-CRRC:
1. Four-stroke commercial outboard.
2. Propel the E-CRRC to 18 kts while transporting 2400 lbs, excluding boat, motor, fuel, and all Stock List-3 (SL-3) equipment, such as the outboard engine maintenance and on-board diagnostic kit.
3. Immersion and rollover protection.
    - Capable of immersion to depths of 3’ while inverted for 30 minutes.
    - Start within six minutes after immersion.
4. Capable of horizontal transport/storage.
5. Integrated carrying handle(s).
6. Non-reflective black exterior.
7. Not to exceed double hearing protection at the operator’s station per MIL-STD-1474E.
8. Weight not to exceed 270 lbs.
9. Starting method: Electric start, primary; Manual start, secondary
10. 20” shaft length.
11. Midsection transom clamps.

PHASE I: Define and develop a concept for a modified four-stroke outboard engine capable of meeting the Marine Corps requirements. Technical approaches may include purchase of multiple four-stroke outboard makes and models to determine technical feasibility of modifications required to achieve the results. Deficiency areas where four-strokes are unable to meet the requirements will be identified and concepts to overcome the deficiencies presented. Multiple approaches may be assessed in Phase I, but a prototype hardware solution (or multiple solutions) must be achievable within the time and funding scope of Phase II. Phase I proposals shall include but are not limited to: discussion of the performer’s experience and knowledge of relevant technologies and their application to marine outboard engines; the performer’s capability to develop and rapidly prototype hardware required to modify a four-stroke outboard engine; ability to estimate the cost and weight of the final concept; the performer’s ability to transition from concept and prototype to manufacturing. Supplementary material should include recent examples of the performer’s ability to develop, refine, and qualify relevant systems for use in military and marine operational environments and produce systems in significant quantities, utilizing either internal resources or via teaming or licensing agreements. Phase I proposals may include preliminary concepts which demonstrate understanding of the relevant trade spaces.

PHASE II: Develop and deliver at least one prototype four-stroke outboard engine suitable for demonstrating the required speed and payload performance, immersion capability, carry handles, and horizontal transport/storage. The prototype system is not expected to be optimized for physical and integration requirements, i.e., color and E-CRRC integration/attachment at this time, however, concepts to achieve all physical and integration requirements shall be presented. An estimate for the final carry weight should be provided with high confidence that the final weight requirement is achievable. Provide a plan for both low and full rate production, describing proposed fabrication capabilities and teaming or licensing agreements, if applicable. Provide a cost estimate for non-recurring initial resources and facilities as well as production of the final outboard engine solution.

PHASE III DUAL USE APPLICATIONS: Transition the prototypes to manufacturing and production of the final system suitable for use on the Marine Corps E-CRRC. Field the final system to Marine Corps units utilizing the E-CRRC.

There is potential for other military service interest in the engine or scaling of the technology and modifications development for larger or smaller outboards on multiple platforms.

REFERENCES:

1. Outboard – Outboard motor – Wikipedia https://en.wikipedia.org/wiki/Outboard_motor
2. Combat Rubbing Raiding Craft – Combat rubber raiding craft – Wikipedia https://en.wikipedia.org/wiki/Combat_rubber_raiding_craft
3. Marine Corps Systems Command, Program Manager Infantry Combat Equipment, COMBAT RUBBER RECONNAISSANCE CRAFT
4. Product Information Sheet. https://www.marcorsyscom.marines.mil/Portals/105/PfM/GCES/PM_ICE/Info%20Sheets/Amphibious%20Reconnaissance%20Team/CRRC_PIS_2018.pdf?ver=2018-05-16-110754-940
5. Marines.mil. 31st MEU BLT 2/4 Small Boat Sustainment, Video posted 13 Dec 2024 https://www.cdi.marines.mil/MOC/Draft-Main-Page/videoid/946975/

KEYWORDS: Four-Stroke; Outboard; Engine; Boat; Speed; Payload

Version 4

N252-076 TITLE: Augmented Reality Vehicle Maintenance & Training System

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Sustainment

OBJECTIVE: Develop an Augmented Reality (AR) capability that aids a maintainer with replacing the brake pads, brake disk, fuel filter, and water separator on a Ultra Light Tactical Vehicle (ULTV). The AR capability shall provide a reach back function in which the video feed of the maintainer's perspective can be wirelessly live streamed from one Electronics Maintenance Support System (EMSS) to another EMSS. The EMSS shall be the host system for the AR capability. The Prototype Device shall be designed with a platform agnostic architecture, in which the capability can be expanded to maintenance actions on multiple Marine Corps platforms.

DESCRIPTION: The problem at hand that the prototype device is intended to solve is a lack of ground vehicle maintenance training and overall maintenance experience. Vehicle maintainers in the Marine Corps have a wide range of varying experience, with some younger Marines having little to no experience with vehicle maintenance. To improve upon this knowledge gap, the Marine Corps is exploring AR to act as a real-time visual aid while conducting vehicle maintenance. The AR capability prototype device is intended to automatically detect vehicle components that are integral to the maintenance task at hand. The corresponding authoritative technical manuals and relevant schematics will be presented to the maintainer automatically upon component detection.

Another important objective of the AR capability prototype device is to provide a reach back capability in which an inexperienced Marine can contact a Subject Matter Expert (SME) in real-time. The end goal is to be able to have the SME be in an entirely different location than the maintainer, be able to see the live visual feed of the maintainer and provide resolution to the issue at-hand. This capability will directly support the Marine Corps approach of Expeditionary Advanced Base Operations (EABO).

The following is a detailed list of the scope of this SBIR effort:

- Develop an AR capability that:
  - can visually aid the user with replacing the brake pads, brake disk, fuel filter, and water separator on a ULTV, using ULTV schematics and technical manuals as the authoritative source.
  - can replace all four ULTV components: brake pads, brake disk, fuel filter, and water separator (Threshold (T)); replace all four ULTV components listed in the threshold, as well as any additional maintenance actions beyond those four (Objective (O)).
  - integrate with the EMSS hardware and software configuration.
  - is capable of providing a live video feed from the AR device to a different EMSS via wireless transmission.
- Augmented Reality Visual Aid Size:
  - Nominal hindrance to maintenance tasks while wearing a noticeable but not uncomfortable device (T); minimum hindrance to maintenance tasks while wearing a device with a small footprint and does not obscure the maintainer from the task at hand (O).
- Augmented Reality Visual Aid Device:
  - Wired operation between the device and EMSS (T); wireless operation between the Prototype Device and EMSS (O).
- Battery Life:
  - One-hour battery life while in operation (T); two-hour battery life while in operation (O).
- System Weight:

  - Two person portable with all accessories (62 lbs.) in accordance with MIL-STD-1472H (T); one person portable with all accessories (31 lbs.) in accordance with MIL-STD-1472H (O).
- Ruggedized:
  - Not ruggedized (T); ruggedized in accordance with MIL-STD-810H (O).

Note: The replacement cost should be considered during prototype device development. If the prototype device is not ruggedized, it is more likely to break and be in need of replacement.

PHASE I: Identify a technically capable solution for providing an AR capability within the hardware and software constraints of the EMSS. The required Phase I deliverables shall be a report which outlines the hardware and software approach of the AR capability. The report is expected to include the following:

- Analysis of AR hardware solutions, with a definition of pros, cons, tradeoffs, and a recommended prototype hardware solution
- Analysis of AR software solutions, with a focus on ease of development, software supportability, integration into EMSS, cyber compliance considerations, and future integration/expandability of the capability.
- Analysis of existing ULTV models and schematics, with a determination on if the existing Technical Data Package is sufficient to produce a prototype solution. If the models provided are insufficient, the proposal shall include a concept for creating the models utilizing 3D scanning methods. A ULTV will be made available to the prototype developer to accomplish this effort.

PHASE II: Produce and validate an AR capability that can assist a maintainer with changing out the brake pads, brake disk, fuel filter, and water separator on a ULTV; and provides a wireless video feed reach back capability (O). The prototype device shall be capable of identifying components that are integral to performing the steps needed to change the ULTV brake pads, brake disk, fuel filter, and water separator. Once a component is identified, the prototype device shall visually display the corresponding Technical Manual and schematic information to the user.

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the final product into its EMSS.

The commercial automative industry has started to integrate AR maintenance aids into their processes and practices. The government prototype device solution will be open source, which means that commercial automative industry companies that have not adopted this capability could utilize the prototype device solution.

This solution also could be utilized in the civilian environment for personal maintenance actions. The older generation is more knowledgeable on conducting car maintenance than the younger generation. This solution could be sold as a training aid to schools or vehicle hobbyists that need a way to teach and learn vehicle maintenance.

Additionally, this capability could have uses in different commercial sectors, such as electronics, home repair, plumbing, etc.

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KEYWORDS: Augmented; Reality; Training; Aid; Maintenance; Decision; Artificial Intelligence

Version 4

N252-077 TITLE: Tactical Sled Loader (TSL) Material Handling System and Cargo Sled

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Develop a Tactical Sled Loader (TSL) to provide the Marine Corps with the capability to rapidly load/offload munitions and cargo with an emphasis on speed to avoid enemy detection and targeting cycles plus the capability to transfer equipment and supplies with a variety of dimensions and weights to and from aircraft and ground transportation assets.

DESCRIPTION: The need for a newly developed TSL originated from a Deliberate Universal Needs Statement (DUNS) from INDOPACOM. The DUNS was vetted through the capabilities portfolio integration board and received a decision memo to pursue and develop solutions. The TSL consists of a Sled Loader hereafter referred to as the Loader and a Cargo Sled hereafter referred to as the Cargo Sled. The Cargo Sleds shall vary in size (primarily their width) to accommodate the various aircraft Cargo cabin dimensions (MV-22 Osprey, CH-53K, C-130). The Loader shall be one size and able to be transported in an MV-22 Osprey (the smallest transport aircraft).

The TSL system shall be equipped with a Loader designed to facilitate the efficient loading and unloading of cargo-laden Sleds to and from assault support aircraft, including but not limited to the MV-22, CH-53K, and C-130.

Tactical Sled Loader (TSL) Requirements

1. Shall employ a combination of manually operated winches and friction-reducing devices (such as rollers) to enable operators to smoothly transfer the loaded Cargo Sled between the Loader and the aircraft's Cargo drop down hatch.
2. Shall have an adjustable height between 25" and 45", ensuring compatibility with various assault support aircraft mentioned above by accommodating differences in Cargo hatch height and facilitating the loading and unloading of the aircraft (MV-22 lowest point and C-130 highest point) and allowing for independent modification of the Loader's pitch and roll. (Note: The platform of the Loader must remain integrated and move uniformly during adjustments. Any height adjustments must be performed manually, requiring continuous user input, and must not damage the Cargo. All hydraulic systems used for this purpose shall employ braided steel lines.)
3. Shall be designed for easy towing and repositioning, both when loaded and unloaded, utilizing standard tow vehicles or by hand.
4. Shall be equipped with tie-down points and parking brakes to ensure stability and safety during loading and unloading operations. (Note: These parking brakes must be sufficiently robust to securely hold a fully loaded TSL, preventing any risk of tipping or slipping when stationed on a flight deck or in a hangar. Some sort of visual indicators that can warn an operator that the load is reaching an unsafe position (e.g., dial indicator) should be considered.
5. Shall contain an integrated, adjustable guide system incorporated into the Loader design to prevent the Cargo Sled from deviating off course during loading and unloading procedures, ensuring a secure and controlled operation; to enable the Loader to expand by a minimum of 8 inches on each side to accommodate Cargo Sleds designed for use in larger aircraft such as C-130 and CH-53K; that is easily retractable to accommodate smaller Cargo Sleds used on MV-22 osprey aircraft; to prevent any lateral movement of the Cargo Sled, ensuring it remains securely on the Loader during operations; to avoid any damage upon contact between the Cargo Sled and the Loader; and to adjust to accommodate Cargo Sleds ranging from 63 to 85 inches in width.
6. Shall feature a versatile four-wheel steering configuration, designed for both towing behind military vehicles and manual pushing or pulling.

7. Shall interface seamlessly and provide adequate support to the small Cargo Sleds developed specifically for MV-22, and large Cargo Sleds designed specifically for C-130, and CH-53K aircraft, collectively referred to as the aircraft, during both loading and unloading operations.
8. Shall be constructed robustly to able to withstand the stresses, shocks, vibrations, and other conditions associated with towed movement and rail, truck, and air transport.
9. Must be compatible for towing with a variety of small vehicles typically found at air terminals, in addition to military vehicles. [Ref 4].
10. Must enable controlled transfers of loaded Cargo Sleds to and from the aircraft Cargo area with all equipment necessary for these transfers, such as winches, rollers, and guide rails, permanently integrated into the Loader. (Note: Leveraging the aircraft's winch system for transferring Cargo Sleds from the Loader to the aircraft is permissible.)
11. Shall incorporate a manually operated winch system, potentially comprised of multiple winches, that must facilitate the movement of the Cargo Sled from the aircraft's Cargo area to the Loader; and should provide pulling capabilities at the front-center, front-right, and front-left of the Cargo Sled. (Note: Simultaneous pulling from the front-right and front-left shall be equivalent to a central pulling load. The intent is to reduce the personnel required to operate this system for cargo movements over legacy methods.)
12. Must not exceed dimensions of 140 inches in length, 65 inches in width, and 25 inches in height in its storage configuration; the height shall be adjustable from 25" to 40" and the width shall be expandable from 65" to 85".
13. Shall not exceed 3,500 lbs in weight, inclusive of all operating fluids and hydraulic reservoir at operational levels.
14. Must be capable of handling loads up to 10,000 pounds (i.e., the "rated load").
15. Must maintain stability when lifted via forklift in an unloaded state, and should allow lifting from both sides, adhering to section 3.3.4.2 of MIL-S-8512D. (Note: Reinforced forklift tine guides, preferably pockets, must be provided for safe and positive insertion, ensuring no damage occurs during lifting. The center of gravity of the Loader, when unloaded, shall be clearly marked on the Loader itself.)
16. Lift/Tie down points conforming to MIL-STD-209K, Interface Standard for Lifting and Tie-down Provisions for all forms of transportation, which serve multiple purposes: lifting the Loader, tying down the Loader, and securing the Cargo Sled to the Loader. (Note: The Loader must not be lifted in a loaded state but should be secured adequately when loaded.)
17. Must be capable of operating in temperatures ranging from -25° to +125°F, capable of operating in 6" Threshold, 24" Objective and near salt water, and compatible with amphibious and commercial shipping.
18. Must be capable of operating in day and night conditions and be capable of providing adequate operator-adjustable lighting for night operations.
19. Must have axle weights (front and rear) that meet the weight restrictions of a C-130, CH-53K, and MV-22 military aircraft.
20. Must be equipped with service brakes capable of stopping and holding the TSL with rated load either forward or backward on a 30-degree slope; and a parking brake that must restrain the TSL with rated load on at least a 30-degree slope.
21. Must be capable, with rated load, of negotiating both an angle of approach and an angle of departure of at least 20 degrees for negotiating the ramp of a landing craft and shall be able to negotiate a longitudinal slope of at least 45 percent in both forward and reverse at a speed of not less than 2 mph; and of negotiating at a 15 percent cross slope in full circle operation in both directions at maximum steer angle without any tire leaving the ground.
22. Must be capable, with rated load, of traversing uneven and unprepared surfaces (rough terrain) to include sand, snow, and mud.
23. Must be capable, with rated load, of traveling forward on flat firm ground at speeds of at least 8 mph and of executing a 360-degree turn.

24. Must be equipped with pneumatic tires that meet the load requirements of the TSL.
25. Must be corrosion resistant (i.e., must meet or exceed current ground equipment protection standards) during unprotected storage and from the effects of a salt-water environment.
26. Must be equipped with a 24-volt negative-ground electrical system or at a minimum have a 12/24 dual system with 24-volt alternator, 24 starter and battery equalizer. (Note: A NATO standard electrical slave receptacle shall be provided.)
27. Must contain commercial standard diagnostic capabilities. (Note: Design shall be such that maintenance and repair can be conducted with common tools wherever possible. No unusual mechanical skill should be required at any echelon of maintenance.)
28. Must be equipped with a rear axle that can oscillate above and below the horizontal axle centerline.
29. Must be equipped with integral tie-down brackets, a rear towing pintle and appropriate lifting eyes. (Note: The towing pintle hook shall be accessible for removal or adjustment.)
30. Must meet applicable DoD human engineering, health, and safety standards and employ user-friendly features for personnel operating in or near the TSL throughout its life cycle. (Note: The TSL design and engineering must be able to achieve certification to handle conventional ordnance and transportability onboard MV-22, CH-53K, and C-130 aircraft.)
31. Must be capable of being embarked and disembarked, without disassembly, aboard all roll-on/ roll-off (RO/RO) ships and amphibious shipping/craft.

For Chemical, Biological, Radiological, and Nuclear, Biological, (CBRN) Survivability, only Chemical, Biological, and Radiological contamination survivability is required. Nuclear survivability is not required. Two types of Cargo Sleds will be developed to accommodate the dimensions of the supported aircraft: a small Cargo Sled designed for use with MV-22 and a large Cargo Sled developed for use with CH-53k and C-130 aircraft.

Cargo Sled Requirements

1. Shall have side, forward, and aft fork pockets to enable lifting from all sides.
2. Shall have a weight capacity of 10,000 lbs. (i.e., “rated load”).
3. Must interface with the aircraft Cargo deck rollers and floor locking systems of aircraft.
4. Must have adequate tie down points to accommodate a range of loads.
5. Will have weapons interface adapter positions for the install of bomb saddles and or missile cradles to support the All Up Round Transport of Weapons as with the MHU-110 munitions trailers.

PHASE I: Develop concepts for a TSL and Cargo Sleds that meet the requirements described above. Demonstrate the feasibility of the concepts in meeting Marine Corps requirements. Establish that the concepts can be developed into a useful product for the Marine Corps. Feasibility will be established by material testing and analytical modeling, as appropriate. Provide a Phase II development plan with performance goals and key technical milestones, and that will address technical risk reduction.

PHASE II: Deliver two working TSL prototypes and three Cargo Sleds for evaluation to determine their capability in meeting performance goals defined in the Description above. Demonstrate technology performance through prototype evaluation and modeling over the required range of parameters. Evaluation results will be used to refine the prototype into an initial design that will meet Marine Corps requirements; and for evaluation to determine its effectiveness in an operationally relevant environment approved by Government. Prepare a Phase III development plan to transition the technology to Marine Corps use. The technology should reach Technology Readiness Level (TRL) 6/7 at the conclusion of this phase.

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the technology for Marine Corps use. Support the Marine Corps for test and validation to certify and qualify the system for Marine Corps use. The prototypes shall be TRL 8 at the conclusion of testing.
Commercial applications may include, but not be limited to, airline industry, logistics, and warehouse/manufacturing operations.

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KEYWORDS: Material Handling; Munitions; Sustainment; Contested Logistics; Aircraft; Cargo; Transportability; Equipment; Distribution; Multi-Domain

N252-078 TITLE: Night Passive Aiming System

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a low-cost Night Passive Aiming System (NPAS) for individual small arms.

DESCRIPTION: This SBIR effort is to develop a low-cost NPAS for individual small arms engagement at night. Marines currently utilize night vision goggles (NVGs) with weapon mounted, near-infrared (NIR), laser pointer/illuminator systems [Ref 1] to place projectile aimpoints on targets and extend target detection distances when ambient illumination is insufficient for image intensifier (I2) sensor operation. Emissions from laser devices are easily detectable by hostile forces equipped with similar night vision capabilities or modern low-light silicon imaging sensors, as found in cell phones or webcams. The nature of small arms projectile ballistic flight also constrains laser pointer engagement distances as the operator must eventually raise the aimpoint above the target's head to compensate for bullet drop. Currently fielded night vision weapon sights, both I2 and thermal imaging, provide corrected aimpoint reticles for long range engagements, but are not at a price point amenable for issuance to every infantry Marine. Their relatively narrow field of view also inhibits the user's ability to rapidly acquire and engage close range targets when compared to NVG/laser engagements.

Technical approaches should emphasize both low cost and light weight (no greater than contemporary small arms laser pointer devices). Multiple approaches may be assessed in Phase I, but a prototype hardware solution (or multiple solutions) must be achievable within the time and funding scope of the base Phase II effort. In all phases of the effort, the proposer shall provide target and environmental modeling assumptions and sensor/optical parameters where applicable.

The prototype system is not expected to be optimized for power consumption, nor to meet full military operational environment requirements; however, it shall be suitably robust for use outdoors in temperate climates. The prototype system should be capable of live fire demonstration on the M27 IAR. Batteries, if applicable, shall be removable by the operator. The prototype should include an external power capability to operate on 120VAC power via an adapter and without internal batteries installed, if applicable.
For NVIPM modeling, the recommended relevant parameters are: 0.75 meter target characteristic dimension, V50 (detection) = 1.5 cycles, V50 (recognition) = 2.2 cycles, 2 Kelvin target contrast for thermal band imaging, and 25% target contrast for reflectance band imaging.

The solution concept of operation shall have sufficient field of view to observe projectile impacts on target, rapidly search for and detect targets, and maintain situational awareness of friendly forces approaching perpendicularly to the target area before they enter the cone of fire. Preferred solutions should not preclude the use of the SBNVG for continuous local situational awareness and dismounted movement at night.

The performer shall provide a plan for both low and full rate production, describing proposed fabrication capabilities and teaming or licensing agreements, if applicable. The performer shall provide a cost

estimate for non-recurring initial resources and facilities as well as production of sight systems based on step ladder pricing.

NPAS solutions:

1. shall be operable with the M27 Infantry Automatic Rifle (IAR), equipped with the M83 Squad Common Optic (SCO) and a muzzle-mounted suppressor (Threshold).
2. shall not preclude the use of NIR laser pointer/illuminator devices on the rifle forward handguard rails (note: positioning of laser pointer devices is flexible to accommodate operator preference).
3. shall assume all operators will also be equipped with the AN/PVS-31D Squad Binocular Night Vision Goggle (SBNVG), which has the ability to rotate monoculars individually away from viewing position if needed.
4. should accommodate, but not be dependent upon, a limited number of operators being equipped with clip-on night sights along the topside handguard rail (ex., AN/PAS-35 Squad Thermal System/Family of Weapon Sights-Individual) and/or supplemental thermal imaging capability for the SBNVG via the AN/PAS-29B Enhanced Clip-On Thermal Imager (E-COTI) (Objective).
5. may utilize or interact with components already issued to all operators, such as the SBNVG. If the solution includes a dedicated night imaging sensor or optically interacts with other sensors, it is recommended that proposers utilize the U.S. Army Night Vision Integrated Performance Model (NVIPM) for sensor range predictions. A copy of NVIPM software can be provided as Government Furnished Information upon contract award, however proposers are expected to have prior proficiency in use as training will not be provided by the Government.
6. shall provide the operator a 70% probability of detecting and engaging personnel targets [Ref 2] at no less than 300 meters in clear air starlight conditions (~1 millilux), without the use of active illumination sources.
7. should provide target recognition and engagement capability at no less than 500 meters in overcast starlight conditions (~100 microlux).
8. should be capable of performing the stated task in dirty battlefield and adverse weather conditions at no less than one-third the clear air range. (Note: Recognition shall be evaluated by the ability of the operator to detect and correctly count the number of upright personnel within a group with no more than 50% line of sight positional overlap between individuals presenting a frontal aspect.)

Phase I proposals shall include, but are not limited to, discussion of the performer's experience and knowledge of relevant technologies and their application to optical systems for small arms; the proposer's ability to model the size, weight, power, cost, and range performance of optical systems for small arms applications; and the proposer's capabilities for rapid prototyping and relevant prior examples. Supplementary material should include recent examples of the performer's ability to develop, refine, and qualify relevant systems for use in military operational environments and produce systems in significant quantities, utilizing either internal resources or via teaming or licensing agreements. Phase I proposals may include preliminary concepts that demonstrate understanding of the relevant trade spaces.

PHASE I: Define and develop a concept for a NPAS. Establish the feasibility of the concept. Prepare a Phase II plan.

PHASE II: Develop and deliver at least one hardware system prototype suitable for demonstrating the range performance, operator employment, and approximate size and weight of the preferred concept on relevant small arms systems. Prepare a Phase III commercialization/transition plan.

PHASE III DUAL USE APPLICATIONS: Further refine the sight system for optimization of size, weight, power, and manufacturing cost and for survivability in the conditions associated with weapon firing shock and the military operational environment. Deliver sufficient representative sight systems to allow qualification and Marine user evaluation for refinement prior to full rate production.

Dual use applications include law enforcement and civilian self-defense or hunting, subject to ITAR and local government restrictions. Related applications may include augmented reality and machine vision systems.

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KEYWORDS: Sensors; Optics; Fire Control; Sights; Small Arms; Weapons; Targeting

N252-079 TITLE: Binary-Level Automated Vulnerability Detection and Patching without Source Code

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop innovative approaches to automatically find and fix software vulnerabilities in binaries without source code. The capability should be robust enough not only to identify zero day exploits and vulnerabilities, which can be weaponized for offensive purposes but, also implants (for defense against supply chain attacks) and malware.

DESCRIPTION: Due to the prevalence of programmers copying and pasting code into their projects, or the inclusion of libraries of unknown origin or quality, the security of the software that underpins critical systems is always in question. Because of this, methods to quickly secure new and existing critical software used in the Fleet is needed for all Program Management Activities and Program Executive Offices. Current techniques to secure software involve manual vulnerability discovery and remediation using subject matter experts (SME) and typically requires access to the source code. However, the source code is usually not available for analysis especially for legacy applications, weapon systems, control systems, and communication systems whose software is proprietary. For this SBIR project, the small business awardee will develop novel approaches to automatically perform security assessments on compiled binaries of multiple instruction set architectures to detect known and unknown vulnerabilities (greater than 90% success rate) and automatically develop patches for any found vulnerabilities.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Determine the technical feasibility of binary level automated vulnerability detection and patching without source code including:
1. Determination of the major challenges and preliminary feasibility of software algorithms.
2. Development of an initial concept design that supports binary level automated vulnerability detection and patching without source code.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate a prototype for binary-level automated vulnerability detection and patching without source code. The prototype deliverables should include:
1. Design and development the algorithms required to perform binary-level automated vulnerability detection and patching without source code.
2. Demonstrate the ability of the prototype to harden vulnerable binary software.
3. A technical roadmap that takes the program through Phase III must be part of the final delivery for Phase II.

Work in Phase II may become classified. Please see note in Description paragraph.

PHASE III DUAL USE APPLICATIONS: Complete final testing, perform necessary integration and transition for use in monitoring operations/applications with appropriate platforms and agencies, and future combat systems under development.
Commercially, this product could be used to enable security monitoring.

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KEYWORDS: Source code; binary; vulnerability; Artificial Intelligence; Machine Learning; AI/ML; software; cybersecurity

N252-080 TITLE: Multi-Spectral Quantum Dot Array

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber;Quantum Science

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design a multi-spectral Quantum Dot (QD) array with a Read-in Integrated Circuit (RIIC) to provide the capability for system design and integration of a high-fidelity Screen Projector (SP) to support testing and evaluation (T&E) of a multi-spectral sensor and seeker. Each array pixel will be populated with between two to four multi-spectral quantum dot emitters and can demonstrate narrow band spectral output in either the ultra-violet (UV), visible, Near Infrared (NIR), Mid-Wave IR (MWIR) band 4a and b, or Long-wave IR (LWIR).

DESCRIPTION: This SBIR topic will investigate multispectral narrow-band QD arrays with emission wavelengths centered for chemical and remote sensing, as well as seeker and sensor modeling and simulation. Multispectral emission from UV to LWIR is potentially useful for chemical spectral analysis, remote sensing, and SP threat engagement simulation as part of a scene projection system. A multispectral band QD array with a narrow-band QD emitter providing electromagnetic radiation in multiple spectral bands is highly beneficial for DoD applications. The narrow-band emission of QDs will enable more precise spectral emissions for chemical composition analysis; provide the core technology for smaller, lighter, and more versatile projection systems; and provide the capability for better wavelength selection for remote sensing applications.

The Navy desires a multispectral 1024 X 1024 emitting array with multiple QDs designed onto each array pixel. Each QD will have a narrow band emission of less than 100 nanometers for better wavelength precision of application-specific operations on chemical and remote sensing. The individual QDs integrated onto each pixel will demonstrate apparent temperatures above 1500K for improved brightness levels needed for SP and remote sensing applications. Multispectral QD arrays will improve the T & E capability of modern and future seeker and sensor projection systems for more effective Hardware-in-the-loop (HITL) and live-virtual-constructive (LVC) testing of multispectral threat detection and warning systems. The RIIC and QD emitter array must provide up to 5 orders of magnitude dynamic range with at least 16-bit resolution. The crosstalk between QD wavelength-specific emitters of less than 1% is needed between all wavelengths but is particularly important for the MWIR bands 4a and 4b. These attributes improve spatial and spectral resolution for chemical and remote detection and modeling projection systems for improved warfighter battlefield survivability.

The proposed approaches will include designing, fabricating, and characterizing the multispectral QD arrays to match Navy-defined in-band emitter wavelength and ranges. An electronically multiplexed QD array suitable for high-fidelity hardware-in-the-loop will have a Phase I LED array approach designed and a Phase II demonstration. It will allow the test programs to tailor flight test scenarios based on HITL test results, reduce flight hour requirements, and improve overall test efficiency. For chemical and remote sensing, this capability will support the warfighter's need to analyze and detect biological or chemical agents. The Navy needs enhanced multispectral scene projectors that are smaller and lighter for placement on HITL and flight line T & E.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for a multi-spectral QD array structure. Write a final report on the design and feasibility of a multi-spectral QD array structure, including the RIIC concept. The Phase I effort will include prototype plans to be developed in Phase II.

PHASE II: Develop a 1024 X 1024 multi-spectral array structure with independently controlled pixels by a RIIC. Demonstrate the multi-spectral narrow spectral band (
Work in Phase II may become classified. Please see note in Description paragraph.

PHASE III DUAL USE APPLICATIONS: Integrate a 1024 X 1024 multispectral QD array with a RIIC and develop a multi-spectral scene projector. Support transition of the Infrared Scene Projector (IRSP) developed in Phase III to Navy T&E laboratories.
A multi-spectral emitting array has potential application for industrial chemical sensing and safety protection. A multi-spectral scene projector has application for both firefighter and medical scenario training.

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KEYWORDS: Quantum; Read-in Integrated Circuit; RIIC; Near Infrared; NIR; Mid-wave Infrared; MWIR; Long-range Infrared: LWIR; multi-spectral scene projector; Electro-Optical and Infrared; EO/IR

N252-081 TITLE: Intelligent Radio Access Network for Beyond 5G Resilient Tactical Networks

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;FutureG;Integrated Network Systems-of-Systems

OBJECTIVE: Research and develop cost-effective communication systems for use in Anti-Access and Area Denial (A2AD) environments via platform-assisted Intelligent Radio Access Network operating as a Beyond 5G (B5G) base station or relay in Resilient Tactical Networks.

DESCRIPTION: Tactical operations are often set in remote locations where cellular infrastructure such as 5G are absent. Military operations such as intelligence, surveillance, and reconnaissance (ISR) can benefit immensely from high speed, low latency 5G communication links. The high-capacity links can enhance command and control (C2) with the high data rate 5G links enabling situation awareness between deployed troops. Another aspect to consider in practical tactical scenarios is user mobility, i.e., the soldiers/tactical vehicles with handheld 5G radios user equipment (UE) are not stationary. Mobility of users result in varying link conditions causing coverage holes in the network. The aerial 5G base station positioning must consider the dynamic link conditions due to mobility and other propagation effects such as Weather Disruptions, Jamming, and Distance/Fading loss due to signal strength. A traditional lookup table-based approach is not adaptive to learn from the dynamic UE environment.

This SBIR topic seeks the development of an inexpensive platform B5G communication system that can operate in contested/congested environments with challenging and dynamic realistic B5G operating conditions in both stationary and on-the-move (OTM) conditions. The platform 5G base station will extend 5G Radio Access Network (RAN) coverage and enable high data rate beyond Line of Sight (LoS) communication among the UEs. To provide this capability, a new intelligent link adaptation subsystem to the B5G RAN architecture is required to improve link resiliency by choosing the appropriate Modulation and Coding Scheme (MCS) scheme for the UE-to-base station link. Thus, the dynamic platform 5G-base station enables large-scale network coverage, UE mobility support, enabling networking to on-the-move for highly dynamic tactical mission consequently providing next generation of assured robust connectivity. The intelligent link adaptation module in conjunction with the platform 5G base station for multi-cell connectivity is envisioned to provide a significant performance leap in terms of UE mobility management, Quality of Service (QoS) specific high data rate guarantee, and sustain reliable connectivity in support of dynamic, remote tactical missions. The proposed scheme must be scalable and can easily accommodate multiple platform 5G-base stations and platform 5G relayers for very large network scenarios. The proposed approach must be able to perform multi-cell connectivity via multiple Unmanned Air Vehicles (UAV) equipped with 5G base stations and UAV 5G relayers and includes the following capabilities:

- High-Level Decision Maker—adaptive 5G UE support for efficient network routing strategy that minimizes 5G network control overhead ("Network control overhead" refers to the amount of additional 5G information required to maintain network services)
- Director—multifunction optimization and conflict resolution to accommodate 5G UE communication tasks,
- 5G Mesh Network Optimization and Learning Engine—dynamic, context-based learning recognizing a 5G disruption event, classifies the 5G disruption event then based on the type of 5G disruption, suggests a mitigation strategy for strengthening the 5G network topology and improving 5G network performance
- Weight Adjuster—discerning critical factors contributing to an effective 5G UE mesh network solution
- Compliant interfaces—seamless 5G UE connection to internal platform communication subsystems and external 5G systems

- Multi-objective reasoning in dynamically changing A2AD environments to enable transfer large data sets in minimal time in synchronous and asynchronous modes among UEs, using either or all the following: data throttling, data compression, Data Error Detection & Correction
- Context-based access management to minimize 5G UEs communication detection, intercept and targeting
- Efficient 5G system switching and mesh capability resource allocations in an A2AD environment
- Reinforcement learning framework that overcomes uncertainty and avoids reliance on static 5G MATLAB data flow model,
- Scalable across multiple different 5G transceivers to operate in A2AD dynamic contested environments
- Robust to different cyber environments through context-based authentication
- Software to sustain 5G link connectivity and if lost reacquisition of connections in A2AD environments
- Vendor-agnostic 5G equipment integration with UAVs and their respective systems and subsystems
- Hybrid decentralized approach for local decisions to support multiplatform 5G UEs collaboration in A2AD environments
- Near real-time mission feedback with reduced 5G system processing times
- Lightweight 5G signaling in a hierarchical command and control (C2) structure supporting battlefield applications with multiple 5G UE distributed platforms
- Negating radio frequency (RF), cyber takeover of unmanned and manned air vehicles
- Compliance with IEEE 802.15.3-2023: Standard for Wireless Multimedia Networks (Threshold (T))
- Compliance with IEEE 802.11ac: Wi-Fi 5 Standard (T)
- Compliance with IEEE 802.11ad: Multiple Gigabit Wireless System (MGWS) (Objective (O))
- Compliance with IEEE 802.11ax: Wi-Fi 6 Standard (O)
- Compliance with FCC CFR 47 Part 15 and Annex K regulations (T)

Proposers may use the multi-path transmission control protocol (MPTCP) published by the Internet Engineering Task Force (IETF). The MPTCP is implemented in layer 4 (transport).
Proposers may use vehicle-to-vehicle (V2V) or vehicle-to-everything (V2X) communication protocols specified by the 3GPP standards for 5G cellular networks.

While 5G networks provide significant benefits and opportunities in terms of high bandwidth and connectivity, traditional security models have major weaknesses in protecting the 5G network infrastructure. The unique characteristics of 5G networks include: (a) seamless connectivity of heterogeneous devices, (b) high network mobility, and (c) distributed network assets enabling access to cost-effective resources for all devices. While 5G networks enable new applications, the use of untrusted network elements arises major security concerns for DoD operations. Traditional network security models assume a known network perimeter as the critical resources to be protected against unauthorized access. The distributed resources and assets of 5G networks make the identification of network perimeter challenging – even impractical. Further, the high mobility of 5G networks requires a dynamic security and risk assessment. To address this issue, zero trust (ZT) principles must be part of the proposed B5G communication system. ZT principles include the trust of a subject requesting access to critical resources continually and dynamically assessed during the entire period of the access. In a 5G network specifically, the mobility of heterogenous devices in a varying environment calls for a dynamic model of the network state that provides information about the risks of accessing a particular resource by a given subject from a particular location.

PHASE I: Perform a trade study between system size, architecture, operating frequency, operating bandwidth, efficiency enhancement schemes, modulation format, data rate, power consumption, anti-jamming capabilities, Low Probability of Intercept (LPI), and Low Probability of Detection (LPD), and zero trust (ZT). Develop a system level model of a Unmanned Aerial Vehicle (UAV) 5G base station transmitter/receiver based on optimized parameters selected from the trade study. Develop a small set of 6–8 independent intelligent yet cooperating UAV 5G systems, which will coordinate their activities to perform some limited-scope 5G communication scenarios and missions. Validate a system level model of a 6–8 UAV 5G base station transmitter/receiver based on optimized parameters in an Extendable Mobile Ad-hoc Network Emulator (EMANE). Required Phase I deliverables, in addition to the standard Contract Deliverables described in the BAA instruction, will include a report with a demonstration plan, UAV 5G system hardware and software designs, and performance goals. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Fabricate and demonstrate 6–8 5G prototype hardware and associated software kits to be integrated and operated on 6–8 UAVs as prototype systems. Make available these UAV 5G prototype systems for evaluation to determine their capability in meeting the performance goals defined in the Phase I report and to perform to the Government's criteria for holding operational evaluation and capability demonstrations. Construct and demonstrate the operation of a TRL 6/7 5G UAV 5G prototype system-assisted Intelligent Radio Access Network operating as a B5G base station or relay in a Resilient Tactical Networks operating in a relevant over-the-air outdoor maritime environment. Incorporate the ZT principles and techniques developed in Phase I into the UAV 5G prototype systems. Deliver these UAV 5G prototype systems to the Government with an associated user manual, interconnect diagram, and a report documenting the results of the Phase II effort. Provide any virtual and/or on-site support requested by the Government during 5G prototype UAV systems' operational evaluation and capability demonstrations.

PHASE III DUAL USE APPLICATIONS: Finalize software development of the manned-unmanned directional mesh networking system with compatibility to open architectures and address any unique requirements for Manned-Unmanned interoperability with a particular data link(s), perform a more formal systems integration task to provide effective software interfaces to particular naval assets, perform operational testing, and participate in integrated demonstrations of Manned-Unmanned networking systems operations.

Demonstrate a field-ready software system with mature implementation in an open relevant operational environment. Perform technology insertion and program integration using engineering model of proposed product/platform or software. Develop and provide a full report of development, capabilities and measurements (showing specific improvement metrics). Provide a user's guide and other documents as necessary for Navy to recreate and use the demonstration capability or software/hardware component(s). Identify and summarize opportunities and plans for potential commercialization.

Military Application: Increase the number and types of platforms able to connect into the UAV network. Improve the throughput, latency, and reliability of existing airborne networking technologies. 5G transceivers provide high data throughput, frequency agility, and covert operation in A2AD environments and will support a variety of military applications, including wireless networks of multiple manned or unmanned ground vehicles supporting ground V2V, vehicle-to-infrastructure, UAS-to-UAS, and UAS-to-ground vehicle/infrastructure communication links.

Commercial Application: Results from this work have applicability to cellular telephone and data networks, to vehicular networks, and to WiFi networking technologies. Civilian applications of 5G technology includes mmWave long-range links such as last mile, small cell networks, mobile backhaul,

Internet of Things (IoT) wireless networks, and 5G networks that can benefit from the reduced interference and improved reliability for high-speed datalinks. In addition to commercial Unmanned Aircraft Systems (UASs) and ground vehicles mirroring the military applications, an emerging Vehicle-to-everything (V2X) market will benefit from the proposed transceiver capabilities, especially with the wide adoption of autonomous vehicles.

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KEYWORDS: Tactical Airborne Networks; Mesh Networks; Directional datalink; Backward Compatibility; Time Division Multiple Access; TDMA; Experimentation

Version 4

N252-082 TITLE: Adaptive Bypass Valve for Solid Fuel Ramjet Missile

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment;Trusted AI and Autonomy

OBJECTIVE: Research and design a state-of-the-art airflow bypass valve between the inlet and the combustor in a solid fuel ramjet missile to control the amount of airflow by positioning at any degree of opening from off to on and on to off, thus allowing the control of thrust, resulting in potential range extension.

DESCRIPTION: Ramjets are ideally suited for supersonic flight within the atmosphere. Among the different ramjet configurations, the Solid Fuel Ramjet (SFRJ) is the simplest and safest design with the highest density-specific impulse. However, the uncontrollability of the combustion process in the SFRJ engine is still the main challenge in this technology. Complex issues include engine unstart, fuel regression complexity, and flame extinction.

This SBIR topic is soliciting innovative proposals for researching, designing, and prototyping a state-of-the-art airflow bypass valve between the inlet and the combustor in supersonic SFRJ missiles.
The electromechanical valve precisely controls the airflow in the combustion system by splitting the inlet airflow into the combustor stream and the annular bypass duct stream around the combustor. Bypassing air around the fuel grain is a promising technique to regulate thrust and improve combustion efficiency. The feasibility of this technique is supported by preliminary development of this adaptive valve. The high-speed precision actuator of this valve adjusts the size of the valve opening in response to a signal from a separate control system unit which is not part of this topic. An adaptive valve would control the amount of airflow by positioning at any degree of opening from bypass ratios of zero to the maximum permissible while maintaining ramjet ignition. The modular design of the valve is highly recommended as it promotes its integration on current and future missile systems. If electrical power is required, an independent power supply is preferred. A successful proposal will address these four areas: bypass valve design, actuator design, internal airflow efficiency, and power management. This proposed research project is expected to lead to a state-of-the-art device that will advance supersonic missile technologies and defense capabilities.

Current Specification:

- Mach range: 2-5
- Altitude: 3-70 Kft msl
- Temperature: Up to 2000 F
- Pressure: Up to 400 psi
- Weight: Low for missile
- Power: Independent Battery (if required)
- Materials: Any

PHASE I: Develop innovative approaches to design an adaptive bypass valve, actuator, and power supply for implementation on an SFRJ missile system. Prepare a Phase II plan.

PHASE II: Build and prototype a state-of-the art bypass valve and demonstrate its performance. Mature the transitioning technology during this interim technology maturity phase, depending on the Acquisition Program Office (PMA) requirements.

PHASE III DUAL USE APPLICATIONS: Integrate the Phase II-developed bypass valve prototype on a supersonic SFRJ missile system and flight test the complete system. Transtion the integrated technology to acquisition programs.

Dual uses of the developed technology may include space explorations, martian ramjets, and interstellar ramjets.

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KEYWORDS: Propulsion; Solid Fuel Ramjet; SFRJ; Supersonic Missile; Adaptive Control; Throttling; Weapon Range Extension

Version 4

N252-083 TITLE: Novel High Performance and Multifunctional Sandwich Composite Structures

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials

OBJECTIVE: Develop advanced sandwich composite structures with improved bond strength and multifunctionality via performance driven design customization.

DESCRIPTION: Sandwich composites are high-performance structures due to their lightweight, high strength-to-weight ratio, impact resistance, and high specific energy absorption capabilities. Honeycomb sandwich composite materials are specifically utilized in both aerospace engineering and marine engineering applications. The performance of the sandwich composite structure is dependent on the type and material selection of the core and face-sheets. In conventional sandwich structures, the core provides compressive and shear strength, and the face-sheets provide bending strength and energy absorption capacity. Core and face-sheet interface debonding is a common failure observed in sandwich composite structures. Current fabrication methods are operating within a limited design space that cannot provide a rational load path for complex geometries subjected to multiaxial loading.

The Navy seeks development of advanced sandwich composite structures with improved bond strength and multifunctionality via performance driven design customization. Innovative core and joining method designs are essential to increase the load-carrying capability without penalizing weight, cost, and fabrication time. Existing 3D printing technology can produce complex core topologies but increasing bonding strength between the face-sheets and core is still an obstacle. The design of the core with improved interface bonding provides a unique challenge and opportunity to develop novel sandwich composites structures.

PHASE I: Develop high-performance sandwich composite structures that have improved bond strength. and additively manufacture different cellular cores with optimized designs. Demonstrate at least one multifunctional capability of the developed structure. The core materials to be used should have comparable properties to aluminum or Nomex cores. Conduct numerical modeling for the optimization of process parameters such as laser power, laser scan speed, and powder feed rate. Perform mechanical tests like edge wise compression, flat wise tension, climbing drum peel, impact, open-hole compression, and compression after impact to demonstrate the desired performance reliability of the sandwich structure. Also, evaluate the structural performance using unit-cell based modeling and simulation. Incorporate a digital technique like machine learning to develop a model to optimize structural performance and process parameters.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Optimize the structural performance, design, and process parameters. Manufacture curved and complex sandwich composites based on the optimized cellular configuration. Conduct mechanical tests for performance evaluation. Expand the unit-cell modeling and the developed machine learning model. Compare the functionality and durability of the multifunctional prototype with a conventional baseline part.

PHASE III DUAL USE APPLICATIONS: Develop and manufacture a representative sandwich composite for naval aircraft and conduct testing to demonstrate its multifunctional capability and that the mechanical performance of the component meets or exceeds that of a conventional sandwich structure. Coordinate with industry partners that are manufacturing sandwich composites to facilitate the utilization and transition of the proposed technology.

The proposed technology can be integrated into rotorcraft and fixed wing aircraft with sandwich structures as direct replacements. These sandwich composites will not only have increased performance compared to conventional sandwich structures but will supplement their known benefits with additional multifunctional capabilities.

This technology will allow aircraft manufacturers to utilize and apply the benefits of this developed additive manufacturing process to critical and complex aircraft sandwich composites.

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KEYWORDS: Multifunctional sandwich composites; Cellular composite core; Composite core bonding; Additive manufacturing; Numerical modeling; Mechanical testing

Version 4

N252-084 TITLE: 400 Gigabit High Speed Data and Video Communications

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems;Integrated Sensing and Cyber

OBJECTIVE: Increase the speed of military aircraft digital data links to 100 Gigabits (Gbts) per second (Gbps) per lane and demonstrate with a 400 Gbps link over one fiber made up of 4 x 100 Gbps per lane.

DESCRIPTION: The growing use of direct digitization receivers and transmitters on military aircraft is leading to high-speed data rate requirements that are outpacing current aircraft datalink development efforts. Presently, naval aircraft fiber datalinks utilize Vertical Cavity Surface Emitting Laser (VCSEL) technology over multimode fiber, which is both cost-effective and temperature-insensitive at low-data rates. However, as data rates rise, the effects of modal dispersion over temperature are causing VCSEL technology development to reach its maximum achievable performance. Recent NAVAIR receiver designs that employ multiple channels of direct digitization are demanding data transport rates in excess of 700 Gbps. To meet this requirement, the existing 10 Gbps/fiber datalinks would need to be bundled with 70 fibers, while current development of 50 Gbps data links would necessitate 14 fibers for one link. The objective of this SBIR topic is to develop and demonstrate a 4 x 100 Gbts solution that is hardened to meet the environmental requirements for military aircraft. Ideally the technology would have the potential to achieve 200 Gbts in future development programs. This would fulfill a 700 Gbps requirement with only 2 fibers and would generate technology margin for future data speed increases up to terabit data rates. The current state-of-the-art commercial data link solutions use 100 Gbps data lanes with ongoing research and development into 200 Gbts data lanes; however, the primary application is indoor data center operations. These devices are unable to meet the extreme temperature and vibration requirements for military aircraft.

Key Performance Parameters for this topic are:
Demonstrate a maximum BER of 10^-12 between two TX/Rx transceivers under the following conditions:
Data Rate: 400Gbt using 4 x 100 Gbt data lanes
Operating Environment:
Operational Temperature: -55°C to 70°C Continuous, +85°C for 10 min
Thermal Shock: 70°C to -55°C at a rate of 35°C/min
Data transport: distance: 50 m Power Budget: > 15 dB
Vibration: To be provided after award

Technical challenges include:

1. The rate of modal dispersion in multimode fiber increases significantly with temperature, posing a difficulty in maintaining a 100 Gbt link performance at extreme temperatures. VCSEL technology has been the leading choice for aircraft datalinks due to its minimal size, weight, and power (SWaP) and wide temperature operation. However, as the demand for higher speed at high temperatures grows, better performance requires adding increasingly complex and power-hungry digital equalization to achieve marginal speed improvement. As a result, other technologies are sought that meet current and future performance requirements in a similar SWaP to VCSL transceivers and offer a clear performance growth potential to adding additional lanes and faster lanes (200 Gbts).
2. 100 Gbt waveforms require a higher signal to noise ratio (SNR) than slower waveforms. However, due to the presence of multiple bulkhead connections in an aircraft datalink, the power budget must be designed to > 15 decibels (dB). The higher SNR requirements of 100 Gbt combined with the power budget may be difficult to achieve with existing VESCL technology.

3. Data links on current aircraft are exclusively multimode; there is a legacy of deployed support equipment for multimode fiber that makes multimode solutions more likely to be adopted but does not preclude single mode solutions with a compelling performance improvement.

PHASE I: Design and model the link, and if possible, demonstrate the key technologies that will enable the data link to function over temperature. Develop and demonstrate the feasibility of a 4 x 100 Gbts solution that is hardened to meet the environmental requirements for military aircraft and ideally with the potential to achieve 200 Gbts in future development programs. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and deliver six transceivers. Test two packaged transceivers and use these transceivers to demonstrate acceptable performance over the full range of thermal shock and vibration.

PHASE III DUAL USE APPLICATIONS: Support the DoD in transitioning the proposed receiver to include working with a program office to develop a final packaging design that meets the platform SWaP and environmental requirements and developing systems specifications for the associated analog photonic links.

Development of this receiver has widespread commercial applications for high-speed commercial networks in stressing environments.

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KEYWORDS: Transceiver; Ethernet; 100 Gbt; Network; Fiber; Multi-mode; Single mode

N252-085 TITLE: Development of Safe, Robust, Sodium-Ion Battery for Naval Aviation

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Renewable Energy Generation and Storage

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design and develop a sodium-ion (Na-ion), secondary, battery solution for use across manned and unmanned aircraft that meets the environmental, operating, and storage conditions of Naval Aviation battery systems; provides increased safety over current lithium-ion battery solutions; and leverages domestic materials supply chains.

DESCRIPTION: U.S. Navy and Marine Corps currently fields lead-acid (Pb-acid), nickel-cadmium (NiCd), and limited lithium-ion (Li-ion) batteries for use across aviation platforms. Pb-acid and NiCd batteries have significantly lower energy density and service life than currently fielded Li-ion batteries used in the commercial electric vehicle market, resulting in negative impacts to aircraft performance and readiness. While Li-ion batteries are significantly more energy dense than legacy chemistries, the potential for thermal runaway events, high cost, and supply chains reliant on foreign entities of concern (FEOC) create safety, cost, and availability risks to programs. Na-ion batteries offer improvements to energy density comparable to Li-ion solutions while utilizing materials abundant domestically and abroad and may offer significant improvements to safety due to their wide thermal operating envelope. Na-ion batteries are currently in development and fielded commercially by FEOCs abroad.

The objective is to design and develop a sodium-ion (Na-ion), secondary, battery solution for use across manned and unmanned aircraft that meets the environmental, operating, and storage conditions of Naval Aviation battery systems, provides increased safety over current lithium-ion battery solutions, and leverages domestic materials supply chains. Therefore, the intent of this SBIR topic is to develop a Na-ion battery that can meet the harsh environmental operating and storage conditions of Naval Aviation battery systems while providing weight and energy savings to the platform with safety improvements and leveraging the materials required in the domestic supply chain.

PHASE I: Define and develop a concept for feasible battery designs including requirements compliance matrices and defined trades. Determine the feasibility of a Na-ion battery design that leverages requirements from multiple, Navy-led, battery commonality efforts that are traceable to systems engineering models of the MIL-PRF-29595 General Specification for Aircraft Rechargeable Lithium Batteries [Ref 3], its associated slash sheets, and small unmanned aerial systems (sUAS) battery designs. Develop a preliminary design review (PDR) level design and recommended validations leveraging the S9310-AQ-SAF-010 Navy Lithium Battery Safety Program [Ref 4] and MIL-PRF-29595 performance specification modified for a Na-ion battery systems.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop five (5) prototype Na-ion batteries to validate improvements to performance, including weight savings, across operational and storage environmental envelopes. Demonstrate use of

the domestic materials supply chain by providing relevant bill of materials (BOMs) for prototypes, as well as outlook on future domestic materials supply chain required for Na-ion battery components. Provide report of validation findings and supply chain analysis to Navy customers and program offices that demonstrates benefits of Na-ion over commercial Li-ion solutions.

PHASE III DUAL USE APPLICATIONS: Develop thirty-one (31) batteries for qualification testing to the S9310-AQ-SAF-010, MIL-PRF-29595 and designated slash sheet, MIL-STD-461, MIL-STD-704, MIL-STD-901, and MIL-STD-167 specifications and instructions for successful aircraft and ship integration. Develop four (4) additional batteries for use during flight test and demonstration for transition to the application platform identified in Phase I.

To date, no Na-ion main ship aircraft battery has been developed for commercial aviation. The Na-ion battery benefits of weight savings while providing robust performance and safety across a wider envelope than Li-ion solutions may be transferrable to commercial aviation. Additionally, if Na-ion solutions cannot meet space, weight and power (SWaP) requirements for use in aircraft, the technology can be transitioned to efforts supporting electrification of Ground Support Equipment (eGSE), Uninterruptable Power Supplies (UPS), and grid energy storage domains.

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KEYWORDS: Sodium; Sodium-ion; Battery; Main Ship Battery; Lithium; Lithium-ion

N252-086 TITLE: Multimode Fiber Optic Multiplexer/Demultiplexer

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;FutureG;Integrated Network Systems-of-Systems

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and package uncooled fiber optic multiplexers and demultiplexers for integration into digital fiber optic transmitters, receivers, transceivers, fiber optic backplanes, OpenVPX circuit cards, and light source and power meter test equipment.

DESCRIPTION: Current airborne military (mil-aero) core avionics, electro-optic, communications, and electronic warfare (EW) systems require ever-increasing bandwidths while simultaneously demanding reductions in space, weight, and power (SWaP). The effectiveness of these systems hinges on optical communication components that realize high per-lane throughput, low latency, and large link budget, and are compatible with the harsh avionic environment.

As digital avionics fiber optic transmitter rates increase from 10 Gbps to 100 Gbps and higher, new fiber optic multiplexer/demultiplexer devices will be required. Key enabling components for use in transmitters, receivers, optical backplane, OpenVPX circuit cards, and fiber optic maintenance and troubleshooting test equipment are 50 micrometer core compatible fiber optic multiplexer/demultiplexer devices. The multiplexer/demultiplexer will be integratable with transmitter and receiver optical subassemblies and fiber optic test equipment optical subassemblies. The optical subassemblies are expected to operate over a -40° to +95° Centigrade temperature range and be less than or equal to one cubic centimeter in volume. The test equipment optical subassemblies will provide a launch condition to enable repeatable optical loss measurements on fiber optic links installed on military aircraft.

PHASE I: Develop design concepts for multimode fiber compatible multiplexer/demultiplexer devices for military digital fiber optic communication and test equipment applications. Demonstrate the feasibility of the multiplexer/demultiplexer designs that meet the objectives described in the Description through modeling, simulation, and analysis, showing a path toward meeting Phase II goals. The Phase I Option, if exercised, will include initial design specifications and capabilities descriptions to build prototype solutions in Phase II. Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Design and develop prototype multiplexer/demultiplexer devices optimized using the results from Phase I. Build and test the multiplexer/demultiplexer devices in transmitter, receiver, and test equipment optical subassemblies. Show operation over the full avionics transmitter and receiver temperature range, and the test equipment temperature range. Perform highly accelerated life testing showing a path toward achieving Technology Readiness Level (TRL) 6.

PHASE III DUAL USE APPLICATIONS: Further advance the technology to higher TRLs and demonstrate on flight and support equipment hardware.

The developed technology will be applicable to data center and computer networks requiring wide temperature range fiber optic communications.

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KEYWORDS: Digital fiber optic communications; Avionics; Short Wavelength Division Multiplexing; SWDM; Transceiver; Packaging; 100 gigabits per second and higher.

Version 4

N252-087 TITLE: Repair Solution for Optically Transparent Canopies and Windscreens

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

OBJECTIVE: Develop a repair solution to address scratch and pitting damage to acrylic transparencies or their transparent coatings that is usable within optically critical zones, is executable on-aircraft by intermediate or organizational level maintainers, and decreases aircraft downtime compared with component replacement or depot-level overhaul.

DESCRIPTION: Scratches and pits to the acrylic outer ply of canopies and windscreens cause optical distortion as well as structural vulnerability, and these types of damage cause significant readiness degradation. Currently, available repair methods are costly, time-intensive, and extremely limited in the scope of damages that can be addressed, such that damages greater than 10 mils (measurement of unit thickness) in depth are essentially unrepairable. Furthermore, the optical requirements of canopies and windscreens are exacting, and repairs to transparencies must meet the original optical requirements of the components without inducing even minor distortion. The requirements for the baseline material can be seen in the military specification, MIL-PRF-25690 [Ref 2]. Novel repair materials and processes are needed to facilitate optically acceptable repairs larger than the current state of the art. Candidate repair concepts will be compatible with acrylic substrates, executable on aircraft, optically transparent in accordance with baseline aircraft requirements, able to endure in aval aircraft environments, and capable of being inspected for acceptability to existing standards using existing or novel inspection techniques.

PHASE I: Define and develop a repair material or concept that meets the requirements listed in the Description. Perform analysis to determine feasibility of the concept.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate a prototype kit and/or process that meets the requirements listed in the Description. Prototype testing and demonstration will be performed at the canopy shop at Fleet Readiness Center Southwest – North Island.

PHASE III DUAL USE APPLICATIONS: Perform final testing at organizational and intermediate level repair locations, and transition for use on H-53 and/or F/A-18. The developed technology will also be applicable to commercial aviation and general aviation.

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KEYWORDS: Transparency; Acrylic Outer Ply; Canopy; Window; Windscreen; Repair; Optical Zone; Scratch and Pit Repair

Version 4

N252-088 TITLE: Advanced Artificial Intelligence/Machine Learning-based Intelligent Agent for Finite Element Modeling of Aerospace Structures

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Sustainment;Trusted AI and Autonomy

OBJECTIVE: Develop an advanced Artificial Intelligence/Machine Learning (AI/ML)-based intelligent agent to automate the generation, prediction, and optimization of finite element models, with the ability to accurately account for model errors, enhance modeling fidelity, and reduce user input bias.

DESCRIPTION: Finite Element Analysis (FEA) is a critical computational tool used across a spectrum of engineering disciplines, including, but not limited to, automotive, aeronautical, civil engineering, and biomedical engineering. FEA enables the prediction of the behavior of materials and systems in response to various physical effects such as mechanical stress, strain, heat transfer, fluid flow, and electrostatics. This computational method has empowered engineers with the ability to develop safer and more efficient designs, optimize systems, and predict failure points, significantly reducing the need for physical prototypes and expensive testing procedures. However, the process of building a Finite Element Model (FEM) is subject to multiple sources of errors, including discretization error, errors from geometrical approximation, errors due to assumptions in material modeling, element formulation selection, and errors from inaccurately represented boundary conditions. The iterative process related to model development helps to better understand the physics and mechanical behavior in the actual assembled system, and better understanding is the purpose of FEM and FEA. The intelligence gained through the iterative modeling process often reveals complexities and system effects that can and should be added into the model to achieve accurate physical behavior or acceptable calibration to the physical system. While conventional error mitigation strategies such as mesh refinement techniques, manual error checking, and model validation against experimental data do exist, these methods can be time-intensive, require extensive human intervention, and may still result in biased results due to user subjectivity. In recent years, AI/ML methods such as Generative Adversarial Networks (GANs), Deep Reinforcement Learning, Machine Vision, and Artificial Neural Networks (ANNs) have demonstrated significant potential to revolutionize the FEM/FEA fields. These advanced computational methods offer the ability to automate model generation, accurately predict and mitigate modeling errors, and streamline the process, thereby significantly reducing human intervention and the associated subjectivity. However, a comprehensive, integrated framework utilizing these AI/ML methods for finite element modeling, accuracy modeling impact assessment, and model optimization is lacking.

The Navy seeks to develop a comprehensive AI/ML software toolkit that can transform an input geometry list to generate the finite elements and nodes used in the FEM input deck automatically. The toolkit will estimate the accuracy impact, optimize the model parameters for enhanced fidelity, ensure the meshing process matches the problem, be efficient, and be free from user input bias. The Navy seeks a software toolkit that can refine the ML models based on the outcomes of these tests. The goal is to improve the system's predictive accuracy and error mitigation advice. With constant learning and adjustment, the system will progressively improve and adapt to handle more complex and varied finite element models. The successful completion of Phase II will provide an advanced prototype system capable of automatically generating FEM input decks from input geometry lists, assessing accuracy impact, and providing strategies for error mitigation, thus enhancing the fidelity of finite element modeling processes. The envisioned AI/ML software toolkit will be capable of handling complex geometries and boundary conditions, accurately representing material behaviors, and robustly accounting for various physical phenomena. Furthermore, the toolkit will provide a user-friendly interface, streamline the workflow of finite element modeling, and effectively communicate results to the user, enabling them to make informed decisions. The toolkit will encapsulate modern AI/ML techniques such as GANs, Deep Reinforcement Learning, Machine Vision, and ANNs.

The system's capabilities will include detecting and analyzing the source of errors in the FEMs, assessing these errors, and offering insights into how to mitigate them. A comprehensive series of tests will be conducted to assess the performance of the prototype system. These will include various scenarios and geometries to ensure the system can handle a broad spectrum of FEM tasks.

To validate the designed system, a basic prototype is needed to demonstrate the core functionalities. This prototype will facilitate the automation of simple finite models' generation from existing CAD data and demonstrate the potential of AI/ML techniques in predicting and mitigating modeling errors. An example of this would be the impact of element size, type, and transition on accuracy.

Additionally, the small business awardee will develop a detailed verification and validation (V & V) test plan, which will define clear, measurable metrics and benchmarks that can be used to quantitatively assess the toolkit's performance and effectiveness. The plan will also aid in identifying areas for potential improvements and modifications in the following phases.

PHASE I: Develop a concept for an AI/ML-driven software toolkit. Demonstrate technical feasibility of the proposed concept for automating finite element meshing, creating nodes and elements, predicting potential accuracy impact, and optimizing models for improved accuracy and fidelity. Prove feasibility of the proposed concept by first performing in-depth study of the current state of finite element modeling processes and the inherent error sources, including those not addressed by meshing alone (i.e., post-processing methodology, boundary condition errors, misrepresentation of the structural behavior, etc.). This study would guide the design and development of the AI/ML-driven toolkit, ensuring that the toolkit robustly accounts for the most significant error sources. This toolkit will encapsulate modern AI/ML techniques such as GANs, Deep Reinforcement Learning, Machine Vision, and ANNs. The focus would be on creating a road map for how these AI/ML techniques can be integrated and utilized to automate the model generation process, predict potential modeling accuracy impact, and optimize the model parameters.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Devise an AI/ML software toolkit. Design and develop a prototype intelligent support system utilizing the AI/ML software toolkit. Focus on effectively integrating these methods into an automated software system that can handle geometric transformations for FEM input deck generation and offer a standardized process without user input bias. Ensure that the toolkit will address the problem of model uncertainty prediction in mixed fidelity FEMs.

PHASE III DUAL USE APPLICATIONS: Transition validated AI/ML modeling toolkit to integrate with existing FE engineering analysis tools.

FEA is widely used in aerospace, automotive, trucking, heavy equipment companies, medical reconstruction in a vast plethora of private sectors. The benefits to the private sector would be confidence in FEA solutions in a variety of domains including structural mechanics, fluid flow analysis, heat conduction, additive manufacturing, electrical and electronics field, bio-engineering, and so forth. The reduction in cost in these fields makes this topic highly beneficial to the private sector. Dropping the costs and turn-around time for analyses will allow additional opportunities for analysis arising from the decreased cost threshold. This toolkit will improve analysis availability across the entire domain of manufacturing.

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KEYWORDS: Finite Element Analysis; Artificial Intelligence / Machine Learning; Solid Structural Mechanics Stress; Preprocessing and Post Processing; Manual Effort and Manual Review; Shape Function and Numerical Integration Points; Particular Stress Regions

Version 4

N252-089 TITLE: Virtual Agent for Data Fusion and Understanding

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

OBJECTIVE: Conceive and develop a novel artificial intelligence (AI) system—virtual agent—capable of sophisticated data fusion and comprehension, adapted for the Navy's diverse data ecosystem. This virtual agent will be competent to process, explain, and generate actionable intelligence from heterogeneous data sources; thereby augmenting situational awareness and decision-making acumen within the dynamic maritime theater.

DESCRIPTION: The Navy is inundated with data emanating from myriad multiform sources: sensor data from maritime and aerial platforms, intelligence dossiers, maintenance logs, environmental metrics, communications intercepts, and so forth. Human analysts are presented with complex radar and sonar returns mapping physical spaces and threats, detailed textual and visual intelligence reports necessitating advanced linguistic and visual analytics, operational and maintenance data indicative of asset readiness, and critical oceanographic and meteorological data conditioning strategic operations.

The colossal volume and diversity of these data pose considerable challenges in terms of real-time processing, comprehensive understanding, and the distillation of actionable intelligence. Prevailing approaches anchored in generative AI have exhibited only limited success in natural language and image production and are ultimately wrecked on the shoals of complexity that human-level/human-style intelligence uniquely can comprehend. Generative AI models are constitutionally defective by their inability to explain—and not merely predict—patterns in data, which in addition disables them from generalizing across disparate data types.

To transcend these limitations, a paradigm shift towards a hybrid AI approach—synergizing human-style machine learning (ML) with human-style symbolic AI into a neurosymbolic hybrid—is imperative. Symbolic AI, using knowledge graphs, ontologies, and rule-based systems, can endow the virtual agent with domain knowledge and reasoning faculties. This consilience of ML and symbolic AI will empower the agent to integrate and interpret a wide spectrum of data, discern explicit causality from latent correlations, and generate robust, actionable insights. The envisioned virtual agent must thus leverage both advanced non-Large Language Models (non-LLM) ML and symbolic AI to actualize comprehensive data fusion and deep understanding.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Conceive and develop a neurosymbolic hybrid virtual agent. This initial phase comprises the architectural design of the system and the selection of data types (including, but not limited to radar and sonobuoy data). The objective is to construct a nascent model that validates the feasibility of integrating these diverse data streams and generating preliminary insights. Additionally, this phase will encompass

the establishment of evaluative metrics of system performance and the formulation of a scalable development plan for subsequent phases. Reference sources to determine trust/confidence in the system will include external validated knowledgebases and domain-specific training.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Build upon the foundational insights of Phase I, advance toward the comprehensive development, rigorous testing, and empirical validation of the virtual agent. Build and refine the algorithms and models, integrate additional data sources, and enhance the prototype system's real-time processing capabilities. Test the virtual agent within realistic Navy operational scenarios to assess its efficacy in terms of accuracy, alacrity, and resilience. Demonstrate the agent's prowess in delivering holistic situational awareness, anticipating potential threats, and proffering actionable strategic recommendations. Ensure that the virtual agent is a deployable system that substantially augments the Navy's data fusion and understanding capabilities, thereby elevating operational efficacy and strategic decision-making.

Work in Phase II may become classified. Please see note in Description paragraph.

PHASE III DUAL USE APPLICATIONS: Complete final testing. Perform necessary integration and transition for use in operational applications with appropriate platforms and agencies, and future combat systems under development.

Commercially, this product could be used to enable security monitoring, smart city operations center, power grid monitoring, and wherever large amounts of sensors or inputs are utilized.

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KEYWORDS: Data fusion; artificial intelligence; machine learning; AI/ML; virtual agent; neurosymbolic agent; decision-making; symbolic AI

Version 4

N252-090 TITLE: Second Order Acoustic Vector Sensors

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Network Systems-of-Systems;Integrated Sensing and Cyber;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate an increase in acoustic performance via second order vector sensors for use in advanced sonobuoy arrays to track and localize stealthier targets within a larger search area to increase success in future U.S. naval anti-submarine warfare operations.

DESCRIPTION: The application of second order acoustic sensors is employed to great effect in commercial acoustic products such as hearing aids. The U.S. Department of the Navy seeks to explore the application of similar second order directional receivers in sonobuoys to increase overall system performance in air Anti-Submarine Warfare (ASW) operations.

Historically, sonobuoys utilized a variety of approaches to achieve first order directional channels, including acceleration, pressure gradient, and multi-mode sensors. The main advantage of a higher order directional acoustic channel is increased directivity index, which also results in a more complex beam pattern compared to the conventional first order channel. Together, these attributes provide an opportunity to increase overall array gain in complex real-world acoustic environments.

The advantages offered by higher order directional receive channels are fundamentally enabled by increasing the number of receiving channels and then combining them in a beamformer based on the expected acoustic propagation for the sensor configuration. As is true in most acoustic products, sonobuoys must balance these potential performance gains against the targeted unit production cost and limited available volume. Other challenges such as build variances, channel self-noise level, and signal processing power consumption must be addressed to form higher order directional channels. This SBIR topic is interested in developing equally spaced vertical line arrays of second order vector sensors with a center frequency at 1000Hz and a bandwidth of 400 Hz, and with a center frequency at 8kHz and a bandwidth of 1.5kHz. This SBIR topic will determine the number of second order vector sensors that can be packaged in an A-size sonobuoy. The self-noise of each individual second order cardioid will be equal to or less than 30 dB. The array self-noise will be reduced by 10 log of N where N is the number of elements in the array. Approaches must allow for continuous beam steering in in azimuth.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified

material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop, design, and demonstrate feasibility of candidate second order directional sensor configurations and associated beamforming approaches that offer performance advantages for sonobuoys in one or more aspects of the overall air-ASW mission. Predict overall performance and offer mitigations to various design challenges. Compare the candidates and down select to a notional design to build and test in Phase II. Demonstrate that chosen candidates can eventually be scaled to fit into an A-size sonobuoy package with considerable effort.

Perform in-depth modelling during the study and identification of possible candidates. Conduct some initial testing to demonstrate the raw material and/or hardware needed to build the candidates for Phase II possess the characteristics used for modeling or that at least the component has the characteristics and performance required for the system to achieve the objectives.

The Navy seeks proven candidate(s) that have been demonstrated and validated to meet the objectives and can be scaled down to fit within the A-size constraints. The small business awardee must provide a full report where their models have been performed with sufficient accuracy and depth to produce meaningful results to meet the objectives. In addition, demonstrate that preliminary testing has been performed on the raw material and/or hardware showing that it possesses the characteristics used for modeling, or that at least the component has the characteristics and performance required for the system to achieve the objectives. The small business awardee must also prove the models and results provided in the report were originated by them.

The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and test prototype(s) of the second order vector sensor to verify Phase I performance predictions. Quantify key performance metrics through a combination of laboratory and open water testing. Extrapolate the expected performance for the intended mission(s).
Work in Phase II may become classified. Please see note in Description paragraph.

PHASE III DUAL USE APPLICATIONS: Integrate the second order vector sensor technology into an A-size sonobuoy, leveraging the prototype fabricated in Phase II. Conduct a demonstration and characterize mission performance under real world conditions.

The new approaches developed in this SBIR topic could enhance acoustic sensing technologies for harbor and port security, marine conservation, and other research or monitoring applications.

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KEYWORDS: Undersea sensor; Acoustic; Vector sensor; Second order beamforming; Sonobuoy; Arrays

Version 4

N252-091 TITLE: Holistic Inflight Variable Employment Suite (HIVES)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Integrated Network Systems-of-Systems

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Augment existing End-User Development (EUD) based Situational Awareness (SA) tools to support continental United States (CONUS) planning/training and outside the continental United States (OCONUS) planning/mission execution matching existing commercial tools (ForeFlight/Aero) with minimal impact when transitioning from both employment scenarios.

DESCRIPTION: At this time, Commercial Civil Aviation tools to streamline mission plan generation, submission, and mission execution are physically separate from military SA tools used during mission execution. This results in duplication of hardware, increased training requirements, and significant workflow for training vs employment missions.

Existing civil planning tools lack detailed models for military aircraft. Leveraging existing Aircraft, Weapons, and Electronics (AWE) and aircraft Joint Mission Planning System (JMPS) models may allow rapid addition of legacy military aircraft to robust flight planning tools. Layering tactical use of terrain and threat avoidance adds significant complexity to the final output of civil planning tools, tradeoffs for complexity, tactical utility, and flight safety must be explicitly addressed for phase 2 implementations. The objective of this SBIR topic is to augment existing EUD-based SA tools to support CONUS planning/training and OCONUS planning/mission execution matching existing commercial tools (ForeFlight/Aero) with minimal impact when transitioning from CONUS/training to OCONUS/employment scenarios. The tool will encompass pre-flight planning, submission/ingestion of planning data, takeoff/landing performance, mission execution, and mission modification. The resultant mission planning augmentation will combine the best capabilities of both civil and military ecosystems. Successful implementations will integrate International Civil Aviation Organization (ICAO) flight plan submission and JMPS AWE capabilities. Data ingestion from both Automatic Dependent Surveillance Broadcast (ADS-B), Traffic Information Service Broadcast (TIS-B)/Flight Information Service Broadcast (FIS-B), and military data links to provide SA in all scenarios will enhance common functionality across the spectrum of employment. The ability to submit flight plans to civil and military authorities as well as tactical threat warnings will highlight the power of a single tool supporting CONUS and OCONUS implementation. Leveraging FIS-B and DoD GRIB2 weather data will minimize the disparity in environmental SA between deployed and garrison missions. DoD GRIB2 is a standard format for sharing gridded binary data used by the department of defense. The ability to import commercially available approach plates and National Geospatial-Intelligence Agency (NGA) sourced data seamlessly in a single interface will approach the goal of a single carry-on device for both training and operational environments.

Blending DoD and civil data sources will provide robust and timely data for aircraft personnel. While commercial repositories of approach plates may surpass the fidelity of NGA sources, emergent,

contingency airfields will only be available through DoD sources. Providing a single, common interface to both sources will streamline all aspects of aircraft employment.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Identify minimum common civil capabilities to integrate into selected SA systems. Identify current DoD processes matching mission plan generation/submission in civil environments. Identify mission data sets available from civil and DoD sources with pathways to integration for each required data set in both peacetime and wartime employment. Develop a consensus for required functionality across disparate aircraft to gauge the difficulty of the overall effort as well as additional "per minimum data set" workload to support disparate aircraft across the fleet. Integration points on both Civil aviation and DoD. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Augment selected SA systems to meet the minimum requirements identified in the Phase I effort.
Work in Phase II may become classified. Please see note in Description paragraph.

PHASE III DUAL USE APPLICATIONS: Focus on the deployment, sustainment, operational readiness, and long-term support for the capability/product developed in Phase II.
This project is a more holistic approach to flight/mission planning in a hybrid environment/battlefield.

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KEYWORDS: Advance Mission Planning; Electronic/Virtual Flight Bag; Electronic Kneeboard; Flight Management; Mission-critical Flexibility; Common interface

Version 4

N252-093 TITLE: Ambient Acoustic Noise Cancelling for Military Boom Microphones

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Network Systems-of-Systems

OBJECTIVE: Design device(s) that will provide cancellation of ambient acoustic noise in the electrical signal from a military boom microphone.

DESCRIPTION: Microphones operating in high-noise environments transmit a significant amount of ambient acoustic noise to the ear in addition to the desired speech signal. This results in higher, and potentially damaging, at-ear noise levels and reduced speech intelligibility.
Reducing acoustic noise carried in the electrical microphone signal will lower at-ear noise levels and improve speech intelligibility.

There is currently no capability in military aircraft to prevent undesired acoustic noise from entering the boom microphones along with the desired speech signal. The noise is transmitted as part of the electrical signal to the communication system and on to the ear. Noise cancellation technology exists for low-noise environments, such as call centers. Technology needs to be developed for the communications systems and high-noise environments of military aircraft. Solution(s) may include an add-on item in line with the existing microphone or microphone and preamplifier and/or replace the existing microphone or microphone and preamplifier.

Solution(s) must be suitable for military aircraft operational use and hardened for the shipboard EMI environment. Noise cancellation is expected to cover the entire operational frequency range of a standard military microphone and preamplifier.

Device(s) must be compatible with existing aircraft communication systems, listed in terms of priority, which provide either:

1. 28 VDC to preamplifier (MIL-A-23595D or similar) and 5 Ohm dynamic microphone (MIL-PRF-26542-2E or similar)
2. 10 VDC to integrated 150 Ohm electret microphone-preamplifier (MIL-M-49199A or similar)
3. No power; 5 Ohm dynamic microphone (MIL-PRF-26542-2E or similar) connected directly to communication system (Solution may be submitted for a subset of these options)

It is preferred that the device(s) be powered by existing aircraft power. If battery power is included in the solution, the device must not degrade the communication signal if battery fails.
Increase of weight on the head shall be no more than 50 grams. Priority will be given to designs that minimize weight. There shall be no increase in forward center of gravity offset of the system compared to legacy. Additions or changes to boom microphone shape or size must not degrade user visibility.
This topic aligns with PMA-202 focus area: Improved Aircrew Communication.

Deficient speech intelligibility can lead to errors and omissions in time-critical acts, which jeopardize flight safety and mission success. High intensity ambient noise during aircraft operations, which enters the communications system via the microphones, circumvents hearing protection.
1. Compatible with/adaptable to current flight helmets and communications gear
2. Independent of aircraft communications electronics and radios

Note: NAVAIR will provide Phase I awardees with the appropriate guidance required for human research protocols so that they have the information to use while preparing their Phase II Initial Proposal. Institutional Review Board (IRB) determination as well as processing, submission, and review of all paperwork required for human subject use can be a lengthy process. As such, no human research will be

allowed until Phase II and work will not be authorized until approval has been obtained, typically as an option to be exercised during Phase II.

PHASE I: Design and build a functional prototype that provides ambient acoustic noise cancellation of the transmitted electrical microphone signal. Verify that the design can reduce pink noise with and without tonal components. Determine the noise reduction achieved in each 1/3 octave band. Verify that the speech signal is not degraded. The Phase I effort will include prototype plans to be developed under Phase II.
Note: Please refer to the statement included in the Description above regarding human research protocol for Phase II.

PHASE II: Build and deliver operational prototypes suitable for the U.S. Navy to conduct flight testing. This will include validating the design reduces aircraft noise and determine noise reduction achieved in each 1/3 octave band and at tonal frequencies and conducting speech intelligibility testing relative to M-87 microphone and EMI testing.
Note: Please refer to the statement included in the Description above regarding human research protocol for Phase II.

PHASE III DUAL USE APPLICATIONS: Flight test, finalize design, and deliver production units. Transition to PMA-202 and add devices as approved items in the Aviation-Crew Systems, Aircrew Personal Protective Equipment Technical Manual.

Noise cancellation of a transmitted electrical microphone signal can be used for any head mounted microphone in a high- or extreme-noise environment. Device(s) can be used or adapted for use in general aviation aircraft.

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KEYWORDS: Microphone; Noise cancelling; Noise cancellation; Noise reduction; Speech intelligibility; Communication

N252-094 TITLE: Cue Aggregation Algorithms for Multi-function Receivers

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Invent and develop a set of new algorithms expressed in machine learning (ML) form running on commercial off-the-shelf (COTS) processors that can both aggregate multiple signals into classes of same pattern of emission and locate methods to distinguish look-alike signals from platforms with different spectral signatures or different intentions.

DESCRIPTION: When ML is used to locate new radio frequency (RF) energy associated with specific classes of signals in a wideband receiver architecture from within the digital representation of the electromagnetic (EM) spectrum collected by the antenna, the software module only responds to the signals it was trained for. While that is often a nuisance as locating appropriate training data and the time to retrain remain significant issues within the DoD, it can also be a virtue. Importantly, the software should alert (cue) only to the signals it was trained for.

If two functionally distinct types of signals are closely matched in the normally used Pulse Description Word parameters, it is plausible to expect their feature space occupancies to be separable but also close together. Thus, the user should be free to train the software to cue as a single report when either is present in the current signal environment or reporting the two signals present separately. Two differently trained copies of the cue generator could first locate all copies of either waveform in the wideband data and then inform the user who needs to know which of the two signals was actually present. For the purposes of this SBIR topic, we call cue generators that aggregate multiple distinct signals into a single cue “association or aggregation processors”.

A second example would be if the training data consists of the same waveform centered over a range of center frequencies. The cues ought to arise out of every occurrence of that waveform independent of frequency. Each cue will still report the actual frequency window of occurrence. Hence if a new association processor is devised that accepts multiple cues from such a frequency agnostics cue generator as inputs, it ought to be able to identify the data’s commonality and discover the connections and similarities between the disjoint in time signals and potentially the guiding principles of their waveform alteration (e.g. their frequency hop patterns).

Thirdly, note that the cue reports include the time of onset and end of a given transmission as well as identifying which trained signal classes appeared. If the separately received signals of the same signal class from two spatially separated antennae are supplied to a single cue generator and are confirmed to occupy the same feature space volume, a time difference of arrival measurement can be produced by aggregating the separate cue reports. Differences in the feature space points could indicate the signal is dominated by a corrupted transmission with a different effective antenna pattern compared to the other signals being compared. This mechanism of testing correlation can be used as the basis for determining

the relative distance to transmitters and watching it evolve in time. Note that this application of cue aggregation processors does not need to be limited to two antenna outputs.
Fourth, it may be desirable to train cue generators for each of the main types of radar signals and one aggregation processor that can recognize all as "radar" as a class so that users wishing to do only comms know to deprioritize all of them.

Relatedly, if the signals associated with a single waveform are used to train an ML algorithm with a tightly defined feature space, it should be possible to train the same algorithm to recognize as related but distinct those signals with nearby feature space localizations, one class of anomaly.

The Phase I proposal needs to define at least one new aggregation processor functionality and a means of demonstrating it in a lab setting within the base effort. The Phase I Option effort shall then begin the Phase II effort by defining and demonstrating additional types of processors. The Phase II may become classified depending on functionalities and GFE data sources provided.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVAIR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Realize the Digital Signal Processing technique. Demonstrate it in the lab on synthetic or collected data during the Phase I Base period. Prepare a Phase II proposal. The Phase I Option, if exercised, will discuss the signal classes to be focused on with the Navy TPOC and begin the preliminary work associated with the Phase II plan.

PHASE II: Work in a collaborative fashion with Navy personnel to develop and demonstrate a set of aggregative processors operating on COTS processors cards during the Phase II Base period. In the Phase II Option, if exercised, integrate these new processors into the DSP libraries and architecture of a software defined, ultrawideband receiver prototype, address any concerns over fielded operations, and demonstrate processor capacities to potential transition sponsors via an outdoor but in CONUS exercise.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Transition the new processors into the soft/firmware libraries of a Program of Record. Support the Navy in transition efforts.

In the civilian market, establish a set of low-cost software defined radios of the Internet of Things (IoT) sort based on the prioritized receiver concept and with a modest set of loaded DSP and specific cue generators. The IoT messages will set the class of signal to be processed at any given time and the cues active at any given time would define the channels offered to the user. An example might be the multi-functionality of modern television industries offering streaming services; broadcast, music, Kindle-like services; appliance control; and interaction with medical sensors and conveyance of information to clinicians, a one-stop shop for RF functions.

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KEYWORDS: adaptive waveforms; coordination of disjoint observers; machine learning; synthetic training data; aggregation; feature space signal representation

N252-095 TITLE: AEGIS MK99 T-1348 Radar Transmitter Components Replacements

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop innovative solid-state components replacements for the MK99 T-1348 Transmitter.

DESCRIPTION: The Navy's MK 99 T-1348 Transmitter is comprised of a Radio Frequency (RF) oscillator multiplier, voltage regulator, Control-Monitor subassembly, and other components. The primary function of the T-1348 Transmitter is RF amplification at J and X bands at the required power levels. The T-1348 Transmitter is an electron and microwave tube technology. The voltage regulator is an electron tube device used in the T-1348 Transmitter used to provide regulated body-cathode voltage for the microwave tube amplifier. The RF Oscillator Multiplier is used in the T-1348 Transmitter and is an analog design used in determining the frequencies in support of modern threats, functions, and information requirements of the Fire Control System (FCS) mission. The Control-Monitor subassembly is used in the T-1348 Transmitter and is an analog design used for communication with the Data Converter Cabinet (DCC). The DCC is an external interface to the T-1348 Transmitter. The Control-Monitor provides fault monitoring and control within the T-1348 Transmitter. The Navy seeks solutions that will replace the RF oscillator multiplier, voltage regulator, and Control-Monitor subassembly with modern technologies that maintain or exceed existing functionality and performance parameters for the FCS. Doing so will decrease costs and provide for less manpower in maintaining the system. There are no current commercial technologies that meet these needs.

The solutions must conform to form, fit, and function in the existing enclosure of the Transmitter cabinet. The solution must fit to existing MK99 T-1348 Transmitter installations aboard DDG 51 Class ships, development and tactical sites, and international ships. The solutions must be functionally equivalent to the current design and meet qualifications for shipboard equipment shock and vibration as specified in MIL-STD-810 and airborne noise and electromagnetic interference as specified in MIL-STD-461. Other operating requirements will be furnished upon request. Any improvements beyond current functionality are encouraged and will also be considered.

The solutions must provide for future upgrading in electronic systems and enable organizational level maintenance. They must work with Continuous Wave Illumination Radar technologies and or conversion of microwave/electron tube technologies using a solid-state design. The solutions must also be manufacturable without supply chain issues.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow

contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for MK99 T-1348 Transmitter component replacements and show it feasibly meets the parameters listed in the Description through testing, analysis, and modelling. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver prototype MK99 T-1348 Transmitter components replacements based on the results of Phase I. Demonstrate that they meet the parameters listed in the Description through testing in relevant environments provided by the government.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the prototype components to Navy use. The prototypes will be tested for integration into the AEGIS Weapon System MK99 T-1348 Transmitter such as its interface with the DCC. Requirement traceability will be used to verify design, analysis, and integration into the overall system using systems engineering principles such as Model Based Systems Engineering (MBSE). The integrated system will be validated in its operating environment and its applicability to organizational level maintenance concept. Testing will be accomplished using modeling, simulation, and realistic signals. Describe how the technology will meet Navy needs.
Other dual-use applications include High Power Continuous Wave amplification used in over-the-air radio or TV transmissions.

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KEYWORDS: Microwave tube; Electron tube; Control-Monitor; Solid-State Components; Voltage Regulator; T1348 Transmitter

N252-096 TITLE: Multi-source Data Ingestion for Warfighters

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a capability that mines information from authoritative sources and reduces operator workload and errors.

DESCRIPTION: Operators of theater-level command and control systems, such as the AN/UYQ-100 Undersea Warfare Decision Support System (USW/DSS), must consider a wide range of sources when forming recommendations for Fleet operations. Operators currently manually access a range of authoritative sources in search of the best information upon which to base Fleet operations decisions. Creating the required range of reports involves significant effort that can take hours to complete. Research must be done, then data collected, and finally uniquely formatted reports generated that are provided to a range of consumers. During complex wartime operations, this reliance on manual means to access information introduces an error-prone environment and creates a heavy sailor workload.
Current mission preparation requires operators to spend numerous hours searching Secret Internet Protocol Router Network (SIPRNET) websites and various publications. Current information is stored in a variety of different formats, including hardcopy information, websites, platform status information contained in meta-data communicated over USW-DSS, and embedded recording systems. This information is not stored or tagged in a manner that enables operators to perform rapid searches for data analytics. The Navy seeks a technology to mine authoritative data in various stored formats and provide an intuitive user interface for generating accurate reports. This capability will reduce error-prone situations and time involved to make Fleet operations decisions. The desired technology will allow machine learning to search for vital information, allowing the operator to focus on interpretation of data rather than the mechanics of the search process. There is currently no commercial product that will meet the required technological need.

The technology sought must rapidly access and parse data types across different authoritative sources to produce results within an intuitive user interface. The data to be accessed includes USW-DSS recordings, operational information, authoritative intelligence, lessons learned, estimated probabilities from prior maneuvers, and historical and projected environmental data. The user interface must enable users access to this data as they need it, and in the standardized formats required by USW-DSS and connected systems. The data mining capability and user interface must operate at multiple security levels (e.g., Unclassified up to TS-SCI), supported by an architecture that could allow information at multiple security levels to be ported up to the highest level of security required for a specific effort. The technology will enable data from previous missions to be easily recalled and overlaid with current data to show history of deployments for contacts of interest and activity levels.

The data mining algorithm must provide for authoritative data availability to appropriate access privileges. It must also meet current Information Assurance (IA) specifications for classification security. No known commercial application is capable of meeting the Navy's requirement to mine from authoritative sources across multiple security levels and meet IA specifications.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for a data mining capability and associated user interface that meets the requirements as stated in the Description section above. Demonstrate the feasibility of the concept in meeting Navy needs and will establish that the concept can be feasibly produced by sample testing, modeling and simulation and or analysis. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop, deliver, and demonstrate a prototype data mining capability and associated user interface. Demonstrate the functionality in a series of user designed sprints with fleet operators to refine the user interface for evaluation. The prototype will be evaluated to determine its capability in meeting the performance goals as defined in the Description. System performance would ideally be demonstrated through installation and prototype testing using the cloud-based USW-DSS prototyping infrastructure provided by the government.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the data mining capability and user interface to Navy use. Finalize the software design and algorithm prototype, for evaluation to determine its effectiveness in an operationally relevant environment in USW-DSS. Support the Navy for test and validation in accordance with the IWS 5.0 USW-DSS Peer Review Group.
The technology will have private sector commercial potential for any secure system such as banking and medical information requiring access and analysis of historical information, reports, and trends analysis.

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KEYWORDS: Data mining; AN/UYQ-100 Undersea Warfare Decision Support System; complex wartime operations; intuitive user interface; Information Assurance (IA) specifications; history of deployments

N252-097 TITLE: Battle Rhythm Situational Awareness and Tasking

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an integrated Artificial Intelligence or Machine Learning (AI/ML) tool for battle rhythm management by operators of theater-level command and control during complex Undersea Warfare (USW) wartime scenarios.

DESCRIPTION: Operators of theater-level command and control systems, such as the AN/UYQ-100 Undersea Warfare Decision Support System (USW-DSS), must handle both simple and complex wartime scenarios. Peace-time training and exercises often lean toward simple scenarios, leaving operators under trained to handle the complexity of battle rhythms that occur during wartime.

Daily battle rhythm events often consist of morning and evening command-level updates and synchronization meetings required to execute planned operations. In low-intensity operations, the battle rhythm may be more deliberate, with daily, weekly, and monthly working groups and boards. During high-intensity operations, the battle rhythm will naturally be accelerated, but commanders and staff may be tempted to accelerate the battle rhythm more than would be useful. Battle rhythms developed during peacetime often do not effectively provide the commander and staff with timely information to make decisions. Inappropriately accelerated battle rhythms lead to operator error and degradation in situational awareness as efforts to provide updates displaces time required to perform sufficient analysis.
The Navy seeks a tool incorporating AI/ML that will support operators through the full range of complexity seen in modern warfare. The technology sought will assist USW-DSS operators in reliably and rapidly conducting battle rhythm tasking, whether during relatively routine operations or the most complex wartime scenarios. This tool does not currently exist in commercial industry.

The desired technology will make use of the range of authoritative data available to USW-DSS, to include bathymetry, sound propagation paths, track information for United States Navy (USN), allies, threats, and any commercial vessels, and planned maneuvers for USN and allied forces. From these data, the desired technology will provide situational awareness to support appropriate battle rhythm tasking, to regularize the balance between providing updates and performing the analysis and planning required to support effective courses of action. It is envisioned that the desired technology could allow machine learning to support a battle rhythm that is responsive to conflict complexity without overwhelming staff to the detriment of force effectiveness. The tool will support situation awareness, enable battle rhythm tasking, and provide a user interface to facilitate these capabilities. Finally, the tool will need to be compliant with existing information assurance (IA) and other cybersecurity requirements.

The Navy is not aware of existing commercial products that can be used to perform the requested battle tempo management support across complex, war-time scenarios, given the need to use authoritative data at multiple classification levels in a manner that complies with existing IA and other cybersecurity requirements.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for a battle rhythm tool that meets the requirements as stated in the Description section above. Demonstrate the feasibility of the concept in meeting Navy needs and will establish that the concept can be feasibly produced by sample testing, modeling and simulation and or analysis. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop, demonstrate, and deliver a prototype battle rhythm tool and conduct a series of user designed sprints with fleet operators to refine the prototype for evaluation. The prototype will be evaluated to determine its capability in meeting the defined performance goals System performance will be demonstrated through installation and prototype testing using the cloud-based USW-DSS prototyping infrastructure provided by the government
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the battle rhythm tool to operational builds of USW-DSS provided to Fleet operators. Finalize the software design and algorithm prototype for evaluation to determine its effectiveness in an operationally relevant environment. Support the Navy for test and validation in accordance with the IWS 5.0 USW-DSS Peer Review Group.
The technology will have additional commercialization potential for other military command structures and civilian disaster-response and emergency management organizations.

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KEYWORDS: AN/UYQ-100 Undersea Warfare Decision Support System; complex wartime scenarios; battle rhythm; artificial intelligence or machine learning (AI/ML); effective courses of action.

N252-098 TITLE: Flow-Exposed Conformal Array Acoustic Elements

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a shock-hardened, modular, hydrophone for the exterior of flow-exposed hull surfaces, (e.g., the bow array) for better acoustic detection performance and decreased maintenance.

DESCRIPTION: Submarines and other undersea warfare systems (USW) use passive sensor information to develop track information (bearing, range, course, and speed vs. time) of maritime entities. Elements of the submarine bow array within the AN/BQQ-10 submarine sonar system have traditionally been embedded inside the outer hull within a thick layer of material.

While the material has historically provided performance benefits, the position of hydrophones in the traditional design has meant that individual hydrophones, if failed, would require extensive work to repair or replace. Traditional designs have considered each constituent hydrophone to be an element of the whole. This makes replacing or repairing each individual hydrophone prohibitively costly.
The Navy seeks a hydrophone and modular array design and prototype that is formed modularly with surface-mounted hydrophones. This will minimize the cost and complexity of any repairs that might be required after initial installation. The prototype will undergo Navy testing to determine that the design meets the Navy's requirement as a pre-requisite to making a decision to adopt the new design for future Navy arrays. Currently there is no commercial product that meets this need.

Key elements of such a design would be to retain or improve on the performance associated with current hull-mounted hydrophones. An example is the DT-574 hydrophone. Individual hydrophone elements will be replaceable while maintaining hydrodynamic smoothness across the speed range expected of submarine operations. The design will have a rho of 1.03, making the total array either be neutrally buoyant or even slightly positively buoyant . The thickness of the array would need to be no more than 9.5 inches.

The hydrophone should be able to achieve non-degraded pressure-sensing capabilities while exposed to travel induced flow. In addition, the hydrophone will need to survive shock. For flank and sail locations the hydrophone would need to achieve the threshold Shock Grade B qualification. For bow implementation, the hydrophone would need to achieve Shock Grade A qualification.
The technology architecture will be extensible to multiple types of hull arrays. In addition to requiring that the flow-exposed acoustic elements meet minimum lifetimes of numerous years, the Navy would also be interested in processes to assist operators and maintainers to understand how long each element could be expected to last before replacement would be required.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating

procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for a modular hull array architecture and a replaceable surface-exposed hydrophone that meet the Description parameters. Show feasibility through analysis, modelling, simulation, and testing. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop, demonstrate, and deliver a representative prototype of the solution for both modular array architecture and individual hydrophone performance. Demonstrate the prototype meets the required range of desired performance attributes given in the Description. System performance will be demonstrated through installation and prototype testing conducted by the Navy.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology to Navy use for hull-mounted arrays for future spirals of a submarine class (e.g., Virginia-class submarines) or a future submarine class (e.g., future SSN). Support the Navy in transitioning the technology to Navy use aboard submarines. Demonstrate and report on performance during laboratory testing. The prototype will be integrated into submarine arrays for which IWS 5.0 develops updates.
The technology could be extended to any passive sensor, including sensors aboard surface ships and unmanned vehicles. The technology would be of greatest use in large or complex arrays where individual hydrophone failures carry disproportionate risk of distorting total array performance, such as arrays for ocean exploration in support of the oil and gas industry.

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KEYWORDS: Hydrophone-based passive array; modular array; undersea warfare systems; DT-574 hull array transducers; neutrally buoyant; surface-mounted hydrophones

N252-099 TITLE: Indirect Laser Detection and Characterization Device

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Directed Energy (DE);Integrated Sensing and Cyber;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Design, develop, and demonstrate critical components and elements for a robust, compact, self-powered (rechargeable batteries), advanced optical sensing system for the detection, classification, and tracking of the HEL in a cluttered environment to provide early cueing of self-defense systems in either a man worn, deployed on a vehicle with tiered fixed/mobile networking, and utilizing both unmanned and manned platform concepts. Due to the rapidly escalating threat that high energy lasers present to Armed Forces of the United States, it is desirable to have reliable early warning system for "tip off" alert to lasers being used for sensing or damaging personnel or platforms. This SBIR topic seeks to augment personnel, either a stationary or mobile, or in a tiered capability with a unique optical sensor capable of sensing, warning, identifying, and potentially enabling the automatic use of countermeasures to address threats. The capability also may offer measurement and signature intelligence (MASINT) when mounted on remotely operated or autonomous vehicles.

DESCRIPTION: Due to the rapidly escalating threat that HELs present to the Armed Forces of the United States, it is desirable to have a reliable early warning system for "tip off" alert to lasers being used for sensing or damaging personnel or platforms. This SBIR topic seeks to augment personnel, either stationary or mobile, or in a tiered capability with a unique optical sensor capable of sensing, warning, identifying, and potentially enabling the automatic use of countermeasures to address threats. The capability also may offer measurement and signature intelligence (MASINT) when mounted on remotely operated or autonomous vehicles. Dual sensing sensors, both in visible & near infrared (VIS/NIR), and Short-wave infrared (SWIR) are near term potentials that should be realized, however additional sensing capabilities in the mid-wave and long wave infrared (MWIR/LWIR) wavelengths are also of high interest areas for potential innovation, but believed to be beyond threshold requirements for a man-wearable or man portable system and near-term solution for the initial architecture.

HEL weapons represent a new and disruptive threat to Armed Forces worldwide. The operational attributes of this class of weapon present a unique detection and defense problem. There is a need for advanced sensing to support initial detection ("tipoff"), as well as triggering defensive protective action as well as enabling defensive counter-targeting or countermeasures that inhibit laser weapon effectiveness. A unique attribute of laser weapons is the ability to be silent and potentially invisible to the human eye, resulting in an ability to counter forces from many potential directions, vastly complicating entry or mission capabilities. The potential damage to sensing technologies are problems that increase the potential for mission failure. To be useful, a cost-effective, low false-alarm rate, distributed, early-warning sensing architecture that utilizes “off axis” sensing through coherent light atmospheric scattering of visible, near infrared, short wave infrared, mid-wave infrared, and long-wave infrared is required to

provide "tipoff" to alert Armed Forces of and characterize (targeting, sensing, damaging, or lethal) incoming laser threats.

The attributes of such an architecture include, but are not limited to:

a) a passive sensor with off axis (non-direct illumination) laser classification capability that includes wavelength, pulse repetition frequency, waveform and pulse duration,
b) a capability to store and retrieve a time stamped datagram that includes wavelength, pulse repetition frequency, waveform and pulse duration,
c) a capability to relay communications using standard server like networking queries (e.g., Transmission Control Protocol/Internet Protocol (TCP/IP) or User Datagram Protocol (UDP) remote services),
d) a wide field of view (threshold: 120 (+/-60) deg horizontal, -20 to +90 deg vertical; objective: 190 (+/- 95) degree horizontal, -20 to +120 vertical),
e) Long sensing duration (threshold: 24 hours; objective: 72 hours) per charge,
f) Replaceable batteries (e.g., 18650) and externally powered (objective: Universal Serial Bus Type-C (USB-C) connector),
g) a cost-effective and covert platform,
h) small, compact (less than 500cc's, 30 cubic inches), lightweight (less than 500 grams), with ability to be single man portable or wearable,
i) Waterproof (IP67),
j) Operating Temperature of -40 to +65°C ,
k) Sensor Array (SA) metrics for discrimination, image capture, and determination characteristics include, but are not limited to:
    1) Dual sensing sensors, format: 1024 x 1024 (visible & near infrared (NIR)), and 256 vertical x 512 horizontal (Short-wave infrared (SWIR),
    2) Sensing range of 50 meters off-axis (perpendicular) with detection and classification from either a blue, green or red (405, 532 or 880nm) or a NIR/SWIR (1064 or 1660nm) - 50 milliwatt high energy laser source in a clear, unobstructed viewing path with quiescent, non-turbulent atmospheric conditions,
    3) Sensing time and characterization of no more than 0.5 seconds,
    4) Sensor scanning frequency of no less than 60hz (threshold), and
    5) Sensor false alarm rate of no more than 1 per million, including solar rejection.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Define sensor requirements in terms of power, volume, weight, noise limitations, motion limitations, and so forth. Identify specific configuration(s) to be included, and develop the strategy and design of integration and scale of the device and attachments. Define a prototype design, operation within a system construct to include the requirements of low detectability, software, and communications to allow the integration of a cooperative, networked sensor array, in either wired or wireless networks.

Reach a demonstratable benchtop device with a prototype design and manufacture plan to be implemented under Phase II.

PHASE II: Develop a prototype device that can perceive, identify, and characterize a laser in an off-axis sensing and perform data collection. Further develop a prototype and demonstrate the prototype or unmanned platform. Perform ground- or sea-based trials data collection of individual vehicles in terms of feature identification performance, operational agility, and accuracy. Perform limited field and/or sea trial to collect test data analysis when mounted on ground-based or airborne platforms.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Complete final testing and perform necessary integration and transition for use in counter-laser surveillance and monitoring operations with appropriate current platforms and agencies and future combat systems under development.
Commercially, this product could be used to enable remote laboratory safety systems, and enable satellite monitoring or laser-based communications.

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KEYWORDS: laser, detection electro-optic; surveillance; classification; remote sensing; Machine Learning; Artificial Intelligence; AI/ML

N252-100 TITLE: Efficient Multiphysics Modeling Framework for Rain-Induced Damage and Aerodynamic Effects on Hypersonic Vehicles

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Hypersonics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and validate automated multiphysics computational tools to accurately predict raindrop interactions with hypersonic vehicles, including droplet-shock interactions, breakup dynamics, flow-field modifications, and thermal protection system (TPS) material damage, while minimizing user expertise requirements and computational costs.

DESCRIPTION: Hypersonic vehicles will encounter various weather conditions, including rain. Raindrops can threaten structural integrity and TPS performance, potentially leading to catastrophic failures [Ref 1]. Rain droplet impacts during flight pose substantial risks to the structural integrity and performance of TPS, potentially leading to catastrophic failures [Ref 1]. Interactions between droplet impacts and the surrounding flow field can disrupt vehicle maneuverability, increasing the risk of loss of control. Existing modeling tools, originally developed for blunt cone geometries, are inadequate for addressing the complexities of modern maneuvering hypersonic vehicles [Ref 2]. New tools are required to simulate droplet interactions with hypersonic shock waves and boundary layers, characterize breakup dynamics, and evaluate flow field modifications.

High-speed droplet impingement is a challenging problem involving a wide variety of physical mechanisms, including aero-breakup and phase change. Modeling droplet impingement at high Mach numbers involves a wide range of physical mechanisms, including material and flow discontinuities, interfacial instabilities, phase change, and turbulence [Ref 3]. Additional complexities stem from the limited algorithms available that can handle high Mach number flows while maintaining a high-order of accuracy, as the variable reconstruction and flux differentiation steps can lead to unbounded volume fractions, negative speeds of sound, or partial densities, requiring positivity-preserving numerical schemes. Advancements in multiphase computational fluid dynamics (CFD) methods have improved the simulation of rain droplet interactions with hypersonic vehicles. For example, diffuse interface multiphase methods provide a simple, computationally efficient, and numerically robust approach to model droplet breakup and impingement [Ref 4]. Additionally, Adaptive Mesh Refinement (AMR) allows for efficient tracking of important flow features such as gas/liquid interfaces, shocks and wakes and the target geometries. Diffuse interface methods, coupled with AMR, have enabled modeling of droplet-shock interactions and instabilities, such as Rayleigh-Taylor (RT) and Kelvin-Helmholtz (KH) [Ref 3]. Integration of CFD with structural solvers like peridynamics has enhanced predictions of material damage during high-speed impacts [Ref 5]. GPU-accelerated multiphase solvers have demonstrated up to 300x speed-ups while maintaining fidelity for shock-droplet interactions [Ref 6].

Experimentally, the effects of weather encounters are difficult to simulate through ground testing. Historically, most knowledge of weather encounters has been provided by flight testing, which has numerous shortcomings such as expense, limited data throughput, and uncertain characterization of the environment. Developments in high-speed diagnostics and advances in experimental campaigns,

including electromagnetic-launcher tests, have validated droplet deformation and aerobreakup under hypersonic conditions [Ref 7]. Despite this progress, gaps remain in material modeling, robustness, and aerodynamic impact analysis, limiting current tools.

Challenges persist in accurately modeling advanced composite materials, such as carbon-carbon, including accumulated damage effects like surface roughness evolution and its impact on erosion rates. Debris shielding, where water and material fragments alter subsequent droplet impacts, remains underexplored [Ref 8]. Integrating key physics like phase change and cavitation into robust simulations is critical. Current tools provide limited insight into aerodynamic changes, including forces and moments crucial to vehicle stability. Addressing these gaps requires next-generation tools that leverage GPU acceleration for scalable simulations while fully integrating composite material behavior, shielding effects, and aerodynamic impacts.
The target requirements for this SBIR topic include:

- Robust Multiphysics Integration: Simulate raindrop interactions with hypersonic vehicles, incorporating critical physical phenomena such as droplet-shock interactions, phase change, and arbitrary equations of state, while ensuring computational stability.
- Advanced Material Modeling: Predict the behavior of advanced composite materials, such as carbon-carbon, capturing high-strain-rate responses, crack propagation, and accumulated surface damage. Fully couple the material response with CFD solutions.
- Shielding and Ejecta Tracking: Develop the ability to track ejecta and model shielding effects, including debris impacts on vehicle surfaces and resulting flow field changes.
- Aerodynamic Effects Analysis: Provide predictions of aerodynamic changes, including forces and moments caused by surface deformation, to assess vehicle stability under adverse conditions.
- High Computational Efficiency: Leverage GPU-accelerated architectures and AMR to achieve scalable, high-fidelity simulations for realistic operating conditions.
- Automated Pre- and Post-Processing: Implement workflows for solver setup, grid generation, and data analysis to minimize user intervention.

PHASE I: Develop a prototype multiphysics modeling framework for simulating raindrop impacts on hypersonic vehicles. Incorporate key phenomena such as droplet-shock interactions, breakup, impingement, and material response. Validate using canonical geometries and existing experimental data. Assess accuracy and feasibility. Define a path for platform-independent computation and establish a roadmap for scaling to complex, three-dimensional vehicle geometries. Prepare a Phase II plan.

PHASE II: Develop a fully integrated multiphysics simulation framework for raindrop impacts on hypersonic vehicles. Incorporate advanced material response models, including damage evolution for carbon/carbon. Implement shielding and ejecta modeling, adaptive mesh refinement, and critical physics such as phase change. Validate and demonstrate the framework on realistic hypersonic vehicle geometries using existing experimental data. Ensure efficient operation on emerging computing platforms and implement workflow automation to minimize user intervention, enabling repeatable and high-quality results.

PHASE III DUAL USE APPLICATIONS: Transition the developed simulation framework for raindrop impacts on hypersonic vehicles to practical applications within the Department of Defense (DoD) and commercial sectors. Validate and optimize the framework for a wide range of hypersonic vehicle configurations, materials, and flight conditions. Collaborate with industry partners and DoD agencies to ensure the framework meets deployment standards and operational requirements. Develop comprehensive training programs, user documentation, and technical support resources to enable widespread adoption and effective use by non-expert users.

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KEYWORDS: Hypersonic Vehicles; Multiphase Flows; Droplet Impingement; Shock-Induced Aerobreakup; Weather Erosion; Multiphysics Simulation; High-Strain Rate Response; Adaptive Mesh Refinement (AMR); High-Performance Computing (HPC)

N252-102 TITLE: Hardware-Level, Reverse-Engineering Resistant Logic Designs for Standard Complementary Metal-Oxide-Semiconductor (CMOS) Processes

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Microelectronics

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OBJECTIVE: Develop a hardware-based design for standard Complementary Metal-Oxide-Semiconductor (CMOS) processes that stymies a range of common attacks and limits the ability of third parties to understand the functional behavior of fabricated chips.

DESCRIPTION: With the advances in various technologies, integrated circuits have become increasingly susceptible to reverse engineering. De-layering of an integrated circuit, for instance, reveals the interconnections between transistors and permits discerning the behavior of function blocks. This, in turn, enables eventually reconstructing the netlist and hence understanding the behavior of the full system. Adversaries would thus be free to copy the designs for their own uses. They could also probe them for faults and develop countermeasures. If these adversaries can insert themselves into the supply chain, then they would additionally be able to introduce back-doors, kill switches, and other mechanisms into the designs.

Reverse engineering is an especially significant threat to small embedded systems that operate in contested areas. Both commercial and non-commercial entities have shown that integrated-circuit extraction is possible and that those circuits can be successfully reverse engineered even when non-standard, hardware-level encryption is employed. Embedded, on-chip memories are also no longer safe from analysis.

Due to the threat that reverse engineering poses, several groups have proposed integrated-circuit camouflage approaches. These include the incorporation of dummy contacts, vias, and filler cells along with programmable standard cells. However, each of these has substantial issues. Dummy contacts, vias, and filler cells have no physical connections with the actual circuit. They thus do not hinder the de-layering process and serve to only marginally slow the progress of automated tools. Using programmable standard cells is incredibly costly, as it is a non-standard process at most, if not all, semiconductor foundries. As a consequence of these limitations, research has concentrated on the development of diverse logic families that rely on different threshold devices to conceal the hardware. The physical appearance of these devices is identical for different functionalities and is extremely difficult to discern from imaging and de-layering. However, many of the proposed logic families rely on the absolute values of the threshold voltage due to the single-ended configuration. They require external bias voltages to control the switches for proper orientation. Process-voltage-temperature variations can significantly affect these logic families and ultimately cause them to malfunction. Switching between nodes is also highly complicated. The Office of Naval Research (ONR) seeks the submission of SBIR proposals that outline novel approaches for resisting reverse-engineering efforts. These approaches should focus on hardware-based protection mechanisms that are completely agnostic to the chosen application. Any SBIR proposals that

include solutions focused either primarily or solely on software-based mechanisms will not be considered responsive to this topic.

There are several requirements for a hypothetical hardware-level obfuscation approach that are found in existing offerings. Some additional research is hence needed. Any proposed approach shall be suitable for both synchronous and asynchronous architectures. The approach shall be implementable in any standard CMOS process without the need for custom masks to both demonstrate basic attack-resistant functionality and achieve reasonably high chip yield rates. Custom masks can be used if they exhibit clear advantages with regard to chip power efficiency, performance, and reliability. To ensure that any proposed approach has broad applicability for transition sponsors and commercial partners, newer nodes shall be targeted. These shall include, at the minimum, either a TSMC 28-nm node or a GlobalFoundries 22-nm node during the first two phases of this SBIR and either a TSMC 16-nm node or a GlobalFoundries 12-nm node in the third phase of this SBIR. The change in nodes is necessary to validate that the approach is mostly, if not entirely, process independent. Regardless of the node, the proposed approach shall demonstrate a clear resilience to imaging-based attacks and de-layering. It shall be impervious to optical, laser, and scanning-electron-microscope sensing modalities along with any similar modalities. It shall also be resistant to side-channel attacks based on time, voltage, and current measurements. Any side-channel measurements should have, at most, a five-percent deviation per operation. Some resistance, if not complete resistance, to laser-fault injection should be theoretically demonstrated and empirically validated. Likewise, some resistance to machine-learning (ML) attacks should be empirically demonstrated. The types of ML attacks will be specified in the third phase of the SBIR. Lastly, there are hard requirements on the number of additional transistors that can be used in any proposed approach compared to the non-obfuscated designs. For an n-input, 1-output Boolean function block, the total number of transistors allowed in the obfuscated design should be either less than or equal to n + (n + 1) x 2^(n - 1). The number of stacked transistors, if used, shall be less than or equal to n - 1. These transistor requirements ensure that any proposed approach strikes a reasonable balance between security, silicon area, power, and performance.
Design Requirements: A hardware-based approach that stymies a range of common attacks and prevents, as much as possible, third parties from understanding the functional behavior of chips fabricated using it. This approach shall possess the following traits:

- Cover all Boolean logic within a single functional block with n inputs and a single output, with n being at least four.
- The number of total transistors for an n-input functional block shall be less than or equal to n + (n + 1) * 2^(n - 1).
- The number of stacked transistors shall be less than or equal to n - 1 for an n-input functional block.
- Implementable in any standard CMOS process for a 28-nm node and below.
- Implementable without the need for custom masks.
- Complete application agnosticity.
- Less than five percent side-channel signature variation per logical operation.
- Usable for synchronous and asynchronous architectures.
- Resilience to vision-based attacks that use various sensing modalities to uncover functional-block and system-level behavior.
- Resilience to side-channel attacks based on time, voltage, and current measurements to deduce functional-block and system-level behavior.
- Partial resilience to common machine-learning attacks that attempt to infer functional-block behavior.
- Partial resilience to laser-fault injection attacks.

Technical challenges: Ideally, the proposed approach should be implementable in only a few more gates, compared to non-obfuscated designs, and therefore offer low-power, high-performance capabilities for a given node.

The proposed approach, and chips fabricated using it, may be used for applications in harsh environments not currently considered by the acquisition program. A path forward for extreme-temperature operating conditions, such as those between -40 C and 120 C, shall be established in the design stage, even if it is not implemented in the prototypes. The chips will not be used in environments where direct contact with water is expected. They will also not be used in environments where either high background radiation, high humidity, high pressure, or some combination thereof, is expected. Lastly, they are not expected to be used in biological organisms.

PHASE I: Outline a hardware-level logic obfuscation approach that is resistant to reverse engineering by optical, laser, and scanning-electron-microscope sensing modalities, reverse engineering by analyzing the routing and metal traces along with the logic architectures, and reverse engineering to side-channel signatures, like timing, current, and voltage. Develop a comprehensive simulation for advanced nodes, such as TSMC 28-nm or below and GlobalFoundries 22-nm or below. Outline the functionality, reliability, and expected yield of the approach, which includes Monte Carlo simulations. Prepare a Phase II plan.

PHASE II: Design, fabricate, verify, and test a simple test chip that uses the proposed hardware-level obfuscation approach. This chip could include, for instance, simple logic gates, adders, and registers. A more complex design shall be considered following the success of this first chip. This could include, for instance, either a 128-bit AES engine or an 8-bit microprocessor. Fabricate and evaluate more than fifty samples, ensuring that a majority of the samples exhibit the desired functionality so as to demonstrate that the approach achieves a high yield and prevents against optical-based and side-channel-based attacks. Build a standard digital library that follows the standard digital design flow. Ensure that this library shall be both timing and place-and-route compatible with existing electronic-design-automation (EDA) synthesis tools like Cadence and Synopsys. Provide a full analysis of power, silicon area, speed, and noise margin compared to standard logic designs for the designs made using this library. Ensure that any chips fabricated using this library are operational within the temperature range specified.

PHASE III DUAL USE APPLICATIONS: Investigate and show a resilience of the proposed approach to attacks that use laser-fault injection reverse engineering and machine-learning reverse engineering. For fabricated chip designs, provide a full analysis of the yield rate and compare to standard logic designs. Once such tasks are completed, develop an automated design compiler for converting existing digital designs to attack-resistant designs for processes such as TSMC 16-nm or below and GlobalFoundries 12-nm or below. This compiler may be created in conjunction with larger industry partners. Work with foundries to develop custom masks to improve performance, reliability, and yield of the proposed secure logic framework, if improvements would offer substantial operational increases. Specify plans to either license the designs or facilitate transition to a major design partner, with the aim to use both the design compiler and standard library in multiple programs of record. Pertinent industries to target during this effort would include mobile device manufacturers, medical device manufacturers, automobile manufacturers, and semiconductor manufacturers. Other relevant industries include those that are involved with designing and producing integrated circuits via a decentralized supply chain and where the risk of intellectual-property theft is high.

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KEYWORDS: Reverse Engineering; Camouflaged Gates; Hardware Obfuscation; Logic Obfuscation; Chip Security; Application-specific integrated circuit; Secure ASIC Design; Secure ASIC Fabrication

N252-103 TITLE: Virtual Sea-Shield -- Immersive Content Creation with Generative AI for Automated C5ISR Situational Awareness

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Human-Machine Interfaces;Integrated Sensing and Cyber;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an Artificial Intelligence (AI)-driven 4D content generation environment that efficiently processes all-domain Command, Control, Communication, Computers, Cyber, Intelligence, Surveillance, and Reconnaissance (C5ISR) information into high-fidelity immersive observables for collaborative situational awareness across the desired layers of naval operations.

DESCRIPTION: Virtual Sea-Shield (VSS) is an all-domain multilayered protective environment that utilizes AI-assisted automatic situational awareness to deter threats (i.e., jamming, unmanned vehicle attacks, missiles, etc.) from targeting naval assets. VSS is a decentralized 4D (time and space) digital content generation environment for building a self-healing adaptive multilayered defensive and offensive naval sea shield across various naval operations. VSS utilizes generative AI for automated content creation by processing all-source intelligence, operational objectives, and engagement plans. In autonomous operations, VSS can track, penetrate, and trigger automatic strikes on designated hostile assets, personnel, logistics nodes, ammunition depots, and supply lines – blunting adversaries' advances and disrupting its ability to maneuver, regroup, and launch attacks. Above all, VSS is a critical access infrastructure for a novel Autonomous Sea-Basing to support the deployment of forces, equipment, supplies, and warfighting capabilities.

Today, a multidisciplinary team of decision-makers, analysts, and warfighters conduct extensive wargaming exercises with a considerable investment in time and resources as they sift through vast amounts of data to look for indication and warning (I&W) signs and supporting evidence to anticipate and counter adversarial trends, strategies, and tactics that may have unfavorable impacts on the U.S. National Security interests. This human-centric approach to attain proper visibility into future events is fraught with the risk of costly errors, human biases, and omissions of valuable insights that may influence the assessment of evidence, skew statistical analysis, or color the recognition of cause and effect.
This SBIR topic envisions a systematic approach to integrate AI-based immersive technologies into the C5ISR environment with improved accuracy and agility over the current human-centric paradigm. VSS leverages state-of-the-art immersive technologies such as augmented reality (AR), virtual reality (VR), spatial reality (SR), and mixed reality (MR) as extended reality (XR) to transport humans from observation experience to immersion experience as they stimulate human attention to focus on I&W signs of emerging situations. The immersive technologies engage end-users to enrich collaborative teaming through natural language, dynamic visualization, and sound as they navigate spatial domains (i.e., sea, air, land, and space) and cyber. Generative AI-driven MR overlays provide a true-to-life presence and awareness that go beyond simply displaying pre-programmed digital elements. A successful VSS will generate realistic and adaptive content overlays that openly respond to human interactions with the contents and induce collaboration – during which generative AI tools will learn adversarial tactics and anticipate the evolution of engagement scenarios, and their consequences based on human decision-

making effects. By doing so, VSS brings about the synergy between humans and AI. It enables warfighters to home in on deterrence options, experiment with novel human-to-machine (i.e., unmanned aerial vehicle (UAV), unmanned underwater vehicle (UUV), space assets) or machine-to-machine teaming engagement opportunities, conduct joint operations rehearsal, or perform training exercises. The VSS performance test and demonstration applications for the proof-of-concept (Phase I) and prototyping (Phase II) may include operational scenarios that capture a series of hostile joint military and commercial mobilization and exercise activities to control contested waters such as amphibious landing and sea-lane blockade. With generative AI, the immersive VSS must automatically extract knowledge from multimodal datasets to demonstrate creative teaming of the blue forces required to penetrate an anti-access area-denial environment, where and how, with sustainable persistence while inside and outside a contested space. The quantitative metrics must include timely decision-making, and precise command and control for engagement strategies and tactics supported by the layered deployment of forces, equipment, and supplies that signify adaptive sea basing. The Phase I proof-of-concept will be unclassified. AI-assisted creative teaming and collaborations will be limited to unclassified multimodal contents (e.g., text, images, video, audio, spatiotemporal 4D models, or other data types) over commercial 5G networks. The VSS end-to-end design must consider distributed deployment, fault-tolerant connectivity, optimal configuration throughputs, and scalability.

Note 1: Phase I will be UNCLASSIFIED and classified data is not required.
Note 2: Contractors must provide appropriate dataset release authorization for use in their case studies, tests, and demonstrations, and certify that there are no legal or privacy issues, limitations, or restrictions with using the proposed data for this SBIR project.
Note 3: Work produced in Phase II may become classified. However, the proposal for Phase II will be UNCLASSIFIED. The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.
Note 4: If the selected Phase II contractor does not have the required certification for classified work, ONR or the related DON Program Office will work with the contractor to facilitate certification of related personnel and facility.

PHASE I: Develop a concept for generative AI models to generate VSS context-relevant content representing naval operational domains (i.e., sea surface, undersea, air, space, land, and cyber). Leverage generative mesh methods to produce high-resolution 3D textures and shades for the virtual world including text-to-3D and multimodal 2D imagery to 3D content. Apply the retrieval-augmented generation method to improve the accuracy and reliability of generative AI models with facts from external sources. Develop domain-relevant common large language models (LLMs) for defining, packaging, assembling, and editing 3D data for digital content creation applications and visualization. Identify training tools to speed up LLMs training and reasoning about unforeseen events (i.e., unknown unknowns). The VSS end-to-end design must consider distributed deployment, fault-tolerant connectivity, optimal configuration throughputs, and scalability. Demonstrate VSS can perform semantic extraction for automated analysis of all-source intelligence, live multidomain intelligence, surveillance, and reconnaissance (ISR) signals, and can learn I&W signs. Demonstrate the VSS performance facilitates novel virtual collaborative teaming configurations (e.g., human-machine and machine-machine teams) to

achieve synchronized operational planning and execution at key points in the contested littoral with immersive decision trees that compute engagement plans, options, and risks associated with the ups and downs of operational encounters.

The feasibility study, test, and demonstration must utilize a combination of OSINT datasets, synthetic datasets from DoN, MCML, unclassified AIS maritime traffic, unclassified commercial satellite imagery, or similar. AI-based data mining includes entity extraction (people, places, and objects), relationships, and transactions -- rigorously searched, recognized, differentiated, and organized into groupings of related facts and relations. Accuracy metrics to ingest and classify multimodal data: structured data mining and interpretation - accuracy of 95% over 98% captured content; unstructured data mining and interpretation – accuracy of 90% over 95% captured content. The software validation, verification, and performance consistency must analyze the VSS sensitivity (i.e., true-positive rate), specificity (i.e., true-negative rate), precision (i.e., positive predictive value), miss rate (i.e., false negative rate), false discovery rate, and false omission rate.

Deliverables: end-to-end initial prototype technology. Conduct appropriate T&E of the conceptual system. Prepare a plan for Phase II. Deliver a final report.

PHASE II: Develop a prototype of the immersive VSS from the candidate technologies in Phase I. Test and demonstrate VSS scenarios with representative operational data sources. Demonstrate the immersive human-machine interaction with the scenarios. Validate the VSS execution timelines to meet the Ops-Floor end-to-end operational requirements. Assess prototype's performance against the metrics detailed in Phase I. Conduct an "end-user satisfaction" survey, on a scale of 0 to 5: a) VSS situation understanding that includes events that go dark, disguised activities and maneuvers, dormant or not radiating targets; b) alignment with formal I&W signals; c) alignment with prioritized deterrence and engagement options; d) timeliness for responsive cross-domain decision-making and collaboration; and e) immersive virtual training. Develop a plan for Phase III with a transition path to a program of record (PoR).
Deliver prototype software, systems interface requirements for mobile and stationary devices, design documentation, source code, user manual, and final report.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Advance these VSS capabilities to TRL-7 and integrate the technology into the Maritime Tactical Command and Control PoR or ISR processing platforms at the Marine Corps Information Operations Center.
Once validated conceptually and technically, demonstrate dual-use applications of this technology in the video gaming industry.

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KEYWORDS: Immersive; Artificial Intelligence; Generative AI; Generative Mesh; Virtual Reality; Spatial Reality; Augmented Reality; Mixed Reality; Large Language Models

N252-104 TITLE: Compact Rotary Engine Materials, Coatings, and Architectures for Robust and Reliable Operation in a Marine Environment

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop advanced materials, coatings, and/or material-architectures, supported by experiment, modeling and simulation, to improve the efficiency, durability, and performance of rotary engines that operate in a marine environment.

DESCRIPTION: The U.S. Navy and Marine Corps has a growing interest in compact and efficient propulsion and power solutions, particularly for applications that demand high power density, operational reliability, and fuel flexibility. Rotary engines, with their favorable power-to-weight ratios and simplified mechanical designs, are well suited for Unmanned Aircraft Systems (UAS) and other compact vehicles that need versatile and durable power sources. Conventional rotary engines often suffer from limitations in durability and efficiency when exposed to the marine environment's high humidity and salt concentrations. Sand and dust, commonly encountered along coasts and in the desert, can also be problematic for these engines. Recent innovations, such as the "inside-out" Wankel and rotary vane designs, have introduced new cooling and sealing concepts that could be beneficial for Naval applications, but many challenges still need to be addressed.
Current Challenges
These challenges stem primarily from the high humidity, salt exposure, and abrasive particulates. Key issues include:

- Corrosion: Saltwater and high humidity accelerate corrosion on metallic engine components. Environmental, Galvanic, Crevice, and Pitting corrosion can deteriorate the structural integrity of critical engine components.
- Abrasive Wear: Sand, dust, and salt particles enter the engine through air intakes and cooling systems, causing abrasion on critical surfaces like apex seals, rotors, and housing walls. This wear diminishes efficiency, lowers power output, and shortens component lifetimes.
- Seal Degradation: Apex and side seals experience rapid wear in marine environments, leading to reduced compression and higher oil consumption.
- Heat Management: High humidity and salt exposure can impair heat dissipation. Poor heat transfer can result in the warping of lightweight materials, such as aluminum alloys, and accelerate material fatigue.
- Lubrication Breakdown: Exposure to moisture and salt can emulsify lubricants, reducing their effectiveness. Rotary engines that rely on oil for sealing face additional maintenance and leakage concerns caused by oil degradation.
- Coating Wear: Protective coatings that defend against corrosion and wear can deteriorate more rapidly in a marine environment, reducing their effectiveness.
- Limited Understanding and Modeling Capability: Poor understanding of degradation mechanisms complicates material, coating, and design choices while increasing modeling uncertainty. Derived fatigue and life performance estimates may be inaccurate, negatively impacting reliability, readiness, and capability.

Ultimately, this SBIR topic aims to advance the state of rotary engine materials and tribology, enabling reliable use on Navy and Marine Corps UAS. Proposers should demonstrate a strong understanding of Navy/Marine Corps unique material/tribology challenges and present a viable approach to solving them. Please note that, due to time and budgetary constraints, the authors are less interested in developing completely new engine designs. Priority will be given to efforts that leverage existing engines to demonstrate proposed technology benefits.

PHASE I: Evaluate the technical challenges and feasibility of the proposed material/coating/design solution to enhance the reliability and performance of rotary engines in marine environments.

- Technical Challenges Assessment: Conduct a detailed analysis of the material and tribological challenges associated with operating the target Rotary Engine in a marine environment. Consider specific issues associated with corrosion, wear, and thermal management.
- Solution Feasibility Assessment: Identify and assess material, coating, and architectural modifications that could address the identified challenges. Consider the practicality, cost-effectiveness, and compatibility of the proposed solution(s) with the target rotary engine application.
- Initial Risk Assessments: Identify and document any technical and programmatic risks associated with the proposed solution. Develop a risk mitigation plan that outlines specific mitigation strategies that will be used to address those risks.
- Modeling and Simulation: Use modeling and simulation tools to predict the performance and durability of selected materials, coatings, and design changes under relevant Naval conditions and to quantify how the proposed solution will mitigate issues without the need for extensive physical testing in Phase I.
- Proof of Concept Testing: Conduct proof of concept tests on a small scale to demonstrate the potential effectiveness of the proposed solution. (Note: This step may involve testing of individual components or material coupons to validate initial assumptions.)
- Cost and Schedule Estimation: Develop a detailed cost and schedule estimate for a potential Phase II effort. Include anticipated development, testing, and validation activities needed to mature the technology.

Deliver a final report summarizing the findings, assessing the feasibility of the proposed solutions, identifying technical and cost risks, and recommending next steps for development in Phase II.

PHASE II: Focus on the detailed design, optimization, fabrication, and testing of the materials, coatings, and/or design modifications identified in Phase I, including:

- Detailed Solution Design and Optimization: Develop and refine designs based on Phase I findings, optimizing materials, coatings, and structural modifications to meet marine environment requirements. Include advanced modeling and iterative simulations to ensure the proposed solution addresses challenges effectively and if necessary.
- Prototype Fabrication and Integration: Fabricate prototypes of the optimized components/assemblies for testing. Ensure that the prototypes integrate selected materials, coatings, and design improvements in a manner that closely replicates final application conditions for rotary engines in marine environments.
- Testing and Characterization: Conduct rigorous testing and characterization of the prototypes under simulated marine conditions, including exposure to saltwater, humidity, sand, and representative stress/thermal loads. Perform cyclic corrosion testing, inclusive of representative temperature variations, to demonstrate solution durability under relevant operating conditions. Assess key performance metrics, such as corrosion resistance, wear rate, sealing integrity, and thermal stability, as appropriate. Testing should aim to validate the effectiveness of the solution and identify additional refinements needed to increase the durability/reliability of the proposed solution.

- Performance Evaluation and Optimization: Analyze test data to understand the performance of the proposed solution. Based on test results, optimize the design, materials, and/or coatings as needed to further enhance the durability while reducing maintenance requirements in a marine environment.
- Risk, Schedule, and Cost Updates: Reassess technical/programmatic risks and update the cost/schedule estimates based on testing outcomes and prototype performance.

Deliver a comprehensive report detailing the prototype performance, testing outcomes, solution optimizations, and updated risk/cost/schedule assessments. Recommendations for further development or scaling should also be included, along with any insights gained for potential Navy/Marine Corps/Commercial applications.

PHASE III DUAL USE APPLICATIONS: • Collaborate with Navy stakeholders, aircraft manufacturers, and/or engine original equipment manufacturers (OEMs) to further mature and integrate the technology into Navy/Marine Corps relevant platforms.

- Transition Plan Development: Develop a plan to transition the proposed technology to a relevant Navy/Marine Corps application.
- Further Technology Maturation: Further refine the technology solution in preparation for technology transition.
- Work closely with industry partners to redesign/optimize certain aspects of the technology to meet transition requirements as necessary and to understand military/commercial requirements and formulate a realistic plan to deploy the technology. Maturation efforts that increase the power density, reliability, and overall performance of rotary engines will further increase customer interest in the technology.
- Optimize the design for manufacturability and scale up production to meet customers' needs.

The proposed rotary engine materials/tribology technology may be applicable across multiple industries (e.g., automative, commercial aviation, power generation).

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KEYWORDS: Rotary Engine; Hybrid Electric; High Temperature Materials; APEX Seals; Vane Seals; High Temperature Tribology; High Temperature Wear and Lubrication; High Temperature Ionic Lubrication; Advanced Ceramics; Additive Materials; Digital Engineering; ICME; Co

N252-105 TITLE: Machine Learning Downscaling Capability for Environmental Forecasts

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Sustainment;Trusted AI and Autonomy

OBJECTIVE: Develop a capability to generate skillful, near real-time environmental forecasts (of the atmosphere, ocean, sea ice, and/or ionosphere) at a much higher spatial horizontal resolution (less than 1 km) and vertical resolution (on the order of 10 m, particularly in the atmospheric boundary layer and/or upper ocean) than current machine learning weather prediction (MLWP) techniques using downscaling or similar methodologies for tactical/local scale applications.

DESCRIPTION: While rapid and impressive progress in the development of skillful MLWP and related environmental models over the past several years has been promising for improving near real-time forecasting, missing technological capability requirements must be solved before such tools can be used operationally. Among the biggest gaps in MLWP - as well as in operational numerical weather prediction (NWP) and even climate simulations - is the disconnect between the improved predictability from large/global scale physics and observations and the lack of forecast fidelity at the local/tactical scale (less than 1-km resolution in the horizontal and high-fidelity in the vertical). Some progress can be made rapidly through the use of ML methods ameliorating the intensive computing requirements, but other progress is needed in scientific methodology to attain higher resolution forecast from lower resolution models informed by and/or consistent with physics-based principles and/or models (see references). This SBIR topic seeks to leverage advances in ML methods, environmental prediction, and downscaling/super resolution techniques to develop new capabilities for targeted local and tactical scale short-range to medium-range forecasts. Many uses of environmental predictions (e.g., tropical cyclone wind and surge effects, electro-optical or acoustic propagation in the boundary layer, coastal winds, terrain-induced phenomena, aviation visibility, ice-edge circulations, ocean eddies, navigation through sea ice) require knowledge of small-scale features to properly calibrate the effects of a forecast. Efforts will synthesize various methodologies to take a coarse set of environmental prediction information and utilize additional data, ML methods, and modeling techniques to better inform predictions of tactical/local scale effects. A particular emphasis on validating realistic environmental structures, particularly given the lack of observational data at these scales, will be necessary.

PHASE I: Develop and compare innovative approaches for performing high-resolution (less than 1-km horizontal grid spacing and high-fidelity in the vertical) short-range to medium-range MLWP forecasts from a lower resolution set of historical data, observations, and forecast fields. Perform a technical feasibility survey of state-of-the-science methods for developing high-resolution environmental fields, such as downscaling, nesting, mesh refinement, and super resolution sharpening and upscaling. Inform the final selection(s) of methodology by the theoretical background review as well as targeted cases studies to demonstrate the strengths and weaknesses of these methods. Prepare a final report that must include a summary of strengths and weaknesses of viable approaches, a comparative analysis, and viability for operational use (such as forecast stability, bias/accuracy, and limitations). Additionally, a Phase II plan should be included.

PHASE II: Expand demonstration into an end-to-end prototype high-resolution (horizontal and vertical) MLWP forecasting capability. While initialization, coarse forecast, and post-processing components are necessary, focus will be on improving the skill and capability of the downscaling phase of the production – in particular, addressing the fidelity of components that support resolving smaller scale processes. While explicit use of high-resolution features such as topography, bathymetry, land and sea ice character, vegetation, land surface type, urban character, etc. may not be necessary, ensure that the model should properly account for the effect of these and other local scale influences. Ensure that the downscaling capability is portable and globally relocatable to arbitrary grid sizes and resolutions at any worldwide

location, with the ability to scale available data and quantify potential skill as needed. Develop a workflow that includes robust methods of validation and verification and identifies strengths and weaknesses of the product compared to traditional NWP downscaling and nesting. Perform multiple demonstrations in coordination with field testing (may be required), particularly to validate local scale effects that are not easily verified given current environmental observing network coverage. Submit required Phase II deliverables to include regular reporting, participation in program reviews, technical documentation, and the end-to-end prototype software at the conclusion of the effort.

PHASE III DUAL USE APPLICATIONS: Perform operational hardening and establish utility and trust for real-time application. Craft and demonstrate dynamic analysis software tools that quickly and accurately convey software system health, error logging and debugging, and processing metadata. Develop additional evaluation metrics and diagnostics to facilitate expert forecaster guidance on using the product (and comparing to other forecast tools in workflow). Ensure that the system has a formalized methodology and data/compute needs for model training and a separate, leaner set of requirements for operational runs, and fully documented. (Note: It is essential that a version of this system is able to run skillfully in a forward/limited compute environment.) Ensure that the techniques are generalizable to apply to a variety of environmental modeling use cases such that follow-on work and commercial applications can be addressed.

Dual-use applications will include partnering with other intergovernmental meteorological agencies such as U.S. Air Force, National Oceanic and Atmospheric Administration (NOAA), and National Aeronautics and Space Administration (NASA) as well as commercialization in multiple potential markets with high-resolution decision making requirements based on forecast skill, such as transportation (short time scales) and agriculture (longer time scales).

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KEYWORDS: machine learning; artificial intelligence; ai/ml; meteorology; oceanography; forecasts; high resolution; machine learning weather prediction (MLWP); downscaling; training datasets; transfer learning; data assimilation; parameterization; high resolution

Version 4

N252-106 TITLE: Innovations in Heater Technologies for Optimized Heating during Welding

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Develop and demonstrate reliable preheat and post weld heat treatment technology that can consistently heat and operate at temperatures up to 1000°F in a shipyard environment and in accordance with Navy requirements.

DESCRIPTION: Preheating and post-weld heat treatment are common requirements for the fabrication of Naval platforms. Preheating is required on a variety of naval alloys including high yield (HY) steels, high strength low alloy (HSLA), mild steel, stainless steel, copper nickel, nickel copper, and aluminum. Depending on material and thickness, temperatures can vary up to 1000°F.

In the past, heater bars could be repaired but recently the devices have changed from repairable tools to a consumable product. When heater bars, power streamer cords, or control units are broken, there is no alternative process to preheat the materials. Critical units are forced to stop and wait for equipment to become free to continue the process, leading to delays throughout the entire shipbuilding process. Heater bars and power streamer cords are no longer durable or reliable, and the current products have a lifecycle that is unpredictable. Some common issues with the heater bars are electrical shorts, connector burn out, and critical failure of the heating element. If any part of the heater bar breaks, the entire unit is unusable. Other limits such as their shapes, weights, and lack of technology monitoring impede use for specific applications requiring unique methods to ensure the products are properly heated prior to or during the welding process, which increase costs. Currently no alternative preheating solutions are suited for shipbuilding environments and there is a lack of availability and innovation in the industry to meet this requirement. The current technology is antiquated with little to no advancement in recent years. Current heating methods (induction and resistance) have availability, attachment, control, and monitoring issues that negatively impact process methods and costs.

The goal of this SBIR topic is to develop and pilot an improved technology for pre- and post-steel heating for welding. The project will investigate and identify the technology needs such as temperature control and increased durability. In addition to currently using inductive and resistive heating, other technologies could be explored.

PHASE I: Explore existing and innovative heating technologies and control mechanisms to identify a solution that can meet the Navy requirements found in reference documents such as Tech Pub 278 and Tech Pub 1688 [Refs 1,2]. Demonstrate the feasibility by a breadbox demonstration. In addition to developing the heater the proposal should address reparability, and sustainability, and data monitoring capability of the heater and control unit. Prepare a Phase II plan.

PHASE II: Develop an appropriate hardware to insert the technology in a shipyard environment. Demonstrate the robustness of the technology, reparability of the heating system, and data monitoring and control capabilities of the system.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the system and control unit, if successful, to all OEMs for naval platforms including private shipyards, public shipyards, and various repair and sustainment facilities. This technology would be applicable to all naval facilities that perform welding operations in both acquisition and sustainment.

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KEYWORDS: Preheat; post-weld heat treatment; welding; temperature control; temperature monitoring; data capture; heaters

Version 4

N252-107 TITLE: A Hybrid Digital Twin Framework for Modeling Machine Process Parameters for Automated Fiber Placement-based Manufacturing of Composite Structures

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

OBJECTIVE: Develop a physics-based and artificial intelligence (AI) driven digital twin framework to tailor automated fiber placement (AFP) process to manufacture complex composite structures with improved manufacture rates and quality.

DESCRIPTION: Low-cost composite airframes are strongly demanded by DoD and commercial industries to reduce total life cycle cost through the use of integration of automation and digital manufacturing of complex composite structures. AFP has been used extensively by major aerospace industries such as Boeing and Spirit AeroSystems for the manufacturing of large-scale composite fuselage structures. However, the poor process planning and time-consuming operator-dependent manual inspection can significantly interrupt the manufacturing process and increase the manual rework rate due to the intolerable level of fabrication-induced defects. These interruptions and need for artisan intervention to maintain part quality diminish the benefits of AFP-based composite manufacturing. An increasing amount of research and technology development has been devoted to implementing a data-driven planning, in-process inspection and defects identification, and a machine learning (ML)-based defects ranking and decision-making framework on the corrective actions to improve production rates and quality. However, key technology gaps still exist including the lack of physics models for the process simulation that is driven by the interaction of moving heat source, roller compaction, and strip tension during fiber placement. Thus, the process still requires operator intervention for corrective actions.
The Navy requires an integrated Digital Twin (DT)-based system that can construct a smart AFP (S-AFP) system by using combined sensor and synthetic data to reduce or eliminate false actions. The primary goal is to demonstrate the use of the developed hybrid DT-based framework to enhance the in-process defects monitoring, process control, and rational metric for rework. The secondary goal is to use the system to improve part performance and quality. The delivery of the project will be a software tool, which is validated by producing a representative aircraft component.

PHASE I: Develop a physics-based and AI-driven DT modeling system. Establish the feasibility of the concept. Prepare a Phase II plan.

PHASE II: Build a prototype system and demonstrate it on a composite part with at least one curvature. To demonstrate the fidelity of the algorithm, use it to predict temperature of the composite part during the AFP process. After demonstrating the accuracy of the temperature prediction using collected sensor data, develop an approach towards future implementation of a closed loop to control the heating element for achieving a desired temperature profile required for the high-quality manufacturing. The Phase II effort will include the demonstration of manufacturability and mechanical performance of the part compared with traditional AFP including metrics on rework time and machine layup time as a percentage of total working time. Apply the same modeling approach to the other process parameters including compaction, speed, and tow tension to generate a combined DT model, including a synthetic environment for simulating and analyzing possible part layups and with the ability to communicate in a closed loop fashion with the smart AFP machine to adjust process parameters to improve layup quality. Analyze the effects these parameters have on process-induced defects through successive manufacturing trials. Perform the final demonstration of the effectiveness of the DT on defect reduction by manufacturing a complex panel with at least two forms of curvature. Analyze and report the same metrics used in the initial phase. Deliver a complete software that is machine agnostic and capable of being applied to a variety of AFP machines.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the system to Navy use.

The expected transition targets are the original equipment manufacturers (OEMs) for aircraft.

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KEYWORDS: Composites; fabrication; AFP; Automated Fiber Placement; lightweight composites; virtual manufacturing

Version 4

N252-108 TITLE: Novel Space-based Remote Sensing of Ocean Surface Vector Winds

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Space Technology;Sustainment

OBJECTIVE: Demonstrate novel capabilities to retrieve ocean surface vector winds (OSVW) and related supporting information within environmental regimes including, but not limited to, precipitation (especially in rain rates above 2 mm/hr to and in exceedance of 50 mm/hr), coastal zones, and regions with sea ice - at a horizontal resolution less than 9 km and error less than 9% - from disaggregated space based environmental monitoring (SBEM) platforms.

DESCRIPTION: Satellite remote sensing of earth's atmosphere has provided incalculable benefit to meteorological understanding and forecasting systems. As traditional large SBEM systems are becoming deprecated in favor of smaller, proliferated observing platforms, there is a growing gap to represent key air-sea interaction state variables. In particular, OSVW represents a critical environmental quantity that modulates both atmospheric and oceanic predictability. Previous work (see References) has shown the ability to derive OSVW from a variety of sensor types and algorithmic capabilities - such as via active scatterometry, passive polarimetry, and synthetic aperture radar (SAR). However, the trend for satellite system capabilities over the past decade has generally degraded OSVW retrievals due to spatial and spectral measurement compromises emphasizing other missions. This has particularly affected accuracy of OSVW in high interest areas, such as coastal zones, tropical cyclones, and around sea ice.
This SBIR topic aims to leverage cutting edge developments in Small- and Cube-Sat design, sensor capabilities, and algorithmic development that permit improved performance and lower Size, Weight, and Power (SWaP) compared to legacy observing systems. In particular, the focus of this effort is architecting and prototyping end-to-end software/algorithmic methodologies to improve the retrieval of OSVW and related quantities (such as ocean surface stress, wave and current information, etc.) under a broad variety of weather regimes and surface conditions, ultimately to demonstrating a new capability to observe atmospheric environmental variables.

PHASE I: Outline and determine the end-to-end feasibility of building and operating a small form factor space-based OSVW platform and the novel science needed to better use observed information to derive OSVW data. Detail all essential components of the remote sensor architecture as well as the infrastructure needed to collect, transmit, and process observations. Delineate the parts of the software that would be on board, versus processed after downlink from a ground station. Scope out and provide preliminary testing of an appropriate modeling and simulation (M&S) framework, including a numerical weather prediction model, surface impacts, and line element atmospheric transmittance modeling. Develop a final summary report, including literature review and overall conclusions/recommendations, to be presented at the end of this Phase. Develop a Phase II plan.

PHASE II: Produce and demonstrate a prototype software process, including the full capability to derive OSVW and related variables - potentially with a relevant sensor payload and its testing on a representative observing platform (e.g., aircraft, low earth orbit). Include a plan for a specific test validation opportunity (such as an existing planned field campaign or other data collect event) to harden development and testing timeline, as well as providing the ability to refine error and uncertainty as needed. Algorithmic development should include the framework and test cases for calibration and validation activities from raw remote sensing observations, intermediate calibrated variables, through environmental retrieval quantities. Co-location with other sources of validation, such as buoys, in-situ profilers, and/or ship or aircraft based remote sensing, will be required. Deliver a prototype software suite and a final validation report of sensor performance in a field event compared to M&S testing at the end of this Phase.

PHASE III DUAL USE APPLICATIONS: Conduct maturation efforts into a satellite analysis system: the algorithmic processing should be well aligned with the end-to-end platform integration, mission operations, communication infrastructure, and data processing chain. Commercialization efforts should be able to support/include a data service that supports meteorological data analysis, forecasting operations, and characterization for data assimilation into numerical weather prediction.

With the expansion of governmental commercial data buys (for example, by NOAA, NASA, and DoD), it is expected this project will fill critical need in the global observing system of atmospheric properties. Downstream product development for various specialized environmental applications (such as tropical cyclone intensity analysis, marine boundary layers, and ocean modeling) are also anticipated. Other civil and commercial applications, including cargo shipping and leisure cruises, will benefit from enhanced data streams and meteorological services provided from this unique observing capability.

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KEYWORDS: Ocean surface vector winds; osvw; scatterometry; geophysical model function; space-based environmental modeling; SBEM; cubesat; microwave; radar

N252-109 TITLE: Automated Intelligence Preparation of the Battlespace from Open-Source Information

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Trusted AI and Autonomy

OBJECTIVE: Develop automated tools using artificial intelligence/machine learning (AI/ML) to generate inputs to Intelligence Preparation of the Battlespace (IPB) for the Marine Corps Mission Planning Process.

DESCRIPTION: The Marine Corps Planning Process (MCPP) requires Marines to perform an IPB, which is the "systematic process of analyzing and visualizing the portions of the mission variables of the adversary, terrain, weather, and civil considerations in a specific area of interest and for a specific mission. By applying IPB, commanders gain the information necessary to selectively apply and maximize operational effectiveness at critical points in time and space." [Ref 1].

However, the IPB is a time and manpower intensive process requiring Marines to analyze physical, temporal, cognitive, and virtual considerations [Ref 2]. Open-source information is often valuable input to the IPB given the dynamic nature of many IPB considerations, but this information is frequently the costliest to find and incorporate. In a tactical situation, Marines may not have the time or expertise to perform a full analysis across these domains.

Current IPB inputs are also not easily actionable by AI/ML analytics due to the analog nature of the resulting products. Automated generation of the IPB in structured, computer-readable formats would enable additional AI/ML tools to incorporate and exploit those products in other parts of the MCPP. A tool that solve this problem should scan through open sources of information of IPB-relevant content. Open sources to include scans should be both user-provided and discovered by web scraping. It should then retrieve and parse that information into a structured, computer-readable format usable by other analytics supporting MCPP. IPB content generated by the tool should include references to the original source of the information. The tool should estimate the accuracy of the information based on the source, the correlations between different open sources, and the correlations to authoritative U.S. sources. Marines should also be able to influence rankings of specific generated content or source sites.
The focus in developing the tool should be in discovery of relevant information to the IPB process, computer understanding of that information, AI-assisted data and cleaning transformation to enable rapid, actionable use by other algorithms, and content evaluation (i.e., source quality, accuracy, uncertainty estimates) of the generated information. Previous work in this area used Natural Language Processing (NLP) and Semantic Search techniques but were lacking in both quality and specificity of the generated information. Generally, results from previous techniques still required significant human review and refinement. Recent innovations in Large Language Models (LLMs) and Deep Learning are expected to support significant improvements in the development of a solution.

PHASE I: Determine the technical feasibility of a concept for the automated generation of IPB content. Develop a quality metric for discovered data sources to include accuracy/uncertainty considerations and user preferences. Prepare a Phase II plan.

PHASE II: Develop and evaluate a prototype for the automatic generation of IPB products from open sources. The prototype should use both a pre-defined list and automatically discovered sources. The prototype should generate IPB products targeting a minimum of three IPB considerations (e.g., civil considerations, adversary order of battle, adversary tactics, etc.). Generated products should be stored in data formats optimized to the structure of the underlying information such as JSON, Geo Tag Image File Format (GeoTIFF), and Network Common Data Form (NetCDF., etc. A product quality metric should be

developed as an extension of the data source quality metric. The product quality metric should granularly evaluate information found in data sources for accuracy/uncertainty via correlation with other sources (both other open sources and authoritative sources) or other means. For Phase II authoritative sources need only include trusted public sources (e.g., open access government webpages).) The prototype should support user input to refine source quality and the accuracy of specific information in the system. Produce the following deliverables: (1) a working prototype developed according to the extended Phase II requirements; (2) product quality metric methodology; (3) a test report documenting results of prototype evaluation.

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the technology for Marine Corps use. Develop the software for evaluation to determine its effectiveness in either a formal Marine Corps schoolhouse or other training setting. Incorporate the product into larger AI-enabled mission planning tools such as the Higher Echelon Mission Planner. Generated IPB products will be presented directly to the Marines via the planning tool and used as inputs to other AI/ML analytics support course of action analysis.

As appropriate, focus on broadening capabilities and commercialization plans. Development of affordable, scalable, non-proprietary technologies are needed to accelerate the transition of the Marine Corps to an information age model. The commercial sector is developing some of these AI-enabled technologies, but they often do not deal with critical issues regarding non-existent, limited, or low-quality source data, do not address the diversity of data modalities employed by Naval forces, and often come with prohibitive licensing and usage fees. This technology will have broad application in the commercial sector. Examples of businesses that would be interested in this technology include companies developing generative AI, companies providing search engines, and news media. All of these companies would benefit from the ability to identify low-quality source data to improve the accuracy of the information they provide to their customers. Additionally, companies focused on Generative AI have also been pushing to incorporate non-textual data into their offerings. Research with multi-modal data sources would benefit them as well.

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KEYWORDS: Artificial Intelligence, Machine Learning, Mission Planning, Intelligence Preparation of the Battlespace, IPB, Marine Corps Planning Process, MCPP, AI/ML

Version 4

N252-110 TITLE: Modeling and Simulation for Multi-modal Exercises

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Sustainment

OBJECTIVE: Develop a simulation model of information warfare that can realistically simulate multi-modal information attacks, in particular cyber-attacks and their precursors in social media campaigns. This model would be used for joint live, virtual constructive exercises to more realistically engage the training audience in scenarios in which information conflict plays important roles. This model be capable of developing and improving scenarios for multi-modal information warfare and provide guidance to planners and trainers to author and manage these exercises. The capability to explain, visualize and provide white cell adjudication support is highly desired in the final product.

DESCRIPTION: Social media provides cyber-attackers with the affordances to recruit confederates in cyber-attack, track mission execution, and perform other acts of coordination prior to and during these attacks. Currently little training is available to assist cyber-defenders to identify the social media precursors, coordination efforts, and tracking. Most cyber-defense exercises are "tabletop" efforts that are largely controlled through "white card injects," Participants are told something has happened, then they are tasked to explain how they would respond. These exercises fail to prepare the participant in the role of first responder/cyber defender to "train as they fight" – to experience the many cues and tips that precede a cyber-attack or to rehearse the steps that they would take to discover, counter, and defeat cyber-attacks. Simulation exercises are needed to provide the opportunity for cyber-defenders to experience rehearsal and response in a sandbox environment to these types of hybrid attacks. The Navy seeks a model that brings together simulations of cyber-attack with simulated social media precursors and related information flows (i.e., "social-cyber maneuvers"). This would enable exercises to include cyber-attack together with their social-cyber precursors and counter-arts for live virtual constructive training.
The desired deliverable would develop: (1) a collection of related hybrid cyber and social-cyber data indicative of these hybrid maneuvers to provide the foundation for a realistic, validated augmented generation system for scenario data; (2) a framework for information maneuvers that broadly encompass cyber and social-cyber maneuvers that would support scenario development, synthetic data production, and scenario validation (for example, the MITRE ATT&CK framework); (3) authoring tools and decision aids to guide the development of social-media facilitated cyber-attacks; and (4) a simulation model that brings together the data and the framework to enable exercise planners to develop realistic scenarios and vignettes for social media facilitated cyber-attacks. The desired deliverable would be able to produce realistic scenarios in under 1 month. It is highly desired that scenario updates and vignette changes are possible in 24 hours so that training could be changed, with the scenario "sped up" or "slowed down" based on participant performance and with injects that could be created and launched during the exercise itself.

PHASE I: Collect and validate data relevant to hybrid (cyber and social-cyber) attacks in a particular use case or set of use cases. Determine an initial data synthesis capability (such as a large language model) that can produce synthetic material indicative of an impending cyber-attack. Establish the feasibility of the initial framework for describing relationships, stages, and red flags that suggest cyber-adversaries are active. Prepare a Phase II plan.

For example, a Distributed Denial of Service (DDOS) attack has several stages: the "call to arms" stage in which audiences are enraged and encouraged to support the attack; recruitment of cyber-attackers; the distribution of tools and resources; the identification of targets and the coordination of "fires" in terms of time and targets. This is an example of an initial use case for Phase I development.

PHASE II: Enlarge the use cases from Phase I and collection of data relevant to these use cases for inclusion in a realistic augmentation generation system needed to validate synthetic data and conform to the developed framework. Develop a catalog of use cases and related information needed to guide exercise planners. Mature the Phase I data synthesis capability (possibly a special use large language model) to produce realistic volumes of synthetic data for information warfare exercises. Develop authoring tools to assist exercise planners in developing scenarios by using the framework and catalog of use cases. Create a working prototype of the simulation capability capable of a full technical demonstration in a live, virtual constructive exercise for validation of the system.

PHASE III DUAL USE APPLICATIONS: Support the transition of the simulation model to Navy use. Components of this effort would be useful to cybersecurity companies in developing simulations of cyber-attacks and their precursors for the purpose of training cybersecurity professionals.

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KEYWORDS: Training, exercise, cyber-defense; cyber-attack; information warfare; modeling and simulation

N252-111 TITLE: Dynamic Scheduler for Digital Signal Processing in Software Defined Radios

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a stable control module for a wideband receiver that adapts in real time to changing mission priorities and the signal environment. This software/firmware module will take a list of signals to be processed and schedule the proper digital signal processing algorithms onto commercial off-the-shelf (COTS) processing cards available within the same server and ensure the signal data arrives on actionable timelines.

DESCRIPTION: The Department of Defense (DoD) has historically procured progressively optimized receivers for each distinct class of radio frequency (RF) system functionality. Software Defined Radios (SDRs) have increasingly become a deployed reality and the diversity of signal definitions each transmitter is capable of emitting increases with fewer and fewer limitations. Fortunately, each class of generic hardware processor (i.e., Central Processing Unit (CPU), Graphics Processing Unit (GPU), and Field Programmable Gate Arrays (FPGA)) has also become more capable of being used for hosting multiple copies of different algorithmic techniques running simultaneously. The notion of using only multi-functional receiver systems capable of being reconfigured to function differently for different signal classes reflecting their distinct functional foci is a natural consequence of these trends.
The Navy now desires technology that enables RF system reconfiguration of functionality to be performed dynamically at unit levels in operational environments. If significantly sub-second time scales reprogramming of the processors can be achieved, adaptive and cognitive responses to even densely populated instantaneous signal environments become conceivable. However, the time taken to first optimize an initial Digital Signal Processing (DSP) task list and then to reprogram and/or reconfigure the available DSP processors for the selected set of tasks must be verifiably bounded because no computation can be performed during reconfiguration. Additionally, this reconfiguration time also adds to the processing latency and shortens the time available for cognitive response to the sensor output. Conversely, too short an interval between reprogramming events can both potentially lead to chaotic system behavior and disrupt signal analysis of complex signals mid-stream, leading to un-converged results and a sub-optimal use of processor time.

This SBIR topic solicits vendors with experience in the control of task scheduling/resource management to devise a generic parameterized scheduler for DSP processing on a diverse set of generic COTS processor cards. Innovations that reduce latency in both the optimization of the laydown of DSP techniques onto the processors and the implementation of such laydowns are desired. The topic's goals are optimization of a scheduler for stable operation, for devoting less than 10% of real-time set-up for the next batch processing epoch, and for maximizing the total priority assigned to individual signals whose processing can be accommodated by the decided upon laydown of techniques onto the DSP processors during these epochs. This includes the general tasks of generating software/firmware coding to optimize the processing laydown; initiating the processing code laydown on the processors; coordinating the retrieval of cued In-Phase and Quadrature (I&Q) data; and ensuring timely delivery of data to the

designated processor card/node. (Notice that we assume each time-frequency bounding box worth of new signal data can be associated with a specific signal class and that such cues (i.e., boxes + signal ID) can be manipulated by the scheduler without having to move the actual I&Q data from its temporary memory location until the DSP will begin.) It is also desirable for an output of the cues that will not be processed in the next epoch to be provided, along with a notation in the cue of such status. Either Vita 49.2 extension packets or Protobuf messages may be used for such cue enhancements.

Given the complexity of this topic, proposers may begin with a set of simplifying assumptions to be replaced with the more complex reality in later phases. For example, one initial assumption might be that a given signal's priority will be fixed, independent of how many instances the signal has in a given epoch and in how many epochs it has occurred. Determining the length of an epoch in terms of Vita 49.2 packet durations and execution times of the DSP techniques is one of the underlying challenges that requires careful reasoning and experimental verification. Later methods for the scheduler to learn the processing times of each DSP technique rather than requiring pre-calibration is desirable. Whether any signal is of such high priority that its processing can be continued over multiple epochs in order to be completed must also be decided. A study is needed of the consequences of failing to converge on the optimal technique laydown before beginning processing, as well as how to detect such an event.

The threshold system ingest bandwidth shall be 500 MHz and objective 4 GHz or more per channel, based on Analog to Digital Converters clocking at over 10 GHz. The intent for this topic is to develop contributions in scheduling DSP reconfigurations within an open, modular system. Use of Red Hat Linux and PCIE5 or higher is expected. Review of proposals will include evaluation of whether the proposed technologies will support explicitly open and modular systems.

The work to be performed beginning in Phase I will be considered CUI and ITAR controlled. In the event development becomes classified prior to Phase III, the facilities clearance status of the vendor's workplace and proposed workforce needs to be included in the initial proposal. All Phase I work will involve strictly unclassified waveforms and DSP techniques. The class and manufacturer of processor cards from combinations of CPUs, GPUs, and FPGAs will be left to the vendor. However, during Phase II, if awarded, there is expected to be a task designed to transition the new module's code into a less specific hardware dependent code such as VHDL. The Government will disclose other details of the expected open systems architecture (OSA) receiver as appropriate throughout the period of performance. This software/firmware module will take a list of signals to be processed and schedule the proper digital signal processing algorithms onto COTS processing cards available within the same server and ensure the signal data arrives on actionable timelines.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Design a concept for the scheduler's instruction set, needed inputs, and desired outputs. Conduct a preliminary design review (PDR) with the government sponsor. Describe approaches to both

static and opportunistic reassignment of DSP techniques over COTS processors, then present a notional plan for doing rapid (sub-millisecond) dynamic reprogramming in an operational setting, defined in terms of the required components, with identification of technological component maturation and technical risks involved. Ensure that the technical approach addresses distinct cases of how many SOIs may be present and populated in the electromagnetic environment; for example, cases to consider include where it is not necessary to fully analyze every event of a specific signal class, when the execution time of each technique is not uniform, and how to minimize time wasted by processors being idle when the received environment is inherently sparse for a given signal class. Include a lab demonstration of the planned method on a synthetic sensor processor having one DSP processor and time varying numbers of signals with known/available DSP processing algorithms present. (Note: It is undesirable for the base demo to be limited to the case where the signal density is low enough to accommodate all the signals present. If so, this demonstration shall be extended in the option period, if exercised, to the case where not all the instances of signals present can be accommodated.) If the Phase I Option is exercised, improve upon the initial demonstration in a hardware realization. Prepare a Phase II plan.

PHASE II: Develop the design into a tangible realization and conduct a laboratory demonstration, possibly in simulation, of the desired reconfigurable DSP activity. Work to scale up the number of simultaneous DSP processes that can be stably managed of same and different sorts. Devise a scheme for dealing with waveforms that take a significantly longer time to completely process and how to optimize the system throughput and control the number of instances of a given signal to analyze out of every batch of occurrences. Expand the system to the case containing all 3 classes of mass market DSP processors and where a prioritized list of signals ready to be processed are presented to the scheduler module. (Note: This new module will then work with algorithms having different execution times and processing resource requirements.) Describe a conceptual calibration scheme. Provide an approach for an automated system to learn how much of an idle time margin must be given for maintaining stable system control. If the Phase II Option is exercised, work with the sponsor to define additional issues to address and participate in system level demos at U.S. based test events/ranges. Prepare a Phase III commercialization/transition plan.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Participate in the development of Navy RF systems utilizing the delivered products and others running simultaneously from multiple vendors, consistent with the same system architecture constraints and interface definitions. Computationally intensive applications involving big data, such as voice recognition and speech processing utilized by emerging Artificial Intelligence applications and sensor fusion required for autonomous vehicles, require complex systems controls to prevent chaotic, continual reprogramming of the DSP resources to maximize impact of processing time.

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KEYWORDS: digital signal processing; reconfigurable; Software Defined Radios; SDR; Field Programmable Gate Array; FPGA; Graphics Processing Unit; GPU; Central Processing Unit; CPU; control theory; resource scheduling

Version 4

N252-112 TITLE: Generative Artificial Intelligence for Course and Content Creation and Conversion (GenAI4C)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Human-Machine Interfaces;Trusted AI and Autonomy

OBJECTIVE: Utilize state-of-the-art artificial intelligence (AI) technologies (e.g., Large Language Models) to develop an AI-aided human-in-the-loop instructional systems design agent/coach, support conversion of "legacy" (e.g., PowerPoint and Word) course materials, and assist in generation of content within Marine Corps learning ecosystem (i.e., Moodle) to update/revise or develop new courses.

DESCRIPTION: In January 2023, the Marine Corps released "Training and Education 2030" that begins with the statement "The current [Training & Education] T&E system is not preparing the Marine Corps for the future operating environment." [Ref 1]. Today's training is built on an industrial-era model where students move through a one-size-fits-all production line, progressing through standardized programs of instruction (POIs) comprised of static slides, student handout documents, written exams, and minimal experiential learning. The same report then goes on to state that "better technology integration in our classrooms and courseware can increase production, produce a more highly trained Marine, reduce their overall time-to-train, and limit the burden on our learning infrastructure."

The recent advancement of AI technologies including Large Language Models (LLMs) and image generators such as Stable Diffusion provide an opportunity to aid Marine Corps transition from industrial to information age learning in three ways: (1) instructional systems design assistance (e.g., develop course outlines, map Knowledge, Skills, and Abilities [KSAs] to tests and quizzes, advise personnel on how to best build instructional exercises in their eLearning environment, etc.) to aid course designers, instructors, and others in revising and bootstrapping course outlines and course content to reduce challenges for personnel (e.g., intimidated by the "blank slate" of creating content from scratch); (2) content generation assistance for creation of multimedia and/or interactive learning aids that includes animations and videos, but also goes beyond the creation of quizzes and flashcards that are possible with today's LLM technologies to generate branching scenarios, and other more interactive content; and (3) conversion to bring "legacy" content (e.g., text, slides, images, infographics, complex diagrams, etc.) into the modern learning environment.

The overarching goal of this SBIR topic is not to use AI as a standalone replacement for curriculum developers, POI managers, or instructors, but to enable human-AI teams to develop and manage training for the Marine Corps faster and more effectively than current processes. Technology created from this effort is expected to show efficiency gains in content creation and course generation without negatively impacting learning outcomes. Proposed solutions must go beyond the existing commercial and open-source efforts to integrate AI into learning platforms [see Ref 2 as an example]. Proposed solutions must create desired modern, interactive-content AI models that can accept input from a variety of sources and formats and output not just text, but images, movies, and more that must work together seamlessly.

The end state of this effort is to provide a government-owned suite of AI-enabled software capabilities for use by the Marine Corps Training and Education enterprise to more efficiently and effectively create courses and content as part of modern e-learning systems that reflects a cutting-edge, information-age learning enterprise. A good example are Marine Corps maintenance schoolhouses whose courses currently include a long classroom component (e.g., PowerPoint slides and lecture) with some hands-on practical application time (e.g., performing the maintenance task on a physical system). The desire is for the classroom components to be richer with multimedia and interactive components (e.g., branching scenarios allowing students to interactively troubleshoot issues); however, creating these from scratch or

converting static text or images is time consuming. Additionally, even converting a Program of Instruction into a new, blank Moodle course is daunting.
The end goal is a tool that helps personnel take their current POI, slides, and documents and creates a new course and populates it with content. Personnel will always be in the loop to verify, modify, and add to AI-created content, but AI technologies that can bootstrap the course and content creation and conversion process would be a significant savings of time and effort.

Due to the potential long review times involved, human subject research is discouraged during Phase I. Phase II plans should include key component technological milestones and plans for at least one operational test and evaluation, to include user testing.

PHASE I: Develop early concepts, wireframes, workflows, and requirements for AI-enabled modern training and education software capabilities to include support for instructional systems design processes (e.g., creating course outlines in eLearning systems based off a Marine Corps POI), content (i.e., multimedia and interactive) creation, and "legacy" (i.e., static PowerPoint and Word) content conversion. Show how the human-AI team will work together to produce training content, since the human designer is an essential part of the development process.

Human factor and human subject testing are critical in follow-on Phases of this topic. Please carefully review the requirements of approval for proposals that include testing of human subject and compliance with Institutional Review Board (IRB) [Refs 3, 4].

PHASE II: Conduct an evaluation with Marines (coordination aided by ONR) of AI-enabled capabilities outlined during the Phase I evaluation to include a usability assessment, process improvement demonstration, and effectiveness evaluations where appropriate. Perform additional demonstrations/experimentation that show system extensibility through plug-and-play of new and updated AI models to improve and expand upon overall system capability. Collect impressions of usability and develop objective metrics of time and effort to create and convert courses and content with a relevant Marine Corps population (i.e., including Marine Corps government and civilians in-the-loop of the course and content creation process). Perform all appropriate engineering tests and reviews, including a critical design review to finalize the system design.

Produce the following deliverables: (1) a working prototype of the system that can interact with existing system specifications; (2) evaluation of system usability and efficiency to convert and create content and courses; (3) a system effectiveness evaluation of system capabilities to provide demonstrable improvements (Human Subjects protocol needs to be approved in Phase II Option if needed for this evaluation).

PHASE III DUAL USE APPLICATIONS: Support the Marine Corps in transitioning the technology for Marine Corps use. Develop the software for evaluation to determine its effectiveness in either a formal Marine Corps schoolhouse or other training setting. As appropriate, focus on broadening capabilities and commercialization plans. Development of affordable, scalable, non-proprietary technologies are needed to accelerate the transition of the Marine Corps to an information age training model.

The commercial sector is developing some of these AI-enabled T&E technologies, but they often do not deal with critical issues regarding non-existent, limited, or low-quality source data, do not address encryption and classification requirements, and often come with prohibitive licensing and usage fees. This technology will have broad application in the commercial sector. Specific examples of businesses in the commercial sector include Moodle (which has a commercial learning services component) and Docebo (an online learning platform for enterprises).

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3. Office of Naval Research. "Human Subject Research Documentation Requirements". https://www.onr.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections/human-subject-research
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KEYWORDS: Artificial intelligence; machine learning; AI/ML; content creation; training and education; instructional systems design; large language models

Version 4

N252-113 TITLE: On-Chip Modulator for Cryogenic Electro-Optic (EO/IR) Sensors

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate a novel photonic integrated circuit (PIC)-compatible optical modulator for on-chip integration with cryogenically cooled electro-optical and infrared (EO/IR) sensors.

DESCRIPTION: Emerging military EO/IR sensors are being developed with smaller pixels and larger array sizes to enable high spatial sampling (i.e., pixel pitch and frame size), high temporal sampling (i.e., frame rate), low latency, and greater bit-depth resolution for real time imaging. The data bandwidth required to enable such real-time imaging for the most advanced sensor designs is reaching the limits of conventional copper wire interconnects. Datalinks using optical interconnects offer a unique and commercially mature component solution that can obviate the copper bandwidth limitation, while offering additional advantages of lower power, lower cost, and on-chip integration [Refs 1,2]. However, mature PIC-compatible modulators may not perform well at cryogenic temperatures or lack full characterization at low temperatures. Common methods used for modulation in the photonic domain include thermo-optic and electro-optic mechanisms, but these were largely developed for room temperature telecommunications applications. For example, common disk and ring resonators are very sensitive to temperature variations. Stabilization is achieved using high-speed heating elements located along the resonator surface, which increase power dissipation, introduce jitter noise, and degrade in performance at low temperature, increasing bit error rate (BER) and reducing modulation bandwidth significantly. Moreover, large bandwidth bottlenecks can be imposed by the integration strategy, such as wire-bonding, whereas flip-chip or alternative integration schemes could offer better performance.

Recently, an electro-absorption-based modulator (EAM) demonstrated > 30 Gbps bandwidth at cryogenic temperatures, which was limited by wire-bond integration, but eliminated the need to maintain resonant coupling [Ref 3]. The EAM is based on the Franz-Keldysh effect, which can provide a constant extinction ratio from room temperature down to 5K. Thin-film lithium niobate (TFLN) modulators, alternatives based on the Pockel's effect, are also candidates for PIC-compatible cryogenic modulators, becoming more efficient at low temperatures and being compatible with Si foundry processing lines. While these advances are promising, innovations in cryo-compatible modulator design and integration are needed to increase modulation bandwidth, improve operational stability, and to facilitate automated operation. The goal of this SBIR topic is to develop or advance a PIC-compatible optical modulator that can stably operate within the relevant cryogenic temperature range for EO-IR sensors (~77K-120K) and can enable on-chip integration with EO/IR sensors.

PHASE I: Design a concept for a PIC-compatible photonic modulator that can provide stable high-speed modulation and tuning at cryogenic temperatures to enable high bandwidth readouts for EO/IR sensors. Establish proof of concept. Develop a modulator fabrication process and a modulator test plan for Phase II.

PHASE II: Optimize electrical and optical design and fabricate a packaged modulator prototype. Demonstrate modulator electrical and optical performance for high speed, high frequency range, and high bandwidth cryogenic operation (77K-120K). Demonstrate single-mode fiber pigtailed electro-optic modulator packaging. Prepare a Phase III commercialization/transition plan.

PHASE III DUAL USE APPLICATIONS: Transition the demonstrated modulator technology to EO/IR sensor systems onto Naval Surface and Expeditionary Platforms.
Cryogenic PIC-based modulators can also be applied to quantum sensing/computing, free space optical communication, and electromagnetic warfare (EW) applications.

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KEYWORDS: Optical Modulator, Cryogenic, Electro-Optic Sensors; EO/IR

Version 4

N252-114 TITLE: Logistic Modeling and Simulation for Hypersonics

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Hypersonics;Integrated Network Systems-of-Systems

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a Modeling and Simulation (M&S) tool that is capable of capturing timing windows for different phases of the weapon system's operational lifecycle based on system, subsystem, and component requirements, weapon system Concept of Operations (CONOPS), and dependencies on external systems or institutions. This tool must be able to determine the minimum and maximum allowable times for certain events based on relevant factors including but not limited to heating rates and receipt of critical mission data during operational use.

DESCRIPTION: The M&S tool will need to consider combinations of various physical effects during pre-launch, launch, and post-launch that affect the integrity and health of the weapon system and finds points of failure. For the pre-launch phase, the tool will need to predict the overall internal temperature of the system due to excess heat generated by onboard electronics, determine possible launch windows that fit within the constraints of onboard systems and/or external environmental conditions, and vibrations during deployment but prior to launch. For the launch phase, the tool will need to predict the overall impact of shocks, vibrations, and accelerations on components during launch. For the post-launch phase, the tool will need to track heating rates from external aerodynamics and internal electronics, Concepts of Operations (CONOPS)- driven vehicle reorientations, required operating windows for onboard systems, externally received signals, and computation time to react to new commands during flight. The tool will also need to calculate timing to inform logistics analysis, like mean time between failures (MTBF) for technologies affected by the conditions described for each phase. This tool will be used by integration experts, logistics assessment experts, and mission planning. This will aid in the confidence of the logistic assessment, storage, transportation, and technology requirements. This M&S capability must be able to modularly accept reference information generated by high fidelity physical models, such as equations, look-up tables, or state information. This project will transition to defense contractors for high-speed weapons and space systems. To meet these needs, maturation and packaging of the technology to meet practical size, weight, and power constraints will be required. Extreme environments may require special considerations to conform to airframe shape and shielding from the aerothermal environment.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program

Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Provide a concept for a logistics M&S capability that can capture timing windows for different phases of a weapon system's operational lifecycle based on system. Ensure that the concept addresses the generation of timing predictions for the pre-launch and post-launch phases using approximations of relevant environmental information or operational constraints of systems. Include heating information that can be represented by polynomial functions over time, piecewise or not, or as tabular lookups. Develop heating models that consider sub 300 degrees Celsius (°C) for the pre-launch phase and greater than 1,000°C for the post-launch phase. Demonstrate models of various conditions that may be present pre-launch, during launch, and post-launch. Provide computational calculation times that can be represented by simple equations to determine how long different phases of calculation last, capturing time to receive and process a command, recalculate, and deploy the command with respect to the remaining time in the mission. If the Phase I Option is exercised, include the initial design specifications and capabilities description to build a prototype solution in Phase II. Prepare a Phase II plan.

PHASE II: The Phase II effort will require the ability to perform work in a classified environment. Concepts presented in Phase I will need to work with classified reference values. Develop a prototype M&S tool that can generate parametrically based on mission parameters. Apply and further develop the prototype for specific cases prioritized by the Conventional Prompt Strike (CPS) program and begin using program data.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: If the demonstration in Phase II is deemed to be of high interest to the Government, support the Government in transitioning the technology for Government use to support current and future weapon and space systems, as well as a wide range of other air-, land-, and sea-based systems.
Commercial applications should be considered for transition (i.e., 5G, navigation systems, and tracking systems).

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1. Kasim, B; Çavdar, A.B.; Nacar, M.A. and Çayirci, E. "Modeling and Simulation as a Service for joint military space operations simulation." The Journal of Defense Modeling and Simulation, 18(1), 2021, pp. 29-38. doi:10.1177/1548512919882499
2. Kemp, Jesse A."Modeling and simulation in support of operational test and evaluation for the Advanced Amphibious Assault Vehicle (AAAV)." Naval Postgraduate School, Monterey California Dissertation, 2001. https://core.ac.uk/download/pdf/36703284.pdf

KEYWORDS: Logistics; Modeling; Simulation; Mean Time Between Failure: Integration; Lifecycle; Root Cause Analysis; Risk; System Configuration; System Engineering; Model Based System Engineering

Version 4

N252-115 TITLE: Low Size, Weight, and Power (SWaP) Phased Array For Vehicle Tracking Using Signals of Opportunity

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Hypersonics;Integrated Network Systems-of-Systems;Microelectronics

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a state of the art (SOTA) low size, weight, and power (SWaP) phased array capable of increasing positional accuracy of a vehicle in a hypersonic regime using signals of opportunity.

DESCRIPTION: Phased Array Tracking is currently being used for fifth generation (5G) cell towers. The tracking is done with stationary active electronically scanned arrays (AESAs) and targets slow moving people or networks to send information. The Conventional Prompt Strike (CPS) Program would like to use a passive phased array that tracks signals of opportunity (SoOp) to give real-time position updates/corrections to a hypersonic vehicle. Given a known position of a radiating system, the phased array can track the angle of arrival (AoA) of a radiated signal. With the AoA and known position of the stationary system, the position of the vehicle can be estimated. This allows corrections to the inertial measurement unit's (IMU's) drift error. The phased array may use any low earth orbit (LEO) or stationary high power SoOp. The trade space between range and size of the phased array is a significant consideration for design. This project will transition to defense contractors for high-speed weapons and space systems. To meet these needs, maturation and packaging of the technology to meet practical size, weight, and power constraints will be required. Extreme environments may require special considerations to conform to airframe shape and shielding from the aerothermal environment.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Conduct a feasibility study to show the use of a passive phased array to track a SoOp and give real-time estimations for the hypersonic vehicle's position. The technology shall show improvements in positional accuracy with varying distance and incident angles from the SoOp, resistance to power saturation of noisy environment, while minimizing SWaP. The use of modeling and simulations is recommended with limitations on budget for hardware in Phase I. Technical merit and competence is evaluated based on the use of models and simulations, algorithms, and technical plan of action. The Phase

I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Demonstrate software and algorithms with the use of the passive phased array to locate a vehicle's current position with the use of the SoOp as an external reference point addressed in the Phase I. The Phase II Statement of Work (SOW) should identify a work plan that provides proof of concept that the technology has the potential to meet the military performance goals, highlighted in Phase I. Work should focus on the increase in accuracy consistency, and reliability. A prototype, modeling and simulation, and/or software shall be provided with intent to be integrated into test assest(s) for verification and validation of the technology by the end of the Phase II.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: If the demonstration in Phase II is deemed to be of high interest to the Government, support the Government in transitioning the technology for Government use. The transitioned product is expected to be able to support current and future weapon and space systems, as well as a wide range of other air, land, and sea-based systems.
Commercial applications should be considered for transition (i.e., 5G, navigation systems, and tracking systems).

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KEYWORDS: Phased Array; Passive; Angle of Arrival; Radio Frequency; Radiation; Signals of Opportunity; Hypersonic; Tracking; Fast Tracking; Low Latency; Algorithm; Tracking Algorithm; Passive Electronically Scanned Array; Active Electronically Scanned Array; Signal

N252-116 TITLE: Delay and Denial of Unmanned Underwater Vehicles

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an innovative solution to delay trespassers or deny access to naval assets by Unmanned Underwater Vehicles (UUVs), Autonomous Underwater Vehicles (AUVs), and Remotely Operated Vehicles (ROVs).

DESCRIPTION: The proliferation of UUVs has created a need for the U.S. Navy to increase its ability to delay trespassers and deny access to its assets. Permanent underwater barriers are not considered a viable option due to the interference with our own port operations. Therefore, a more operationally supportable approach is required to achieve the U.S. Navy's aims. The Navy is searching for the capability to block, capture, redirect, destroy, or in any other manner neutralize the threat of an unauthorized UUV that is trespassing into sensitive underwater areas.

The innovative solution will be capable of delaying and denying small and medium sized UUVs travelling at sprint speeds [Ref 2]. The U.S. Navy defines small and medium UUVs as having a diameter no greater than 21 inches. The solution will be capable of or denying these UUV classes at depths up to 800 feet below the surface. Once the solution is deployed, it will allow the UUV to travel no more than 200 m, and ideally no more than 100 m, before being rendered ineffective.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Define and develop a method to deny an UUV access to our assets once it has been identified that meets the requirements in the Description. Define the proposed components of the system, including power sources, electronics, denial tactics, interfaces, and any other design components deemed necessary by the developer. Demonstrate the feasibility of the concept through modelling and simulation.
The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype UUV Delay and Denial system that meets the requirements listed in the Description. Demonstrate the efficacy of the system in an appropriate environment. The prototype shall be delivered by the end of Phase II.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the U.S. Navy while transitioning the technology to Navy use through production, integration, and maintenance. Ensure that the final product includes any systems required for interception and subsequent denial or destruction of unauthorized UUVs. A user-friendly interface for ease of use should also be part of the final product.
Prepare a strategy to utilize the technology for commercial use. Possible uses include protection of commercial ports and shipping, or research into marine life.

REFERENCES:

1. Pullen, John. "The Latest Advancements in UAS." Avionics International, May 2023. https://www.aviationtoday.com/2023/05/04/the-latest-advancements-in-uas/
2. "What is the Best Underwater Drone?" Flying Magazine, May 2024. https://www.flyingmag.com/guides/best-underwater-drone/

KEYWORDS: Delay; Deny; Interdiction; Unmanned; Autonomous; Harbor Protection; Capture; Three Dimension Trajectory; UUV; AUV; Underwater Vehicle

N252-117 TITLE: Interception of Unmanned Underwater Vehicles

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an innovative solution to intercept Unmanned Underwater Vehicles (UUVs), Autonomous Underwater Vehicles (AUVs), and Remotely Operated Vehicles (ROVs).

DESCRIPTION: UUVs are a rapidly advancing technology, with their future progression likely following a similar path to that of Un-crewed Aerial Systems (UASs). Over the past decade, UAVs have significantly improved in navigation and maneuverability, and nations around the world have supported these developments for their own purposes [Ref 1]. Events in recent global conflicts indicate that similar technological advances are well on their way for UUVs. Therefore, an ideal proposal for UUV systems should allow flexibility for future enhancements, particularly the potential to improve the system's capabilities to intercept future threats.

The proliferation of UUVs has created a need for the U.S. Navy to increase its ability to prevent trespassers from accessing its assets. Permanent underwater barriers are not considered a viable option due to the interference with our own port operations. Therefore, a more operationally supportable approach is required to achieve the U.S. Navy's aims. The Navy is searching for the capability to intercept any unauthorized UUV that is trespassing in sensitive underwater areas. Interception is defined as the capability to reach the location of a trespassing UUV with enough accuracy for neutralizing actions to take place.

The innovative solution will be capable of intercepting small and medium sized UUVs travelling at sprint speeds [Ref 2]. The U.S. Navy defines small and medium UUVs as having a diameter no greater than 21 inches. The solution will include the capability to intercept these UUV classes at depths up to 800 feet below the surface.

Modelling the interception of underwater vehicles has additional challenges. UUVs operate in three-dimensional (3D) space, increasing the number of possible trajectories they can take. An interceptor can commit to an intercept path, but if the unauthorized UUV changes its trajectory, the higher density fluid (in contrast with air) means that it will take longer for the interceptor to fully react to the change. If the UUV is faster than the interceptor, the problem is compounded. Furthermore, the additional trajectories allowed in 3D space also limit the interceptor's ability to fully anticipate and prepare for changes in the UUV's motion [Ref 3].

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able

to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Define and develop an interception method that meets the requirements in the Description. Define and develop a method to deny the UUV access to our assets once it has been intercepted and meets the requirements in the Description. Define the proposed components of the system, including power sources, electronics, interception algorithms, denial tactics, interfaces, and any other design components deemed necessary by the developer. Demonstrate the feasibility of the concept through modelling and simulation.

The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype UUV Interception and Denial system that meets the requirements listed in the Description. Demonstrate the efficacy of the system in an appropriate location. The prototype shall be delivered by the end of Phase II.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the U.S. Navy while transitioning the technology to Navy use through production, integration, and maintenance. Ensure that the final product includes any systems required for interception and subsequent denial or destruction of unauthorized UUVs. A user-friendly interface for ease of use should also be part of the final product.
Prepare a strategy to utilize the technology it has developed for commercial use. Possible uses include protection of commercial ports and shipping, or research into marine life.

REFERENCES:

1. Pullen, John. “The Latest Advancements in UAS.” Avionics International, May 2023. https://www.aviationtoday.com/2023/05/04/the-latest-advancements-in-uas/
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KEYWORDS: Interception; Interdiction; Unmanned; Autonomous; Harbor Protection; Capture; Three Dimension Trajectory; UUV; AUV; Underwater Vehicle

N252-118 TITLE: High Sensitivity Piezo-Ceramic Materials for Hydrophone Devices

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Integrated Sensing and Cyber

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Demonstrate improved piezo-ceramic materials and hydrophone devices with at least 6 dB gain in sensitivity over existing solutions for Navigational Sonar Systems (NSS) applications.

DESCRIPTION: Piezo-ceramic materials provide the heart of nearly all Navy acoustic transmit and receive systems. There are a wide range of transmit materials available for system designers to choose from. However, for passive receive applications, the number of materials available to design around are limited to 2-3 choices. This technology space has not evolved in 20 years and thus sensor design has also stagnated. With the development of textured ceramic processing technology, new possibilities exist for creating enhanced hydrophone materials [Ref 1]. This could offer system designers more freedom in terms of size reduction, shape, configuration, enhanced signal to noise ratio, and a wider supplier base for material procurement.

Prior Navy investments in textured ceramics have focused heavily on materials for transmit applications [Ref 2]. As such, this technology space is maturing rapidly. This SBIR topic seeks to leverage these previous investments and provide new materials for receive-only applications. Specifically materials and manufacturing methods designed to deliver piezoelectric voltage coefficients, g33, g31, and g15, of at least 100 V·m/Nx10-3, 35 V·m/Nx10-3, or 75 V·m/Nx10-3 respectively, with a dielectric constant of at least 500 (unitless) and a dielectric loss tangent of < 0.005 (unitless) (all at 1V drive, 1 kHz frequency) are targeted, as are advanced manufacturing techniques [Refs 2,3,4]. These g-coefficient values represent a 2x increase over the best available materials and are expected to yield the desired minimum improvement in hydrophone sensitivity over a wide range of possible sensor designs. The targeted dielectric constant and loss tangent values are mandated by the need to maintain signal to noise ratio (SNR) performance [Ref 5].

PHASE I: Conduct an initial study of the required material properties of highly sensitive piezo-ceramic materials to include the following items:

- A discussion of how the technology approach will satisfy the advanced material properties requirements. This could include traditional ceramic processing, tape casting, additive manufacturing, or other techniques.
- A discussion of the material chemistry and fabrication processes required to achieve the desired properties including an assessment of risks and risk mitigation strategies.
- Ability to scale processing to part quantities of 1000 should be discussed along with a projected process rate.
- Initial demonstrations of material properties showing feasibility of achieving the desired properties.
- A discussion regarding the material's impact to receive device applications.

- Device modeling and simulation (based on available material properties) to demonstrate free field voltage sensitivity or acceleration sensitivity at frequencies from 1 to 100kHz, although it is not necessary to cover the entire band.

The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build prototypes solution in Phase II.

PHASE II: Produce prototype materials in sizes and shapes allowing for prototypes of relevant Navy sensors to be constructed. Produce and demonstrate on representative test pieces, with the accompanying full matrix of piezoelectric coefficients required for accurate simulation of sensor devices per ANSI/IEEE STD 176-1987 [Ref 6]. Demonstrate new device concepts possible with enhanced material properties at the prototype level. Provide advanced characterization of the materials and devices, including property variation with pre-load stress, property variation with temperature, and material microstructure. Deliver three (3) prototype devices for testing as well as the modeling and simulations for further integration including simulated free field voltage sensitivity (FFVS) and beam patterns for devices over the frequency range of 1 to 100 kHz. These simulations should include an assessment of survival of grade A and grade B underwater explosive shock per MIL-S-901D [Ref 7] with recommendations on how to improve the design for shock tolerance. The prototypes shall be delivered by the end of Phase II.

PHASE III DUAL USE APPLICATIONS: Based on the prototypes developed in Phase II, continue development that must lead to productionization and demonstration of an array of hydrophones leading toward integration within existing submarine housings. Address minimizing lot unit variability. These arrays should be characterized for FFVS and beam patterns. In addition, the arrays should be characterized for survival of grade A and grade B underwater explosive shock per MIL-S-901D test conditions and procedures.
In addition to military application, these hydrophones are applicable for other commercial maritime applications. These areas include commercial shipping vehicles, underwater unmanned vehicles (UUV), and submarines.

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KEYWORDS: Piezoelectric voltage coefficient; Textured ceramic; Piezoceramic; Hydrophone; Sensor; Ceramic processing

N252-119 TITLE: Novel Detection Methods for Unmanned Underwater Vehicle

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems;Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an innovative solution beyond the traditional active and passive sonar technologies to detect Unmanned Underwater Vehicles (UUVs), Autonomous Underwater Vehicles (AUVs), and Remotely Operated Vehicles (ROVs).

DESCRIPTION: UUVs are a rapidly advancing technology, with their future progression likely following a similar path to that of Unmanned Aerial Vehicles (UAVs). Over the past decade, UAVs have significantly improved in navigation and maneuverability, and nations around the world have supported these developments for their own purposes [Ref 1]. Events in recent global conflicts indicate that similar technological advances are well on their way for UUVs. Therefore, an ideal solution should allow flexibility for future enhancements, particularly in detecting emerging threats.

The proliferation of UUVs has increased the need for the U.S. Navy to have the capability to detect unauthorized entry into controlled areas. Traditional sonar systems are well-established. Active sonar systems have proven to be our most effective means of locating objects underwater, since they generate their own sound waves and energy source. However, this same wave generation can disclose their position to an observer, and they have been less effective against the smaller vehicles that have proliferated in recent years. Furthermore, active sonar can be very disruptive to marine life when emitting high levels of energy [Ref 2]. Passive sonar systems do not generate their own waves, which keeps their location hidden, but also makes them reliant on an unknown object being noisy enough to stand out from the background. Essentially, passive sonar systems do not emit any sound themselves; instead they listen to the sounds naturally present in the environment.

Alternatives to sonar technologies remain in their infancy. Numerous options have been proposed to improve current capabilities, including co-opting sounds from marine life, high-quality imaging and identification algorithms, and magnetic field sensors [Refs 3,4,5]. Despite their potential, none has yet proved capable of improving our detection capabilities. The U.S. Navy wishes to further investigate the use of novel underwater detection technologies (which may or may not include the aforementioned) for the purpose of protecting its assets.

Proposals for this SBIR topic should present technologies that incorporate principles beyond those of active and passive sonar systems. The innovative solution will be capable of detecting small and medium sized UUVs travelling at sprint speeds. The U.S. Navy defines small and medium UUVs as having a diameter no greater than 21 inches. The system will be capable of detecting these UUV classes up to at least 500, and ideally 1000, yards away. It will also be capable of detecting them at depths up to 800 feet below the surface.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Define and develop a concept that will utilizes a technology that has UUV detection capabilities beyond active or passive sonar systems and meets the requirements in the Description. It. Demonstrate the feasibility of the concept through simulation and modelling. Define the proposed components of the system, including power sources, electronics, detection methods, interfaces, and any other design components deemed necessary by the developer.
The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype UUV detection system. Demonstrate, in an appropriate environment, that the prototype system meets the requirements listed in the Description. The prototype shall be delivered by the end of Phase II.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the U.S. Navy in transitioning the technology to Navy use through production, integration, and maintenance. The final product should include any systems required for detection of unknown underwater vehicles. A user-friendly interface for ease of use should also be part of the final product.
Prepare a strategy to utilize the technology it has developed for commercial use. Possible uses include monitoring of commercial ports and shipping lanes, the oil and gas industries, or research into marine life.

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KEYWORDS: Reduction of Sonic Impact; Underwater Detection; Unmanned Underwater Vehicle; UUV; Autonomous Underwater Vehicle; AUV; Diver; Sonar Alternatives; Underwater Imaging; Marine Life Sounds; Magnetic Field Sensors

N252-120 TITLE: Plasma Modeling and Simulation for Hypersonics

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Hypersonics;Integrated Network Systems-of-Systems

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a plasma modeling and simulation (M&S) tool able to characterize different atmospheric conditions in a hypersonic regime. Evaluate environmental as well as electromagnetic effects such as ablation, transmissions, absorptions, reflections, and other plasma characteristics for the hypersonic vehicle.

DESCRIPTION: Plasma has unique characteristics that in some cases act as a Faraday cage, transparent medium, and/or an absorbent medium. These extremely complex characteristics cause the evaluation and assessment of plasma and electromagnetic interactions difficult to predict. M&S provides high fidelity approximations before full-scale tests. This M&S tool shall be capable of estimating the plasma characteristics based on hypersonic flight creating the plasma via velocity, altitude, species of atmosphere, density of air, material of exterior of hypersonic vehicle, solar radiation, other electromagnetic radiation, etc. Another aspect of the tool shall allow for electromagnetic interactions with the plasma and the effects of the interaction on both the plasma and electromagnetic waves. This includes showing ablation, thermal, and other effects on the surface of any material that could be on the vehicle. The plasma expands as atmospheric pressure changes. This change in plasma density and plasma volume has various effects that may also be modeled. This tool must be user friendly for electromagnetic evaluation for those with limited plasma physics knowledge, such as engineers. This project will transition to defense contractors for high-speed weapons and space systems. To meet these needs, maturation and packaging of the technology to meet practical size, weight, and power constraints will be required. Extreme environments may require special considerations to conform to airframe shape and shielding from the aerothermal environment.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Demonstrate that a model can be constructed that shows the typical shape and spatial form of the plasma sheath. Show that the effects of electromagnetics with the plasma and how the plasma should

change. Show a plan to calculate the effects of interactions. Demonstrate how these calculations will be processed in a low fidelity but high-speed scenario as well as in the high fidelity process. Demonstrate the M&S tool with a fundamental physics based model. Show quantitative ways to perform the model at high fidelity and with the least amount of processing time. Describe/show a user interface that those familiar with plasma physics may use to perform complex high fidelity physics based models and visualize the data efficiently as well as how less familiar users may utilize the tool. Show the plan for the tool being capable of modeling various changes to the plasma with transient responses as the vehicle travels. Show the planned ability to get still shots of the plasma and run various test cases on the still shot of the plasma. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop a more refined model with complex fluid dynamics and higher fidelity advanced physical models of the plasma. This will include the complexities of the electromagnetic interaction and changes in the plasma shape/form as the additional energy is added to the plasma medium. Develop the user interface to incorporate all the features described in the Phase I section . Demonstrate the high fidelity model and all the ways to assess the characteristics of the plasma in space, time, reflections, absorptions, transmission, radiation, species, etc. Deliver the tool to the Government by end of the Phase II or the Phase II Option if exercised.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Government in transitioning the technology for Government use. Ensure that the transitioned product is able to support current and future weapon and space systems, as well as a wide range of other air-, land-, and sea-based systems.
Commercial applications should be considered for transition (i.e., 5G, navigation systems, and tracking systems).

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KEYWORDS: Plasma; Plasmonics; Modeling; Simulation; Hypersonic; Complex Fluid Dynamics; Low Latency; Electromagnetics; Radio Frequencies; Reflection; Absorption; Transmission; Filamentation; Arc Jet; Arcjet

N252-121 TITLE: Green Chemistry Application for Scalable Production of Carbon-Based Advanced Textiles Supporting Hypersonics

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Hypersonics;Sustainment

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a processing method using green chemistry to produce carbon fiber from advanced organic textiles for hypersonic applications. Provide sustainable alternatives to commonly used chemicals or processes by reducing solvent byproducts, hazardous chemical use, and increasing yield that address long-term affordability.

DESCRIPTION: The advent for advancements in Hypersonics has been upon us for the last two decades. Matching the performance metric of continually going faster with each new design iteration, the Thermal Protection System (TPS) plays a major role in a hypersonic flight vehicle's (HFV) performance at high MACH numbers. The TPS operates to protect the substructure of the HFV from high temperature aerodynamic heating. The majority of the TPS structure is composed of carbon fiber, derived from textiles such as polyacrylonitrile (PAN) or rayon [Refs 1,2]. In the production of these textiles, a great deal of chemical waste is produced, creating a large footprint that is a detriment to the environment. This often limits locations for carbon fiber production and U.S.-based access, and drives costs due to increased demand. The supplier base is relatively small and consists of fewer than five companies, which [Ref 2] has led to an overburdened supply chain [Ref 3]. Additionally, the advanced organic textile used in producing carbon fiber is sourced from overseas, where a sudden loss of supply would significantly disrupt DoD and other defense-related programs [Ref 3]. Despite the fragility of the supply chain, the carbon fiber market is estimated to be USD $6.5B in 2022 and $21.7B in 2032, where over 40% of the industrial value is tied to the aerospace or defense market [Ref 2]. There is a potential gap of over 55,000 metric tons by 2026 where it would take several years to add additional capacity [Ref 2]. Sustainable alternative methods to producing these base level textiles would expand market level availability and drive cost margins for the Hypersonics and carbon fiber industry.

This SBIR topic looks for novel approaches that address scalability using the principles of green or sustainable chemistry to reduce waste products and ensure EPA compliance in the production of advanced organic textiles for carbon fiber production. Green chemistry is a field devoted to sustainability and forms a framework of twelve principles: prevent waste, atom economy, less hazardous synthesis, design benign chemicals, benign solvents and auxiliaries, design for energy efficiency, use of renewable feedstocks, reduce derivative, catalysis, design for degradation, real-time analysis for pollution prevention, and inherently benign chemistry for accident prevention [Ref 4]. Some of the toxic and hazardous waste products produced from current processes include, but are not limited to, HCN, NH3, CO2, CS2, and H2S [Ref 5].

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and

Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and SSP in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a proof of concept for a sustainable, repeatable chemical process that yields advanced organic textile fibers for the production of carbon fiber with comparable thermomechanical properties to existing textiles for TPS applications in hypersonic environments. Provide materials thermomechanical characterization analysis, including, but not limited to, density, moisture content, and break strength of the advanced organic textile. Demonstrate that this process could result in a reduction of toxic and hazardous waste and volatile compounds. Provide a detailed plan for full scale production of the advanced organic textile.

The Phase I Option, if exercised, will include the initial design specification and capabilities to build a prototype solution in Phase II.

PHASE II: Mature the process by testing the appreciability of the conversion of advanced organic textiles to carbon fiber. Provide materials thermomechanical characterization analysis, including, but not limited to, char yield, density, moisture content, and break strength after carbonization. Demonstrate the feasibility of production through processing the advanced organic textiles through a carbon fiber conversion process, either traditional or novel.
It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: If the demonstration in Phase II is deemed to be of high interest to the Government, support the Government in transitioning the technology for Government use. Ensure that the transitioned product is able to support current and future weapon and space systems, as well as a wide range of other air-, land-, and sea-based systems. Continue to scale the green/sustainable chemistry process to enable the continuous production of advanced organic textiles for carbon fiber production based on the prototypes developed in Phase II. Integrate the converted carbon fiber into a woven part to be “thermomechanically” characterized.
Integrate the technology into a flight experiment to validate the efficacy of the produced carbon-based TPS part. Carbon fiber is used in commercial space applications as well as many Navy/DoD components, such as Navy Strategic Systems Programs (SSP), Navy Conventional Prompt Strike (CPS), and Primes developing TPS. Sustainable, green chemistry-developed advanced organic textiles would be of interest to the aforementioned programs.
This project should transition to defense contractors for high-speed weapons and space systems.

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KEYWORDS: Green Chemistry; Sustainable Chemistry; Carbon Conversion; Hypersonics; Textiles; Thermal Protection Systems; Rayon and Polyacrylonitrile Carbon Fiber; PAN

Version 2

DEPARTMENT OF THE NAVY (DON)
25.2 Small Business Innovation Research (SBIR)
Direct to Phase II (DP2) Announcement and Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to Direct to Phase II (DP2) SBIR topic only:**
  - **N252-D09 through N252-D11**

- Submitting small business concerns are encouraged to thoroughly review the DoD SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoD Program BAA is located at: https://www.defensesbirsttr.mil/SBIR-STTR/Opportunities/#announcements. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

- The information provided in the DON Proposal Submission Instructions takes precedence over the DoD Instructions posted for this BAA.

- A submitting small business concern **MUST** use the DP2 Phase I Feasibility proposal template for Volume 2. This template is specific to DON DP2 topics and meets DP2 submission requirements. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.

- Proposing small business concerns that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF) or any combination of these are eligible to submit proposals in response to DON topics advertised in this BAA. Information on Majority Ownership in Part and certification requirements at time of submission for these proposing small business concerns are detailed in the section titled ADDITIONAL SUBMISSION CONSIDERATIONS.

- DON provides notice that Basic Ordering Agreements (BOAs) or Other Transaction Agreements (OTAs) may be used for Phase II awards.

- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing small business concern.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DON. More information on the programs can be found on the DON SBIR/STTR website at www.navysbir.com. Additional information on DON's mission can be found on the DON website at www.navy.mil.

The Department of Defense (DoD), including the Department of the Navy (DON), may issue an SBIR award to a small business concern under Phase II , without regard to whether the small business concern received a Phase I award for such project. Prior to such an award, the head of the agency, or their designee, must issue a written determination that the small business concern has demonstrated the scientific and technical merit and feasibility of the technology solution that appears to have commercial potential (for use by the government or in the public sector). The determination must be submitted to the Small Business Administration (SBA) prior to issuing the Phase II award. As such, DON issues this portion of the BAA in accordance with the requirements of the Direct to Phase II (DP2) authority. Only those proposing small business concerns that are capable of meeting the DP2 proposal requirements may participate in this DP2 BAA. No Phase I awards will be issued to the designated DP2 topic.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	**When**	**Contact Information**
Program and administrative	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoD SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
	BAA Open	DoD SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
Electronic submission to the DoD SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMAND (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
N252-D09 to N252-D10	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil

N252-D11	Mr. Shadi Azoum	Naval Information Warfare Systems Command (NAVWAR)	info@navwarsbir.com

Each DON SBIR DP2 topic requires documentation to determine that Phase I feasibility, described in the Phase I section of the topic, has been met.

The DON SBIR DP2 is a two-step process:

> STEP ONE: Prepare and Submit a Phase I Feasibility Proposal (instructions and link to template provided below). The purpose of the Phase I Feasibility Proposal is for the proposing small business concern to provide documentation to substantiate that both Phase I feasibility and the scientific and technical merit described in the topic have been met. The Phase I Feasibility Proposal must: demonstrate that the proposing small business concern performed Phase I-type research and development (R&D) and provide a concise summary of Phase II objectives, work plan, related research, key personnel, transition/commercialization plan, and estimated costs. Feasibility documentation MUST NOT be solely based on work performed under prior or ongoing federally funded SBIR/STTR work. The government will evaluate Phase I Feasibility Proposals and select small business concerns to submit a Full DP2 Proposal. Demonstrating proof of feasibility is a requirement for a DP2 award. The small business concern must submit a Phase I Feasibility Proposal to be considered for selection to submit a Full DP2 Proposal.
>
> STEP TWO: If selected, the cognizant SYSCOM Program Office will contact the small business concern directly to provide instructions on how to submit a Full DP2 Proposal.

DON SBIR reserves the right to make no awards under this DP2 BAA. All awards are subject to availability of funds and successful negotiations. Proposing small business concerns must read the topic requirements carefully. The Government is not responsible for expenditures by the proposing small business concern prior to award of a contract. For 25.2 topics designated as DP2, DON will accept only Phase I Feasibility Proposals (described below).

DP2 PROPOSAL SUBMISSION REQUIREMENTS

The following section details requirements for submitting a compliant DON SBIR DP2 Proposal to the DoD SBIR/STTR Programs.

(NOTE: Proposing small business concerns are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoD SBIR/STTR Innovation Portal (DSIP). Proposing small business concerns are required to submit proposals via the DoD SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoD SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing small business concerns submitting through DSIP for the first time will be asked to register. It is recommended that proposing small business concerns register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DON.

Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoD SBIR/STTR Program BAA for further information.

Eligibility. Each proposing small business concern must:

- Have demonstrated feasibility of Phase I-type R&D work
- Have submitted a Phase I Feasibility Proposal for evaluation
- Meet Offeror Eligibility and Performance Requirements as defined in the Proposal Fundamentals section of the DoD SBIR/STTR Program BAA
- Comply with primary employment requirements of the principal investigator (PI) during the Phase II award including, employment with the small business concern at the time of award and during the conduct of the proposed project. Primary employment means that more than one-half of the PI's time is spent in the employ of the small business concern
- Register in the System for Award Management (SAM) as defined in the Certifications and Registrations section of the DoD SBIR/STTR Program BAA. To register, visit https://sam.gov/

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoD SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - A submitting small business concern MUST use the DP2 Phase I Feasibility proposal template for Volume 2. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.

      This template is specific to DON DP2 topics and meets DP2 submission requirements:
      - Not to exceed 30 pages, regardless of page content; Phase I Proof of Feasibility portion not to exceed 20 pages, Snapshot of Proposed Phase II Effort portion not to exceed 10 pages
      - Single column format, single-spaced typed lines
      - Standard 8 ½" x 11" paper
      - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
      - No font size smaller than 10-point

  - Additional information:
    - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing small business concerns are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).** The text fields related to costs for the proposed effort must be answered in the Cost Volume of the DoD Submission system (at https://www.dodsbirsttr.mil/submissions/), however, proposing small business concerns DO NOT need to download and complete the separate cost volume template when submitting the DON SBIR Phase I Feasibility Proposal. Proposing small business concerns are to include a cost estimate in the Order of Magnitude Cost Estimate Table (example below) within the Snapshot of Proposed Phase II Effort portion of the Technical Volume (Volume 2). Please refer to Table 3 below for guidance on cost and period of performance. Costs for the Base and Option are to be separate and identified on the Proposal Cover Sheet and in the Order of Magnitude Cost Estimate Table in the Technical Volume (Volume 2).

Order of Magnitude Cost Estimate Table			
Line Item – Details	**Estimated Base Amount**	**Estimated Option Amount**	**Total Estimated Amount Base + Option**
Direct Labor (fully burdened) – Prime			
Subcontractors/Consultants			
Material			
Travel & ODC			
G&A			
FCCM			
Fee/Profit			
TABA (NTE $25K, included in total amount)			
Total Estimated Costs			

TABLE 3: COST & PERIOD OF PERFORMANCE

Topic Number	Base		Option		Total (NTE)
	Cost (NTE)	POP (NTE)	Cost (NTE)	POP (NTE)	
N252-D09 to N252-D10	$1,000,000	30 mos.	$300,000	12 mos.	$1,300,000
N252-D11	$750,000	18 mos.	$1,000,000	24 mos.	$1,750,000*

* Step Two: for the Full Phase II submission, if selected, N252-D1X will require the Phase II Option 1 and Phase II Option 2 to be detailed separately:

- Phase II Option 1: Cost $500,000, Period of Performance 12 months
- Phase II Option 2: Cost $500,000, Period of Performance 12 months

- Additional information:
  - For Phase II a minimum of 50% of the work is performed by the proposing small business concern. The percentage of work requirement must be met in the Base costs as well as in the Option costs. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of work for the proposing small business concern the sum of all direct and indirect costs attributable to the proposing small business concern represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The subcontractor percentage is calculated by taking the sum of all costs attributable to the subcontractor as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator. **NOTE:** G&A, if proposed, will only be attributed to the proposing small business concern.
  - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
  - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
  - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3.

- **Company Commercialization Report (Volume 4)**. DoD collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may or will require to process a proposal, if selected, for contract award.

  All proposing small business concerns must review and submit the following items, as applicable:
  - — **Majority Ownership in Part.** Proposing small business concerns which are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DON topics advertised within this BAA. Complete the certification as detailed under ADDITIONAL SUBMISSION CONSIDERATIONS.

  - Additional information:
    - — Proposing small business concerns may include the following administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing small business concern may want to include in Volume 5:
      - Additional Cost Information to support the Cost Volume (Volume 3)
      - SBIR/STTR Funding Agreement Certification
      - Data Rights Assertion
      - Disclosure of Information (DFARS 252.204-7000)
      - Prior, Current, or Pending Support of Similar Proposals or Awards
      - Foreign Citizens
    - — Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
    - — Do not include documents or information to substantiate the Technical Volume (Volume 2) (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
    - — A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing small business concerns are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoD requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

DP2 EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I Feasibility proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoD and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing small business concern has met eligibility requirements and followed the instructions for Proposal Cover Sheet as specified in the DoD SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I Feasibility proposals using the evaluation criteria specified in the Method of Selectin and Evaluation Criteria section of the DoD SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the evaluation process; the DON will only do a compliance review of Volume 3. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

  The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:
  - A submitting small business concern MUST use the DP2 Phase I Feasibility proposal template for Volume 2. The DP2 Phase I Feasibility proposal template can be found at https://navysbir.com/links_forms.htm.
    This template is specific to DON DP2 topics and meets DP2 submission requirements:
    - Not to exceed 30 pages, regardless of page content; Phase I Proof of Feasibility portion not to exceed 20 pages, Snapshot of Proposed Phase II Effort portion not to exceed 10 pages
    - Single column format, single-spaced typed lines
    - Standard 8 ½” x 11” paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point, except as permitted in the instructions above.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will undergo a compliance review to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base and Option (refer to Table 3).
  - Must meet minimum percentage of work; a minimum of 50% of the work is performed by the proposing small business concern. The percentage of work requirement must be met in the Base costs as well as in the Option costs.

- **Company Commercialization Report (Volume 4).** The CCR (Volume 4) will not be evaluated by the DON nor will it be considered in the award decision. However, all proposing small business concerns must refer to the DoD SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing small business

concern has included items in accordance with the DP2 SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoD SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing small business concerns to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of Defense, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by small business concerns seeking a Federally funded award. Please review the Certifications and Registrations section of the DoD SBIR/STTR Program BAA for details on how DoD will assess security risks presented by small business concerns. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DON to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing small business concerns may request, in their Cost Volume (Volume 3), to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). The amount proposed for TABA cannot include any profit/fee by the proposing small business concern and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the SBIR proposing small business concern. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. A TABA Report, detailing the results and benefits of the service received, will be required annually by October 30.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.
- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service
- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must NOT:
- Be subject to any indirect costs, profit, or fee by the SBIR proposing small business concern
- Propose a TABA provider that is the SBIR proposing small business concern

- Propose a TABA provider that is an affiliate of the SBIR proposing small business concern
- Propose a TABA provider that is an investor of the SBIR proposing small business concern
- Propose a TABA provider that is a subcontractor or consultant of the requesting small business concern otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:

- Phase II:
  - DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:

- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

If a proposing small business concern requests and is awarded TABA in a Phase II contract, the proposing small business concern will be eliminated from participating in the Navy SBIR Transition Program (STP), the DON Forum for SBIR/STTR Transition (FST), and any other Phase II assistance the DON provides directly to awardees.

All Phase II awardees not receiving funds for TABA in their awards must participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing small business concern shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). A small business concern whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Majority Ownership in Part. Proposing small business concerns that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, **are eligible** to submit proposals in response to DON topics advertised within this BAA.

For proposing small business concerns that are a member of this ownership class the following must be satisfied for proposals to be accepted and evaluated:

a. Prior to submitting a proposal, proposing small business concerns must register with the SBA Company Registry Database.
b. The proposing small business concern within its submission must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://navysbir.com/links_forms.htm. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
c. Should a proposing small business concern become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposing small business concern must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification, which can be found on https://navysbir.com/links_forms.htm.

System for Award Management (SAM). It is strongly encouraged that proposing small business concerns register in SAM, https:// sam.gov, by the Close date of this BAA, or verify their registrations are still active and will not expire within 60 days of BAA Close. Additionally, proposing small business concerns should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. A small business concern selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.204-7019, in order to be considered for award, a small business concern is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. If the use of human, animal, and recombinant DNA is included under a DP2 proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing small business concerns within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct cost analysis, confirm eligibility of the proposing small business concern, and to take other relevant steps necessary prior to making an award.

Contract Types. In addition to the negotiated contract award types listed in the section of the DoD SBIR/STTR Program BAA titled Additional Considerations, for Phase II awards the DON may (under appropriate circumstances) propose the use of an Other Transaction Agreement (OTA) as specified in 10 U.S.C. 4021/10 U.S.C. 4022 and related implementing policies and regulations. The DON may choose to use a Basic Ordering Agreement (BOA) for Phase I and Phase II awards.

Contract Deliverables. Contract deliverables are typically progress reports and final reports. Required contract deliverables must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm

that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

Version 2

Navy SBIR 25.2 Direct to Phase II
Topic Index

N252-D09	DIRECT TO PHASE II: F0-Wideband Acoustic Receiver and Source (F0-WARS) Sonobuoy
N252-D10	DIRECT TO PHASE II: High-Density Energy Storage System
N252-D11	DIRECT TO PHASE II: Advanced Interference Mitigation (AIM) Techniques

N252-D09 TITLE: DIRECT TO PHASE II: F0-Wideband Acoustic Receiver and Source (F0-WARS) Sonobuoy

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate an updated and evolved air-deployable source and receiver combination sonobuoy that can collect target strength in the F0 Frequency range.

DESCRIPTION: The U.S. Navy requires a new F0-Wideband Acoustic Receiver and Source (F0-WARS) Sonobuoy to facilitate the collection of acoustic information on enemy targets from Navy anti-submarine warfare (ASW) platforms. A new F0-WARS Sonobuoy will provide enhanced operational capability with improvements in directional information and enhanced echolocation in 3-dimensional (3D) space with incorporation of the directional and depth estimation features. Beyond the F0-WARS Sonobuoy mission lies its potential application to tactical ASW with the use of 3D tracking, shallow water echo parameter estimation, and Roll-On/Roll-Off Mission Processing. Together, a new F0-WARS Sonobuoy will provide significant capability not currently satisfied by legacy, enhanced legacy or other WARS activity, bringing significant tactical capability to the fixed- and rotary-wing ASW missions. As part of the development, the F0-WARS Sonobuoy will be enhanced with the incorporation of NATO-compatible upper electronics, enhanced in-buoy signal processing and associated circuit board consolidation, in a true A-size form factor. End-to-end testing will be performed in water. Environmental testing and over-the-side and air-drop testing of the F0-WARS Sonobuoy will be performed in a representative environment.

The capabilities of current frequency transmitter/receiver sensors do not provide a calibrated, coherent source/receiver combination tailored for target strength collection. Innovative sensor technologies are sought with enhanced electromechanical property ceramics for the F0 frequency and responsiveness gaps for the transmitter and receiver elements that are capable of transmitting, collecting, and processing surveillance information. The Navy and other DoD agencies need to collect active target strength data from enemy targets to achieve key ASW measurement capabilities at F0. The output power source levels are indicated in the classified Annex for the sonobuoy specification. Selected firms need a clearance to view this information after the contract is awarded.

The objective of this SBIR topic is to develop and demonstrate an updated and evolved air-deployable source and receiver combination sonobuoy that can collect active target strength data in the F0 Frequency range. The unit must be capable of both shallow and deep-water operations deploying the active and passive sensing elements through 500 ft (152.4 m) and have a minimum one-hour life (or 50 pulse seconds). Coherent signals of interest are in the High-Frequency range, to include but not be limited to, continuous waveforms (CW) and frequency modulation (FM) waveforms. Communication between the aircraft and sensor unit will be compliant with the NATO digital uplink format, STANAG 4718.
This expendable sensor solution must be low power and sized to fit within an "A" size sonobuoy. A-size sonobuoy standards are: dimensions of 4.88 in. (12.39 cm) diameter x 36 in. (91.44 cm) length and weight of 40 lb. (18.14 kg) or less.

PHASE I: For a Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements of the stated objective above. The below actions would be required to satisfy the requirements of Phase I:
The small business awardee interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the small business has completed development of technology as stated above. Documentation should include all relevant information including, but not limited to technical reports, test data, prototype designs/models, and performance goals/results. Work submitted within the feasibility documentation must have been substantially performed by the offeror and/or the principal investigator (PI). Read and follow all of the DON SBIR 25.2 Direct to Phase II Broad Agency Announcement (BAA) Instructions. Phase I proposals will NOT be accepted for this topic.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and Phase I feasibility described in Phase I above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Develop and fabricate an over-the-side prototype unit(s) required to span the F0 Frequency range and demonstrate in both acoustic facilities and the ocean environment. Throughout this development phase, emphasize a comprehensive evaluation of the prototype's performance under high-ambient and low-source level conditions, ensuring its adaptability and resilience in diverse acoustic settings. Collaborate with similarly focused domain experts and utilize feedback from preliminary tests to further refine and optimize the system at F0 Frequency. Explore integration pathways with existing Navy Maritime Patrol and Reconnaissance infrastructure to maximize system collaboration. Demonstrate the prototype's ability to attain desirable ASW measurement capabilities at High Frequency and provide a roadmap for iterative improvements and integration based on and feedback. Finalize the concept design and make recommendations for Phase III production-oriented designs, detailing potential challenges and solutions for scalable manufacturing.

PHASE III DUAL USE APPLICATIONS: Transition over-the-side prototype unit(s) into an air deployable sonobuoy system. Ensure that the sensor meets A-size packaging requirements specified in the PMA-264 Production Sonobuoy Specification. Perform required testing to verify that the sensor passes all required environmental, structural, and operational tests to include, but not be limited to, Environmental Exposure, Air Certification, Hazards of Electromagnetic Radiation to Ordnance (HERO), and Office of Naval Intelligence (ONI) certification. Note: Upon successful testing, Low-Rate Initial Production (LRIP) will need to be successful for transition to the platform.

This technology/topic can benefit any entity such as the Office of Naval Intelligence (ONI) that requires calibrated active target strength measurements within the underwater environment.

REFERENCES:

1. Urick, R. J. "Principles of underwater sound for engineers (3rd ed.)." Peninsula Publishing. 1983. https://www.worldcat.org/title/8688952

2. Holler, R. A.; Horbach, A. W. and McEachern, J. F. "The ears of air ASW: a history of US Navy sonobuoys." Navmar Applied Sciences Corporation, 2008. https://www.worldcat.org/title/720627294
3. Ultra Electronics Maritime Systems. "Sonobuoys." Ultra Electronics, 2009. https://web.archive.org/web/20110419103757/http:/www.ultra-uems.com/sonobuoys.html
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5. "Standardization agreement: STANAG 4718: Sonobuoy digital telemetry (Ed. 1)." North Atlantic Treaty Organization, The NATO Standardization Office (NSO), 4 November 2020. https://nso.nato.int/nso/nsdd/main/standards?search=4718

KEYWORDS: Anti-Submarine Warfare; Sonobuoy; Low Frequency; Navy Underwater Active Multiple Ping; NUAMP; Acoustics; Intelligence

Version 2

N252-D10 TITLE: DIRECT TO PHASE II: High-Density Energy Storage System

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials; Directed Energy (DE); Renewable Energy Generation and Storage

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a high-density energy storage system (HESS) for use with an existing medium voltage motor drive system.

DESCRIPTION: The Navy requires a +/- 1000 VDC, split bus HESS to deliver energy to a medium voltage motor drive. The system must store at minimum 1.1 megajoule (MJ) (Threshold) of energy, 5.5 MJ (Objective) and have a system capacitance that exceeds 2.3 Farad (F) (Threshold (T)), 11 F (Objective (O)). The energy storage system must be capable of both delivering power to a motor via a three-level neutral point clamped (NPC) inverter and accepting power from regenerative braking of the motor. Energy is supplied to the motor for approximately 0.5 seconds, with regenerative braking occurring over a span of 3-5 seconds. Then, energy is supplied to the motor for approximately 20 seconds, the bus is recharged, and the cycle repeats with variable time delay between cycles.

The HESS must be modular, meaning you can add or remove additional HESS units to change overall system capacitance. An individual unit should be no larger than 93in x 45in x 81.5in. Military standards should be referenced for shock (MIL-DTL-901E [Grade A]) [Ref 5], vibration (MIL-STD-167-1A [Type 1]) [Ref 6], electromagnetic interference (MIL-STD-461G) [Ref 4], and environmental factors (MIL-STD-810H) [Ref 3] since the system must be rugged to be viable.

The energy storage system must also interface with a charging power supply. Existing charging power supplies are capable of outputting 90 kW to each bus at 30 amperes; however, this likely will not be sufficient to charge a highly energy-dense bus in the time required. Therefore, an alternate charging system design, intermediate power electronics between the existing charging power supply and energy storage system, or a hybrid energy storage approach (e.g., using components with different characteristics to promote fast charging during initial motor start-up and maximum energy capture during regenerative braking) are all acceptable approaches and considered within scope. If a new charging system is proposed, it must accept 440 Volts (V), 3 phase AC power. The bus must be charged in 60 seconds (T) or 10 seconds (O) at start up. The charging system can be contained within the HESS cabinet, or a separate cabinet that should be no larger than 48in x 63in x 38in.

The HESS must have a locally operable disconnect switch that can be monitored. The system must be capable of being discharged to 0 V via an existing energy dump (resistor bank), must be maintainable, and must not prohibit maintenance of connected equipment. It must have a means of verifying that discharged components have a voltage value less than +/- 20 VDC.

The energy density of the system must surpass the limits of typical capacitors. The volumetric density of the system must be at least 615 kJ/mm^3, and the gravimetric density must be at least at 0.1 Whr/kg. The system must aim to minimize weight and volume. If applicable, a battery management system, or

equivalent for alternate technologies, must be incorporated to monitor, control, balance, collect data, facilitate safe use of the system, and extend its life. Mean time between failures (MTBF) of the HESS must be greater than or equal to 27,000 operational hours that can be demonstrated via modeling. Capacitors, as a commonly used method of energy storage, may be limited in energy density and ability to quickly store generated energy. Advances in supercapacitors, ultracapacitors, batteries, hybrid energy storage solutions, and other related technologies associated with high-density energy storage may be relevant. A modular or scalable approach is preferred to promote applicability for additional military and commercial use cases.

The proposed technology should also ensure that the prototype device can be:

1. integrated seamlessly in place of, or in addition to, an existing energy storage system with minimal modifications to upstream/downstream power equipment and controls.
2. scalable for smaller or larger applications in the long term.
3. manufactured at a cost-effective price point at scale.
4. maintainable by sailors.
5. safe to operate/manage.

PHASE I: For a Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort and developed a concept for a workable prototype or design to address, at a minimum, the basic requirements of the stated objective above. The below actions would be required to satisfy the requirements of Phase I:

1. Developed a conceptual design, workable prototype or scalable solution for a high-density energy storage system (HESS) capable of meeting the requirements outlined in the description.
2. Demonstrated that the proposed HESS technology offers a higher energy density and more efficient energy storage capability compared to existing capacitor-based solutions.

FEASIBILITY DOCUMENTATION: Offerors interested in participating in Direct to Phase II must include in their response to this topic Phase I feasibility documentation that substantiates the scientific and technical merit and Phase I feasibility described in Phase I above has been met (i.e., the small business must have performed Phase I-type research and development related to the topic NOT solely based on work performed under prior or ongoing federally funded SBIR/STTR work) and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed development of technology as stated in Phase I above.

PHASE II: Develop a subscale prototype HESS. Validate and demonstrate that the proposed HESS technology meets requirements for charging, storage, delivering energy in a medium voltage motor drive system, and receiving energy generated by the motor. Develop plans for how the technology can be scaled to meet full-scale system requirements.

Assess the prototype focused on energy storage capacity, energy efficiency, heat dissipation, safety, maintainability, and integration compatibility with other system components. Scalability and cost-effectiveness of the proposed technology will also be explored and evaluated.

PHASE III DUAL USE APPLICATIONS: Develop a full-scale HESS design and integrate it into the existing medium voltage motor drive system, test the new system, and prepare for acquisition into the corresponding program of record.

Energy storage for vehicles and renewable energy storage are potential commercial markets for this technology.

REFERENCES:

1. Raza, W.; Ali, F.; Raza, N.; Luo, Y.; Kim, K. H.; Yang, J.; Kumar, S.; Mehmood, A. and Kwon, E. E. "Recent advancements in supercapacitor technology." Nano Energy 52, 2018, pp. 441-473. https://www.sciencedirect.com/science/article/pii/S2211285518305755
2. Jagadale, A.; Zhou, X.; Xiong, R.; Dubal, D. P.; Xu, J. and Yang, S. "Lithium ion capacitors (LICs): Development of the materials." Energy Storage Materials 19, 2019, pp. 314-329. https://www.sciencedirect.com/science/article/pii/S2405829718315174
3. "MIL-STD-810H w/Change 1: Department of Defense test method standard: Environmental engineering considerations and laboratory tests". MIL-STD-810 Working Group, 18 May 2022. https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=35978
4. "MIL-STD-461G: Department of Defense interface standard: Requirements for the control of electromagnetic interference characteristics of subsystems and equipment." U.S. Air Force, 11 December 2015. http://everyspec.com/MIL-STD/MIL-STD-0300-0499/MIL-STD-461G_53571/
5. "MIL-DTL-901E: Detail specification: Shock tests, H. I. (High-Impact) shipboard machinery, equipment, and systems, requirements for." Naval Sea Systems Command, 20 June 2017. http://everyspec.com/MIL-SPECS/MIL-SPECS-MIL-DTL/MIL-DTL-901E_55988/
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KEYWORDS: Energy storage; High power density; Capacitors; Supercapacitors; Energy; Power; Batteries; HESS

Version 2

N252-D11 TITLE: DIRECT TO PHASE II: Advanced Interference Mitigation (AIM) Techniques

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Network Systems-of-Systems; Integrated Sensing and Cyber; Space Technology

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop and demonstrate innovative receive-side sensitivity enhancements that improve signal acquisition, noise suppression, and interference mitigation for Multifunctional Information Distribution System (MIDS) terminals. This effort focuses on maximizing receiver performance independent of transmission power, leveraging advanced adaptive filtering, diversity techniques, and low-noise reception strategies to ensure reliable communication in contested and degraded environments. Solutions must be Field Programmable Gate Array (FPGA)-compatible, maintain low latency (=10µs), and integrate seamlessly into existing MIDS systems without degrading performance.

DESCRIPTION: The MIDS Program Office (MPO) is responsible for advancing tactical data link (TDL) communications, including Link-16 and Tactical Targeting Network Technology (TTNT). As modern battlefields become increasingly congested with electromagnetic interference, adversarial jamming, and spectrum competition, the ability to maintain a high-sensitivity, interference-resistant reception is critical to mission success.

This effort focuses on achieving significant sensitivity gains, enhancing the receiving capability of MIDS terminals by separating and improving the receive-side performance independent of transmitter-based enhancements. Current limitations in receiver sensitivity restrict the ability to capture and process weak signals, particularly in contested environments where interference mitigation and adaptability are crucial.

To address this, the Navy is seeking innovative solutions that integrate:

- Advanced signal processing for adaptive filtering and interference suppression
- Optimized low-noise reception to amplify weak signals while minimizing distortion
- Novel antenna configurations that enhance signal gain through spatial, frequency, or polarization techniques

These enhancements will fundamentally improve the resilience and efficiency of MIDS terminals, enabling more robust signal acquisition and improved signal-to-noise ratio (SNR) under degraded conditions. Unlike traditional solutions that focus on boosting transmission power, this effort is uniquely centered on maximizing the effectiveness of the receiver itself, ensuring clear, uninterrupted reception even in challenging operational environments.

The Navy seeks mature solutions (Technology Readiness Level (TRL) 5 or higher) that can be implemented in FPGA architectures, allowing for real-time adaptation and rapid deployment. Proposed solutions should be capable of:

- Demonstrating =10dB improvement in receive sensitivity
- Maintaining low-latency processing (=10µs) for real-time applications
- Efficiently utilizing FPGA resources (=10% overhead)
- Ensuring compatibility with existing MIDS systems without degradation of performance

Of particular interest are adaptive diversity techniques, such as maximal ratio combining (MRC), spatial diversity, and multi-path fading mitigation, which can independently improve receive performance in highly dynamic environments. These innovations will provide a separate and robust receive-side enhancement, ensuring a more resilient and high-sensitivity reception capability for tactical data links. The successful transition of these technologies will provide independent, receiver-focused improvements that support long-range, interference-resistant communication. This SBIR topic seeks vendors who have already demonstrated initial feasibility and validation of sensitivity enhancement techniques, with the goal of integrating these advancements into the MIDS terminal for operational deployment.

Work produced in Phase II may become classified . Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVWAR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Feasibility documentation MUST NOT be solely based on work performed under prior or ongoing federally funded SBIR/STTR work. Demonstrating proof of feasibility is a requirement for a Direct to Phase II award.

For this Direct to Phase II topic, the Government expects that the small business would have accomplished the following in a Phase I-type effort:

- Assessed the feasibility of developing, integrating, and demonstrating diversity gain and interference mitigation techniques. Their prior work should include theoretical analysis, simulation studies, and experimental validation, as well as feasibility assessments and identification of key challenges.
- Conducted a survey of existing algorithms and established baseline figures of merit for selecting algorithms implementable in a FPGA.
- Developed simulations to quantify improvements in SNR or signal reliability achieved through diversity techniques, specifying the additional SNR gained under defined channel conditions.

FEASIBILITY DOCUMENTATION: Offerors interested in proposing to this Direct to Phase II topic must include in their response Phase I feasibility documentation that substantiates the scientific and technical merit; proof that Phase I feasibility (described in Phase I above) has been met (i.e., the small business must have performed Phase I-type research and development related to the topic, but feasibility documentation must not be solely based on work performed under prior or ongoing federally funded SBIR/STTR work.); and describe the potential commercialization applications. The documentation provided must validate that the proposer has completed Phase I-type development of technology as stated above. Documentation should include all relevant information including, but not limited to: technical reports, test data, prototype designs/models, and performance goals/results. Work submitted within the feasibility documentation must have been substantially performed by the offeror and/or the principal investigator (PI).

Version 2

PHASE II: The contract type for topic N252-D11 will be Cost-Plus-Fixed-Fee.

The selected small business will design, develop, and prototype receive-side sensitivity enhancements aimed at improving the signal acquisition and interference mitigation capabilities of MIDS terminals. The primary focus is to implement and validate high-sensitivity reception techniques that operate independently of transmitter-based improvements, ensuring more reliable communications in contested and degraded operational environments:

1. Algorithm Development & FPGA Integration:
   - Design, develop, and implement advanced signal processing algorithms for adaptive filtering, diversity gain, and low-noise reception.
   - Ensure that algorithms can be efficiently integrated into a FPGA-based architecture with minimal processing overhead (=10% FPGA resource utilization).
   - Maintain low latency (=10µs) to support real-time tactical applications.
2. Prototype Development & SDR Testing:
   - Develop a software prototype implementing the proposed sensitivity enhancement techniques for evaluation.
   - Conduct initial software-in-the-loop (SIL) simulations to assess the performance of new receiver algorithms in varied signal environments, including high-interference scenarios.
   - Transition from software testing to hardware implementation by integrating the algorithms into an SDR-based prototype.
3. Performance Validation & Testing:
   - Evaluate the prototype's ability to achieve a =10 dB improvement in receive sensitivity through controlled laboratory testing and simulation.
   - Validate algorithm performance against real-world modulations and emulated threat signal sets, assessing:
   - SNR improvements
   - Interference suppression and adversarial jamming resilience
   - Long-range reception capability
   - Perform extensive testing to ensure that the receive enhancements do not degrade link integrity or negatively impact existing MIDS system performance.
   - Government may provide threat signal data for additional testing and may also conduct independent verification at a Government facility.
4. Prototype Hardware Demonstration:
   - Implement and test the prototype within a MIDS Joint Tactical Radio System (JTRS) laboratory environment.
   - Assess integration feasibility with existing MIDS terminals, ensuring modular, scalable deployment without adverse effects on system operations.
   - Evaluate final FPGA resource utilization, latency, and computational efficiency to ensure compliance with Navy requirements.
5. Phase III Transition Planning:
   - Develop a detailed technology transition and commercialization strategy for Phase III implementation.
   - Engage with MIDS prime vendors and key stakeholders to ensure seamless integration into operational systems.
   - Identify any additional system modifications or optimizations needed for full deployment within Navy platforms.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

Version 2

PHASE III DUAL USE APPLICATIONS: Transition the developed technology for Navy deployment:

1. Refine and validate final algorithms for full-scale implementation.
2. Ensure compliance with Information Assurance (IA) standards for Navy systems.
3. Support SDR integration and full mission capability testing.
4. Provide comprehensive training and documentation for seamless transition.
5. Partnership with MIDS prime vendors is encouraged to support final production and deployment.

Advanced interference mitigation techniques have applications beyond military systems, including:

- 5G/6G wireless communications
- Satellite and deep-space communications
- Secure networking in high-interference commercial environments

REFERENCES:

1. D. Wang, J. Wang, X. You, Y. Wang, M. Chen, and X. Hou, "Spectral efficiency of distributed MIMO systems," IEEE Journal on Selected Areas in Communications, vol. 31, no. 10, pp. 2112–2127, 2013. https://ieeexplore.ieee.org/document/7564722
2. S. Sedighi and E. Ayanoglu, "Bit-interleaved coded multiple beamforming in millimeter-wave massive MIMO systems," IEEE Transactions on Communications, vol. 68, no. 10, pp. 6174–6185, 2020. https://ieeexplore.ieee.org/document/9133524
3. D. Yu, S. Xu, and H. H. Nguyen, "Diversity gain of millimeter-wave massive MIMO systems with distributed antenna arrays," EURASIP Journal on Wireless Communications and Networking, vol. 54, pp. 1–13, 2019. https://jwcn-eurasipjournals.springeropen.com/articles/10.1186/s13638-019-1366-8

KEYWORDS: Receive sensitivity; diversity gain; beamforming; Multiple Input Multiple Output; MIMO

Defense Health Agency (DHA)
2025.2 Small Business Innovation Research (SBIR)
Proposal Submission Instructions

INTRODUCTION

The Defense Health Agency (DHA) SBIR Program seeks small businesses with strong research and development capabilities to pursue and commercialize medical technologies.

Proposers responding to a topic in this BAA must follow all general instructions provided in the Department of Defense (DoD) SBIR Program BAA. The instructions below are important to review, as they include agency-specific requirements that add to or deviate from the DoD Program BAA. Proposals that do not address agency-specific requirements may be considered non-responsive. Only Government personnel will evaluate proposals submitted under this solicitation cycle.

Proposers are encouraged to thoroughly review the DoD Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The DoD Program BAA is located at: Active Solicitations (dodsbirsttr.mil). Be sure to select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: Sign Up form (campaigner.com).

Specific questions about the DHA SBIR Program and agency-specific instructions included below should be directed to: DHA SBIR Program Management Office (PMO) Email: usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP. Proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided within the DoD SBIR Program BAA.

Technical Volume (Volume 2)

The technical volume is not to exceed **20 pages** and must follow the format and content requirements provided in the DoD SBIR Program BAA. Do not duplicate the electronically-generated Cover Sheet or insert information associated with the Technical Volume into other sections of the proposal, as these will count toward the 20-page limit.

Only the electronically-generated Cover Sheet and Cost Volume are excluded from the 20-page limit. Technical Volumes that exceed the 20-page limit will be deemed non-compliant and will not be evaluated.

Cost Volume (Volume 3)

The Phase I Base amount must not exceed $250,000 over a 6-month period of performance. Costs must be clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. DHA will occasionally accept deviations from the POW requirements with written approval from the Funding Agreement Officer.

Travel must be justified and should relate to direct Research Development Test & Evaluation (RDT&E) Technology Readiness Level (TRL)-increasing task costs. Travel costs must include the purpose of the trip(s), number of trips, origin and destination, length of trip(s), and number of personnel.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD SBIR Program BAA for full details on this requirement. Information contained in the CCR will be considered by DHA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoD Program BAA for more information.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoD Program BAA.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees unless a topic is otherwise marked. Phase II is the demonstration of the technology found feasible in Phase I. Specific due date, content, and submission requirements of the Phase II proposal will be provided by the DHA SBIR PMO; typically during month five of the Phase I contract.

Due to limited funding, the DHA SBIR Program reserves the right to limit awards under any topic. Only proposals considered to be of superior quality will be funded. Small businesses submitting a proposal are required to develop and submit a Commercialization Strategy describing feasible approaches for transitioning and/or commercializing the developed technology in their Phase II proposal, including plans to pursue appropriate seed-stage investment. This plan shall be included in the Technical Volume.

The Cost Volume must contain a budget for the entire 24-month Phase II period in an amount not to exceed the maximum dollar amount of $1,300,000. Budget costs must be submitted using the Cost Volume format (accessible electronically on the DoD submission site) and shall be presented side-by-side on a single Cost Volume Sheet.

DHA SBIR Phase II proposals have seven volumes: Proposal Cover Sheets; Technical Volume; Cost Volume; Company Commercialization Report; Supporting Documents; Fraud, Waste, and Abuse; Disclosures of Foreign Affiliations or Relationships to Foreign Countries.

The Technical Volume has a 40-page limit including: table of contents, pages intentionally left blank, references, letters of support, appendices, technical portions of subcontract documents (e.g., statements of

work and resumes) and any attachments. Technical Volumes that exceed the 40-page limit will be deemed non-compliant and will not be evaluated.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
The DHA SBIR Program does not participate in the Technical and Business Assistance (formerly the Discretionary Technical Assistance Program). Contractors shall not submit proposals that include Technical and Business Assistance.

The DHA SBIR Program offers a Commercialization Readiness Program providing assistance to small businesses that receive Phase I and Phase II awards.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoD SBIR Program BAA.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the BAA. Non-selected companies may request feedback within 15 calendar days of the non-select notification. The Corporate Official identified in the firm's proposal shall submit the feedback request to the SBIR Office at usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil. Feedback is provided in an official PDF via email to the Corporate Official identified in the firm proposal within 60 days of receipt of the request. Requests for oral feedback will not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the feedback request.

NOTE: Feedback is not the same as a FAR Part 15 debriefing. Acquisitions under this solicitation are awarded via "other competitive procedures". Therefore, offerors are neither entitled to nor will they be provided FAR Part 15 debriefs.

Refer to the DoD SBIR Program BAA for procedures to protest the Announcement. Protests after Award, as further prescribed in FAR 33.106(b), FAR 52.233-3, should be submitted to:

Ms. Samantha L. Connors SBIR/STTR Chief, Contracts Branch 8
Contracting Officer
U.S. Army Medical Research Acquisition Activity
Email: samantha.l.connors.civ@health.mil

AWARD AND CONTRACT INFORMATION
Phase I awards will total up to $250,000 for a 6-month effort and will be awarded as Firm-Fixed-Price Purchase Orders.

Phase II awards will total up to $1,300,000 for a 24-month effort and will typically be awarded as Firm-Fixed-Price contracts. If a different contracting type is preferred, such as Cost-Plus, the rationale as to why must be included within the proposal.

Phase I/Phase II awardees will be informed of contracting specialist, contracting officer, and Technical Point of Contact/Contract Officer Representative upon award.

ADDITIONAL INFORMATION

RESEARCH INVOLVING HUMAN SUBJECTS, HUMAN SPECIMENS/DATA, OR ANIMAL RESEARCH

The DHA SBIR Program highly discourages offerors from proposing animal or human use research during Phase I due to the significant lead time required to prepare documentation and secure approval, which could substantially delay the funding and performance of the Phase I award.

Prior to contract award when an IRB is indicated, proposers must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human subjects, human specimens, or research with animals. If necessary, approvals are not obtained within two months of notification of selection, the decision to award may be terminated.

Offerors are expressly forbidden to use, or subcontract for the use of, laboratory animals in any manner without the express written approval of the U.S. Army Medical Research and Development Command (USAMRDC) Animal Care and Use Review Office (ACURO). Written authorization to begin research under the applicable protocol(s) proposed for this award will be issued in the form of an approval letter from the USAMRDC ACURO to the recipient. Modifications to previously approved protocols require re-approval by ACURO prior to implementation.

Research under this award involving the use of human subjects, to include the use of human anatomical substances or human data, shall not begin until the USAMRDC's Office of Human Research Oversight (OHRO) provides formal authorization. Written approval to begin a research protocol will be issued from the USAMRDC OHRO, under separate notification to the recipient. Written approval from the USAMRDC OHRO is required for any sub-recipient using funds from this award to conduct research involving human subjects. If the Offeror intends to submit research funded by this award to the U.S. Food and Drug Administration, Offerors shall propose a regulatory strategy for review.

*NOTE: Exempt animal or human research use shall also reflect 'yes' on the proposal coversheet for USAMRDC ACURO and OHARO records.

Non-compliance with any provision may result in withholding of funds and or termination of the award.

For more information please visit the MRDC Office of Human Research Oversight website here:
USAMRDC: USAMRDC: Office of Human Research Oversight (OHRO) (health.mil)

FEDERAL FACILITY USE

The DHA SBIR Program highly discourages small business concerns (SBCs) from subcontracting to a federal facility and/or utilizing for testing due to the significant lead time required to secure approval, which could substantially delay the performance of the award.
Use of federal facilities is prohibited without an approved waiver from the DHA SBIR/STTR Office.

An SBC whose proposed work includes federal facility use is required to provide a written justification, uploaded to the Supporting Documents (Volume 5), that includes the following information:

1. An explanation of why the SBIR/STTR research project requires the use of the federal facility, including data that verifies the absence of non-federal U.S. facilities, in support of the overall mission and research area.
2. Evidence that there is no applicable U.S. facility that has the ability or expertise to perform the specified work.
3. Why the Federal Agency will not and cannot fund the use of the Federal facility or personnel for the SBIR/STTR project with non-SBIR/STTR funds.

The DHA SBIR Program has the right of refusal. Companies that fail to meet requirements specified above will be at risk of delay to award or funding.

If the proposal is selected, the U.S. Army Medical Research Acquisition Activity (USAMRAA) will assist in establishing the waiver for DHA SBIR/STTR Office approval. If approved, the proposer will subcontract directly with the federal facility and not a third-party representative.

Transfer of funds between a company and a Military Lab must meet the following APAN 15-01 requirements (the full text of this notice can be found at https://usamraa.health.mil/SiteAssets/APAN%2015-01%20Revised%20Feb%202018.pdf):

(1) The DoD Intramural Researcher must obtain a letter from his/her commanding officer or Military Facility director authorizing his/her participation in the Extramural Research project. This letter must be provided to the Extramural Organization for inclusion in the proposal or application.

(2) The DoD Intramural Researcher must also coordinate with his/her local RM office (or equivalent) to prepare a sound budget and justification for the estimated costs. Where there are no DoD-established reimbursement rates [e.g., institution review board (IRB) fees, indirect cost rates, etc.], the Military Facility's RM office (or equivalent) must provide details of how the proposed rates were determined. The DoD Intramural Researcher must use the budget and justification form enclosed in APAN 15-01 when developing the estimated costs and provide it to the Extramural Organization for inclusion in the proposal or application.

(3) The Extramural Research proposal or application must include a proposed financial plan for how the Military Facility's Intramural Research costs will be supported [i.e., directly funded by DoD, resources (other than award funds) provided by the Awardee to the Military Facility, or award funds provided by the Awardee to the Military Facility (in accordance with the requirements below)].

(4) The DoD Intramural Researcher should also coordinate with his/her technology transfer office.

INTERNATIONAL TRAFFIC IN ARMS REGULATION (ITAR)

For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain nonprofit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases, such as Phase III, if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

System for Award Management (SAM)

As stated in FAR Provision 52.204-7 System for Award Management, proposing Small Business Concerns must be registered in SAM, https://sam.gov, at the time of submission and shall continue to be registered until time of award, during performance, and through final payment of any contract, basic agreement, basic ordering agreement, or blanket purchasing agreement resulting from this solicitation.

DHA SBIR 25.2 Topic Index

DHA252-001	Auxiliary Microbial Detection and Clinical Feature Identification for Medical Services Support

DHA252-001 TITLE: Auxiliary Microbial Detection and Clinical Feature Identification for Medical Services Support

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Military Infectious Disease

OBJECTIVE: To develop a self-contained, low-cost, and rapid detection system for use in large-scale combat operations. This system should enable medical personnel to quickly detect and identify clinically relevant microbial organisms from clinical isolates in situations where advanced assets or high-complexity techniques are unavailable or degraded.

DESCRIPTION: Future combat operations are expected to involve intense, division-on-division conflicts, resulting in high casualty volumes and prolonged medical intervention times. In this environment, medical Providers at the point of injury or in forward-deployed medical facilities will face significant challenges in diagnosing and managing microbial-mediated diseases and wound infections. Current diagnostic tools are often too complex, requiring specialized training and expertise that may not be available in austere or resource-constrained settings. Furthermore, these devices often require regular maintenance, calibration, and resupply of reagents and consumables, which may be difficult or impossible to obtain in a contested logistics environment. The consequences of this capability gap would be severe, with delayed or incorrect diagnoses potentially resulting in increased morbidity, mortality, and long-term health consequences for Warfighters. The development of a portable and reliable backup diagnostic tool is critical to ensuring that medical Providers can detect microbial pathogens quickly and accurately, even in resource-constrained environments, and provide timely and effective treatment, ultimately saving lives and improving patient outcomes.

This topic seeks innovative solutions for massively scalable and distributive (i.e., affordable) rapid diagnostic tools. Existing tools, such as biochemical test strips and sugar utilization testing reagents, are useful but have several limitations, including a short shelf life, cumbersome inventory, cold-chain requirements, poor limit of detection, and difficulty detecting antimicrobial resistance, which render them of limited utility as a reliable backup diagnostic tool for resource-limited clinical labs. Additionally, the proposed solution should be able to replace or supplement existing diagnostic tools and have a clear potential for civilian use and marketplace. The goal of this request is to develop a highly integrated, simplified inventory and quality control, cost-effective, and rapid paper-based (or similar matrix) microbial identification system suitable for use in medical support of large-scale combat operations at multiple rear echelons.

The proposed solution should enable rapid detection and identification of clinically relevant microbial organisms from cultured isolates, providing broad and relevant results rapidly. The system should be designed to test cultured isolates, distinguishing between Gram-positive and Gram-negative bacteria and fungi, and provide unambiguous primary output (CLIA-Waived or Moderate Complexity designation). To support use in resource-constrained environments, the system should require minimal logistical support, have a minimum 12-month shelf life at room temperature (without refrigeration), and be compatible with extreme environments. By meeting these clinical and operational requirements, the proposed solution can provide a self-contained, low-cost, and compact detection system for use in large-scale combat operations. This proposal aims to replace traditional, labor-intensive, and often inaccurate methods of microbial identification, such as biochemical test strips and sugar utilization testing reagents, with a rapid, reliable, and user-friendly diagnostic system that can accurately identify pathogens in resource-constrained environments.

Note: this proposal will exclude all technologies involving electronic components, batteries, complex interfaces, …etc.

PHASE I: Given the short duration of Phase I, this phase should focus on a feasibility study or proof of concept that is tested. The goal is to investigate the potential for a self-contained, low-cost, and rapid detection system for use in large-scale combat operations. The study should combine or merge relevant assays to address the identification requirement for clinically relevant isolates. For example, selecting one Gram-positive bacterium, one Gram-negative bacterium, and one fungus from one set of sample types for demonstration is an acceptable plan. The results of the study should demonstrate feasibility, ease of operation, and proof-of-concept, and establish reasonable qualitative pathogen signal identification and confirmation. The study should combine or merge relevant assays to address the identification requirement for clinically relevant isolates. The study should provide empirical confidence and accuracy data, including controls, false positive and negative rates, and an estimate of the feasible limit of detection (LOD). The results should indicate next steps and provide a clear plan for prototype development in Phase II, including a plan for pilot clinical validation and testing with several select organisms to demonstrate the scientific principles behind the detection mechanisms. This phase will down-select promising designs for expansion in Phase II.

PHASE II: During this phase, the prototype should be developed and tested to demonstrate earlier and more robust detection of diverse sets of microbes from diverse sample types and reduce the downstream need for incubators or specialized equipment in analysis. The technology should provide broad and relevant results in less than 24 hours without requiring additional steps, such as coagulation tests, urease activity, sugar utilization, catalase, etc. The testing should be conducted in controlled and rigorous conditions, evaluating the qualitative and quantitative outcomes of the product with regards to quantification, identification of pathogenic organisms, and characteristics such as reagent reactivity and antibiotic resistance. The system's design should also prioritize compactness, with dimensions not exceeding 4 x 9 inches and a thickness of less than 0.15 inches, and ease of use, with minimal training required. The results of the testing should indicate next steps and provide a clear plan for further development and commercialization. The company should focus on optimizing the design for manufacturability, with an emphasis on affordability, to influence the Government's ability to become a future customer. The offeror should consider early interactions through Q-submission (Q-SUB) process with FDA for guidance on planned pre-clinical testing. The offeror should provide a comprehensive regulatory strategy outlining details of proposed pre-clinical & pivotal testing along with specific regulatory plans, as well as a CLIA designation letter.

PHASE III DUAL USE APPLICATIONS: The ultimate goal of this phase is to secure FDA approval and commercialize the technology. The company should develop partnerships that adhere to Quality Management System (QMS) requirements to demonstrate and commercialize the technology in civilian-relevant settings, such as disaster response and rural or low-resource healthcare, where there is a significant need for rapid and accurate diagnostic capabilities. The point-of-care diagnostics market is a large and growing market, with a potential value of over $72 billion, and the company should provide evidence of customer pull and interest in this technology. The company should also investigate potential opportunities with other government agencies, such as Congressionally Directed Medical Research Programs (CDMRP) and Joint Warfighter Medical Research Programs (JWMRP), that may be interested in supporting the development and commercialization of this technology. If the technology is successful, it may be considered for future acquisition programs, and the company should be prepared to work with the Government to explore potential transition paths. The company should develop a plan for making the product available to potential military and civilian users and should be aware of the potential for future government funding opportunities to support the technology's development and deployment.

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KEYWORDS: MDO, Infectious disease, Microbiology, diagnosis, monitoring, trauma, prolonged care

Defense Threat Reduction Agency (DTRA)
DoD 2025.2 Small Business Innovation Research (SBIR) Program
Proposal Submission Instructions

INTRODUCTION

The Defense Threat Reduction Agency (DTRA) mission is to enable the DoD, the U.S. Government, and International Partners to counter and deter Weapons of Mass Destruction (WMD) Chemical Biological, Radiological, Nuclear) and Improvised Threat Networks. The DTRA SBIR program is consistent with the purpose of the Federal SBIR/STTR Program, i.e., to stimulate a partnership of ideas and technologies between innovative small business concerns and through Federal-funded research or research and development (R/R&D).

The approved FY25.2 topics solicited for the Defense Threat Reduction Agency (DTRA) Small Business Innovation Research (SBIR) Program are included in these instructions followed by the full topic description. Offerors responding to this Broad Agency Announcement (BAA) must follow all general instructions provided in the related Department of Defense Annual Program BAA and submit proposals by the date and time listed in this release. Specific DTRA requirements that add to or deviate from the DoD Annual Program BAA instructions are provided below with references to the appropriate section of the DoD document.

The DTRA Small Business Innovation Research (SBIR) Program is implemented, administered, and managed by the DTRA SBIR/STTR Program Office. Specific questions pertaining to the administration of the DTRA SBIR program, and these proposal preparation instructions should be directed to:

Ms. LaTricia Price
DTRA SBIR/STTR Program Manager
LaTracia.T.Price.civ@mail.mil
Tel: 571-616-6583

Defense Threat Reduction Agency
8725 John J. Kingman Road
Stop 6201
Ft. Belvoir, VA 22060-6201

For technical questions about specific topic requirements during the pre-release period, contact the DTRA Technical Point of Contact (TPOC) for that specific topic. To obtain answers to technical questions during the formal BAA open period, visit: https://www.dodsbirsttr.mil/submissions/login. For questions regarding the Defense SBIR/STTR Innovation Portal, contact DSIP Support at: dodsbirsupport@reisystems.com.

Proposals not conforming to the terms of this announcement will not be considered. DTRA reserves the right to limit awards under any topic, and only those proposals of superior scientific and technical quality as determined by DTRA will be funded.

DTRA reserves the right to withdraw from negotiations at any time prior to contract award. The Government may withdraw from negotiations at any time for any reason to include matters of national security (foreign persons, foreign influence or ownership, inability to clear the firm or personnel for security clearances, or other related issues).

Please read the entire DoD announcement and DTRA instructions carefully prior to submitting your proposal as there have been significant updates to the requirements.

Proposers responding to a topic in this BAA must follow all general instructions provided in the Department of Defense (DoD) SBIR Program BAA. DTRA requirements in addition to or deviating from the DoD Program BAA are provided in the instructions below.

Proposers are encouraged to thoroughly review the DoD Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The DoD Program BAA is located at: Defense SBIR/STTR - Funding Opportunities (defensesbirsttr.mil). Be sure to select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the administration of the DTR SBIR Program, and these proposal preparation instructions should be directed to: **Ms. LaTricia Price, DTRA SBIR/STTR Program Manager**; (LaTracia.T.Price.civ@mail.mil).

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoD SBIR Program BAA.

Technical Volume (Volume 2)

The technical volume is not to exceed a 20-page limit and must follow the formatting requirements provided in the DoD SBIR Program BAA. Any pages in the technical volume over 20 pages will not be considered in proposal evaluations.

Content of the Technical Volume

A Phase I Proposal Template is available to provide helpful guidelines for completing each section of your Phase I technical proposal. This can be found at Training Materials (dodsbirsttr.mil)

Offerors should follow the DoD SBIR Program BAA guidelines regarding Technical Volume content.

Cost Volume (Volume 3)

The Phase I Base amount must not exceed $200,000. For the Cost Volume, The Defense Threat Reduction Agency requires the use of a Microsoft excel spread sheet which is available on the DSIP portal. Note: The DTRA Cost Volume template will be accessible once the Cost Volume is initiated.

Important: when completing the cost volume, enough information should be provided to allow the agency to understand how you plan to use the requested funds if a contract is awarded. Itemized costs of any subcontract or consultant should be provided to the same level as for the prime small business. If an unsanitized version of costs cannot be provided with the proposal, the Government may request it during negotiations if selected. Refer to the instruction provided in the DoD Annual SBIR program BAA for additional details on the content of the Cost Volume.

Note: Cost for travel funds must be justified and related to the needs of the project.

The Phase I Base amount, notwithstanding the amount allocated for **TECHNICAL AND BUSINESS ASSISTANCE (**TABA), must not exceed $200,00.00. All costs must be clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3. [DTRA requires the use of an excel spreadsheet for the Cost Volume. The cost template becomes visible to the offeror when the Cost Volume is initiated.]

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. For SBIR projects, DTRA normally will not accept any deviation to the POW requirements however if discovered during review in Contracting the offeror may be allowed to revise the cost proposal to be in line with the POW requirements.

Page Limit, Cost and Duration:

Project Phase	Technical Vol Page Limit	Cost	Duration
Phase I	20 pages	$200,000.00	7 Months
Phase II	40 pages	$1,300,000.00	24 Months

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD SBIR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DTRA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoD Program BAA for more information.

DIRECT TO PHASE II PROPOSAL GUIDELINES
The Defense Threat Reduction Agency does not participate in the Direct to Phase II (DP2) proposal submission program.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees.
Those small business concerns submitting a Phase II proposal should plan to submit a fully developed proposal into the DSIP proposal system within thirty (30) days after the end of the Phase I period of performance. The small business concern may or may not be automatically notified of the recommended proposal due date.

The Phase II proposal Technical Volume should generally follow the outline and structure of the Phase I to include benefits or lessons learned from the Phase I effort.

DTRA plans on a Phase II project not to exceed $1,300,000.00 notwithstanding TABA, and two (2) years duration.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
In accordance with the Small Business Act (15 U.S.C. 632), DTRA will authorize the recipient of a Phase I or Phase II STTR award to purchase Discretionary Technical & Business Assistance services, such as access to a network of scientists and engineers engaged in a wide range of technologies, or access to technical and business literature available through on-line data bases, for the purpose of assisting in areas such as:

- making better technical decisions concerning such projects,
- solving technical problems which arise during the conduct of such projects,
- minimizing technical risks associated with such projects,
- developing/ commercializing new commercial products/processes resulting from such projects; and meeting cyber security requirements.

If you are proposing use of Discretionary Technical and Business Assistance (TABA), you must provide a cost breakdown in the Cost Volume under "Other Direct Costs (ODCs)" and provide a one-page description of the vendor you will use and the Technical and Business Assistance you will receive. For the Phase I project, the amount for TABA may not exceed $6,500 per award. For the Phase II project, the TABA amount may be less than, equal to, but not more than $50,000 per project. The description should be included in Volume 5 of the proposal.

Approval of Discretionary Technical and Business Assistance is not guaranteed and is subject to review of the contracting officer.

For Discretionary Technical and Business Assistance, small business concerns may propose one or more vendors. Additionally, business-related services aimed at improving the commercialization success of a small business concern may be obtained from an entity, such as a public or private organization or an agency or other entity established or funded by a State that facilitates or accelerates the commercialization of technologies or assists in the creation and growth of private enterprises that are commercializing technology.

EVALUATION AND SELECTION

All proposals will be evaluated in accordance with the evaluation criteria listed in the DoD SBIR Program BAA.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the BAA. DTRA has a single Evaluation Authority (EA) for all proposals received under this solicitation. The EA either selects or rejects Phase I and Phase II proposals based upon the results of the review and evaluation process plus other considerations including limitation of funds, and investment balance across all the DTRA topics in the solicitation. To provide this balance, a lower rated proposal in one topic could be selected over a higher rated proposal in a different topic. DTRA reserves the right to select all, some, or none of the proposals in a particular topic.

Notifications.

Following the EA decision, the DTRA SBIR/STTR office will release notification e-mails of selection or non-selection status for a Phase I award within 90 days of the closing date of the BAA. The E-mails will be sent to the addresses provided for the Principal Investigator and Corporate Official.

Offerors may request a debriefing of the evaluation of their not selected proposal and should submit this request via email to: dtra.belvoir.RD.mbx.sbir@mail.mil and include **"SBIR 25.2 / Topic No. XXX Debriefing Request" in the subject line**. Debriefings are provided to help improve the offeror's potential response to future solicitations. Debriefings do not represent an opportunity to revise or rebut the EA decision.

For selected offers, DTRA will initiate contracting actions which, if successfully completed, will result in contract award. DTRA Phase I awards are issued as fixed-price purchase orders with a maximum period of performance of seven-months. DTRA may complete Phase I awards without additional negotiations by the contracting officer or without opportunity for revision for proposals that are reasonable and complete.

DTRA Support Contractors

Select DTRA-employed support contractors may have access to contractor information, technical data or computer software that may be marked as proprietary or otherwise marked with restrictive legends.

Each DTRA support contractor performs under a contract that contains organizational conflict of interest provisions and/or includes contractual requirements for nondisclosure of proprietary contractor information or data/software marked with restrictive legends. These contractors require access while providing DTRA such support as advisory and assistance services, contract specialist support, and support of the Defense Threat Reduction Information Analysis Center (DTRIAC). The contractor, by submitting a proposal or entering into this contract, is deemed to have consented to the disclosure of its information to DTRA's support contractors.

The following are, at present, the prime contractors anticipated to access such documentation: ASRC Federal (contract specialist support); Kent, Campa and Kate, Inc. (contract closeout support), ARServices (Program Management Advisory and Assistance Services A&AS), Systems Planning and Analysis, Inc. (Subject Matter Expertise A&AS), Amentum (A&AS), Polaris Consulting (Small Business Program Support), Seventh Sense Consulting, LLC (Acquisition Support), Savantage Solutions (Accounting and Financial Systems Support); TekSynap Corporation and Kapili Services, LLC (DTRIAC). This list is not all inclusive (e.g., subcontractors) and is subject to change.

Protests.
Refer to the DoD SBIR Program BAA for procedures to protest the Announcement.

As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: (a) Protests, as defined in section 33.101 of the Federal Acquisition Regulation, that are filed directly with an agency, and copies of any protests that are filed with the Government Accountability Office (GAO), shall be served on the Contracting Officer (addressed to Mr. Herbert Thompson, Contracting Officer, as follows) by obtaining written and dated acknowledgment of receipt from (if mailed letter) Defense Threat Reduction Agency, ATTN: AL-ACQ (Mr. Herbert Thompson), 1991 Wyoming Boulevard, SE, Kirtland AFB, NM 87117. If Federal Express is used for the transmittal, the appropriate address is Defense Threat Reduction Agency, ATTN: AL-ACQ (Mr. Herbert Thompson), 8151 Griffin Avenue SE, Building 20414, Kirtland AFB, NM 87117-5669.

AWARD AND CONTRACT INFORMATION
DTRA plans on Phase I projects for a seven (7) month period of performance with six months devoted to the research and the final month for the final report. The award size of the Phase I contract is no more than $200,000.00 notwithstanding a maximum of $6,500.00 for Discretionary Technical and Business Allowance (TABA). For a Phase II project, DTRA plans on a 24-month period of performance. The award size of a Phase II contract is no more than $1,300,000.00 notwithstanding a maximum of $50,000.00 for Discretionary Technical and Business Allowance (TABA) for the entire project.

ADDITIONAL INFORMATION
The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, will apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. More information is available at https://www.pmddtc.state.gov/ddtc_public.

The technology within some DTRA topics is restricted under export control regulations including the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR). ITAR controls the export and import of listed defense-related material, technical data and services that provide the United States with a critical military advantage. EAR controls military, dual-use and commercial items not listed on the United States Munitions List or any other export control lists. EAR regulates export-controlled items based on user, country, and purpose. **The offeror must ensure that their firm complies with all applicable export control regulations.**

NOTE: Export control compliance statements found in these proposal instructions are not meant to be all inclusive. They do not remove any liability from the submitter to comply with applicable ITAR or EAR export control restrictions or from informing the Government of any potential export restriction as fundamental research and development efforts proceed.

Cyber Security
Any Small Business Concern receiving an SBIR award is required to provide adequate security on all covered contractor information systems. Specific security requirements are listed in DFARS 252.204.7012, and compliance is mandatory.

Feedback
To encourage participation in, and improve the overall SBIR award process, offerors may submit feedback on the SBIR solicitation and award process to: dtra.belvoir.RD.mbx.sbir@mail.mil for consideration for future SBIR BAAs.

DTRA SBIR 25.2 Topic Index

DTRA252-001	Plume Localization Using Machine-learning for Environmental Sensing (PLUMES)
DTRA252-002	Predicting Force Protection Threats against DTRA from Foreign Adversarial Messaging
DTRA252-003	Displacement Damage Modeling for Satellite Component Design, Evaluation, and Validation

DTRA252-001 TITLE: Plume Localization Using Machine-learning for Environmental Sensing (PLUMES)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: DTRA seeks an innovative AI-driven solution for detecting, tracking, and guiding an unmanned aerial system (UAS) through airborne plumes generated during explosive tests. The system should integrate data from onboard systems and ground systems to include LIDAR and radar to autonomously identify plume characteristics, maintain situational awareness in changing environmental conditions, and provide precise steering capabilities to optimize data collection. The software should be platform and sensor suite agnostic. The innovation seeks to improve data collection during testing with potential expansion to operational use cases requiring plume tracking. Recent advances in AI/ML and computer vision have significantly improved autonomous tracking and classification capabilities, making this a feasible solution for real-time operations in hazardous environments [1,2].

DESCRIPTION: Current methods for tracking plumes in explosive test environments rely on manual UAS operation, which is inefficient and susceptible to human error. Explosive tests incorporate numerous other sensor systems such as LIDAR and radar, which are currently unused for UAS operations. Furthermore, rapidly changing weather conditions and unpredictable plume behavior make consistent data collection challenging. Advances in AI/ML and sensor fusion techniques allow for real-time plume detection and classification, leveraging computer vision algorithms trained on EO/IR, LIDAR, and hyperspectral imaging data [3,4].

The proposed system should:

- Provide a platform and sensor agnostic algorithm to fuse multiple data inputs quickly to inform UAS operations.

- Utilize AI/ML to distinguish relevant plumes from background interference (e.g., dust, clouds, smoke) [5].

Enable autonomous UAS flight control for real-time plume tracking and sampling.

- Provide robust data security and transmission capabilities for real-time operational use.
- Be adaptable to various UAS platforms and modular for integration into existing explosive test frameworks.

PHASE I: PHASE I – Proof of Concept

Create a proposed technical roadmap for integrating AI/ML-driven plume tracking capabilities with UAS systems. Additionally, the proposal should have a computer vision model for real-time plume identification with the ability to discern similar items, such as clouds or ground terrain. The framework will explain integration of other sensor data (LIDAR, radar) to inform UAS operations. A defined security data framework is needed to protect collected information, which may be used later for model development. Initial developments can be based on simulation testbed activities to validate AI-driven tracking algorithms. Deliverables will include a proof-of-concept report, feasibility study, preliminary AI/ML model, and system architecture plan. A plan outlining the approach for scaling the system to meet Phase II requirements should also be submitted.

PHASE II: Phase II - Prototype Development:

Demonstrate a scaled-down working prototype that integrates AI/ML algorithms, on board and ground sensors, and real-time UAS control. This should be done through live-scale testing in a controlled environment, thus demonstrating tracking capabilities in various plume conditions. Metrics related to AI performance and optimization are required with proposed timelines and approaches to improvement. Considerations for different environments and weather conditions will be presented. Last, data collection and transmission from UAS to the ground control systems, as well as the data management for model training, will be demonstrated. Deliverables will include a fully functioning prototype (using UAS of performer choice), live demonstration results, detailed AI model evaluation, and sensor integration report. A design plan should also be submitted outlining the plans for scaling the system to meet Phase III requirements.

PHASE III DUAL USE APPLICATIONS: Demonstrate system capabilities in a full-scale test event at either a DTRA explosive test site or in a suitably similar test site and environment. Showcase finalized system integration for seamless operation within existing test range infrastructures and UAS equipment. Develop commercialization pathways, including applications in hazardous material detection, industrial emissions monitoring and emergency response. Deliverables: fully operational system, final validation report, commercialization and transition plan.

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KEYWORDS: predictive analytics, information environment, force protection

DTRA252-002 TITLE: Predicting Force Protection Threats against DTRA from Foreign Adversarial Messaging

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

OBJECTIVE: Develop predictive models to determine when online messaging is going to breakout into a different part of the online, foreign information environment or into the physical world as activities directed against DTRA personnel or facilities.

DESCRIPTION: Foreign adversaries have employed online messaging to target and undermine DTRA's programs, personnel, or overall reputation. A capability is needed to predict when such messaging may "break out" of its originating source(s) in the foreign information environment (news sources, social media, and websites) and move to other parts of the foreign information environment, where it may lead to the threatening or incitement of physical action against DTRA personnel or facilities. The output of predictive models can help DTRA posture from force protection and public affairs perspective.

PHASE I: PHASE I – Proof of Concept

Conduct and document a review of prior work to predict physical threats from online messaging (e.g., prior work in counterterrorism, brand management, etc. [1]). Develop proof-of-concept model(s) that use publicly available information (PAI) for a country (or region) in which DTRA operates to predict transitions between parts of the foreign information environment and which of those transitions may lead to threatening physical activity. Note the focus here is not on predicting which individuals or groups that may engage in these behaviors but rather gauging the overall threat level to DTRA based on the content and spread of the messaging. Validate models using historical case studies and subject matter experts. Model will be scoped to specific foreign populations, specific parts or topics of the information environment, and/or a small set of potential threats.

PHASE II: Phase II - Prototype Development:

Objective: Refine model(s) to include explanatory capabilities and demonstrate generality of the models to other foreign populations, specific parts or topics of the information environment, and potential threats. Assess model accuracy against ground truth data and improve model transparency, explainability, and portability to different applications.

PHASE III DUAL USE APPLICATIONS: Apply prototype models to a broader variety of online information environments and media types to predict a broader range of threats. Dual use applications for industrial security are envisioned.

REFERENCES:

1. Renshaw, S. L., & Carley, K. M. (2024). Linking online activity to offline behavior: A meta-review of three decades of online-to-offline scholarship with future implications for AI. Emerging Trends in Drugs, Addictions, and Health, 4, 100154. https://doi.org/10.1016/j.etdah.2024.100154 Access, 9, 87663-87685.

KEYWORDS: predictive analytics, information environment, force protection

DTRA252-003 TITLE: Displacement Damage Modeling for Satellite Component Design, Evaluation, and Validation

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Microelectronics; Space Technology

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: DTRA seeks specialized time-dependent computing code which can model displacement damage effects in critical semiconductor components of satellites.

DESCRIPTION: Because satellites have many semiconductor components that are sensitive to displacement damage, they experience performance and functionality degradation following particle radiation exposure in both natural and nuclear environments. While proton, electron, and heavy ion irradiation all contribute to displacement damage, neutron irradiation is of particular interest owing to neutrons' exceptional ability to penetrate deeply. In exoatmospheric conditions, neutrons travel much longer distances before experiencing a collision than in terrestrial environments, further increasing the risk of long-term damage to a system.

To better understand how a satellite will perform in nuclear environments and better predict how radiation damage will impact longevity, DTRA seeks to develop a specialized time-dependent computing code which can model displacement damage in critical semiconductor components of satellites with a final objective of predicting changes to device and circuit performance following radiation exposure. The ideal code will allow the user to define the radiation source, materials, and the environment of interest. Radiation sources will minimally include fast neutrons with considerations for energy, uniformity (isotropic versus anisotropic sources), and radiation strength measurements (fluence, dose, and/or dose rate), though protons, heavy ions, and α-particles may be included. Additionally, β-particles may be included, though utility will be limited to scenarios relating to hardening only. Materials of interest are those associated with critical components such as silicon bipolar technologies, bipolar transistors and integrated circuits, diodes, silicon metal-oxide-semiconductor (MOS) technologies, silicon visible imaging arrays, silicon particle detectors, light emitting diodes (LED), laser diodes, solar cells, and infrared detectors. In addition to silicon, these materials might specifically include germanium, silicon germanium, silicon carbide, gallium nitride, and gallium arsenide. Materials considerations should also include dimensionality, doping, and polycrystalline substances. Environments of interest include those that are static and time-varying and account for contributions to changes to conductivity in a material (i.e., temperature). User-defined environments should also account for plural surrounding materials (i.e., voids, electronics housing, etc.) and associated material interfaces to account for their contributions to displacement damage. Computational processes must account for radiation transport, energy deposition, and displacement damage modeling. Associated outputs should include but are not limited to number of primary knock-on atoms (PKA) per unit volume (inclusion of multiple models for determining displacements per PKA with associated corrections is anticipated), atomic displacement rate density, atomic displacement rate, ray tracing visualization, and relevant statistics to describe model performance and/or fidelity. Displacement damage modeling outputs should include numeric descriptions and visualizations for particles involved in collisions and associated damage cascades. Such outputs could describe the number of collisions a particle experiences, energy transfer associated with such collisions, concentration and identity of displaced particles in a particular sector, and time- and

temperature-dependent changes associated with annealing. Visualizations should graphically depict where particles could be expected to come to rest following a collision or involvement in a damage cascade or following annealing.

PHASE I: Conduct a proof of concept to identify models, processes, and algorithms most suitable for computationally modeling radiation transport, including calculating energy deposition and visualizing ray tracing. The proof of concept should also identify processes and algorithms most suitable modeling displacement damage and associated numeric descriptions and visualizations, including outputs for the number of PKA per unit volume, atomic displacement rate, atomic displacement rate density, other visual representations, and statistics that describe model performance and/or fidelity. The phase I deliverable is a report and a preliminary proof of concept demonstration that details advantages, disadvantages, and limitations of proposed methods to include confidence metrics..

PHASE II**:** Prototype Development – Develop mature models, processes, and algorithms to improve model performance and fidelity. Phase II deliverables include at least one demonstration, the developed code, associated documentation, and a report that details pertinent technical and computational details and the development approach. Some level of certification and/or validation and verification is required.

PHASE III DUAL USE APPLICATIONS: Finalize and commercialize software for use by customers (e.g., Government, satellite designers, etc.). Although additional funding may be provided through DoD sources, the awardees should look to other public or private sector funding sources for assistance with transition and commercialization. Dual use applications may include broadening the scope beyond exoatmospheric environments and semi-conductor materials. Additionally, any particle radiation not already included may be included for such applications.

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KEYWORDS: nuclear; radiation; survivability; radiation transport; energy deposition; displacement damage; modeling and simulation; satellite; semiconductor; strategic system

Office of the Secretary of Defense (OSD)
National Geospatial-Intelligence Agency (NGA)
25.2 Small Business Innovation Research (SBIR)
Proposal Submission Instructions

INTRODUCTION

NGA is a Department of Defense (DoD) combat support agency and a member of the U.S. Intelligence Community (IC). NGA develops imagery and map-based intelligence solutions for U.S. national defense, homeland security, and safety of navigation. NGA's mission is to "provide timely, relevant, and accurate geospatial-intelligence in support of national security." Today, NGA manages the National System for Geospatial-Intelligence (NSG), which provides the foundation for correlating U.S. intelligence activities to the location of the Earth.

Geospatial intelligence, or GEOINT, is the exploitation and analysis of imagery and geospatial information to describe, assess and visually depict physical features and geographically referenced activities on the Earth. GEOINT consists of imagery, imagery intelligence and geospatial information. Additional information pertaining to the NGA mission and high-level course can be obtained by viewing the agency's website at https://www.nga.mil and NGA's strategy documents at https://www.nga.mil/about/strategy.html.

NGA Research supports the NSG and National Security Strategy by solving hard defense and intelligence problems for the IC and DoD in three broad areas: Foundational GEOINT; Advanced Phenomenologies; and Analytic Technologies. NGA Research works with customers on early concepts through to advanced developments in operating systems and environments.

Proposers responding to a topic in this BAA must follow all general instructions provided in the Department of Defense (DoD) SBIR Program BAA. NGA requirements in addition to or deviating from the DoD Program BAA are provided in the instructions below.

Proposers are encouraged to thoroughly review the DoD Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The full DoD Program BAA is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the administration of the NGA SBIR Program and these proposal preparation instructions should be directed to: sbir@nga.mil.

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoD Program BAA.

Proposal Coversheet (Volume 1)

The proposal coversheet is prepared on DSIP and in accordance with the DoD Program BAA.

Technical Volume (Volume 2)
The technical volume is not to exceed 20 pages and must follow the formatting requirements provided in the DoD SBIR Program BAA. Technical volumes exceeding 20 pages will be deemed non-compliant and will not be evaluated.

Content of the Technical Volume
The content of the Technical Volume follows the Technical Proposal Template provided in the DoD Program BAA. In the facilities and equipment section (section 9) please identify how Controlled Unclassified Information will be protected in accordance with NIST SP 800-171 and provide your Supplier Performance Risk System (SPRS) score.

Cost Volume (Volume 3)
The Phase I Base amount must not exceed $100,000 up to 6 months period of performance. Costs for the Base must be clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. NGA will not accept any deviation to the POW requirements.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD Program BAA for full details on this requirement. Information contained in the CCR will not be considered by NGA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoD Program BAA for more information.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoD Program BAA.

DIRECT TO PHASE II PROPOSAL GUIDELINES

Proposal Coversheet (Volume 1)
The proposal coversheet is prepared on DSIP and in accordance with the DoD Program BAA.

Technical Volume (Volume 2)
The technical volume consists of two parts: Part A – Feasibility Documentation, and Part B – Technical Proposal. Each part is not to exceed twenty (20) pages, for a technical volume maximum page count of 40 pages. The Government will not consider pages in excess of these

page count limitations. Number all pages of your proposal consecutively and follow the formatting requirements provided in the DoD SBIR Program BAA.

Content of the Technical Volume
The offeror shall not propose option period(s). The content of the Technical Volume follows the Technical Proposal Template provided in the DoD Program BAA. Additional items that need to be addressed are listed below:

(6) Commercialization Strategy. In addition, the Commercialization Strategy shall also address Section 508 compliance as noted below:

Section 508 Compliance

The contractor shall ensure that all systems, hardware, software, software engineering, and information technology associated with this effort is made in a manner that is accessible for people with the standards for people with disabilities as directed in the NGA Instruction 8400.4 and Section 508 of the Rehabilitation Act of 1973 as amended in 1998 (Section 508). Specifically, all Information and Communications Technology (ICT) associated with this contract, may use the Web Content Accessibility Guidelines (WCAG) 2.1 to comply with the Section 508 or use alternative designs or technologies which result in substantially equivalent or greater access to and use of the product for people with disabilities. Furthermore, the contractor shall pursue human centered design and usability guidelines to ensure that all services associated with this Topic Area are accessible by as many users as possible and to drive modernization, innovation, and enhance mission support.

As part of the offeror's proposal, the offeror should include an outline of specifically how Section 508 compliance will be achieved in the design of the ICT product. The proposal for Phase II should provide an explicit, detailed description of the approach, indicate what is planned, how and where the work will be carried out, a schedule of major events, how the solution will be Section 508 Compliant, and the final product to be delivered. The methods planned to achieve each objective or task should be discussed explicitly and in detail. If a determination is made that a Section 508 exception request is justified, the rationale for the exception request must be made and submitted as a part of the proposal.

(7) Key Personnel. In addition, please list the clearance level held by each of the personnel, as we will be operating in a classified environment.

(9) Facilities and Equipment. In addition, the Facilities and Equipment shall address the following as noted below:

Please identify how Controlled Unclassified Information will be protected in accordance with NIST SP 800-171 and provide your Supplier Performance Risk System (SPRS) score.

Since this work will require working in a classified environment, please identify what your Facility Clearance Level is, Safeguarding Level for each facility, CAGE code for each facility and the associated physical addresses.

Cost Volume (Volume 3)
The Phase II amount must not exceed $1,000,000 for up to a twenty-four (24)-month period of performance. Costs must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. NGA will not accept any deviation to the POW requirements.

Company Commercialization Report (CCR) (Volume 4)
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PHASE II PROPOSAL GUIDELINES

Phase II proposals may only be submitted by Phase I awardees.

Proposal Coversheet (Volume 1)
The proposal coversheet is prepared on DSIP and in accordance with the DoD Program BAA.

Technical Volume (Volume 2)
The technical volume is not to exceed 40 pages and must follow the formatting requirements provided in the DoD SBIR Program BAA. Technical volumes exceeding 40 pages will be deemed non-compliant and will not be evaluated.

Content of the Technical Volume
The offeror shall not propose option period(s). The content of the Technical Volume follows the Technical Proposal Template provided in the DoD Program BAA. Additional items that need to be addressed are listed below:

> **(6) Commercialization Strategy.** In addition, the Commercialization Strategy shall also address Section 508 compliance as noted below:

Section 508 Compliance

The contractor shall ensure that all systems, hardware, software, software engineering, and information technology associated with this effort is made in a manner that is accessible for people with the standards for people with disabilities as directed in the NGA Instruction 8400.4 and Section 508 of the Rehabilitation Act of 1973 as amended in 1998 (Section 508). Specifically, all Information and Communications Technology (ICT) associated with this contract, may use the Web Content Accessibility Guidelines (WCAG) 2.1 to comply with the Section 508 or use alternative designs or technologies which result in substantially equivalent or greater access to and use of the product for people with disabilities. Furthermore, the contractor shall pursue human centered design and usability guidelines to ensure that all services associated with this Topic Area are accessible by as many users as possible and to drive modernization, innovation, and enhance mission support.

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(7) Key Personnel. In addition, please list the clearance level held by each of the personnel, as we will be operating in a classified environment.

(9) Facilities and Equipment. In addition, the Facilities and Equipment shall address the following as noted below:

Please identify how Controlled Unclassified Information will be protected in accordance with NIST SP 800-171 and provide your Supplier Performance Risk System (SPRS) score.

Since this work will require working in a classified environment, please identify what your Facility Clearance Level is, Safeguarding Level for each facility, CAGE code for each facility and the associated physical addresses.

Cost Volume (Volume 3)
The Phase II amount must not exceed $1,000,000 for up to a twenty-four (24)-month period of performance. Costs must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. NGA will not accept any deviation to the POW requirements.

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DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
NGA will not be providing TABA assistance.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoD Program BAA. In addition to the DoD Program evaluation criteria, we will evaluate a vendors ability to be able to perform controlled work (CUI/Classified) as part of their technical evaluation score.

NGA may use SETA support to assist with the programmatics of executing the evaluation process but will not use SETA support to evaluate proposals.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the BAA.

Refer to the DoD Program BAA for procedures to protest the Announcement.
As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: Joylene Burton, Joylene.U.Burton@nga.mil.

AWARD AND CONTRACT INFORMATION
Unless otherwise stated in the individual topic announcement:

- Phase I awards are capped at $100,000 each over a maximum six (6)-month period of performance.
- Phase II awards are capped at $1,000,000 each over a maximum 24-month period of performance.

NGA caps sequential Phase II contracts (those proposed near the completion of the initial Phase II or Direct to Phase II contract) at the then current Small Business Administration (SBA) “without seeking

SBA approval" ceiling over a maximum 24-month period of performance. See https://www.sbir.gov/about for the most recent information.

NGA typically provides a firm fixed price contract for its awards within 180 days of the proposal due date. The type of contract is at the discretion of the Contracting Officer.

ADDITIONAL INFORMATION

Security

Classified Work Performance Security Requirements

Contractors must abide by the DD Form 254 - Contract Security Classification Specification (referred to as DD254 hereafter), the National Industrial Security Program Operating Manual (NISPOM 32 CFR Part 117), and all other applicable security policies and regulations including operational security (OPSEC) requirements.

Some personnel may need to obtain access to Special Access Program (SAP), and/or Control Access Program (CAP) and/or Alternate Compensatory Control Measures (ACCM) Information as required for the execution of NGA's sensitive program missions. Access will not be granted without written permission from the COR.

Defense Counterintelligence and Security Agency (DCSA) remains the Cognizant Security Agency (CSA) for individuals performing work with collateral (SECRET or TOP SECRET) security clearances. As such, Security Executive Agent Directive (SEAD) 3 reporting should continue to be provided to the Contractor's Facility Security Officer (FSO) for reporting to DCSA via Defense Information Security System (DISS). DCSA will also dictate mandatory security training as the CSA for the collateral cleared community.

Personnel Security Requirements

Contractor personnel performing Top Secret/Sensitive Compartmented Information (TS/SCI) work on the contract are required to have active TS/SCI clearances for access to TS/SCI facilities, when performing duties within TS/SCI environments, and for access to TS/SCI computer systems for the performance of this contract. Contractors are subject to a Counterintelligence-Scope Polygraph examination, as requested by NGA. As a condition of employment and assignment, contractors who have not successfully completed polygraph testing within the last five years must immediately schedule a polygraph examination and must complete the process, in no more than three test sessions, within 90 days.

All Contractor personnel shall possess a current Top Secret Personnel Security Clearance and be eligible for favorable NGA adjudication for SCI access. The government will be responsible for verifying security clearances/SCI eligibility of the Contractor personnel in accordance with (IAW) SEAD 7: "Reciprocity of Background Investigations and National Security Adjudications". If needed, NGA will sponsor SCI accesses at its sole discretion. NGA will provide badges, Common Access Cards (CAC) and other items (e.g. parking hangtags) when applicable, for required Contractor personnel.

Cleared Contractor personnel will be enrolled into Director of National Intelligence (DNI) Continuous Evaluation System (CES) throughout the lifecycle of the contract while at NGA IAW SEAD 6: "Continuous Evaluation". Cleared Contractor personnel must self-report security

issues, foreign contacts, and other applicable information IAW SEAD 3: "Reporting Requirements for Personnel with Access to Classified Information or Who Hold a Sensitive Position" and NGA Instruction 5205.3, "Security Reporting Requirements". In addition, cleared Contractors will be required to submit electronic fingerprints and be enrolled into NGA's Report of Arrest and Prosecution Background (RAPBack) program, which supports CES.

Cleared Contractor personnel, who are designated as Tier 3 Privileged Users or holding enhanced access through a Special/Controlled Access Program (SAP/CAP), will be required to participate in NGA's annual Security Financial Disclosure Program (SFDP) as directed by D/NGA in the 20 November 2020 memorandum titled "Designation of NGA Personnel Required to Participate in the Security Financial Disclosure Program".

The Contractor shall inform NGA as soon as practicable when its employees will no longer support the contract (see DD254). NGA desires notification prior to the day the individual no longer supports the contract, but requires notification no later than the day that contract support ends. If Contractor personnel will no longer be supporting NGA via an NGA contract, NGA shall take all appropriate action, up to and including debriefing the contractor from SCI access and terminating logical and physical access to NGA IT systems and facilities.

Upon notifying NGA that a Contractor will no longer support the contract, the Contractor must return all NGA Badges, CAC, hangtag, and any other government furnished property. These items can be returned to the NGA Workforce Support Center, NGA Site Security Office, or the Contracting Officer Representative (COR) as soon as practicable, but no later than four (4) business days from the date of Contractor departure.

When Contractor personnel are unable to turn these items into the NGA Workforce Support Center, NGA Site Security Office, or COR then it is the responsibility of the Contractor's FSO to collect the items from the individual and return them to NGA as soon as practicable, but no later than four (4) business days from the date of Contractor departure.

If the FSO debriefs their employee, the FSO shall provide a copy of the debriefing statement, plus all applicable Government-furnished items (e.g. NGA Badge, CAC, Courier Card, parking hangtags, etc.) to the NGA Site Security Office or the NGA Workforce Support Center as soon as practicable, but no later than four (4) business days from the date of Contractor departure.

Physical Security Requirements

All classified work performed at a non-NGA facility must be approved by the COR. Any classified work performed at contractor sites must be performed in either an NGA accredited SCIF, Other Government Agency (OGA) SCIF, or an approved secure collateral space that has either a Memorandum of Agreement (MOA), Memorandum of Understanding (MOU), Joint Use Agreement or Co-Use Agreement with NGA for this contract. SCIF or secure facility accreditation shall be at least the level commensurate with the level of safeguarding required by the contract DD254.

Pursuant to "IC Technical Specifications for Construction and Management of Sensitive Compartmented Information Facilities Version 1.5.1; IC Tech Specs for ICD/ICS 705", the NGA accrediting official, or designee, shall conduct periodic security inspections/reviews to ensure the efficiency of NGA accredited SCIF operations, identify deficiencies, and document corrective actions taken. Inspections/reviews of NGA accredited SCIFs shall be conducted based on threat,

facility modifications, sensitivity of programs, past security performance, or at least every five years. Technical Surveillance Countermeasures (TSCM) activities in NGA accredited SCIFs will only be conducted by NGA TSCM teams consisting of NGA military, civilian, or contractor personnel who have successfully completed approved TSCM training. Authorized inspectors shall be admitted to a SCIF without delay or hindrance when inspection personnel are properly certified to have the appropriate level of security clearance and SCI indoctrination for the security level of the SCIF.

Contractor personnel are forbidden from bringing prohibited or unauthorized items into any NGA installation or other secure facility covered under this contract. These items include weapons, cell phones, cameras, two-way pagers, laptops, recording devices of any kind, flash drives, or any other removable media. Exceptions may be granted by NGA Security upon the Contractor's request. If granted, Contractor personnel must bring documentation showing approval prior to entering with the item(s). Violations may subject the Contractor and its offending personnel to civil and/or criminal liability IAW applicable laws and regulations governing access to secure Government facilities.

Unclassified Work Performance Security Requirements

If applicable, and after discussion with the COR, and the COR has provided written approval, uncleared Contractor personnel are authorized to work on this contract at the Unclassified level, with access up to DoD Controlled Unclassified Information (CUI) at the Contractor site without the requirement of a security clearance.

Uncleared Personnel Security Requirements

If applicable, and after discussion with the COR, and the COR has provided written approval, any Contractor personnel working with CUI information must receive a favorable HSPD-12 and/or HSPD-12 Tier 1 adjudication prior to accessing CUI information. Contractor personnel who require access to CUI for 60 days or less must receive a favorable HSPD-12 adjudication. Contractor personnel who require CUI access for more than 60 days must receive a favorable HSPD-12 Tier 1 adjudication. NGA will sponsor the HSPD-12 and HSPD-12 Tier 1 background investigation for Contractor personnel as needed.

Foreign nationals are not permitted to perform unclassified work under the terms of this contract without prior written approval from the Contracting Officer (CO) or the COR. Should NGA identify the use of unauthorized personnel, the CO may direct the Contractor, at its own expense, to remove and replace any unauthorized Contractors, Subcontractors, or other personnel performing on the contract. Such action may be taken at the NGA's discretion without prejudice to its rights under any other contract provision, e.g., termination for default.

Uncleared Contractor personnel visiting NGA facilities or other sites may receive the appropriate visitor badge and be escorted, as appropriate. The Contractor will return the visitor badge at the end of each visit day.

Information Security Requirements

NGA shall have the sole authority to determine whether, and to what extent, protected information shall be provided to the Contractor under the terms of the contract. Access to classified information will be pursuant to the security requirements established in the DD254. The

Contractor shall not access, download, print, or further disseminate any classified information outside the execution of the defined contract requirements without the guidance and written permission from the COR.

When the Contractor receives protected information under the terms of this contract, the Contractor will comply with all applicable NGA, DoD, and IC information security policies (including the Consolidated NGA (CoNGA) Security Classification Guide) for the proper marking, handling, processing, storing, and safeguarding of classified and unclassified material. The Contractor will ensure that document markings are given the lowest possible security classification to maximize dissemination while still maintaining the information's confidentiality and integrity as necessary.

Access to CUI will be pursuant to DFARS clause 252.204.7012. The prime contractor shall include this clause, including paragraph (m), in subcontracts, or similar contractual instruments, for operationally critical support or for which subcontract performance will involve covered defense information, including subcontracts for commercial items, without alteration, except to identify the parties. The Contractor shall determine if the information required for Subcontractor performance retains its identity as covered defense information and will require protection under this clause. Contractor personnel shall not release any unclassified information, regardless of medium (e.g., film, tape, document), pertaining to any part of this contract or any program related to this contract, unless the COR has given prior written approval or in performance of a project that has been scoped and negotiated by NGA.

NGA will assess and authorize Contractors to process classified information on government-owned information systems, conduct oversight of such information system processing, and provide information system security guidelines IAW with FAR clause 52.204-21 and other information system security control policies, standards, and procedures. Contractor will be responsible for executing specific provisions under the contract for the accreditation of classified information systems, and upon request, shall provide the COR a written statement confirming accreditation compliance with the minimum information systems requirements established in the DD254. The Contractor shall include the substance of FAR clause 52.204-21 in subcontracts under this contract (including subcontracts for the acquisition of commercial products or commercial services, other than commercially available off-the-shelf items), in which the Subcontractor may have federal contract information residing in or transiting through its information system.

In accordance with FAR Part 4.7 and FAR clause 52.215-2, the Contractor may retain one archival copy of any Government-furnished information provided to the Contractor during the course of performance of the contract, subject to Contractor following relevant data safeguarding practices and CUI restrictions. The disposition of such Government-furnished information is otherwise solely at the discretion of the Government. As directed by the NGA CO or COR, the Contractor will destroy or return to NGA all Government-furnished information upon contract termination or when no longer required for contract performance.

In situations when electronic transmission is not practicable, Cleared Contractor personnel may hand-carry contract-related classified information when authorized by the COR in writing. Approved Contractor personnel must obtain NGA courier authorization prior to hand-carrying of classified contract-related information. Only classified information related to the performance of this contract may be hand-carried. Contractor personnel will be limited to hand-carrying classified information between approved Contractor facilities and US Government facilities

unless the COR approves other circumstances in writing, when applicable, on a case-by-case basis.

Cybersecurity Requirements

The Contractor must, at a minimum, implement 10 U.S.C. Sections 391 and 393 along with National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171, "Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations" requirements.

If the Contractor intends to use an external cloud service provider to store, process, or transmit any covered defense information in performance of this contract, the Contractor shall require and ensure that the cloud service provider meets security requirements equivalent to those established by the Government for the Federal Risk and Authorization Management Program (FedRAMP) Moderate baseline (https://www.fedramp.gov/resources/documents/)

In the event that the Contractor discovers a cyber incident that affects a covered Contractor information system or covered defense information residing therein, or that affects the Contractor's ability to perform the requirements of the contract that are designated as operationally critical support and identified in the contract, the Contractor shall—

1. Notify the DoD Cyber Crime Center (DC3) and the COR in writing within 72 hours of incident discovery

2. Conduct a review for evidence of compromise of covered defense information, including, but not limited to, identifying compromised computers, servers, specific data, and user accounts. This review shall also include analyzing covered Contractor information system(s) that were part of the cyber incident, as well as other information systems on the Contractor's network(s), that may have been accessed as a result of the incident in order to identify compromised covered defense information, or that affect the Contractor's ability to provide operationally critical support.

In the event the Contractor discovers and isolates malicious software in connection with a reported cyber incident, the Contractor shall submit the malicious software to DC3 in accordance with instructions provided by the COR. The Contractor should NOT send the malicious software to the COR.

Insider Threat Requirements

Contractor will establish and maintain an insider threat program to gather, integrate, and report relevant and available information indicative of a potential or actual insider threat, consistent with E.O. 13587 "Structural Reforms to Improve the Security of Classified Networks and the Responsible Sharing and Safeguarding of Classified Information" and Presidential Memorandum "National Insider Threat Policy and Minimum Standards for Executive Branch Insider Threat Programs." As soon as practicable, Contractor shall report to the COR events that may have an effect on the status of the entity's or an employee's eligibility for access to classified information; report events that indicate an insider threat to classified information or to employees with access to classified information; report events that affect proper safeguarding of classified information; and report events that indicate classified information has been, or is suspected to be, lost or compromised.

Clauses

52.204-7 System for Award Management.

As prescribed in 4.1105(a)(1), use the following provision:

SYSTEM FOR AWARD MANAGEMENT (OCT 2018)

(a) Definitions. As used in this provision—

"Electronic Funds Transfer (EFT) indicator means a four-character suffix to the unique entity identifier. The suffix is assigned at the discretion of the commercial, nonprofit, or Government entity to establish additional System for Award Management records for identifying alternative EFT accounts (see subpart 32.11) for the same entity.

Registered in the System for Award Management (SAM) means that–

(1) The Offeror has entered all mandatory information, including the unique entity identifier and the EFT indicator, if applicable, the Commercial and Government Entity (CAGE) code, as well as data required by the Federal Funding Accountability and Transparency Act of 2006 (see subpart 4.14) into SAM

(2) The offeror has completed the Core, Assertions, and Representations and Certifications, and Points of Contact sections of the registration in SAM;

(3) The Government has validated all mandatory data fields, to include validation of the Taxpayer Identification Number (TIN) with the Internal Revenue Service (IRS). The offeror will be required to provide consent for TIN validation to the Government as a part of the SAM registration process; and

(4) The Government has marked the record "Active".

Unique entity identifier means a number or other identifier used to identify a specific commercial, nonprofit, or Government entity. See www.sam.gov for the designated entity for establishing unique entity identifiers.

(b)

(1) An Offeror is required to be registered in SAM when submitting an offer or quotation, and shall continue to be registered until time of award, during performance, and through final payment of any contract, basic agreement, basic ordering agreement, or blanket purchasing agreement resulting from this solicitation.

(2) The Offeror shall enter, in the block with its name and address on the cover page of its offer, the annotation "Unique Entity Identifier" followed by the unique entity identifier that identifies the Offeror's name and address exactly as stated in the offer. The Offeror also shall enter its EFT indicator, if applicable. The unique entity identifier will be used by the Contracting Officer to verify that the Offeror is registered in the SAM.

(c) If the Offeror does not have a unique entity identifier, it should contact the entity designated at www.sam.gov for establishment of the unique entity identifier directly to obtain one. The Offeror should be prepared to provide the following information:

(1) Company legal business name.

(2) Tradestyle, doing business, or other name by which your entity is commonly recognized.

(3) Company physical street address, city, state, and Zip Code.

(4) Company mailing address, city, state and Zip Code (if separate from physical).

(5) Company telephone number.

(6) Date the company was started.

(7) Number of employees at your location.

(8) Chief executive officer/key manager.

(9) Line of business (industry).

(10) Company headquarters name and address (reporting relationship within your entity).

(d) Processing time should be taken into consideration when registering. Offerors who are not registered in SAM should consider applying for registration immediately upon receipt of this solicitation. See https://www.sam.gov for information on registration.

(End of Provision)

52.204-27 Prohibition on a ByteDance Covered Application.

As prescribed in 4.2203, insert the following clause:

PROHIBITION ON A BYTEDANCE COVERED APPLICATION (JUN 2023)

(a) *Definitions.* As used in this clause—

Covered application means the social networking service TikTok or any successor application or service developed or provided by ByteDance Limited or an entity owned by ByteDance Limited.

Information technology, as defined in 40 U.S.C. 11101(6)—

(1) Means any equipment or interconnected system or subsystem of equipment, used in the automatic acquisition, storage, analysis, evaluation, manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information by the executive agency,

if the equipment is used by the executive agency directly or is used by a contractor under a contract with the executive agency that requires the use—

(i) Of that equipment; or

(ii) Of that equipment to a significant extent in the performance of a service or the furnishing of a product;

(2) Includes computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the central processing unit of a computer, software, firmware and similar procedures, services (including support services), and related resources; but

(3) Does not include any equipment acquired by a Federal contractor incidental to a Federal contract.

(b) *Prohibition*. Section 102 of Division R of the Consolidated Appropriations Act, 2023 (Pub. L. 117-328), the No TikTok on Government Devices Act, and its implementing guidance under Office of Management and Budget (OMB) Memorandum M-23-13, dated February 27, 2023, "No TikTok on Government Devices" Implementation Guidance, collectively prohibit the presence or use of a covered application on executive agency information technology, including certain equipment used by Federal contractors. The Contractor is prohibited from having or using a covered application on any information technology owned or managed by the Government, or on any information technology used or provided by the Contractor under this contract, including equipment provided by the Contractor's employees; however, this prohibition does not apply if the Contracting Officer provides written notification to the Contractor that an exception has been granted in accordance with OMB Memorandum M-23-13.

(c) *Subcontracts*. The Contractor shall insert the substance of this clause, including this paragraph (c), in all subcontracts, including subcontracts for the acquisition of commercial products or commercial services.

(End of clause)

5X252.204-7000-90 PUBLIC RELEASE OF INFORMATION (MAR 2023)

(a) Except as provided in paragraph (b) of this clause, information pertaining to this contract shall not be released to the public unless authorized by the Contracting Officer in accordance with DFARS 252.204-7000, Disclosure of Information. Requests for approval to release information pertaining to this contract shall be submitted to the Contracting Officer by means of NGA Form 5230-1, National Geospatial-Intelligence Agency Request for Clearance for Public Release.

(b) The contractor may provide past performance information regarding this contract, without completing an NGA Form 5230-1 and without Contracting Officer approval, when submission of such information is to the Office of the Director of National Intelligence (ODNI), the Central Intelligence Agency (CIA), the National Reconnaissance Office (NRO), the National Security Agency (NSA), the Defense Intelligence Agency (DIA), and NGA to support source selections at those agencies. The contractor is responsible for the proper classification and handling of such information, and shall provide a copy of the information provided to the Contracting Officer.

(End of Clause)

5X52.209-9003: PROTECTION OF INFORMATION AND NONDISCLOSURE AGREEMENTS (JUN 2009)

(a) Definitions. As used in this clause only:

(1) Protected Information and Computer Software means, unless specifically excluded by paragraph below, all information and computer software, in any form or media, that in the course of performing work under this contract are disclosed to the Contractor, its subcontractors, or their employees, or to which those persons otherwise are given access to, by:

(i) NGA,

(ii) Other government agencies,

(iii) Foreign governments, or

(iv) Other contractors while directly supporting NGA, which is accompanied by written legends identifying use or disclosure restrictions or disclosed under circumstances that the Contractor knows are subject to use or disclosure restrictions established in writing by the Government.

(2) Protected Information and Computer Software does not include information that:

(i) Has been released to the general public through no action of the undersigned in breach of this agreement or through no action of any other party in breach of any other obligation of confidentiality owing to the Government or the owner of the protected information or computer software;

(ii) Has been lawfully obtained by the recipient outside the course of the performance of this contract;

(iii) Has been properly licensed or provided directly by the owner (or other authorized source) of the information or computer software to the recipient to the extent so licensed or provided;

(iv) Is owned by the recipient or was developed independently of the disclosure hereunder; or

(v) Has been disclosed to the recipient by the Government with explicit authorization to use or disclose the information for another purpose, to the extent so authorized.

(b) Use and disclosure restrictions. The Contractor shall use and disclose Protected Information and Computer Software only as necessary for the performance of the requirements of this contract. Protected Information and Computer Software may not be used or disclosed for any other purpose, including bid or proposal preparation or business marketing, without the written approval of the Contracting Officer. Furthermore, unless otherwise directed by the Contracting Officer, the Contractor shall comply with all restrictions set forth in any legends, licenses or instructions provided to the Contractor or accompanying Protected Information and Computer Software or other written directives of the Government known to the Contractor. The use and disclosure obligations imposed by this paragraph shall expire as follows:

(1) There shall be no expiration date for the following Protected Information and Computer Software:

(i) Technical data or computer software containing Limited Rights, Restricted Rights, Government Purpose Rights, Special License Rights, or Unlimited Rights legends;

(ii) information or software marked Limited Distribution (LIMDIS);

(iii) information or software marked Source Selection Information;

(iv) contract proposal information marked pursuant to FAR 52.215-1(e) limiting its use for proposal evaluation purposes only;

(v) information and computer software marked Contractor Proprietary or a similar legend;

(vi) data known by the Contractor to be protected by the Privacy Act; and

(vii) information and software marked Controlled Unclassified Information (CUI) or For Official Use Only (FOUO).

(2) For other information or software accompanied at time of disclosure by a written legend identifying use or disclosure restriction time periods, the expiration date shall be as stated in or derived from the legend.

(3) For all other Protected Information and Computer Software, the expiration date shall be 3 years from the date the information or software is first disclosed to the Contractor.

Notwithstanding the above obligations, the Contractor is not in breach of this agreement if the Contractor uses or discloses Protected Information and Computer Software in response to an order of a court or administrative body of competent jurisdiction, but only to the extent permitted by that authority and only if the Contractor gives the Contracting Officer, to the extent practical, notice of the tribunal's order before the use or disclosure is made that allows NGA a reasonable time to object to the order.

(c) Unauthorized Use or Disclosure. The Contractor shall immediately notify the Contracting Officer of any unauthorized use or disclosure known by the Contractor of Protected Information and Computer Software in violation of the obligations contained in this clause.

(d) Disposition. At the conclusion of performance of work under this contract, the Contractor shall immediately return to the Government all Protected Information and Computer Software in its possession. Furthermore, if an employee of the Contractor who has had access to Protected Information and Computer Software is terminated or reassigned and thus is no longer performing work under this contract, the Contractor shall immediately return all Protected Information and Computer Software in the employee's possession. Moreover, if a Contractor's employee is dedicated to support a specific NGA Office or Directorate or NGA program under this contract, but is subsequently reassigned to support another NGA Office or Directorate or NGA program under this contract, the Contractor shall immediately return all Protected Information or Computer Software in the employee's possession previously furnished by the prior NGA Office or Directorate or NGA program. In lieu of returning Protected Information and Computer Software, the Contracting Officer or Contracting Officer's representative may authorize the destruction of the information or the transfer of the information to another employee of the Contractor working under the contract. Finally, this clause shall not be interpreted as preventing

the Contractor from retaining records required by statutes or other clauses of this contract, such as FAR 52.215-2 Audit and Records--Negotiations.

(e) Third party beneficiaries. This clause is executed for the benefit of the Government and the owners of Protected Information and Computer Software. The Government and the owners of Protected Information and Computer Software (and their delegatees, successors and assignees) are third party beneficiaries of the obligations contained in this clause who, in addition to any other legal rights they may have, are intended to have the rights of direct action against the Contractor or any person to whom the Contractor has disclosed or released Protected Information and Computer Software, to seek damages from any breach of this clause, or to otherwise enforce this clause.

(f) Duration. The above obligations imposed by this clause shall survive the termination or completion of this contract.

(g) Classified Information. This clause is in addition to and in no manner abrogates requirements, obligations or remedies regarding the protection of classified information and does not supersede the requirements of any laws, regulations, other directives or nondisclosure agreements regarding classified information.

(h) Other Restrictions. This agreement does not abrogate any other obligations currently placed upon the Contractor or which may be imposed upon the Contractor in the future by the Government or other persons; or remedies afforded those persons regarding those obligations.

(i) Nondisclosure agreements. The Contractor shall require and ensure that each of its employees who may receive or be given access to Protected Information and Computer Software signs the nondisclosure agreement provided by attachment to this contract prior to the employee performing work under this contract covered by the nondisclosure agreement. The Contractor shall maintain copies of signed nondisclosure agreements for a period of at least three years after final payment under this contract. At the direction of the Contracting Officer, the Contractor shall make those agreements available for inspection by the Contracting Officer and will furnish the Contracting Officer copies of those agreements at no additional cost to the Government if requested by the Contracting Officer.

(j) The Contractor shall include the substance of this clause in all subcontracts under this contract in which subcontractors may be disclosed or granted access to Protected Information and Computer Software.

(End of Clause)

5X52.227-9000 UNAUTHORIZED USE OF NGA NAME, SEAL AND INITIALS (JUN 2006)

(a) As provided in 10 U.S.C. Section 425, no person may, except with the written permission of the Director, National Geospatial-Intelligence Agency, knowingly use the words “National Geospatial-Intelligence Agency”, “National Imagery and Mapping Agency” or “Defense Mapping Agency”, the initials “NGA”, “NIMA” or “DMA”, the seal of the National Geospatial-Intelligence Agency, National Imagery and Mapping Agency, or the Defense Mapping Agency, or any colorable imitation of such words, initials, or seal in connection with any merchandise, retail product, impersonation, solicitation, or commercial activity in a

manner reasonably calculated to convey the impression that such is approved, endorsed, or authorized by the Director, NGA.

(b) Whenever it appears to the U.S. Attorney General that any person is engaged or about to engage in an act or practice which constitutes or will constitute conduct prohibited by paragraph (a), the Attorney General may initiate a civil proceeding in a district court of the United States to enjoin such act or practice. Such court shall proceed as soon as practicable to hearing and determination of such action and may, at any time before such final determination, enter such restraining orders or prohibition, or take such other action as is warranted, to prevent injury to the United States, or to any person or class of persons whose protection the action is brought.

(End of Clause)

5X52.232-9000: Submission of Invoice-Federal Payment Center (FPC) (OCT 2017) – For use in contracts paid by the FPC vendor pay office.

(a) The contractor shall prepare each invoice in accordance with the Prompt Payment Act and email one copy of the invoice to the DOD/FPC Scott AFB, IL at FMFOINSP@nga.mil. The DOD/FPC at Scott AFB, IL requires an email copy, but will accept a hard copy that is mailed to Federal Payment Center, P.O. Box 25767, Scott AFB, IL 62225.

(b) At the same time of submission of the invoice to the FPC vendor pay office, the contractor shall fax or email one copy to **[Contracting Officer]**, and one copy to **[Contracting Officer Representative]**. The contractor shall ensure that the invoice submitted to the payment office is the same invoice that is submitted to the CO and the COR without alteration.

(c) Upon receipt of the invoice, the COR will complete the receiving report and submit via the RRPT database tool. A copy of the completed receiving report shall also be provided to the Contracting Officer shown on the face of this contract/order.

(d) Contractors wishing to check the payment status of their vouchers may do so by calling FPC Vendor Support at 636-321-5251. In addition, questions may be directed to the Contracting Officer's Representative (COR). In the absence of a COR, contact the Procurement Contracting Officer (PCO), whose name and contact information appear on the face page of this contract/order.

(End of Clause)

5X52.237-9001 CONTRACTOR IDENTIFICATION (JAN 2012)

The contractor shall ensure that contractor personnel, including their sub-contractor personnel, identify themselves as contractor personnel, by introducing themselves or being introduced as contractor personnel when:

(1) attending meetings with Government personnel or contractors performing under a contract awarded to support NGA requirements,

(2) answering government telephones,

(3) providing any type of written or electronic mail correspondence, and

(4) working in any other situation where their actions could be construed as an official Government act or representation of the Government.

The contractor shall ensure that contractor personnel possess and properly display Government-issued identification badges when on NGA property or when attending NGA meetings not located on NGA property.

The contractor will ensure that contractor personnel, when performing in a contractor capacity, refrain from using their retired or reserve component military rank or title in all written and verbal communications.

The Government may include the results of the contractor's ability to adhere to this clause in quality assurance surveillance plans and award fee plans as part of the overall administration of this contract.

(End of Clause)

5X52.37-9000 Contractor Employee Data for Access to NGA Facilities or Sensitive Systems (OCT 2005)

1. This clause defines the contractor's responsibilities for providing accurate contractor data, and providing updates to that data, for NGA's Human Capital Management System (HCMS). NGA requires that all contractors provide initial and timely updates to HCMS data for all personnel performing under this contract who have access to NGA facilities or sensitive systems, as determined by the contracting officer.

2. The Contractor shall:

a. Provide the Contracting Officers Representative (COR) a Point of Contact (POC) for providing and maintaining contractor personnel data for the HCMS database. The POC shall be provided to the COR, in writing, within 10 days of contract award (or modification inserting this clause). For contracts with an on-site Project Lead or Program Manager, this person shall serve as the POC.

b. Provide the COR initial HCMS data for their personnel within 10 days of contract award or modification. The information that is to be provided for HCMS shall include: persons full legal name, social security number, citizenship status, NGA contract number, prime contractor name, NGA location and organization where the person will be working, and a 24/7 emergency contact point for the contractor.

c. Notify the COR of all contractor data changes within 10 days of the change. Changes include new or departing contractor personnel and any change to information provided in paragraph b above. If the contract number under which a contractor or its personnel work changes, the POC for the contract receiving the personnel shall notify the COR within 10 days of the change.

d. Provide response to all inquiries made by NGA as to the validity and completeness of contractor data records in the HCMS database within two weeks of date of request.

e. Ensure all employees attend in-processing and out-processing briefings.

(End of Clause)

5X52.246-9000 - Contractor Compliance with all applicable National Geospatial-Intelligence Agency (NGA) and U.S. Government installation regulations, directives,

instructions, rules, policies and procedures. (MAR 2023)

A. The Contractor shall comply with, and shall ensure that its personnel, to include subcontractors, comply with all applicable NGA regulations, directives, instructions, rules, policies and procedures. The Contractor may request copies from the Contracting Officer's Representative (COR), the Contracting Officer or their designated representative(s).

B. The Contractor shall comply with, and shall ensure that its personnel, to include subcontractors, comply with regulations, directives, instructions, rules, policies, procedures and other applicable requirements issued by the U.S. Government Installation Commander where NGA is a tenant activity, including, but not limited to, those relating to force protection, security, health and safety. The Contractor may request copies from the Contracting Officer's Representative (COR), the Contracting Officer or their designated representative(s).

C. The Contractor shall institute and implement an effective program to ensure their employees and subcontractors, comply with all applicable requirements in accordance with paragraphs A and B above as well as paragraphs E and H below.

D. The specific requirements covered in paragraphs A, B, E, and H may be specified in the Performance Work Statement, elsewhere in the contract, or in NGA and/or Government installation regulations, directives, instructions, rules, policies and procedures. Specific requirements may include, but are not limited to categories such as:

- Security in/out processing
- Personnel in/out processing
- Facility access, parking, and in/out processing
- Information technology access and in/out processing
- PeopleSoft access, updates and in/out processing
- Periodic and special training requirements

E. Facility Access and Badging. The following criteria must be met in order to be issued an NGA IC badge and to gain access to an NGA-controlled facility:

(1) NGA IC badges will only be issued to those contractors who provide direct charge support on an active TS/SCI NGA contract, even when seated at corporate locations outside of NGA Government facilities. Green badges must be used at least once during a one-month period at an NGA Government facility. Failure to use an NGA IC badge may result in suspension or termination of the badge for lack of activity. The badges will expire at the end of the supported contract. Note: NGA IC badges will not be issued to any contractor who does not need access to an NGA facility, i.e., Corporate VIPs, etc. Infrequent visitors must report to the Visitor's Center.

(2) Notwithstanding the above, NGA badges will be issued to contractors, who are required by contract, to gain access to an NGA facility in the event of an emergency or when after-hours access is required. The Government POC or Contracting Officer's Representative (COR) coordinates the submission of an application for an NGA badge; establishment of a PeopleSoft record of SCI accesses; completion of NGA Form 5212- 7A, "Request for Identification/Building Access Picture Badge"; and submission of the application to the appropriate Site Security Office for approval. (Reference NGA Instruction 5210.8, Physical Security Program)

F. The Contracting Officer may direct the Contractor, at its own expense, to remove and replace any Contractor personnel who fail to comply with or violate applicable requirements of this clause. Such action may be taken at the Government's discretion without prejudice to its rights under any other provision of this contract, including the Termination for Default clause.

G. The Contracting Officer may include the results of the Contractor's ability to adhere to this clause in past performance reports, quality assurance surveillance plans and award fee plans as part of the overall administration of this contract.

H. NGA Inspector General.

(1) The contractor must report to the NGA Inspector General (IG), DoDIG, or Intelligence Community IG any and all possible violations of federal law or illegal intelligence activities related to this contract by individuals charging directly or indirectly to this contract.

(2) The IG shall have access to any individual charging directly or indirectly to this contract whose testimony is needed for the performance of the IG's duties. In addition, the IG shall have direct access to all records, reports, audits, reviews, recommendations, documents, e-mails, papers, or other material that relate to this contract with respect to which the IG has responsibilities. Failure on the part of any contractor to cooperate with the IG shall be grounds for administrative action by the Director, Office of Contract Services, including contractual remedies.

(3) NGA contractors and contractor personnel may report suspected instances of improper conduct through the NGA IG Hotline. Contractors shall make their employees aware of this Hotline: 571-557-4849, secure 578-4849, or toll free 1-800-380-7729 or by contacting the OIG at IG@nga.mil or secure at IG@nga.ic.gov.

(4) The contractor agrees to include the substance of this clause in all subcontracts exceeding the simplified acquisition threshold except those for commercial products or commercial services and those where the NGA association must be protected.

(5) This requirement is supported in the Federal Acquisition Regulation (FAR) clause 52.203-13, which requires timely disclosure to the Government of credible evidence of violation of law, and timely and complete response to OIG requests for documents and access to employees and information.

(6) This requirement is supported in NGA policy. NGAI 7410.1 requires all personnel, to include contractors, to cooperate fully with NGA OIG audits, inspections, and investigations.

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OSD252-D02	Image Labeling for Sparse and Rare Targets in Satellite Imagery for Computer Vision Applications
OSD252-D03	Tools for Test & Evaluation on Incompletely-Labeled Satellite Imagery Datasets

OSD252-001 TITLE: Capabilities for the Detection or Classification of Physical Damage using Remote Sensing

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

OBJECTIVE: Develop a physical damage detector for overhead imagery to prioritize the collection of additional imagery and the allocation of damage assessment analysis resources.

DESCRIPTION: **NGA has designated the automated damage assessment and change detection algorithm developed for this topic as Controlled Unclassified Information (CUI). The technical solution shall include a plan for protecting the weights and technical specifications associated with the automated damage assessment or change detection algorithm from unauthorized access and theft. DFARS clause 252.204-7012 applies.**

This announcement seeks technologies that automate physical damage assessments using commercial remote-sensing data. Assessing physical damage of infrastructure, buildings, etc., over large geographic areas due to human conflict or natural disasters requires a substantial amount of human labor. Timely damage assessments are necessary for the effective allocation of resources, such as for humanitarian assistance and disaster recovery (HADR), and providing situational awareness to military commanders and policy makers. Overhead commercial satellites and airborne sensors are frequently used by geospatial and imagery analysts to assess physical damage [1, 2]; however, few commercial sensors have associated automated damage assessment capabilities fine-tuned to that sensor with clearly defined performance bounds across geographic areas, types of damage, and types of structures. The aim of this effort is to design and develop capabilities that reliably identify physical damage to buildings and infrastructure that meet the following criteria:

1. 1 .Solutions apply to at least one commercial overhead sensor, where proposers shall identify their own source of remotely sensed data, i.e., the government will not necessarily furnish the data.
2. Output of the solution, e.g., polygons and associated metadata, can be ingested into geographic information system (GIS) software platforms using commonly used data formats, such as KML [3].
3. Damage estimates are accompanied by meaningful and human-interpretable confidence values, e.g., like the expressions of likelihood detailed in Intelligence Community Directive (ICD) 203 [4].
4. If more than two levels of physical damage are inferred, as opposed to the binary classes of undamaged vs. damaged/destroyed or change detection, then the damage levels align with common production standards (e.g., CJCSI 3162.02 for Department of Defense customers [5]).
5. Solution and sensor operating bounds (e.g., geographic area limitations of the solution or sensor, types of damage or effect, collection geometries, time of day) clearly defined, along with confidence estimates of those bounds, where qualitative estimates could be acceptable with proper justification.
6. Evaluation measures of performance demonstrated on publicly available data. If only proprietary data are available for evaluation, then the evaluation data must be thoroughly described to make the comparison of performance across commercial vendors possible. In either case, the data must be sufficiently described to understand the scope of the evaluation and whether there is sufficient evidence that a solution would be expected to generalize to other areas or damage types.
7. If a two-stage approach of building segmentation, or more generally feature segmentation, followed by damage inference is used or proposed, then proposers are strongly encouraged to use commercially or openly available feature polygons. A strong justification will be needed if not using a commercially or openly available feature polygon data set, e.g., no data sets exist for a given test area. If a feature polygon data set of buildings, non-building structures, etc., is created

as part of this effort, then the National System for Geospatial Intelligence (NSG) Application Schema definitions (NAS) [6] and GEOINT Content Extraction Specification (GCES) [7] should be used, at least as a baseline. Deviation from the NAS and GCES is acceptable if the proposer articulates where the differences are with sufficient justification.

A reasonable assumption for usage of these capabilities is that the output will be used to identify potential physical damage for further human review, where the goal is to reduce the amount of human labor required to perform damage assessments. The assumption that outputs will be used in human-machine teams can be used to determine how the criteria in the topic description are addressed, including how to communicate operating bounds, target evaluation measures, etc.

PHASE I: Research and design of an automated damage assessment or change detection solution related to physical damage following the criteria in the topic description. Proposals must demonstrate an innovative approach, whether that be enabling an underutilized sensor or phenomenology and therefore demonstrating a strong understanding of the trade space, proposing a novel solution that advances the state of the art, or measurably and substantially reducing human labor required for damage assessments over wide geographic areas. Deliver a written report on the design, any preliminary results, and a Phase II plan.

This topic is accepting both Phase I and Direct to Phase II (DP2) proposals. Proposers interested in submitting a DP2 proposal must provide documentation to substantiate that the scientific and technical merit and feasibility described above has been met and describe the potential commercial applications. Documentation should include, reference, or summarize all relevant information including, but not limited to technical reports, test data, prototype designs/models, and performance goals/results.

PHASE II: Implement an automated damage assessment or change detection algorithm related to physical damage following the criteria in the topic description and Phase I expectations. Effective implementations for all criteria are expected. Deliver a written report detailing results.

PHASE III DUAL USE APPLICATIONS: If proposed solutions are developed with or derived from Open-Source code or data, the Open-Source licenses of those original sources must allow for commercialization and the proposed use documented in the SBIR proposal. Data used to train an algorithm should include proper citations and sourcing, including licensing and releasability.

REFERENCES:

1. D. Palumbo, A. Abutaleb, P. Cusiac, and E. Rivault, “At least half of Gaza’s buildings damaged or destroyed, new analysis shows,” BBC, January 30, 2024. [Online]. Available: https://www.bbc.com/news/world-middle-east-68006607. [Accessed Jan. 16, 2025].
2. Y. Asi, D. Mills, P. G. Greenough, D. Kunichoff, S. Khan, J. Van Den Hoek, C. Scher, S. Halabi, S. Abdulrahim, N. Bahour, A. K. Ahmed, B. Wispelwey, and W. Hammoudeh, “’Nowhere and no one is safe’: spatial analysis of damage to critical civilian infrastructure in the Gaza Strip during the first phase of the Israeli military campaign, 7 October to 22 November 2023,” Conflict and Health, vol. 18, no. 24, 2024, https://doi.org/10.1186/s13031-024-00580-x.
3. “KML,” Open Geospatial Consortium. [Online]. Available: https://www.ogc.org/publications/standard/kml/. [Accessed Jan. 27, 2025].
4. “Analytic Standards, Intelligence Community Directive 203 Technical Amendment,” Office of the Director of National Intelligence, 2023. [Online]. Available: https://www.dni.gov/files/documents/ICD/ICD-203.pdf. [Accessed Jan. 16, 2025].
5. “Methodology for Combat Assessment,” United States Joint Chiefs of Staff, 2021. [Online]. Available:

https://www.jcs.mil/Portals/36/Documents/Library/Instructions/CJCSI%203162.02A.pdf. [Accessed Jan. 16, 2025].
6. Building, NSG Standards Registry Identifier 100083, National Geospatial-Intelligence Agency, Mar. 2007. [Online]. Available: https://nsgreg.nga.mil/as/view?i=100083
7. Building, Foundation GEOINT Standards Data Product Specification Identifier 100083, National Geo-Spatial Intelligence Agency. [Online]. Available: https://fgs-dps.gs.mil/content/100083

KEYWORDS: physical damage assessment; PDA; battle damage assessment; BDA; volumetric change; change detection

OSD252-D02 TITLE: Image Labeling for Sparse and Rare Targets in Satellite Imagery for Computer Vision Applications

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

OBJECTIVE: Refine an existing image labeling regime to accelerate the production of datasets containing rare and diverse targets.

DESCRIPTION: **All work will be conducted on classified data at the TS/SCI level.**
Computer vision detector algorithms are developed using machine learning from massive quantities of labels corresponding to relevant objects in overhead satellite imagery. Machine-assisted image labeling is a technical approach that can quickly create and supply the algorithm with largely annotated datasets, however, this method tends to strengthen highly-represented samples at the expense of examples in the tail of the distribution. Furthermore, machine-assisted labeling methods struggle when applied to challenging objects of interest (OOIs) such as those below:

- Scarcity – the OOI may have only one example per image or may have been observed in only a limited number of geographic locations, such that the sample size is too small to initiate a machine-assisted process.
- Lack of Distinctiveness
  - The OOI has few distinguishable features such as shipping crates.
  - The OOI is easily confused with similar objects, such as derivatives of a more common object.
- Occlusion - The OOI may only be partially showing in the image due to occlusion or intentional concealment.
- Natural Obfuscation - The OOI may be hard to distinguish from naturally occurring earth features. Examples include disturbed earth, construction activity, trenching, and berm building.

In all the rare and challenging OOI situations above, either there are too few object representations, only partial representations, or inconsistent representations to initiate a detector or feature extractor-based method for machine-assisted approaches. This topic seeks to improve the efficiency and cost effectiveness of labeling overhead imagery for rare or challenging OOIs. Potential methods of interest include but are not limited to advances in machine-assisted labeling, improved tooling and human-labeling workflows, automated search and alignment of additional imagery that may find additional OOI examples.

PHASE I: Demonstrate a basic model based labeling methodology to accelerate accurate label production on rare and otherwise constrained OOIs using the Microsoft COCO (JSON) metadata format. The combined human and tool interactions should be optimized for efficiency and production throughput with minimized human labor, and employ innovative approaches to quality control for sparse datasets. Conduct verification of the integrated prototype for quality assurance. Prepare software system for installation on classified computer systems.

PHASE II: Develop a labeling system to operate on rare and challenging OOIs in panchromatic satellite imagery to streamline the creation of labeled datasets in NGA GeoCOCO metadata format extension of the Microsoft COCO format. The approach should significantly decrease the time required for conventional manual labeling methods without significantly increasing mislabeled or missed OOIs. Establish metrics that capture annotation time and quality. Collaborate with other performers to resolve unanticipated challenges and to deploy the system on a classified network for mission use. Once deployed, conduct verification and validation of the integrated product for quality assurance. Perform at practical demonstrations of dataset creation for OOIs specified by NGA. Deliver two separate documents,

one consisting of lessons learned and the other detailing the technical architecture and functionality. Performers must possess TS/SCI clearances and conduct work at the same level.

PHASE III DUAL USE APPLICATIONS: Enhance the methodology and tooling from Phase II to reduce to the greatest extent possible human in the loop interactions. Collaborate with NGA mission partners to apply labeling system to mission problem sets. Capture all metrics, lessons learned, and opportunities for improvement in delivered report including overall system performance and data quality metrics.

REFERENCES:

1. Sager, C., Jamiesch,C., & Zscheh, P. A survey of image labelling for computer vision applications. Journal of Business Analytics 4(2), 1-20; doi:10.1080/2573234X.2021.1908861 (2021);
2. Haider, T. & Michahelles, F. Human-machine collaboration on data annotation of images by semi-automatic labeling. In Proceedings of Mensch und Computer 2021 (MuC '21), 552–556; doi: 10.1145/3473856.3473993 (2021);

KEYWORDS: human-machine teaming; human-computer interaction; image annotation; image labeling; semi-automated image labeling; computer vision.

OSD252-D03 TITLE: Tools for Test & Evaluation on Incompletely-Labeled Satellite Imagery Datasets

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy

OBJECTIVE: Develop tooling for rapid performance analysis of diverse AI/ML models on overhead imagery datasets with label noise, over wide ranges of environmental conditions.

DESCRIPTION: ** All work will be conducted on classified data at the TS/SCI level.**
Deployment and maintenance of AI/ML models at scale requires frequent, reliable test and evaluation. The trustworthiness of such evaluation depends on two factors: well-defined performance metrics and well-defined test data. Performance metrics are extensively studied, as are techniques to split datasets into training and test partitions, so the emphasis of this topic is the development of tools to support the use of such test partitions.

A key, unresolved issue in the use of test data is the presence of label noise, such as missing or incomplete annotations, poorly sized, oriented or aligned localizations (bounding boxes), or mislabelled annotation names (misclassifications). Such noise is accepted to be present in most AI/ML datasets, and it has been determined to have potentially significant impacts on evaluation quality for a range of tasks and fields [1, 2], but few methods exist to mitigate these impacts.

Tools to identify and correct label noise are available, but require significant human effort for large-scale data, making them insufficient for most applications. Forgoing a labeled test set entirely and estimating performance based on training data or unlabeled data may also be viable [3, 4, 5], but this has not been extensively studied, particularly in the domain of overhead imagery where errors may be highly correlated with environmental conditions (for example, difficulties in annotating cloudy imagery).
A flexible system to reliably estimate true AI/ML model performance without a need for perfect test data would fill a significant gap in capability for organizations working with large numbers of models and large amounts of data and (such as NGA) as well as organizations without the time or human capital to exhaustively assess datasets used for AI/ML development (such as academic labs and smaller commercial entities). Such a system would greatly enhance organizations' confidence in post-deployment model performance without incurring additional cost and delay.

The primary interest of this topic is in tools for evaluation of object detection and automated target recognition tasks, but other computer vision tasks such as image classification, semantic segmentation, change detection, etc. are within scope.

PHASE I: Demonstrate a basic software package for the estimation of AI/ML model performance on computer vison detector tasks for overhead satellite imagery. Methods should be robust to noise or absence of test data. Prototype metrics to quantify the accuracy of the developed methodology (that is, the gap between observed performance on imperfect data and the actual performance of the model). Methods should generalize to multiple real overhead imagery datasets over multiple architectures. Deliver all software packages and a summary report of performance investigations.

PHASE II: Apply methods and tooling of Phase I to overhead imagery. Provide more detailed analyses of performance, such evaluations for various combinations of collection geometries and environmental conditions. Develop techniques to extrapolate performance estimates to novel data which may be unlabeled, partially labeled, and/or labeled under a different ontology. Software from Phase II should be compatible with multiple overhead imagery datasets over multiple model architectures. Demonstrate tooling and methodologies on novel detector development without preexisting large datasets on at least three challenge problems developed in consort with NGA sponsor. Deliver all software packages and a

summary report of performance investigations. Performers are expected to accomplish this work at the TS/SCI level.

PHASE III DUAL USE APPLICATIONS: Enhance software from Phase II to function with minimal human intervention on large-scale datasets and AI/ML models in line with commercial applications. Extensively document usage and functionality of all developed software. Deliver all software packages and a summary report of performance investigations.

REFERENCES:

1. Northcutt, Curtis G., Anish Athalye, and Jonas Mueller. "Pervasive label errors in test sets destabilize machine learning benchmarks." arXiv preprint arXiv:2103.14749 (2021).
2. Wang, Yueye, et al. "Impact of Gold-Standard Label Errors on Evaluating Performance of Deep Learning Models in Diabetic Retinopathy Screening: Nationwide Real-World Validation Study." Journal of medical Internet research 26 (2024): e52506.
3. Deng, Weijian, and Liang Zheng. "Are labels always necessary for classifier accuracy evaluation?." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021.
4. Okanovic, Patrik, et al. "All models are wrong, some are useful: Model Selection with Limited Labels." arXiv preprint arXiv:2410.13609 (2024).
5. Białek, Jakub, et al. "Estimating Model Performance Under Covariate Shift Without Labels."

KEYWORDS: test and evaluation; machine learning; artificial intelligence; computer vision; trusted ai.

Appendix A
TECHNICAL PROPOSAL TEMPLATE (VOLUME 2)

INSTRUCTIONS

These instructions and template apply to DoD SBIR/STTR Phase I topics and provide general guidelines for completing the Phase I Technical Volume. Information provided in the Service/Component-specific instructions for the topic of interest take precedence over any instructions listed below.

The template (beginning on the following page) is the format model that may be used to prepare the Phase I Technical Volume. Do not include the instructions provided on this page or any bracketed [ ] guidance in the template.

Disclosure

Offerors that include in their proposals data which they do not want disclosed to the public for any purpose, or used by the U.S. Government except for evaluation purposes, must:

(1) Mark the first page of each Volume of the Submission with the following legend:
"This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of-or in connection with-the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]";

(2) Mark each sheet of data it wishes to restrict with the following legend:
"Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

Format

The Technical Volume shall meet the following requirements:

- Please refer to Service/Component-specific topic instructions for the page limit and how a technical volume is handled if the stated page count is exceeded. It is the proposing firm's responsibility to verify that the Technical Volume does not exceed the page limit after upload to DSIP. Unless otherwise noted, all content in the Technical Volume will count toward the limit.
- Single column format, single-spaced typed lines.
- Standard 8 ½" x 11" paper format.
- Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
- **The header on each page of the Technical Volume should contain your company name, topic number, and DSIP-assigned proposal number.**
- No font smaller than 10-point. For headers, footers, imbedded tables, figures, images, or graphics that include text, a font size of smaller than 10-point is allowable, though proposers are cautioned that the text may be unreadable by evaluators.

Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

Delete this instruction page and begin the Technical Volume starting with the following page.

[Title]

Volume 2: Technical Volume

[Note: Remove the disclosure statement below if not applicable to your proposal. Refer to Instructions.]

This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed – in whole or in part – for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of – or in connection with – the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages <insert numbers or other identification of sheets>.

1. Identification and Significance of the Problem or Opportunity.

[Define the specific technical problem or opportunity addressed and its importance.]

2. Phase I Technical Objectives.

[Enumerate the specific objectives of the Phase I work, including the questions the research and development effort will try to answer to determine the feasibility of the proposed approach.]

3. Phase I Statement of Work (include Subcontractors and/or Research Institutions).

(a) [Provide an explicit, detailed description of the Phase I approach. If a Phase I option is required or allowed by the Component (refer to Component-specific instructions for topic of interest), describe appropriate research activities which would commence at the end of Phase I base period should the Component elect to exercise the option. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail. This section should be a substantial portion of the Technical Volume.

(b) The BAA may contain topics that have been identified by the Program Manager as research or activities involving Human/Animal Subjects and/or Recombinant DNA. If Phase I performance includes performance of these kinds of research or activities, please identify the applicable protocols and how those protocols will be followed during Phase I. Please note that funds cannot be released or used on any portion of the project involving human/animal subjects or recombinant DNA research or activities until all of the proper approvals have been obtained. **SBCs proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal in order to avoid potential delay of contract award.**]

4. Related Work.

[Describe significant activities directly related to the proposed effort, including any conducted by the principal investigator, the proposing firm, consultants, or others. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. The Technical Volume must persuade evaluators of the proposer's awareness of the state of the art in the topic. Describe any previous work not directly related but similar to the proposed effort. Provide the following: (1) a short description, (2) the client for which work was performed (including the Government Point of Contact to be contacted including e-mail address and phone number), and (3) date of performance including project completion.]

5. <u>Relationship with Future Research or Research and Development</u>.

(a) [State the anticipated results of the proposed approach if the project is successful.

(b) Discuss the significance of the Phase I effort in providing a foundation for a Phase II research or research and development effort.

(c) Identify the applicable clearances, certifications and approvals required to conduct Phase II testing. Outline the plan for ensuring timely completion of stated authorizations in support of a Phase II research or research and development effort.]

6. <u>Commercialization Strategy.</u>

[Describe in approximately one page the SBC's strategy for commercializing this technology in DoD, other Federal Agencies, and/or private sector markets. Provide specific information on the market need the technology will address and the size of the market. Also include a schedule showing the quantitative commercialization results from the project that your company expects to achieve.]

7. <u>Key Personnel</u>.

[Identify key personnel who will be involved in the Phase I effort including information on directly related education and experience. A concise technical resume of the principal investigator, including a list of relevant publications (if any), must be included (Please do not include Privacy Act Information). All resumes will count toward the page limit for Volume 2, as specified in the Component-specific instructions.]

[Principal Investigator Name]
[School, Degree, Year]

Relevant Experience
[A concise description of the principal investigator's relevant technical experience and its application to this topic.]

Relevant Awards or Patents
[List any awards received or patents granted or applications submitted for work related to this topic.]

Relevant Publications
[List any publications relevant to this topic.]

[Repeat this format as necessary to address the qualifications of all key personnel.]

8. <u>Foreign Citizens</u>.

[Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. The proposal may be deemed nonresponsive if the requested information is not provided. You may be asked to provide additional information during negotiations in order to verify the foreign citizen's eligibility to participate on a SBIR/STTR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)). Note: If no foreign nationals will be involved in proposed work, the word "None" can be substituted for the table.]

Name [include direct employees, subcontractors, and consultants]	Foreign National (Yes/No)	Country of Origin	Type of Visa or Work Permit	Level of Involvement (Role)

9. Facilities/Equipment.

[Describe available instrumentation and physical facilities necessary to carry out the Phase I effort. Justify equipment purchases in this section and include detailed pricing information in the Cost Volume. State whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name), and local Governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.]

10. Subcontractors/Consultants.

[Propose efforts as applicable to either SBIR or STTR as follows:

SBIR. Involvement of a university or other subcontractors or consultants in the project may be appropriate. A minimum of two-thirds of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be carried out by the proposing small business firm, unless otherwise approved in writing by the Contracting Officer. SBIR efforts may include subcontracts with Federal Laboratories and Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories and FFRDCs; however, proposers must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoD SBIR/STTR Broad Agency Announcement for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.)

STTR. Involvement of a Research Institution in the project is required. A minimum of 40 percent of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be conducted by the proposing small business firm, and a minimum of 30 percent of the research and/or tasks in Phase I, as measured by direct and indirect costs, must be conducted by a single Research Institution. STTR efforts may include subcontracts with Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories, but they do not qualify as a Research Partner; proposers may only subcontract to Federal Laboratories within the remaining 30 percent and must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoD SBIR/STTR Broad Agency Announcement for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.]

11. Prior, Current or Pending Support of Similar Proposals or Awards.

[If a proposal submitted in response to this BAA is substantially the same as another proposal that was funded, is now being funded, or is pending with another Federal Agency, another or the same DoD Service/Component, you must disclose this on the proposal cover sheet and provide the following information:

a) Name and address of the Federal Agency(s) or DoD Component to which a proposal was submitted, will be submitted, or from which an award is expected or has been received.

b) Date of proposal submission or date of award.

c) Title of proposal.

d) Name and title of principal investigator for each proposal submitted or award received.

e) Title, number, and date of BAA(s) or solicitation(s) under which the proposal was submitted, will be submitted, or under which award is expected or has been received.

f) If award was received, provide contract number.

g) Specify the applicable topics for each proposal submitted or award received.

Note: If this does not apply, state in the proposal "No prior, current, or pending support has been provided for proposed work."]

12. <u>Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software.</u>

The Offeror asserts for itself, or the persons identified below, that the Government's rights to use, release, or disclose the following technical data or computer software should be restricted:

Technical Data or Computer Software to be Furnished with Restrictions	Basis for Assertion	Asserted Rights Category	Name of Person or Organization Asserting Restrictions
[(LIST)]	[(LIST)]	[(LIST)]	[(LIST)]

[Completion of this table and submission of the proposal constitutes signature for the information listed in the table above.]

[**ADDITIONAL INFORMATION/INSTRUCTION:** Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this BAA generally remain with the contractor, except that the Government obtains a royalty-free license to use such technical data only for Government purposes during the period commencing with contract award and ending twenty years after completion of the project under which the data were generated. This data <u>must</u> be marked with the restrictive legend specified in DFARS 252.227-7018 Class Deviation 2020-O0007. Upon expiration of the twenty-year restrictive license, the Government has unlimited rights in the SBIR data. During the license period, the Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) For evaluation purposes; (2) As expressly permitted by the contractor; or (3) A use, release, or disclosure that is necessary for emergency repair or overhaul of items operated by the Government. See DFARS clause 252.227-7018 Class Deviation 2020-O0007 "Rights in Noncommercial Technical Data and Computer Software – Small Business Innovation Research (SBIR) Program."

If a proposer plans to submit assertions in accordance with DFARS 252.227-7017 Class Deviation 2020-O0007, those assertions must be identified and assertion of use, release, or disclosure restriction MUST be included with your proposal submission. The contract cannot be awarded until assertions have been approved. Please note that only the table is included in the page limitation; any supporting data

concerning the contract/grant number and awarding agency, as well as planned use or need of the data asserted, can be provided in Volume 5, Supporting Documents.

The following instructions apply to the fields in the table above (Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software).

1) For technical data (other than computer software documentation) pertaining to items, components, or processes developed at private expense, identify both the deliverable technical data and each such item, component, or process. For computer software or computer software documentation identify the software or documentation.

2) Generally, development at private expense, either exclusively or partially, is the only basis for asserting restrictions. For technical data, other than computer software documentation, development refers to development of the item, component, or process to which the data pertain. The Government's rights in computer software documentation generally may not be restricted. For computer software, development refers to the software. Indicate whether development was accomplished exclusively or partially at private expense. If development was not accomplished at private expense, or for computer software documentation, enter the specific basis for asserting restrictions.

3) Enter asserted rights category (e.g., Government purpose license rights from a prior contract, rights in SBIR/STTR data generated under another contract, limited, restricted, or government purpose rights under this or a prior contract, or specially negotiated licenses).

4) Corporation, individual, or other person, as appropriate.

Enter “none” when all data or software will be submitted without restrictions.]

Appendix B
DEFINITIONS

The following definitions from the SBA SBIR/STTR Policy Directive, the Federal Acquisition Regulation (FAR) and other cited regulations apply to this BAA.

Commercialization

The process of developing products, processes, technologies, or services, and the production and delivery (whether by the originating party or others) of the products, processes, technologies, or services for Federal Government or commercial markets purchase or use.

Cooperative Research and Development

An SBC and a research institution jointly conduct R&D. For purposes of the STTR Program, the SBC performs 40 percent of the work, and the single research institution performs not less than 30 percent of the work. For purposes of the SBIR Program, this refers to work a research institution conducts as the SBC's subcontractor. The proposing SBC must conduct at least two-thirds of the research and/or analytical work in Phase I.

Covered Individual

An individual who contributes in a substantive, meaningful way to the scientific development or execution of a R&D project proposed to be carried out with a DoD-funded award. DoD has further designated covered individuals as including all proposed key personnel.

Essentially Equivalent Work

Work that is substantially the same research, which is proposed for funding in more than one contract proposal or grant application submitted to the same federal agency or submitted to two or more different federal agencies for review and funding consideration; or work where a specific research objective and the research design for accomplishing the objective are the same or closely related to another proposal or award, regardless of the funding source.

Export Control

The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. Details are available here.

NOTE: Export control compliance statements found in the individual Service/Component-specific proposal instructions are not meant to be all inclusive. They do not remove any liability from the submitter to comply with applicable ITAR or EAR export control restrictions or from informing the U.S. Government of any potential export restriction as fundamental R&D efforts proceed.

Federal Laboratory

In 15 U.S.C. §3703, it means any laboratory, any federally funded R&D center (FFRDC), or any center established under 15 U.S.C. §§ 3705 & 3707 that a federal agency owns, leases, or otherwise uses and the Federal Government funds, whether the U.S. Government or the contractor operates.

Federally Funded Award

A Phase I, Phase II (including Direct to Phase II, sequential Phase II/subsequent Phase II and cross-agency Phase II), or Phase III SBIR or STTR award made using a funding agreement.

Foreign Affiliation

Under 15 U.S.C. § 638(e)(16), foreign affiliation means a funded or unfunded academic, professional, or institutional appointment or position with a foreign government or government-owned entity, whether full-time, part-time, or voluntary (including adjunct, visiting, or honorary). This includes appointments or positions deemed adjunct, visiting, or honorary with research institutions located in a foreign country of concern.

Foreign Country of Concern

In 15 U.S.C. § 638(e)(17), foreign country of concern means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country the Secretary of State determines as a country of concern.

Foreign Entity

Foreign entity means any branch, partnership, group or sub-group, association, estate, trust, corporation or division of a corporation, non-profit, academic institution, research center, or organization that foreign owners, foreign investors, foreign management, or a foreign government establish, direct, or control.

Foreign Government

Foreign government means any government or governmental body, organization, or instrumentality, including government owned-corporations, other than the U.S. Government or U.S. state, territorial, tribal, or jurisdictional governments or governmental bodies. The term includes, but is not limited to, non-U.S. national and subnational governments, including their respective departments, agencies, and instrumentalities.

Foreign National

Foreign National (also known as Foreign Person) under 22 CFR 120.16 mean any natural person who is not a lawful permanent resident as defined by 8 U.S.C. § 1101(a)(20) or who is not a protected individual as defined by 8 U.S.C. § 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society, or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments (e.g., diplomatic missions).

"Lawfully admitted for permanent residence" means the status of having been lawfully accorded the privilege of residing permanently in the United States as an immigrant in accordance with the immigration laws, such status not having changed.

"Protected individual'' means an individual who (A) is a citizen or national of the United States, or (B) is an alien who is lawfully admitted for permanent residence, is granted the status of an alien lawfully admitted for temporary residence under 8 U.S.C. § 1160(a) or 8 U.S.C. § 1255a(a)(1), is admitted as a

refugee under 8 U.S.C. § 1157, or is granted asylum under Section 8 U.S.C. § 1158; but does not include (i) an alien who fails to apply for naturalization within six months of the date the alien first becomes eligible (by virtue of period of lawful permanent residence) to apply for naturalization or, if later, within six months after November 6, 1986, and (ii) an alien who has applied on a timely basis, but has not been naturalized as a citizen within two years after the date of the application, unless the alien can establish that the alien is actively pursuing naturalization, except that time consumed in the Service's processing the application shall not be counted toward the two-year period.

Fraud, Waste and Abuse

a. Fraud includes any false
b. about a material fact or any intentional deception designed to deprive the United States unlawfully of something of value or to secure from the United States a benefit, privilege, allowance, or consideration to which an individual or business is not entitled.
c. Waste includes extravagant, careless or needless expenditure of government funds, or the consumption of government property, that results from deficient practices, systems, controls, or decisions.
d. Abuse includes any intentional or improper use of government resources, such as misuse of rank, position, or authority or resources.
e. The SBIR/STTR Program training related to Fraud, Waste and Abuse is available here. See Section 1.13 for reporting fraud, waste, and abuse.

Funding Agreement

Any contract, grant, or cooperative agreement entered between any Federal Agency and any SBC for the performance of experimental, developmental, or research work, including products or services, Federal Government-funded in whole or in part. DoD Services/Components will only use contracts and other transaction authority (OTA) agreements for all SBIR awards.

Historically Black Colleges and Universities, and Minority-Serving Institutions

Department of Education list for historically Black colleges and universities and minority-serving institutions.

HUBZone Certified Small Business Concern

An SBC with SBA certification under the Historically Underutilized Business Zones (HUBZone) Program (13 C.F.R. § 126) as a HUBZone firm listed in the dynamic small business search (DSBS).

Malign Foreign Talent Recruitment Program

As defined in 42 U.S.C § 19237, the term “malign foreign talent recruitment program” means-

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual-

(i) engaging in the unauthorized transfer of intellectual property, materials, data products, or

other nonpublic information owned by a United States entity or developed with a Federal R&D award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the IP, materials, or data products;

(ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;

(iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal R&D award;

(iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;

(v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a R&D award or required to engage in work that would result in substantial overlap or duplication with a Federal R&D award;

(vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

(vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the R&D award, contrary to the institutional policies or standard terms and conditions of the Federal R&D award;

(viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or

(ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal R&D award; and

(B) a program that is sponsored by-

(i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern;

(ii) an academic institution on the list developed under section 1286(c)(8) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232) ; or

(iii) a foreign talent recruitment program on the list developed under section 1286(c)(9) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232).

Performance Benchmark Requirements

SBCs with multiple SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The purpose of these requirements is to ensure that Phase I SBCs with multiple prior SBIR/STTR awards are making progress towards commercializing the work done under those awards. The Phase I to Phase II transition rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The commercialization benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' applications for more experienced firms. Detailed information on benchmark calculations and increased performance standards for more experienced firms can be found here.

Personal Conflict of Interest

A situation in which an individual has a financial interest, personal activity, or relationship that could impair the employee's ability to act impartially and in the best interest of the government when performing under the contract. (A de minimis interest that would not "impair the employee's ability to act impartially and in the best interest of the government" is not covered under this definition.)

Among the sources of personal conflicts of interest are:

a. The covered employee's, their close family members' or other members of their household's financial interests;
b. Other employment or financial relationships (including seeking or negotiating for prospective employment or business); and
c. Gifts, including travel.

Financial interests referred to in this definition's first paragraph may arise from:

a. Compensation, including wages, salaries, commissions, professional fees, or fees for business referrals;
b. Consulting relationships (including commercial and professional consulting and service arrangements, scientific and technical advisory board memberships, or serving as an expert witness in litigation);
c. Services provided in exchange for honorariums or travel expense reimbursements;
d. Research funding or other forms of research support;
e. Investment in the form of stock or bond ownership or partnership interest (excluding diversified mutual fund investments);
f. Real estate investments;
g. Patents, copyrights, and other IP interests; or
h. Business ownership and investment interests.

Principal Investigator/Program Manager

The principal investigator/project manager is the proposing SBC-designated individual who provides the scientific and technical direction to a funding agreement-supported project.

Proprietary Information

Proprietary information is any information that a SBC considers to be non-public information the SBC owns and is marked accordingly.

Research Institution

Any organization located in the United States that is:

a. A university.
b. A nonprofit institution as defined in Section 4(5) of the Stevenson-Wydler Technology Innovation Act of 1980.
c. A contractor-operated federally funded R&D center, as identified by the National Science Foundation in accordance with the government-wide FAR issued in accordance with the Office of Federal Procurement Policy Act Section 35(c)(1). A list of eligible FFRDCs is here.

Research or Research and Development

Any activity that is:

a. A systematic, intensive study directed toward greater knowledge or understanding of the subject studied.
b. A systematic study directed specifically toward applying new knowledge to meet a recognized need; or
c. A systematic knowledge application toward the production of useful materials, devices, systems, or methods, including design, development, and improvement of prototypes and new processes to meet specific requirements.

Research Involving Animal Subjects

All activities involving animal subjects shall be conducted in accordance with DoDI 3216.01 "Use of Animals in DoD Programs," 9 C.F.R. parts 1-4 "Animal Welfare Regulations," National Academy of Sciences Publication "Guide for the Care & Use of Laboratory Animals," as amended, and the Department of Agriculture rules implementing the Animal Welfare Act (7 U.S.C. §§ 2131-2159), as well as other applicable federal and state law and regulation and DoD instructions.

"Animal use" protocols apply to all activities that meet any of the following criteria:

a. Any research, development, test, evaluation or training, (including experimentation) involving an animal or animals.
b. An animal is defined as any living or dead, vertebrate organism (non-human) that is being used or is intended for use in research, development, test, evaluation or training.
c. A vertebrate is a member of the subphylum Vertebrata (within the phylum Chordata), including birds and cold-blooded animals.

See DoDI 3216.01 for definitions of these terms and more information about the applicability of DoDI 3216.01 to work involving animals.

Research Involving Human Subjects

All research involving human subjects shall be conducted in accordance with 32 C.F.R. § 219 "The Common Rule," 10 U.S.C. § 980 "Limitation on Use of Humans as Experimental Subjects," and DoDI 3216.02 "Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research," as well as other applicable federal and state law and regulations, and DoD Services/Component guidance. Proposing SBCs must be cognizant of and abide by the additional restrictions and limitations imposed on the DoD regarding research involving human subjects, specifically as they regard vulnerable populations (DoDI 3216.02), recruitment of military research subjects (DoDI 3216.02), and informed consent and surrogate consent (10 U.S.C. § 980) and chemical and biological agent research (DoDI 3216.02). Food and Drug Administration regulation and policies may also apply.

"Human use" protocols apply to all research that meets any of the following criteria:

a. Any research involving an intervention or an interaction with a living person that would not be occurring or would be occurring in some other fashion but for this research.
b. Any research involving identifiable private information. This may include data/information/specimens collected originally from living individuals (broadcast video, web-use logs, tissue, blood, medical or personnel records, health data repositories, etc.) in which the identity of the subject is known, or the identity may be readily ascertained by the investigator or associated with the data/information/specimens.

See DoDI 3216.02 for definitions of these terms and more information about the applicability of DoDI 3216.02 to research involving human subjects.

Research Involving Recombinant DNA Molecules

Any recipient performing research involving recombinant DNA molecules and/or organisms and viruses containing recombinant DNA molecules shall comply with the National Institutes of Health Guidelines for Research Involving Recombinant DNA Molecules, dated January 2011, as amended. The guidelines can be found at: https://osp.od.nih.gov/wp-content/uploads/2016/05/NIH_Guidelines.pdf. Recombinant DNA is defined as (i) molecules that are constructed outside living cells by joining natural or synthetic DNA segments to DNA molecules that can replicate in living cells or (ii) molecules that result from the replication of those described in (i) above.

Service-Disabled Veteran-Owned Small Business (SDVOSB)

A service-disabled veteran or service-disabled veterans-owned and controlled SBC defined in Small Business Act 15 USC § 632(q)(2) and SBA's implementing SDVOSB regulations (13 CFR 125).

Small Business Concern (SBC)

A concern that meets the requirements set forth in 13 C.F.R. § 121.702 (available here).

An SBC must satisfy the following conditions on the date of award:

a. Is organized for profit, with a place of business located in the United States, which operates primarily within the United States, or which makes a significant contribution to the United States economy through payment of taxes or use of American products, materials or labor;
b. Is in the legal form of an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust or cooperative, except that if the concern is a joint venture, each entity to the venture must meet the requirements set forth in paragraph (c) below;
c. Is more than 50 percent directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States), other SBCs (each of which is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States), or any combination of these; and
d. Has, including its affiliates, not more than 500 employees. (See here for definition of an affiliate.)

Subcontract

A subcontract is any agreement, other than one involving an employer-employee relationship, including consultants, the funding agreement awardee enters calling for supplies or services for the performance of the original funding agreement.

Subcontractor

Subcontractor means any supplier, distributor, vendor, firm, academic institution, research center, or other person or entity that furnishes supplies or services pursuant to a subcontract, at any tier.

United States

United States means the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

Women-Owned Small Business Concern

An SBC where one or more women own at least 51 percent, or in the case of any publicly owned business, women own at least 51 percent of the stock, and women control the management and daily business operations.

Appendix C
POTENTIAL APPLICABLE FEDERAL ACQUISITION REGULATION, DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT CLAUSES

Note: Green cells are potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Blue cells are potential required FAR and DFARS clauses, when applicable.

Clause Number	Title	Date	When Applicable
52.203-17	Contractor Employee Whistleblower Rights	Nov-23	
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements	Jan-17	
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards	Jun-20	
52.204-13	System for Award Management Maintenance	Oct-18	
52.204-18	Commercial and Government Entity Code Maintenance	Aug-20	
52.204-19	Incorporation by Reference of Representations and Certifications	Dec-14	
52.204-21	Basic Safeguarding of Covered Contractor Information Systems	Nov-21	
52.204-23	Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab and Other Covered Entities	Dec-23	
52.204-25	Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment	Nov-21	
52.204-27	Prohibition on a ByteDance Covered Application	Jun-23	
52.204-30	Federal Acquisition Supply Chain Security Act Orders—Prohibition	Dec-23	
52.209-06	Protecting the Government's Interest When Subcontracting with Contractors Debarred, Suspended, or Proposed for Debarment	Nov-21	
52.209-10	Prohibition on Contracting with Inverted Domestic Corporations	Nov-15	
52.219-06	Notice of Total Small Business Set-Aside	Nov-20	
52.219-08	Utilization of Small Business Concerns	Feb-24	
52.219-28	Post-Award Small Business Program Representation	Feb-24	
52.222-03	Convict Labor	Jun-03	
52.222-19	Child Labor-Cooperation with Authorities and Remedies	Feb-24	
52.222-21	Prohibition of Segregated Facilities	Apr-15	
52.222-25	Affirmative Action Compliance	Apr-84	
52.222-26	Equal Opportunity	Sep-16	
52.222-36	Equal Opportunity for Workers with Disabilities	Jun-20	
52.222-50	Combating Trafficking in Persons	Nov-21	

Clause Number	Title	Date	When Applicable
52.225-01	Buy American-Supplies	Oct-22	
52.225-13	Restrictions on Certain Foreign Purchases	Feb-21	
52.226-07	Drug-Free Workplace	May-24	
52.226-08	Encouraging Contractor Policies to Ban Text Messaging While Driving	May-24	
52.227-01 Alt I	Authorization and Consent - Alternate I (Apr-84)	Jun-20	
52.227-02	Notice and Assistance Regarding Patent and Copyright Infringement	Jun-20	
52.227-11	Patent Rights-Ownership by the Contractor	May-14	
52.227-20	Rights in Data-SBIR Program	May-14	
52.232-11	Extras	Apr-84	
52.232-23	Assignment of Claims	May-14	
52.232-25	Prompt Payment	Jan-17	
52.232-33	Payment by Electronic Funds Transfer - System for Award Management	Oct-18	
52.232-39	Unenforceability of Unauthorized Obligations	Jun-13	
52.232-40	Providing Accelerated Payments to Small Business Subcontractors	Mar-23	
52.232-01	Disputes	May-14	
52.233-04	Applicable Law for Breach of Contract Claim	Oct-04	
52.242-15	Stop-Work Order	Aug-89	
52.243-01 Alt V	Changes-Fixed-Price Alternate V (Apr-84)	Aug-87	
52.244-06	Subcontracts for Commercial Products and Commercial Services	Feb-24	
52.246-09	Inspection of Research and Development (Short Form)	Apr-84	
52.252-02	Clauses Incorporated by Reference	Feb-98	
52.252-06	Authorized Deviations in Clauses	Nov-20	
52.253-01	Computer Generated Forms	Jan-91	
252.203-7000	Requirements Relating to Compensation of Former DoD Officials	Sep-11	
252.203-7002	Requirement to Inform Employees of Whistleblower Rights	Dec-22	
252.204-7000	Disclosure of Information	Oct-16	
252.204-7003	Control of Government Personnel Work Product	Apr-92	
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls	Oct-24	
252.204-7009	Limitations on the Use or Disclosure of Third-Party Contractor Reported Cyber Incident Information	Jan-23	
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting (DEVIATION 2024-O0013)	May-24	
252.204-7016	Covered Defense Telecommunications Equipment or Services—Representation	Dec-19	

Clause Number	Title	Date	When Applicable
252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services—Representation	May-21	
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services	Jan-23	
252.204-7019	Notice of NISTSP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7022	Expediting Contract Closeout	May-21	
252.204-7024	Notice on the use of the Supplier Performance Risk System	Mar-23	
252.227-7016	Rights in Bid or Proposal Information	Jan-23	
252.227-7018	Rights in Noncommercial Technical Data and Computer Software--Small Business Innovation Research (SBIR) Program (DEVIATION 2020-O0007) (Jul 23)	Nov-23	
252.227-7019	Validation of Asserted Restrictions--Computer Software	Jan-23	
252.227-7025	Limitations on the Use or Disclosure of Government-Furnished Information Marked with Restrictive Legends	Jan-23	
252.227-7030	Technical Data--Withholding of Payment	Mar-00	
252.227-7037	Validation of Restrictive Markings on Technical Data	Jan-23	
252.227-7039	Patents--Reporting of Subject Inventions	Apr-90	
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports	Dec-18	
252.232-7006	Wide Area WorkFlow Payment Instructions	Jan-23	
252.232-7010	Levies on Contract Payments	Dec-06	
252.235-7010	Acknowledgment of Support and Disclaimer	May-95	
252.235-7011	Final Scientific or Technical Report	Dec-19	
252.243-7001	Pricing of Contract Modifications	Dec-91	
252.244-7000	Subcontracts for Commercial Items	Nov-23	
52.203-03	Gratuities	Apr-84	Exceeding the simplified acquisition threshold.
52.203-05	Covenant Against Contingent Fees	May-14	Exceeding the simplified acquisition threshold.
52.203-06 or Alt I	Restrictions on Subcontractor Sales to the Government or ALT I	Nov-21	Exceeding the simplified acquisition threshold, Alt I commercial products or commercial services.
52.203-07	Anti-Kickback Procedures	Jun-20	Contracts exceeding $150,000.
52.203-08	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-12	Limitation on Payments to Influence Certain Federal Transactions	Jun-20	Contracts exceeding $150,000.
52.204-02	Security Requirements	Mar-21	May require access to classified information; cost contract (see 16.302) for research and development with an educational institution is contemplated.

Clause Number	Title	Date	When Applicable
52.212-04	Contract Terms and Conditions—Commercial Products and Commercial Services	Nov-23	Commercial products or commercial services.
52.212-05	Contract Terms and Conditions Required to Implement Statutes or Executive Orders-Commercial Items	May-24	Commercial products or commercial services.
52.219-14	Limitations on Subcontracting	Oct-22	Set aside for small business and the contract amount is expected to exceed the simplified acquisition threshold.
52.222-35	Equal Opportunity for Veterans	Jun-20	Contracts exceed $150,000.
52.222-37	Employment Reports on Veterans	Jun-20	If contract contains 52.222-35.
52.222-40	Notification of Employee Rights Under the National Labor Relations Act	Dec-10	Exceeding the simplified acquisition threshold.
52.222-54	Employment Eligibility Verification	May-22	Contracts exceed $150,000.
52.223-03	Hazardous Material Identification and Material Safety Data	Feb-21	Requires the delivery of hazardous materials.
52.229-03	Federal, State, and Local Taxes	Feb-13	A fixed-price contract is contemplated; and the contract is expected to exceed the simplified acquisition threshold.
52.232-02	Payment under Fixed-Price Research and Development Contracts	Apr-84	Fixed Price R&D.
52.233-03	Protest After Award	Aug-96	Exceed the simplified acquisition threshold.
52.242-13	Bankruptcy	Jul-95	Exceed the simplified acquisition threshold.
52.242-17	Government Delay of Work	Apr-84	Supplies other than commercial or modified-commercial products.
52.245-01	Government Property	Sep-21	When property is expected to be furnished.
52.245-09	Use and Charges	Apr-12	When the clause at 52.245-1 is included.
52.246-04	Inspection of Services-Fixed Price	Aug-96	Services, or supplies that involve the furnishing of services, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-16	Responsibility for Supplies	Apr-84	Supplies, services involving the furnishing of supplies, or research and development, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-23	Limitation of Liability	Feb-97	Exceed the simplified acquisition threshold.
52.247-34	F.o.b. Destination	Nov-91	When the delivery term is f.o.b. destination.
252.204-7015	Notice of Authorized Disclosure of Information for Litigation Support	Jan-23	Commercial products and commercial services.
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism	May-19	Contracts value of $150,000 or more.
252.211-7003	Item Identification and Valuation	Jan-23	For supplies, and for services involving the furnishing of supplies.
252.223-7001	Hazard Warning Labels	Dec-91	Requires submission of hazardous material data sheets.
252.223-7008	Prohibition of Hexavalent Chromium	Jan-23	For supplies, maintenance and repair services, or construction.
252.225-7001	Buy American and Balance of Payments Program	Feb-24	Acquisition of commercial products and commercial services.

Clause Number	Title	Date	When Applicable
252.225-7002	Qualifying Country Sources as Subcontractors	Mar-22	(i) 252.225-7001, Buy American and Balance of Payments Program. Use if one or more is included: (ii) 252.225-7021, Trade Agreements. (iii) 252.225-7036, Buy American - Free Trade Agreements - Balance of Payments Program.
252.225-7012	Preference for Certain Domestic Commodities	Apr-22	Commercial products and commercial services.
252.225-7052	Restriction on Acquisition of Certain Magnets, Tantalum, and Tungsten	May-24	Products and commercial services, that exceed the simplified acquisition threshold.
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime	Jan-23	Commercial products and commercial services.
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region	Jun-23	Products utilizing funds appropriated or otherwise made available for any fiscal year.
252.225-7972	Prohibition on the Procurement of Foreign-Made Unmanned Aircraft Systems (DEVIATION 2020-O0015)	May-20	
252.225-7967	Prohibition Regarding Russian Fossil Fuel Business Operations (DEVIATION 2024-O0006, Revision 1)	Feb-24	Exceeding the simplified acquisition threshold.
252.228-7001	Ground and Flight Risk	Mar-23	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.228-7005	Mishap Reporting and Investigation Involving Aircraft, Missiles, and Space Launch Vehicles	Nov-19	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.235-7002	Animal Welfare	Dec-14	Uses live vertebrate animals.
252.235-7004	Protection of Human Subjects	Jul-09	Involving human subjects.
252.243-7002	Requests for Equitable Adjustment	Dec-22	Exceeding the simplified acquisition threshold.
252.245-7003	Contractor Property Management System Administration	Apr-12	Containing the clause at FAR 52.245-1, Government Property.
252.245-7005	Management and Reporting of Government Property	Jan-24	Containing the clause at FAR 52.245-1, Government Property.
252.247-7023	Transportation of Supplies by Sea	Jan-23	Except - those with an anticipated value at or below the simplified acquisition threshold.